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Lawful Interception (LI) implementation guidance;
LI for IMS based STIR/SHAKEN
(3GPP TR 33.929-2 version 19.0.0 Release 19)



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Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

The present document is part of a multi-part TR as described below:

- Part 1: LI for IMS based services. See TR 33.929-1 [10].
- Part 2: LI for IMS based STIR/SHAKEN (present document).
- Part 3: LI for messaging services. See TR 33.929-3 [11].
- Part 4: LI for data in 5G core. See TR 33.929-4 [12].
- Part 5: LI for ID association caching. See TR 33.929-5 [13].
- Part 6: LI for IMS based RCS. See TR 33.929-6 [14].
- Part 7: LI location acquisition capabilities. See TR 33.929-7 [15].
- Part 8: LI for MMS. See TR 33.929-8 [16].
- Part 9: LI for PTC. See TR 33.929-9 [17].

In Part 1, the illustrations of LI for IMS-based services focus on the LI aspects of IMS sessions which include the architecture topologies and the call flows covering basic sessions, redirected sessions, target non-local ID, conferencing, roaming (local break-out and home-routed roaming).

In Part 2 (present document), the illustrations of LI for IMS based STIR/SHAKEN focus on various STIR/SHAKEN related LI reporting scenarios.

In Part 3, the illustrations of LI for messaging services focus on SMS over NAS and SMS over IP.

In Part 4, the illustrations of LI for data in 5G core focus on the LI aspects of data interception focussing on the PDU session related events including the handover scenarios.

In Part 5, the illustrations of LI for ID association caching focus on the capabilities used to provide the temporary identity to permanent identity (and vice-versa) associations known to the CSP network. The illustrations include the architectural concepts and few examples of use-cases encountering various timing scenarios.

In Part 6, the illustrations of LI for IMS based RCS focus on the architecture topologies and the call flows when the RCS service is offered by the CSP or by a Third Party Provider.

In Part 7, the illustrations of LI location acquisition capabilities focus on the conceptual overview and the flow diagrams for location reporting, LALS and location acquisition.

In Part 8, the illustrations of LI for MMS focus on the conceptual overview and the flow diagrams for MMS.

In Part 9, the illustrations of LI or PTC focus on the conceptual overview and the flow diagrams for MCPTT and PoC, the two commonly referred to as Push to Talk over Cellular (PTC).

Introduction

Unlike the previous generation of LI technical specifications, the latest LI technical specifications (TS 33.126 [2], TS 33.127 [3], TS 33.128 [4]) contain only the normative part of the requirements. However, the implementers may need additional information such as architectural diagrams, conceptual scenarios, flow-diagrams and additional background information to better understand the requirements in those specifications. The present document collects the relevant informative annexes from the previous generation of LI technical specifications and then expands them to include the similar level of details as for latest technologies such as 5G.

1 Scope

The present document provides architectural diagrams, conceptual scenarios, flow-diagrams, examples, and other background information which can be useful to implement the LI functions defined in TS 33.126 [2], TS 33.127 [3] and TS 33.128 [4].

The present document covers the LI for IMS based STIR/SHAKEN that focus on illustrating various STIR/SHAKEN related LI reporting scenarios that go along with the LI for IMS based services (TR 33.929-1 [10]).

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
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[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 33.126: "Lawful Interception requirements".

[3] 3GPP TS 33.127: "Lawful Interception (LI) architecture and functions".

[4] 3GPP TS 33.128: "Protocol and procedures for Lawful Interception (LI)".

[5] 3GPP TS 24.147: "Conferencing using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3".

[6] 3GPP TS 23.501: "System architecture for the 5G System (5GS)".

[7] 3GPP TS 24.174: "Support of multi-device and multi-identity in the IP Multimedia Subsystem (IMS); Stage 3".

[8] 3GPP TS 23.228: "IP Multimedia Subsystem (IMS), Stage 2".

[9] 3GPP TR 33.928: "ADMF logic for provisioning Lawful Interception (LI)".

[10] 3GPP TR 33.929-1: "Lawful Interception (LI) implementation guidance; LI for IMS-based services".

[11] 3GPP TR 33.929-3: "Lawful Interception (LI) implementation guidance; LI for messaging services".

[12] 3GPP TR 33.929-4: "Lawful Interception (LI) implementation guidance; LI for data in 5G core".

[13] 3GPP TR 33.929-5: "Lawful Interception (LI) implementation guidance; LI for ID association caching ".

[14] 3GPP TR 33.929-6: "Lawful Interception (LI) implementation guidance; LI for IMS based RCS".

[15] 3GPP TR 33.929-7: "Lawful Interception (LI) implementation guidance; LI location acquisition capabilities".

[16] 3GPP TR 33.929-8: "Lawful Interception (LI) implementation guidance; LI for MMS".

[17] 3GPP TR 33.929-9: "Lawful Interception (LI) implementation guidance; LI for PTC".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Symbols

For the purposes of the present document, the following symbols apply:

None.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

3GPP	3rd Generation Partnership Program
ADMF	Administration Function
AS	Application Server
CC	Content of Communication
CSCF	Call Session Control Function
CSP	Communication Service Provider
eCNAM	Enhanced Calling NAMe
E-CSCF	Emergency Call Session Control Function
HPLMN	Home PLMN
IBCF	Interrogating Border Control Function
I-CSCF	Interrogating CSCF
ID	Identity
IMS	IP Multimedia Subsystem
IRI	Intercept Related Information
LALS	Lawful Access Location Services
LBO	Local Break Out
LI	Lawful Interception
LMISF	LI Mirror IMS State Function
LMISF-IRI	LMISF for the handling of IRI
MCPTT	Mission Critical Push To Talk
NAS	Non-Access Stratum
PASSporT	Personal ASSertion Token
P-CSCF	Proxy Call Session Control Function
PLMN	Public Land Mobile Network
PoC	Push to talk over Cellular
POI	Point Of Interception
PTC	Push to Talk over Cellular
RCD	Rich Call Data
S-CSCF	Serving Call Session Control Function
SHAKEN	Signature-based Handling of Asserted information using toKENs
SMS	Short Message Service
SIP	Session Initiation Protocol
STIR	Secure Telephony Identity Revisited
VPLMN	Visited PLMN

4 STIR/SHAKEN related LI reporting

4.1 General

The subsequent clauses illustrate the STIR/SHAKEN related LI reporting as a part of IMS LI. While both IRI and CC apply for an IMS LI, the STIR/SHAKEN related LI reporting includes just the IRI. The call of target non-local ID applies for STIR/SHAKEN related LI reporting as well.

As defined in TS 33.127 [3], TS 33.128 [4] and illustrated in TR 33.928 [9], the following NFs participate in providing the IRI-POI functions for STIR/SHAKEN related LI reporting:- Telephony AS.

- IBCF.
- P-CSCF.
- LMISF-IRI.

The factors that influence the determination of a NF that provides the IRI-POI functions are illustrated in clause 4.3.

As defined in TS 33.128[4] and illustrated in TR 33.928 [9], reporting of "div" PASSporT information of redirecting party(ies) when the IMS session is redirected later on the signaling path is determined by the parameter ReportDiversionPASSporTInfo provisioned into the IRI-POIs present in the Telephony AS or the IBCF. When the parameter is set to "true", "div" PASSporTs are reported. When parameter is absent or set to "false", "div" PASSporTs are not reported unless the target is a redirecting party whose identity is signed.

4.2 Background

4.2.1 Overview

When STIR/SHAKEN is supported, the caller identity is authenticated and signed by a Signing AS at the originating end of the call, the authenticated/signed identities are sent to the terminating end of the call in the form of "shaken" PASSporT or "div" PASSporT, the latter only when the call is diverted due to call forwarding. The "shaken" and "div" PASSporTs include the attestation of the caller identity ("total trust", "partial trust", or "no trust") and the time of the call. The caller identity may also include signed Rich Call Data (RCD).

At the terminating end of the call, the received "shaken" and "div" PASSporTs (if present) are verified (by a Verification AS), and when the verification result is good, the call is offered to the terminating party with the Validation Result. The RCD information or enhanced calling name information (eCNAM) may also be delivered to the terminating party.

When the verification fails, the call is not offered to the terminating party.

4.2.2 Signing and Verification

4.2.2.1 General

Either the Telephone AS or the IBCF would interact with the Signing AS and Verification AS. The conditions that determine which of the two would interact is illustrated in clause 4.2.3.

4.2.2.2 Calls without the redirection - "shaken" PASSporT

The originating party identity is included in the origination claim of the "shaken" PASSporT and the called party number is included in the destination claim of the "shaken" PASSporT.

Figure 4.2.2.2-1 illustrates interaction with the Signing AS in the originating network.

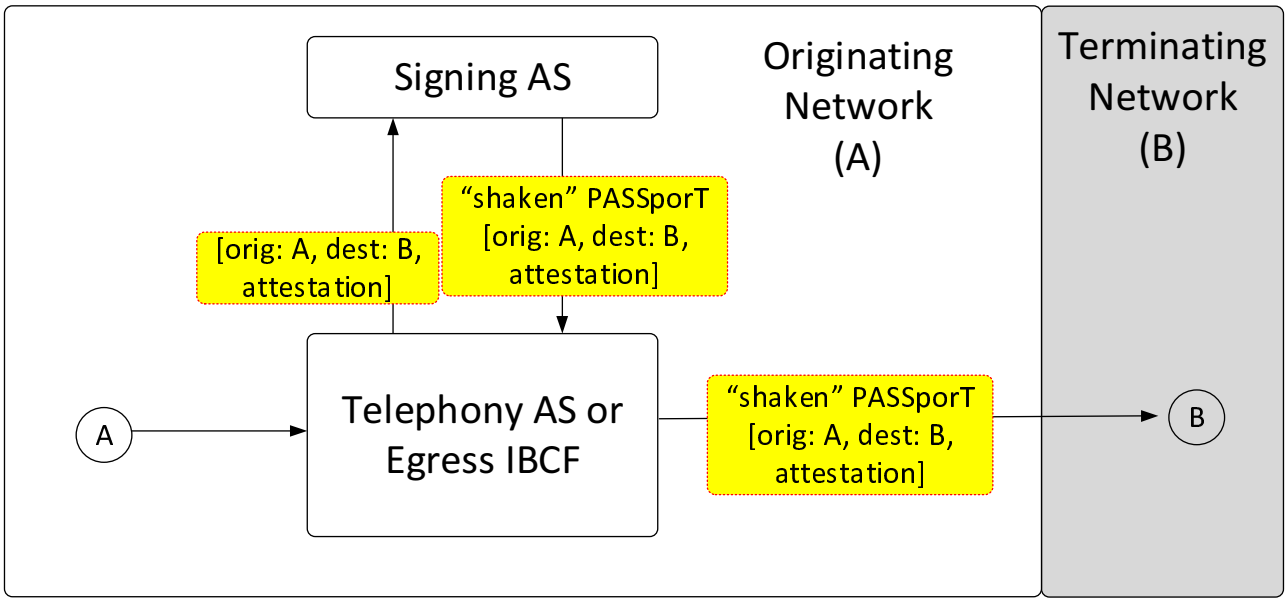


Figure 4.2.2.2-1: Originating party number signature in the "shaken" PASSsporT

In the illustration shown in figure 4.2.2.2-1, A is the originating party and B is the called party. As inferred, either the Telephony AS or the IBCF from the IMS domain in the originating network would interact with the Signing AS.

4.2.2.3 Calls with the redirection - "div" PASSporT

When an incoming call is redirected, the redirecting party information is also signed by the Signing AS, and the signed information of the redirecting party is sent toward the destination of that call as "div" PASSporT.

The "div" PASSporT includes the originating party number in the origination claim, the redirecting party number in the diversion claim and the destination number in the destination claim.

Figure 4.2.2.3-1 illustrates interaction with the Signing AS in the intermediate network.

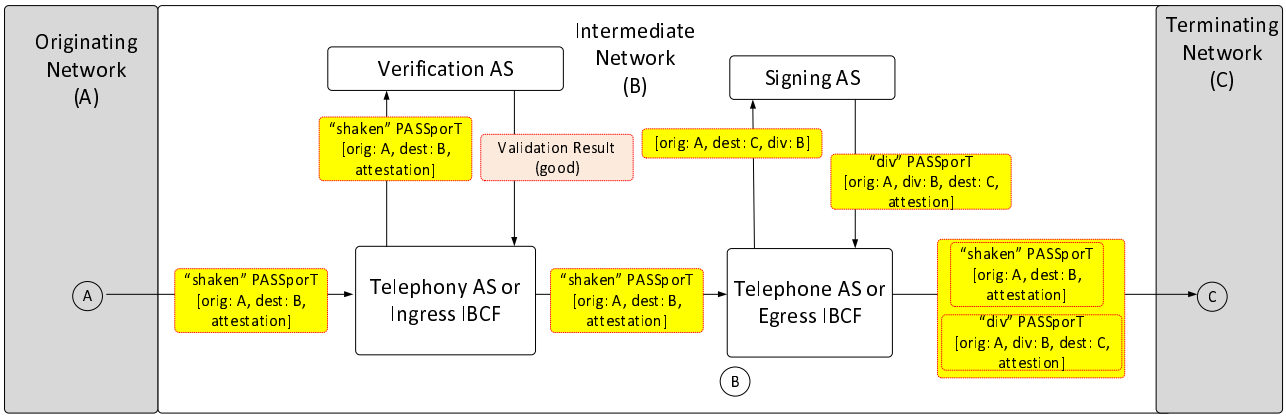


Figure 4.2.2.3-1: Redirecting party number signature in the "div" PASSporT

In the illustration shown in figure 4.2.2.3-1, A is the originating party and B is the redirecting party and C is the terminating party. As inferred, either the Telephony AS or the IBCF from the IMS domain of the intermediate network would interact with the Verification AS and the Signing AS. Note that the call redirection may also happen in the originating network or in the terminating network.

4.2.2.4 Terminating end of the call

At the termination end of the call, the "shaken" PASSporT and the "div" PASSporT (if present) are validated by a Verification AS and if the validation result is good, the call is offered to the terminating party.

Figure 4.2.2.4-1 illustrates interaction with the Verification AS in the terminating network.

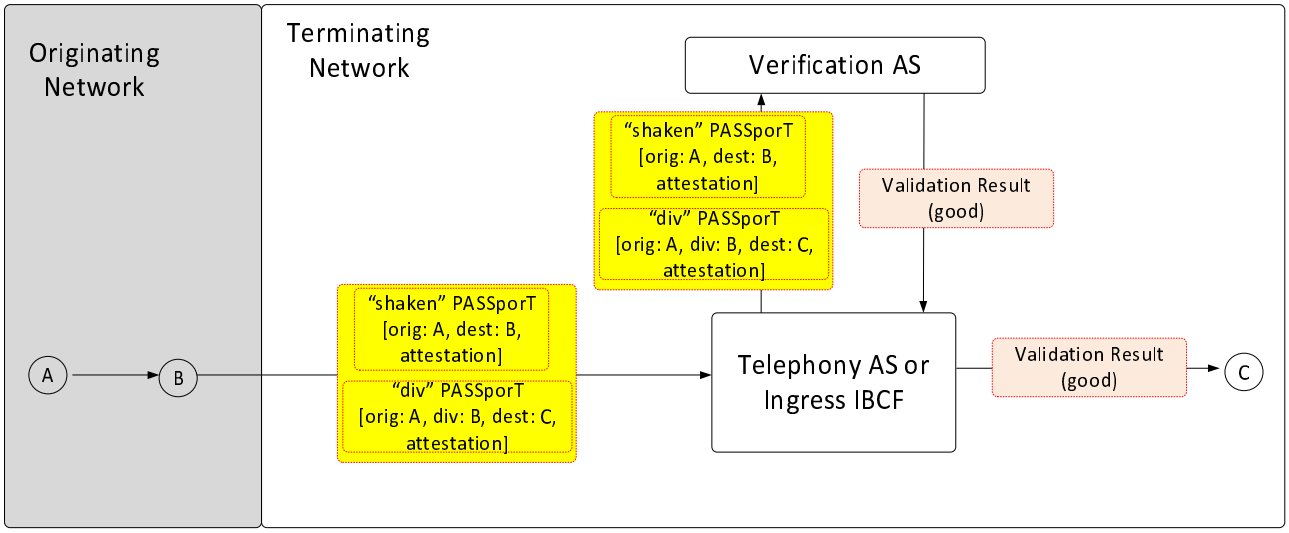


Figure 4.2.2.4-1: Verification of "shaken" PASSporT and "div" PASSporT in the terminating network

In the illustration shown in figure 4.2.2.4-1, A is the originating party and B is the redirecting party (both in the originating network) and C is the terminating party. As inferred, either the Telephony AS or the IBCF from the IMS domain of the terminating network would interact with the Verification AS.

When the validation result shows good, the result is sent to the terminating party C (in figure 4.2.2.4-1) when the call is offered. The call offer may also provide RCD information and (or) enhanced calling name information (eCNAM).

The Telephony AS or IBCF that interacts with the Verification AS would forward the PASSporTs with the attestation to the next hop (which happens to be the S-CSCF). The P-CSCF removes the PASSporTs before offering the call to the terminating party. This is illustrated in figure 4.2.2.4-2.

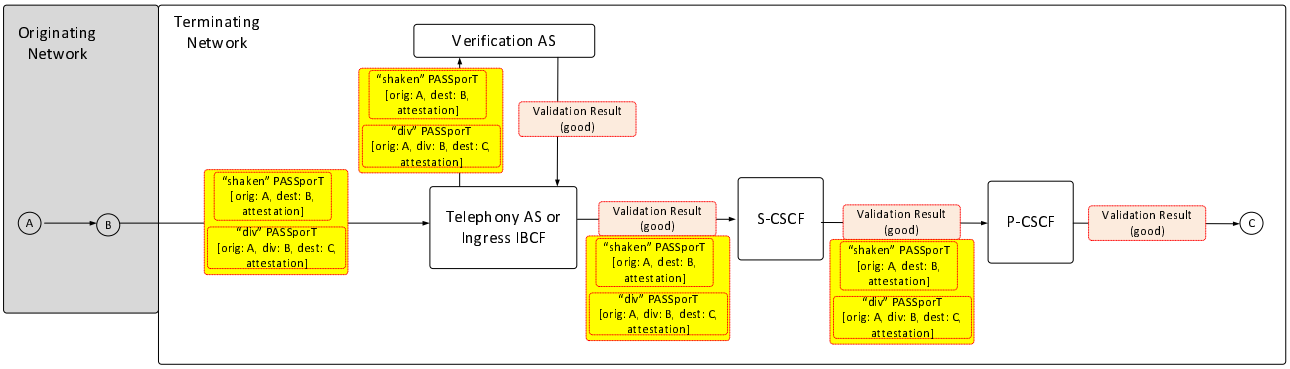
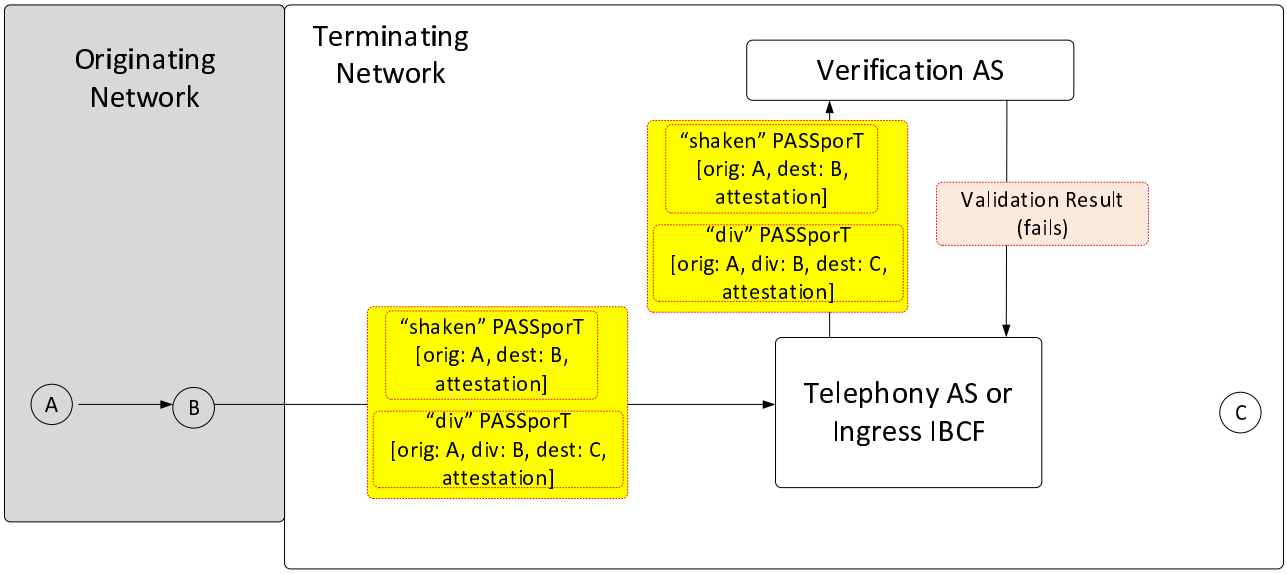


Figure 4.2.2.4-2: PASSporTs are removed by the P-CSCF before the call offer

In the illustration shown in figure 4.2.2.4-2, A is the originating party and B is the redirecting party (both in the originating network) and C is the terminating party. As inferred, either the Telephony AS or the IBCF from the IMS domain of the terminating network would interact with the Verification AS. The P-CSCF removes the PASSporTs before the call is offered to the C. The Validation Result is forwarded to C. Note that C can also be outbound roaming in which case, Validation Result is sent along with the call offer to the VPLMN. With LBO, the P-CSCF in the VPLMN would remove the PASSporTs.

When the verification fails, the call is not offered to the terminating party (see figure 4.2.2.4-3).



4.2.2.4-3: Validation fails

In the illustration shown in figure 4.2.2.4-3, A is the originating party and B is the redirecting party (both in the originating network) and C is the terminating party. As inferred, either the Telephony AS or the IBCF from the IMS domain of the terminating network would interact with the Verification AS. In this illustration, the validation fails and therefore, the call is not offered to the terminating party C.

4.2.3 IMS NFs that interact with the Signing AS and Verification AS

4.2.3.1 General

The Telephony AS or Egress IBCF present on the path of an IMS session would interact with the Signing AS. Likewise, the Ingress IBCF or the Telephony AS would interact with the Verification AS.

The following sub-clauses illustrate the conditions that determine which of those would interact with the Signing AS and Verification AS.

4.2.3.2 Signing AS

For an intra-CSP IMS session, the caller identity (i.e. originating party) may or may not be signed depending on the CSP deployment option.

When the signing of caller identity for intra-CSP session is required or when the Rich Call Data (RCD) is included as a part of calling party information, the Telephony AS would interact with the Signing AS.

When the signing of caller identity for an intra-CSP session is not required, or the RCD is not present, based on CSP choice, either the Telephony AS or the Egress IBCF would interact with the Signing AS. For emergency session, IBCF would interact with the Signing AS.

Figure 4.2.3.2-1 summarizes the above points by illustrating the algorithm that determines the IMS NF that would interact with a Signing AS.

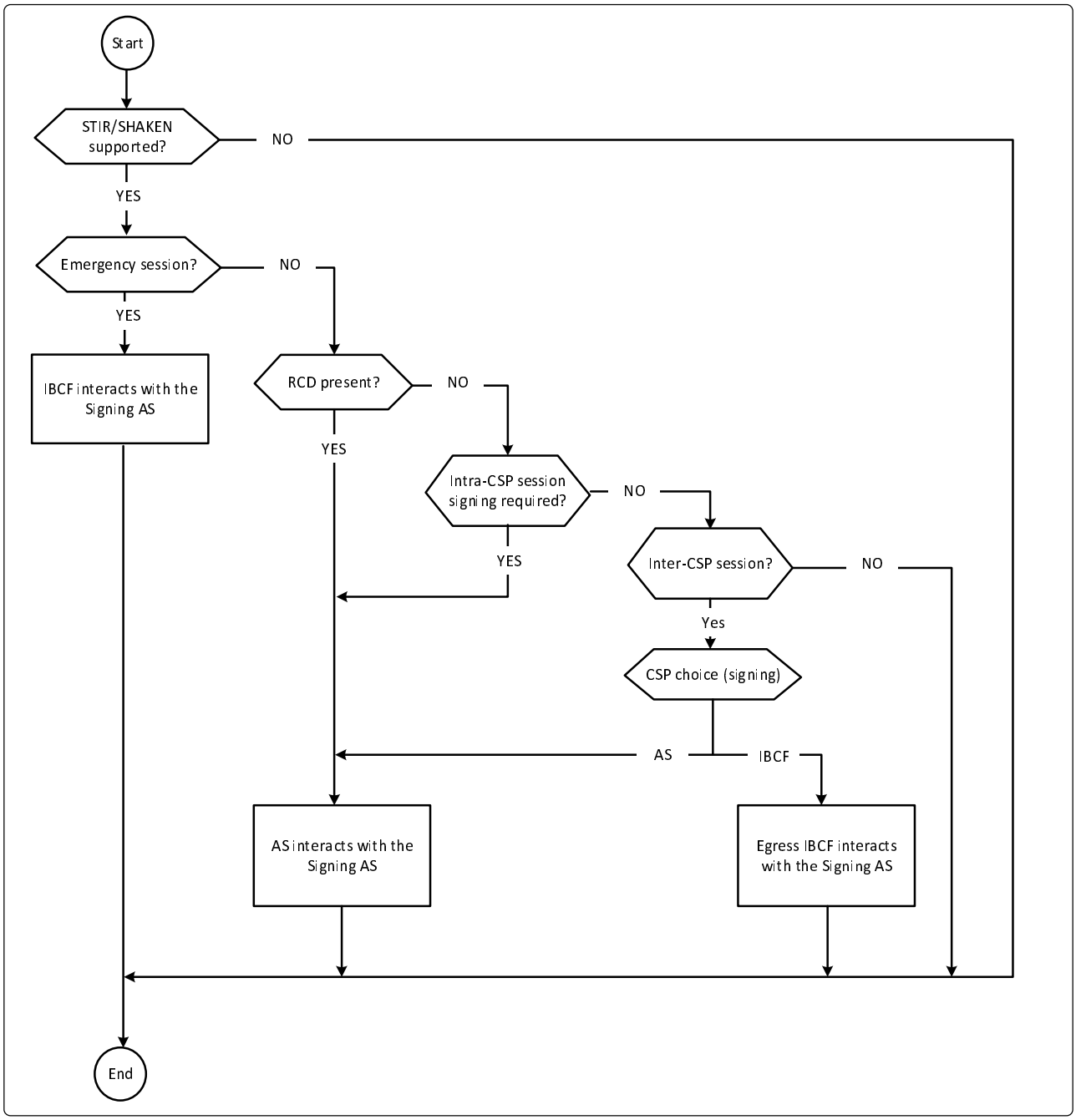


Figure 4.2.3.2-1: Algorithm that determines the IMS NF that interacts with the Signing AS

In figure 4.2.3.2-1, the AS represents the Telephony AS. When the signing of intra-CSP IMS session is not required with the CSP choice for interacting with the Signing AS is Telephony AS, the signing is to be done only for inter-CSP IMS session. However, with a such a choice, the Telephony AS may still interact with the Signing AS to sign the caller identity if it cannot determine the whereabouts of the terminating party.

The verification of IMS intra-CSP session is also done when the signing of intra-CSP session is done.

4.2.3.3 Verification AS

If an intra-CSP IMS session was signed, then the verification would also apply to an intra-CSP IMS session and in this case, the Telephone AS would interact with the Verification AS.

When the intra-CSP IMS session is not signed, the verification will not apply for an intra-CSP IMS session. In other words, when the verification is done only for an inter-CSP IMS session, either the Telephony AS or the Ingress IBCF

would interact with the Verification AS based on the CSP deployment choice. For an emergency callback, when required, the Ingress IBCF or Telephony AS would interact with the Verification AS depending on CSP choice.

Figure 4.2.3.3-1 summarizes the above points by illustrating the algorithm that determines the IMS NF that would interact with a Verification AS.

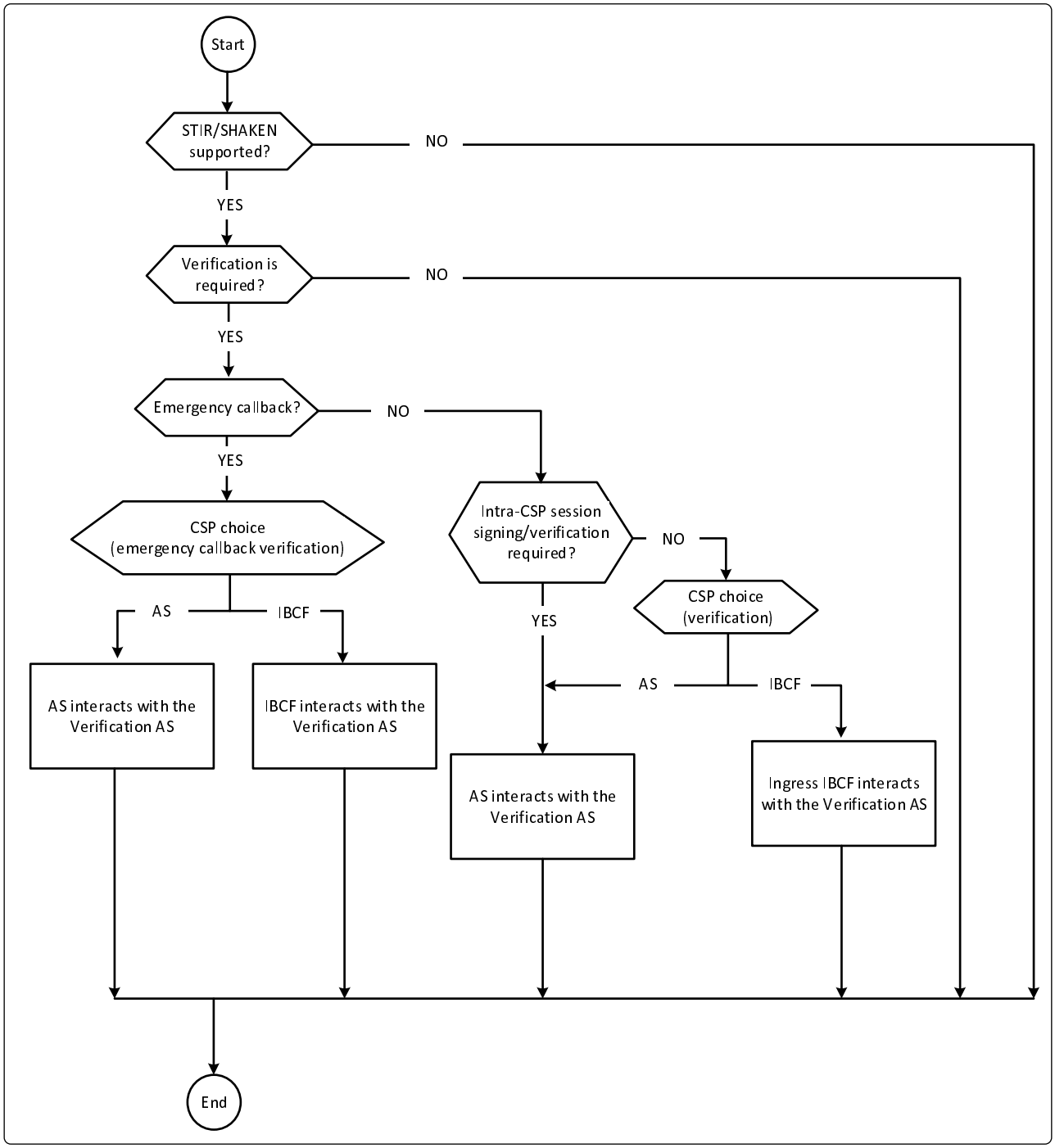


Figure 4.2.3.3-1: Algorithm that determines the IMS NF that interacts with the Verification AS

In figure 4.2.3.3-1, the AS represents the Telephony AS. Even when the signing/verification of intra-CSP IMS session is not required, and the CSP choice is IBCF for verification, the Telephony AS may still interact with the Verification AS if the incoming SIP INVITE includes the PASSporTs. This may happen when an incoming call is redirected and the Telephony AS happens to interact with the Signing AS per the illustrations of clause 4.2.3.2, and the call is terminated locally.

Basically, the Telephony AS and IBCF would use the following logic to determine whether or not to interact with the Verification AS.

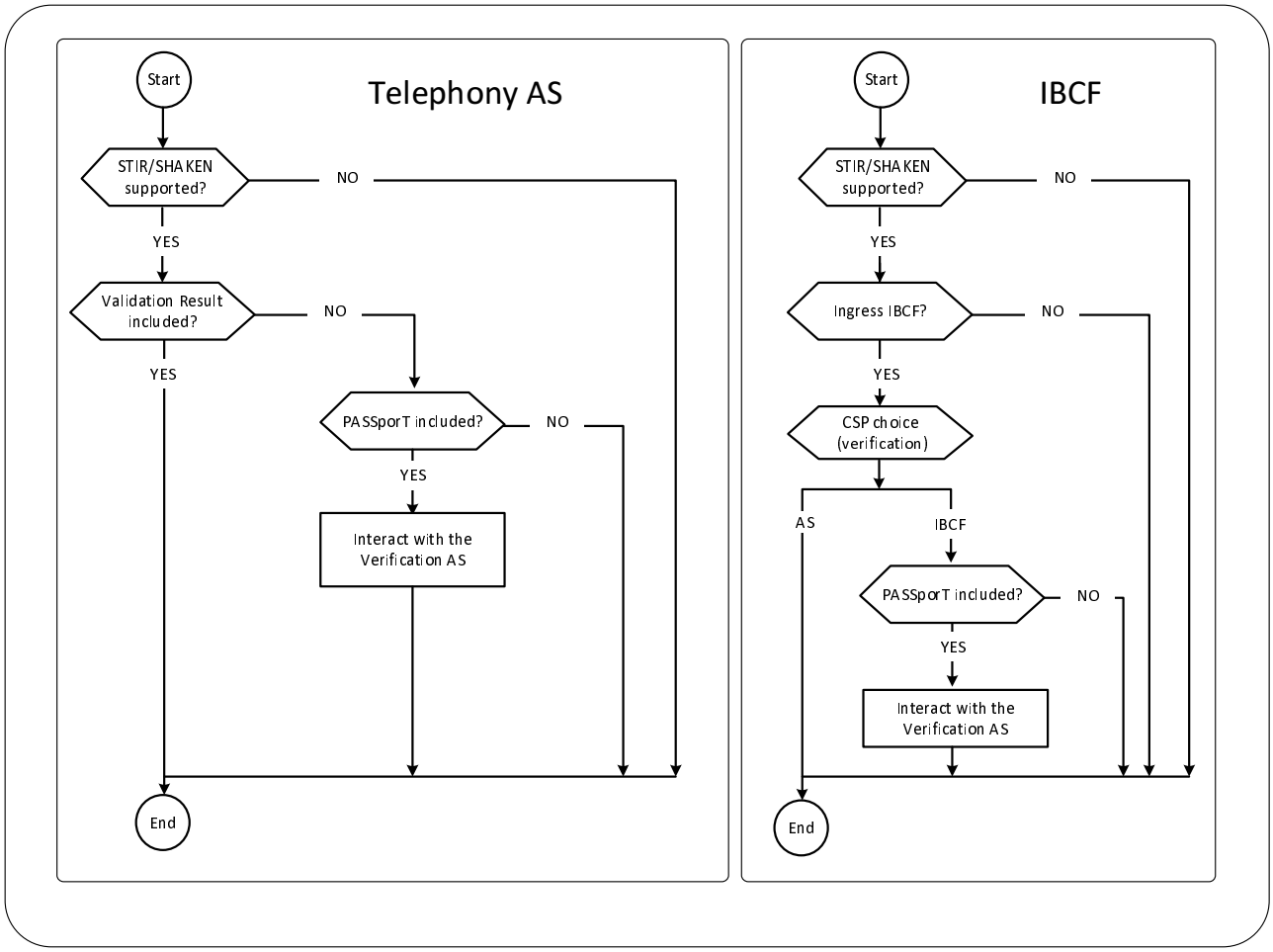


Figure 4.2.3.3-2: Algorithm that Telephony AS and IBCF use to determine for Verification AS interaction

If the Validation Result is present in the SIP INVITE, then there is no need for the Telephony AS to interact with the Verification AS again. If the PASSporT is included without a Validation Result, then the Telephony AS will always interact with the Verification AS to validate the signatures.

Only Ingress IBCF would interact with the Verification AS.

4.3 IMS LI and STIR/SHAKEN LI

4.3.1 General

The IMS NFs that provide the IRI-POI functions for IMS session related LI reporting can be different from the IMS NFs that provide the IRI-POI functions for the STIR/SHAKEN related LI reporting.

A comparison of the two is illustrated in the following clauses for various use-cases observing two deployment choices for IMS LI (default and alternate). In a basic call scenario, Party A (denoted as A) calls Party B (denoted as B) in the subsequent clauses. In a redirected call scenario, Party A (denoted as A) calls Party B (denoted as B) redirected to Party C (denoted as C) in the subsequent clauses.

4.3.2 Originating IMS network – IBCF interacts with the Signing AS

4.3.2.1 General

In the illustrations shown in this clause, the Egress IBCF interacts with the Signing AS. Clause 4.2.3.2 illustrates the conditions under which the IBCF would interact with the Signing AS. In brief, IBCF would interact with the Signing

AS for inter-CSP IMS sessions when signing is not required for intra-CSP IMS sessions and the CSP choice is IBCF for interactions with the Signing AS and when no other conditions demand the Telephony AS to interact with the Signing AS.

As a general concept, the IMS NF that interacts with the Signing AS provides the STIR/SHAKEN related LI reporting of the PASSporTs (i.e. the signature of the calling or redirecting party identities).

4.3.2.2 Simple scenarios

Scenario 1: A calls B, A is the target

A (originating network) calls B (terminating network). A is the target in the originating network.

The S-CSCF provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

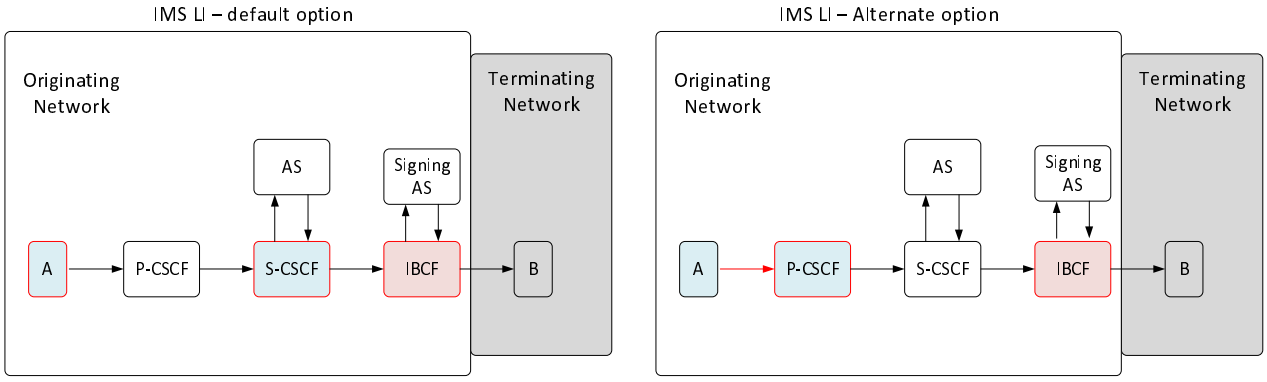


Figure 4.3.2.2-1: Scenario 1 - A (target) calls B

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the Egress IBCF that interacts with the Signing AS to authenticate/sign A's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target A.

Scenario 2: A calls B, B is the target non-local ID

A (originating network) calls B (terminating network). This is an inter-CSP IMS session. B is a target non-local ID in the originating network.

The IBCF provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

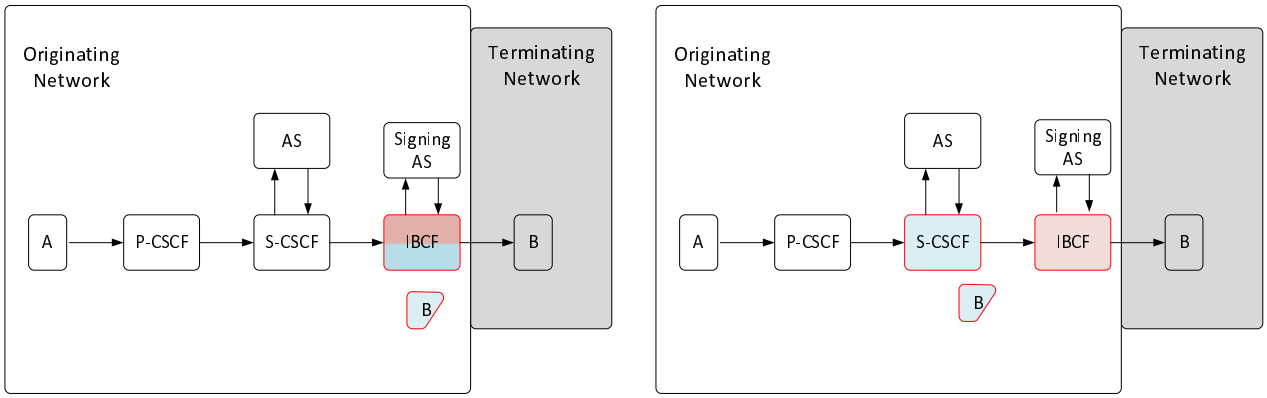


Figure 4.3.2.2-2: Scenario 2 – A calls B (target non-local ID)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the Egress IBCF that interacts with the Signing AS to authenticate/sign A's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target non-local ID B.

4.3.2.3 Complex scenarios

Scenario 1: A (target) calls B, redirected to C (terminating network)

A calls B in the originating network, and B has call forwarding to C in the terminating network. A is the target.

The S-CSCF (A) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (A) provides the IRI-POI functions for IMS LI with the alternate option.

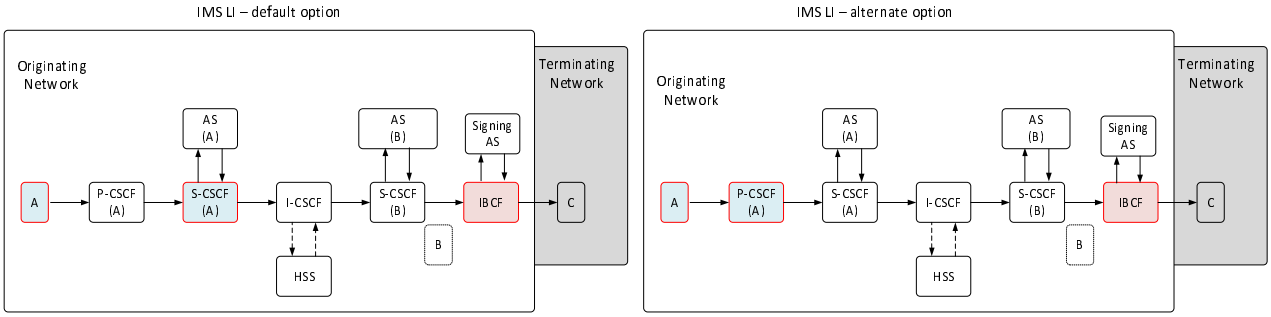


Figure 4.3.2.3-1: Scenario 1 - redirected call – A (target) calls B redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the Egress IBCF that interacts with the Signing AS to authenticate/sign A's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target A.

Scenario 2: A calls B (target), redirected to C (terminating network)

A calls B in the originating network, and B has call forwarding to C in the terminating network. B is the target.

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and IBCF provides the IRI-POI functions for IMS LI with the alternate option.

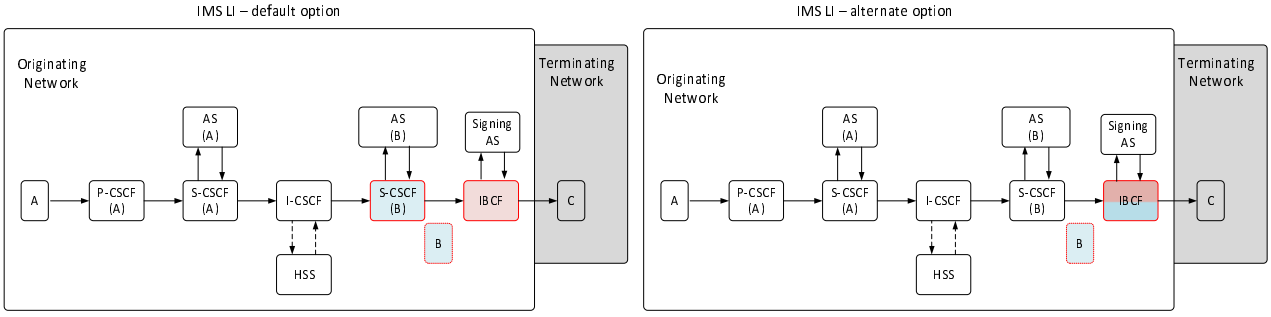


Figure 4.3.2.3-2: Scenario 2 – Redirected call – A calls B (target) redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the Egress IBCF that interacts with the Signing AS to authenticate/sign target's identity (i.e. B's identity) provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target B.

Scenario 3: A calls B redirected to C (target non-local ID)

A calls B in the originating network, and B has call forwarding to C in the terminating network. C is the target non-local ID in the originating network.

The IBCF provides the IRI-POI functions for IMS LI with the default option, and S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

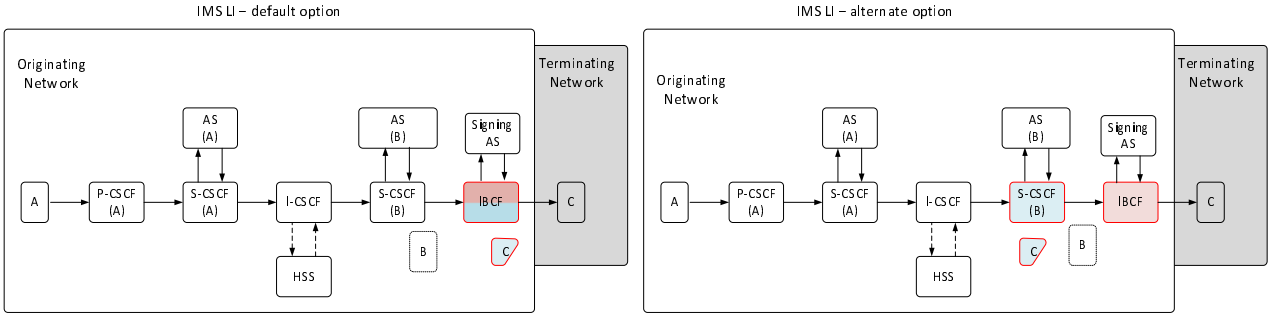


Figure 4.3.2.3-3: Scenario 3 – Redirected call – A calls B redirected to C (target non-local ID)

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the Egress IBCF that interacts with the Signing AS to authenticate/sign A's identity and B's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target non-local ID C.

Scenario 4 : A (target) calls B (target), redirected to C (target non-local ID)

A calls B in the originating network, and B has call forwarding to C in the terminating network. A and B are the targets and C is a target non-local ID in the originating network.

For target A, the S-CSCF (A) provides the IRI-POI functions for IMS LI with the default option, and P-CSCF (A) provides the IMS-LI with the alternate option.

For target B, the S-CSCF (B) provides the IRI-POI functions for IMS-LI with the default option, and IBCF provides the IRI-POI functions with the alternate option.

For target non-local ID C, IBCF provides the IRI-POI functions for IMS LI with default option and S-CSCF (B) provides the IRI-POI functions for IMS-LI with the alternate option. with the alternate option.

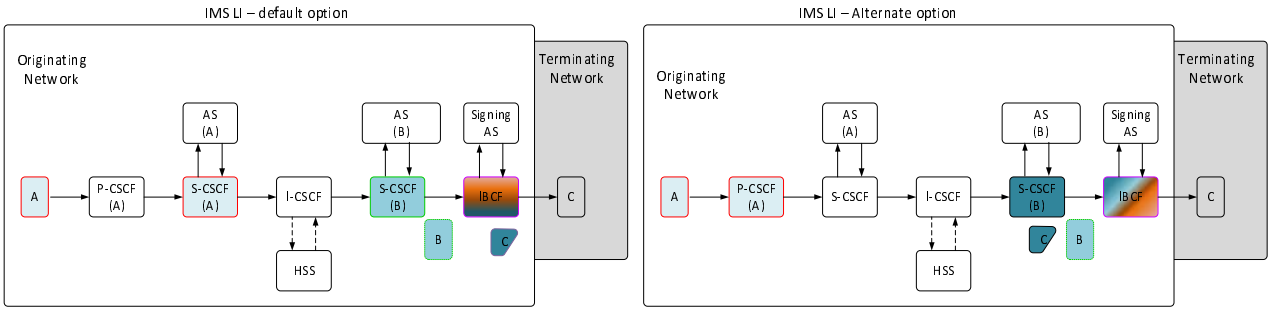


Figure 4.3.2.3-4: Scenario 4 – Redirected call – A (target) calls B (target) redirected to C (target non-local ID)

The left part of diagram shows for IMS LI with default option (S-CSCF (A) for A, S-CSCF (B) for B and IBCF for C) and the right part of the diagram shows for IMS LI with alternate option (P-CSCF (A) for A, IBCF for B and S-CSCF (B) for C). Note that default and alternate options for target non-local ID are mutually independent of the default, alternate options for targets that are not non-local ID.

In all cases, the Egress IBCF that interacts with the Signing AS to authenticate/sign target's identities (i.e. A's identity and B's identity) provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for target A, target B and target non-local ID C.

Intra-CSP IMS session

Scenario 5: Intra-CSP IMS session, A calls B, A is the target

The S-CSCF provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

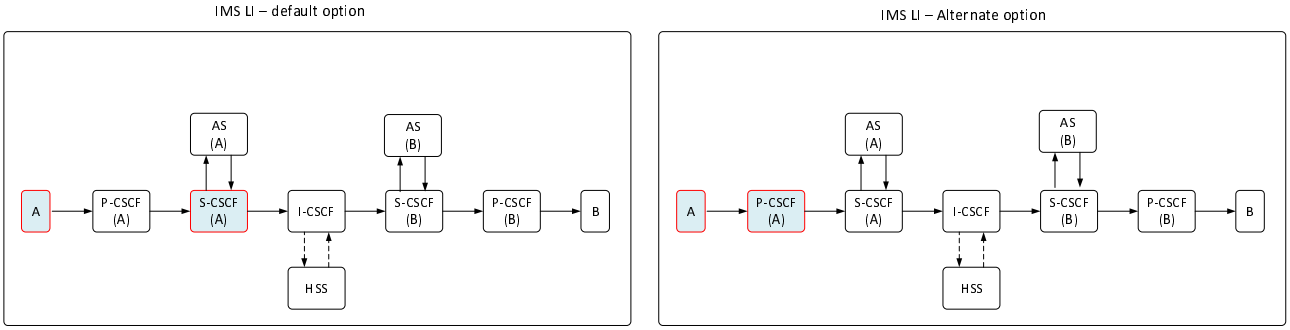


Figure 4.3.2.3-5: Scenario 5 – intra-CSP IMS session, A (target) calls B

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

From STIR/SHAKEN perspective, no STIR/SHAKEN related LI reporting applies to this scenario since signing of A's identity is not performed. As a matter of fact, IBCF is not on the signalling path for this scenario.

Note that IBCF would interact with the Signing AS only when signing of intra-CSP IMS session is not required and the CSP choice is IBCF for interacting with the Signing AS.

4.3.3 Terminating IMS network – IBCF interacts with the Verification AS

4.3.3.1 General

In the illustrations shown in this clause, the Ingress IBCF interacts with the Verification AS. Clause 4.2.3.3 illustrates the conditions under which the IBCF would interact with the Verification AS. In brief, IBCF would interact with the Verification AS for inter-CSP IMS sessions when intra-CSP IMS signing/verification is not required and CSP choice is IBCF for interaction with the Verification AS.

As a general concept, the IMS NF that interacts with the Verification AS provides the STIR/SHAKEN related LI reporting of the Validation Results (i.e. for verification). However, there are scenarios where an IMS NF may provide the STIR/SHAKEN related LI reporting of the Validation Result even if that IMS NF does not interact with the Verification AS (e.g. redirecting scenarios, inbound roaming).

4.3.3.2 Simple scenarios

Scenario 1: A (originating network) calls B (target)

A (originating network) calls B (terminating network). B is the target in the terminating network.

The S-CSCF provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

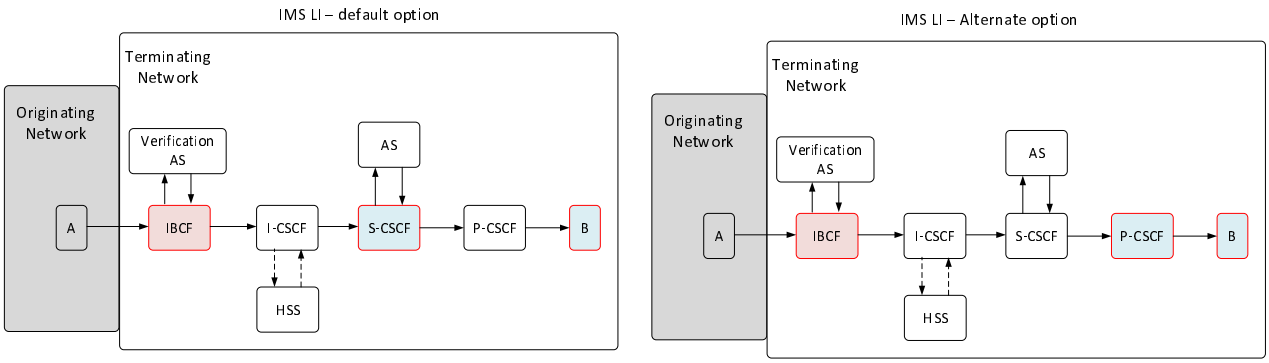


Figure 4.3.3.2-1: Scenario 1 - A calls B (target)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the Ingress IBCF that interacts with the Verification AS to validate the received PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B.

Scenario 2: A (originating network) calls B. A is the target non-local ID

A (originating network) calls B (terminating network). This is an inter-CSP IMS session. A is a target non-local ID in the originating network.

The IBCF provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

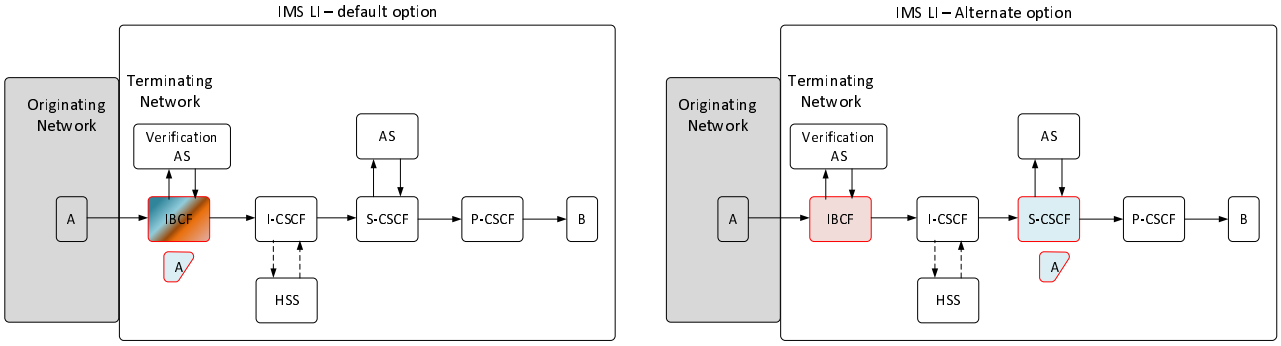


Figure 4.3.3.2-2: Scenario 2 – A (target non-local ID) calls B

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the Ingress IBCF that interacts with the Verification AS to validate the received PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target non-local ID A.

4.3.3.3 Complex scenarios

Redirected call scenarios

Scenario 1: A (originating network) calls B (target), redirected to C

A (originating network) calls B (in the terminating network) and B has an intra-CSP call forwarding to C. B is the target.

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

=

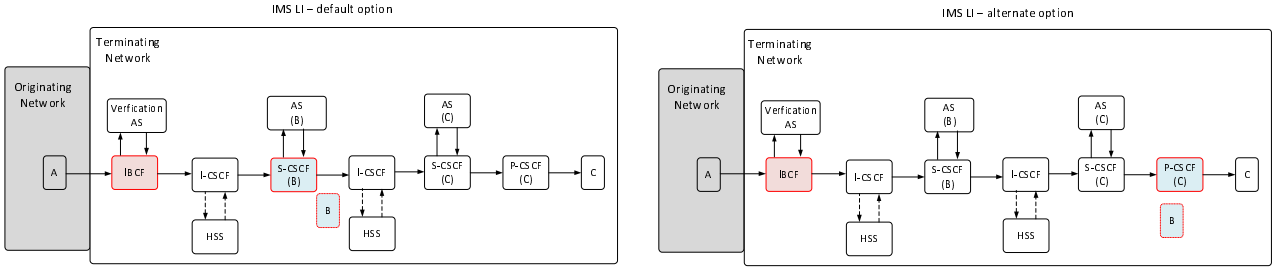


Figure 4.3.3.3-1: Scenario 1 - redirected call – A calls B (target), redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option

In both cases, the Ingress IBCF that interacts with the Verification AS to validate the received PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B.

Scenario 2: A (originating network) calls B, redirected to C (target)

A (originating network) calls B (in the terminating network) and B has an intra-CSP call forwarding to C and. C is the target.

The S-CSCF (C) provides the IRI-POI functions for IMS LI with the default option, and P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

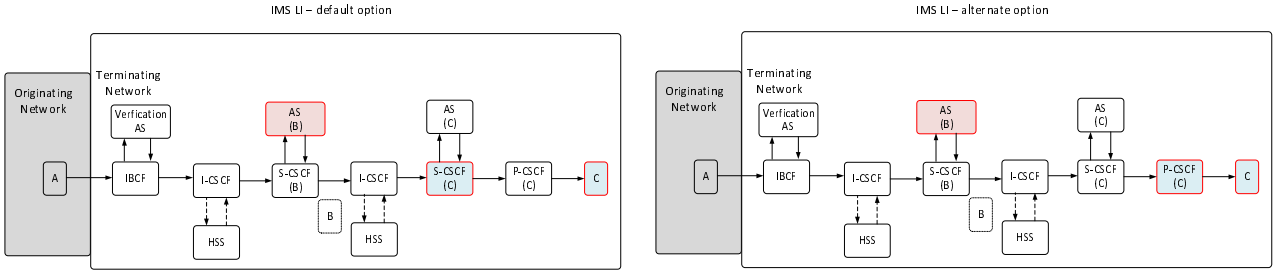


Figure 4.3.3.3-2: Scenario 2 – Redirected call – A calls B, redirected to C (target)

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the Ingress IBCF interacts with the Verification AS to validate the received PASSporT. However, in this case B is not a target and the IBCF does not know that the IMS session is redirected to C (a target). Therefore, IBCF does not provide the IRI-POI functions for STIR/SHAKEN related LI reporting with the Validation Result. Instead, the AS (B) provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target C.

Scenario 3: A (originating network) calls B (target), redirected to C (target)

A (originating network) calls B (in the terminating network) and B has an intra-CSP call forwarding to C. Here, B and C are the targets.

With the default option, the S-CSCF (B) provides the IRI-POI functions for IMS-LI for the target B and S-CSCF (C) provides the IRI-POI functions for IMS LI for the target C. With the alternate option, P-CSCF (C) provides the IRI-POI functions for IMS LI for targets B and C.

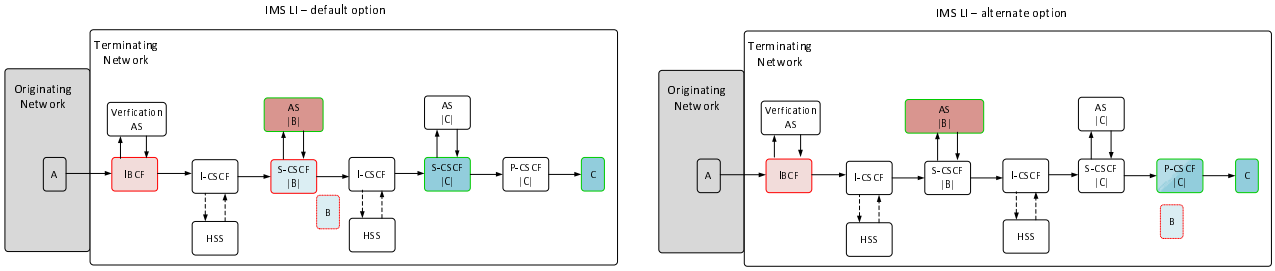


Figure 4.3.3.3-3: Scenario 3 – Redirected call – A calls B (target), redirected to C (target)

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the Ingress IBCF that interacts with the Verification AS to validate the received PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B. However, in this case Ingress IBCF does not know that the IMS session is redirected to C (the second target). Therefore, the AS (B) provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for target C.

Scenario 4: A (originating network) calls B, redirected to C. A is target non-local ID

A (originating network) calls B (in the terminating network) and B has an intra-CSP call forwarding to C and. A is the target non-local ID in the terminating network.

The IBCF provides the IRI-POI functions for IMS LI with the default option, and S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

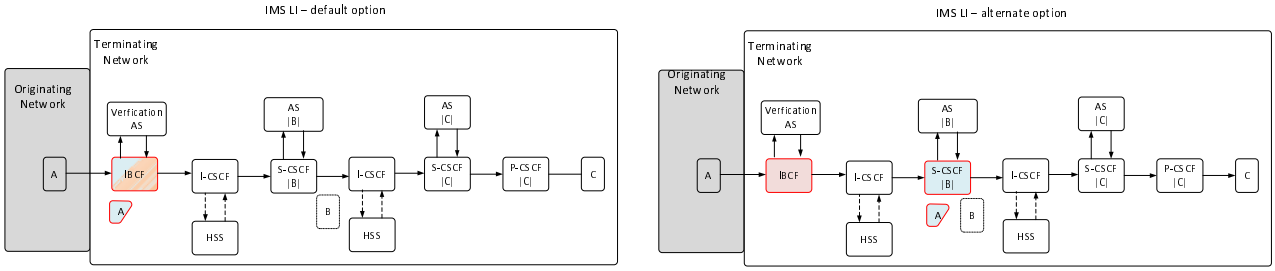


Figure 4.3.3.3-4: Scenario 4 – Redirected call – A (target non-local ID) calls B redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the Ingress IBCF that interacts with the Verification AS to validate the received PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target non-local ID A.

Intra-CSP IMS session

Scenario 5: Intra-CSP IMS session, A calls B, B is the target

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

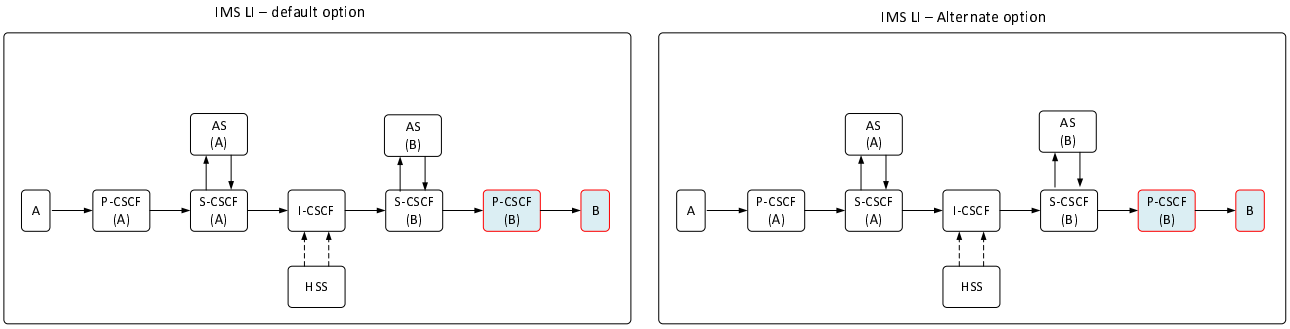


Figure 4.3.3.3-5: Scenario 5 – intra-CSP IMS session, A calls B (target)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

From STIR/SHAKEN perspective, no STIR/SHAKEN related LI reporting applies to this scenario since neither the signing (A's identity) nor the verification is performed. As a matter of fact, IBCF is not on the signalling path for this scenario.

Note that IBCF would interact with the Verification AS only when signing/verification of intra-CSP IMS session is not required and the CSP choice is IBCF for interacting with the Verification AS.

4.3.4 Originating IMS network – Telephony AS interacts with the Signing AS

4.3.4.1 General

In the illustrations shown in this clause, the Telephony AS interacts with the Signing AS. Clause 4.2.3.2 illustrate the conditions under which the Telephony AS would interact with the Signing AS. In brief, Telephony AS would interact with the Signing AS when signing of intra-CSP IMS session is required, or signing of RCD present in the call setup request (i.e. SIP INVITE) is needed, or when signing is not required for intra-CSP IMS sessions and the CSP choice is Telephony AS for interactions with the Signing AS and the session is inter-CSP session.

As a general concept, the IMS NF that interacts with the Signing AS provides the STIR/SHAKEN related LI reporting of the PASSporTs (i.e. the signature of the calling or redirecting party identities).

Within the drawings, AS, AS (A), AS (B), AS (C) represent the Telephony AS.

4.3.4.2 Intra-CSP IMS session signing is required

4.3.4.2.1 Simple scenarios

Scenario 1: Intra-CSP IMS session - A (target) calls B

The S-CSCF (A) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (A) provides the IRI-POI functions for IMS LI with the alternate option.

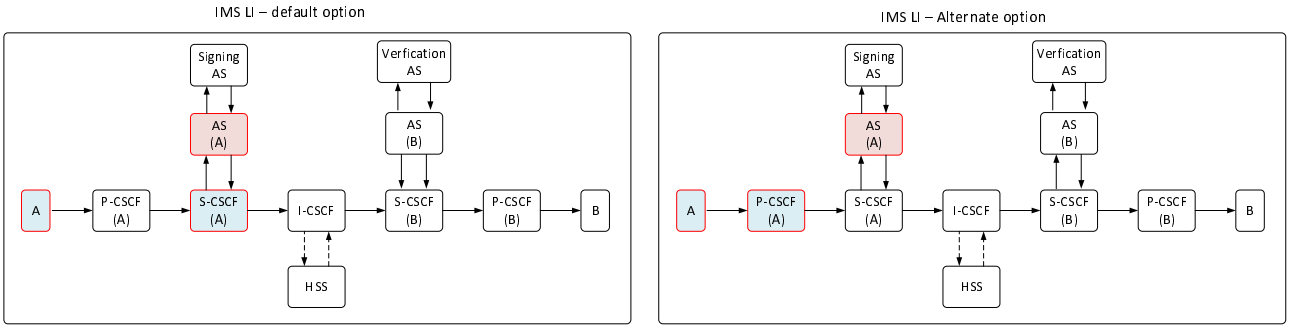


Figure 4.3.4.2.1-1: Scenario 1 – Intra-CSP IMS session, A (target) calls B

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (A) that interacts with the Signing AS to authenticate/signa A's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target A.

Scenario 2: Inter-CSP IMS session - A (target) calls B

A in the originating network calls B in the terminating network.

The S-CSCF (A) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (A) provides the IRI-POI functions for IMS LI with the alternate option.

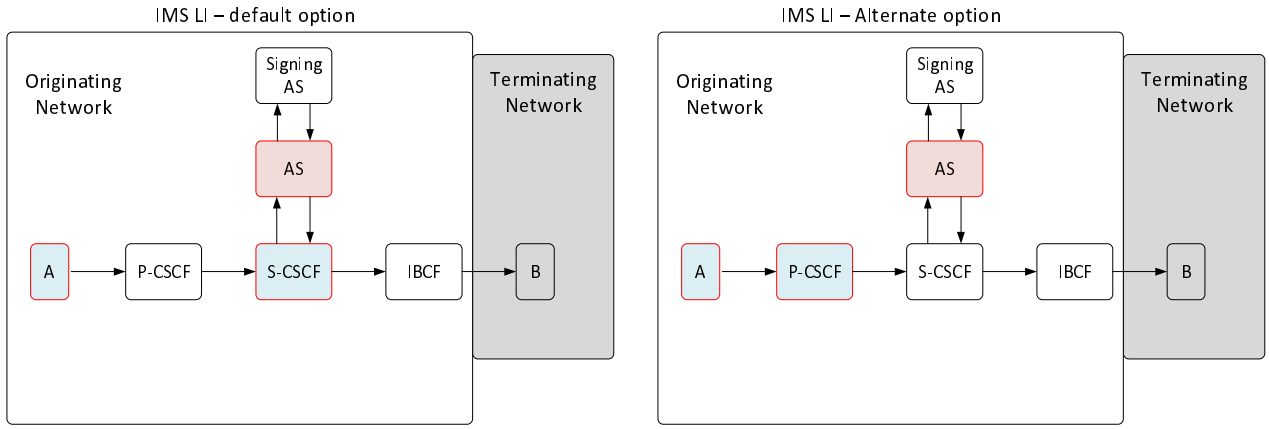


Figure 4.3.4.2.1-2 Scenario 2 – Inter-CSP session, A (target) calls B

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS that interacts with the Signing AS to authenticate/sign A's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target A.

Scenario 3: Inter-CSP IMS session - A calls B (target non-local ID)

A in the originating network calls B in the terminating network. B is the target non-local ID in the originating network.

The IBCF provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

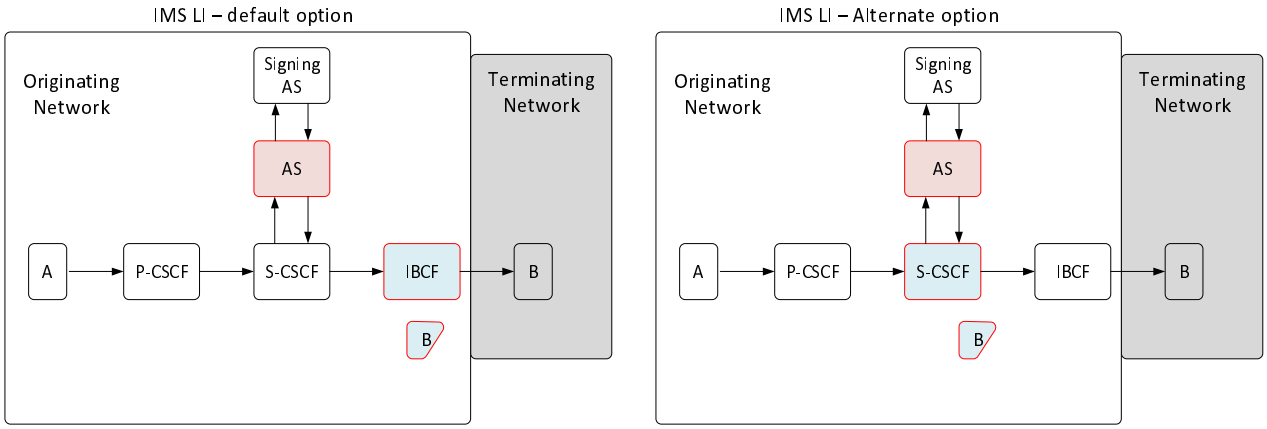


Figure 4.3.4.2.1-3: Scenario 3 - A calls B (target non-local ID)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS that interacts with the Signing AS to authenticate/sign A's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target non-local ID B.

4.3.4.2.2 Complex scenarios

Redirected calls

There are two cases as far as the STIR/SHAKEN related LI reporting:

1. When the IRI-POI is provisioned with the ReportDiversionPASSporTInfo set to false.
2. When the IRI-POI is provisioned without the ReportDiversionPASSporTInfo or set to true.

When the ReportDiversionPASSporTInfo is set to false, the scope of STIR/SHAKEN LI reporting is limited to the reporting of target identities. In other words, "div" PASSporT associated with the redirecting party will not be part of the STIR/SHAKEN LI reporting when the calling party is the target. The "div" PASSporT associated with the redirecting party is delivered as a part of STIR/SHAKEN related LI reporting when the redirecting party is the target.

When the ReportDiversionPASSporTInfo is absent or set to true, the scope of STIR/SHAKEN related LI reporting includes all identities (i.e. including the "div" PASSporT) when the calling party is the target.

Redirected calls, intra-CSP IMS sessions

Scenario 1: A (target) calls B, redirected to C

The S-CSCF (A) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (A) provides the IRI-POI functions for IMS LI with the alternate option.

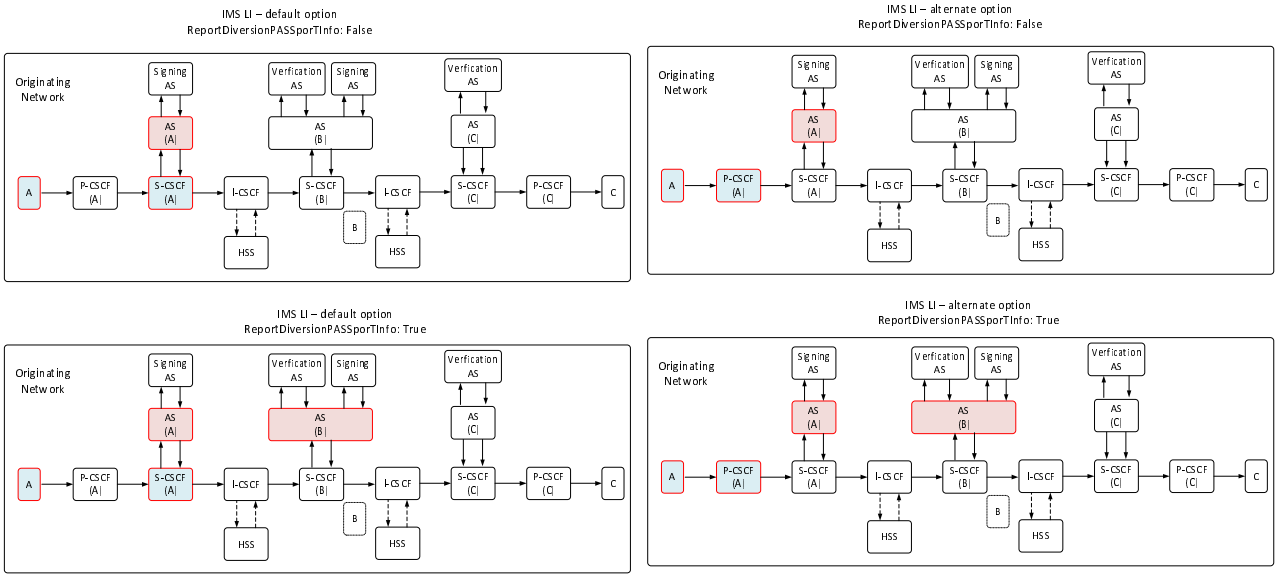


Figure 4.3.4.2.2-1: Scenario 1 – A (target) calls B redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option. The top part of the diagram applies when the ReportDiversionPASSporTInfo is set to false and the bottom part of the diagram applies when the ReportDiversionPASSporTInfo is set to true.

In all cases, the AS (A) that interacts with the Signing AS to authenticate/sign A's identity (i.e. target's identity) provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for target A. When the ReportDiversionPASSporTInfo is set to true, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity will also provide the IRI-POI functions for the STIR/SHAKEN related LI reporting for target A.

Scenario 2: A calls B (target), redirected to C

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

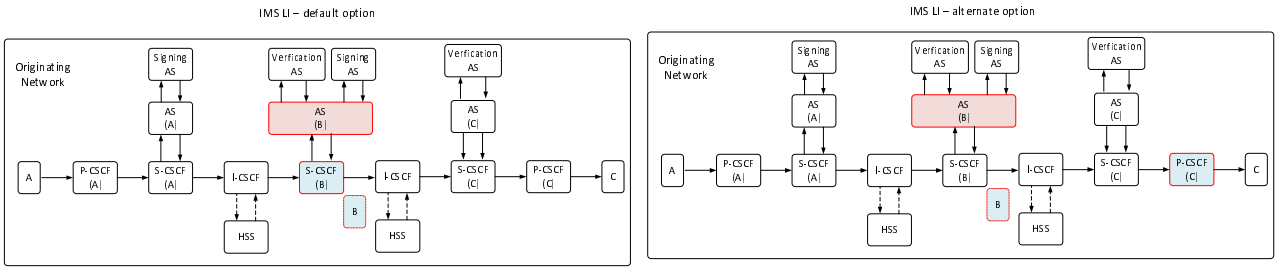


Figure 4.3.4.2.2-2: Scenario 2 – A calls B (target) redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target B.

In addition, the AS (B) that interacts with the Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B. Verification part of STIR/SHAKEN related LI reporting when AS interacts with the Verification AS is further illustrated in clause 4.3.5.

Redirected calls, inter-CSP IMS sessions

Scenario 3: A (target) calls B, redirected to C. C is in the terminating network

A calls B in the originating network, and B has call forwarding to C in the terminating network. A is the target.

The S-CSCF (A) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (A) provides the IRI-POI functions for IMS LI with the alternate option.

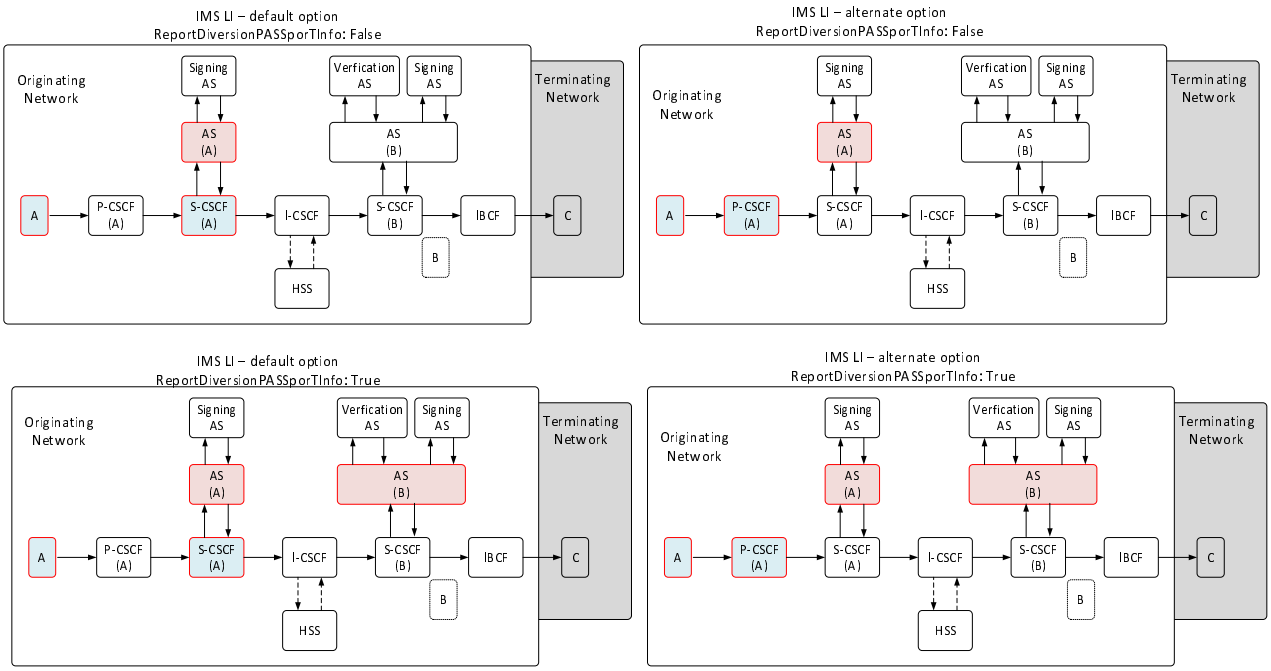


Figure 4.3.4.2.2-3: Scenario 3 – A (target) calls B redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option. The top part of the diagram applies when the ReportDiversionPASSporTInfo is set to false and the bottom part of the diagram applies when the ReportDiversionPASSporTInfo is set to true.

In all cases, the AS (A) that interacts with the Signing AS to authenticate/sign A's identity (i.e. target's identity) provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for target A. When the ReportDiversionPASSporTInfo is set to true, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity will also provide the IRI-POI functions for the STIR/SHAKEN related LI reporting for target A.

Scenario 4: A calls B (target), redirected to C

A calls B in the originating network, and B has call forwarding to C in the terminating network. B is the target.

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the IBCF provides the IRI-POI functions for IMS LI with the alternate option.

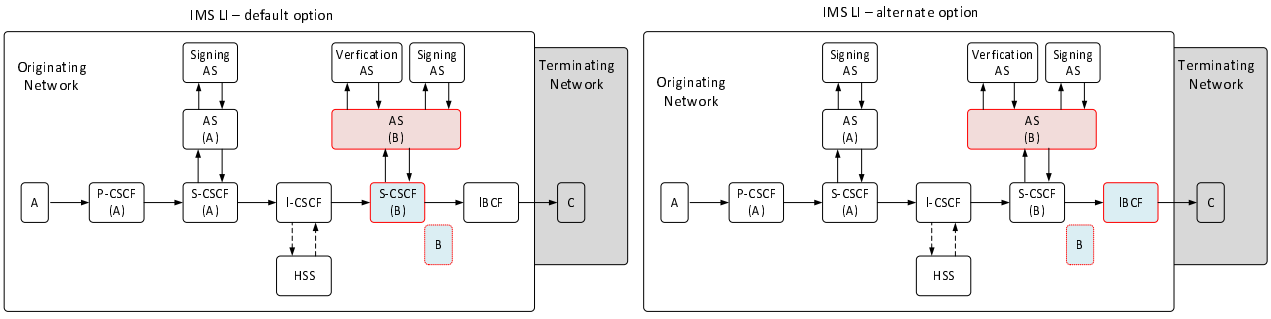


Figure 4.3.4.2.2-4: Scenario 4 – A calls B (target) redirected to C

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target B.

In addition, the AS (B) that interacts with the Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B. Verification part of STIR/SHAKEN related LI reporting when AS interacts with the Verification AS is further illustrated in clause 4.3.5.

Scenario 5: A (target) calls B (target), redirected to C

A calls B in the originating network, and B has call forwarding to C in the terminating network. A and B are the targets.

For target A, the S-CSCF (A) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (A) provides the IRI-POI functions for IMS LI with the alternate option.

For target B, the S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the IBCF provides the IRI-POI functions for IMS LI with the alternate option.

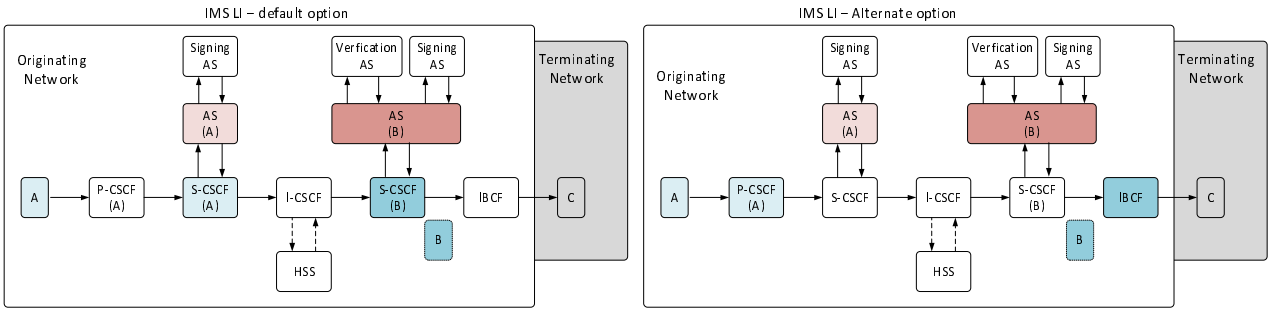


Figure 4.3.4.2.2-5: Scenario 5 – A calls B (target) redirected to C

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (A) that interacts with the Signing AS to authenticate/sign A's identity (i.e. target's identity) provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for target A. When the ReportDiversionPASSporTInfo is set to true, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity will also provide the IRI-POI functions for the STIR/SHAKEN related LI reporting for target A.

In both cases, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target B.

In addition, the AS (B) that interacts with the Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B. Verification part of STIR/SHAKEN related LI reporting when AS interacts with the Verification AS is further illustrated in clause 4.3.5.

Scenario 6: A calls B, redirected to C. C is the target non-local ID in the originating network

A calls B in the originating network, and B has call forwarding to C in the terminating network. C is the target non-local ID in the originating network.

The IBCF provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF(B) provides the IRI-POI functions for IMS LI with the alternate option.

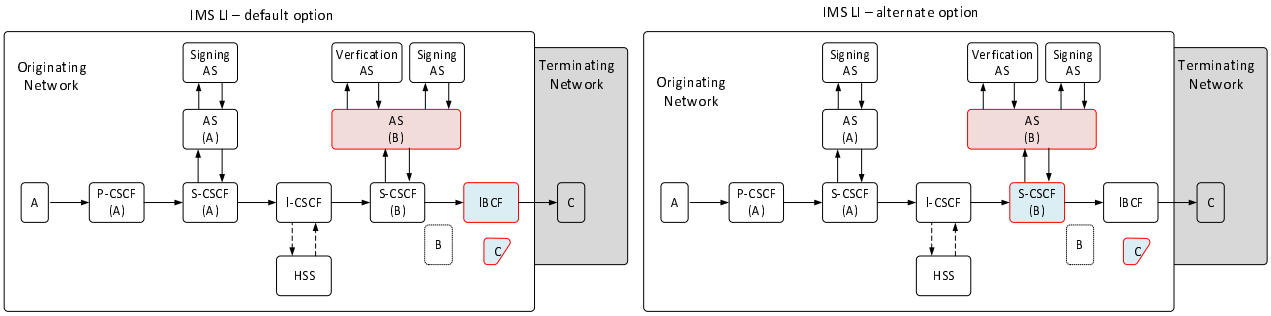


Figure 4.3.4.2.2-6: Scenario 6 – A calls B redirected to C (target non-local ID)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for target non-local ID C. The signatures related to both A's and B's identities are provided as a part of STIR/SHAKEN related LI reporting.

4.3.4.3 Intra-CSP IMS session signing is not required

4.3.4.3.1 General

Even when the intra-CSP IMS session signing is not required, the illustrations shown in clause 4.3.2.2 apply for all the scenarios when the RCD is present. Therefore, in the illustrations in the following scenarios, it is presumed that RCD is not present.

4.3.4.3.2 Simple scenarios

Scenario 1: A (target) calls B.

A (originating network) calls B (terminating network). A is the target in the originating network.

The S-CSCF provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

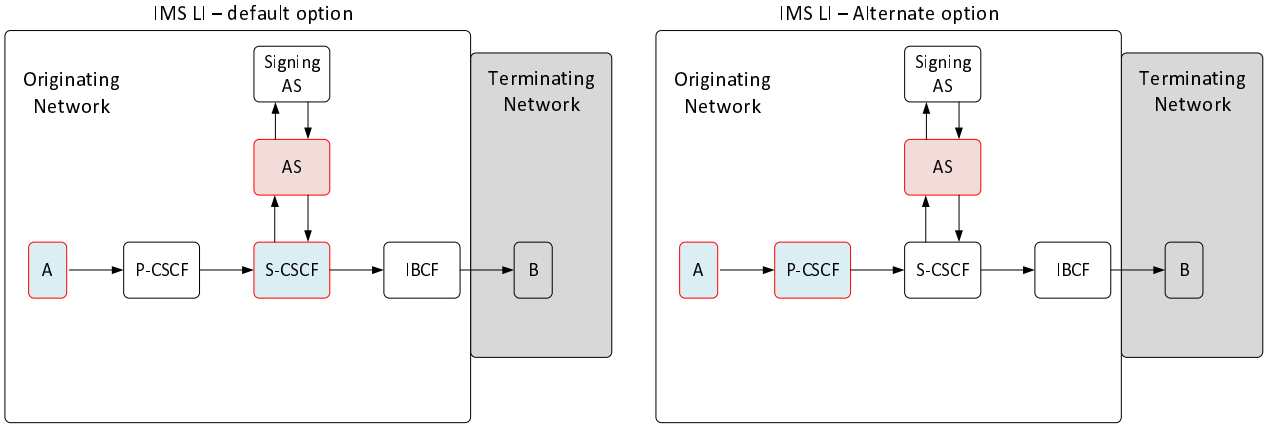


Figure 4.3.4.3.2-1 Scenario 1 - A (target) calls B

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS that interacts with the Signing AS to authenticate/sign A's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting.

Note that this scenario same as the scenario 2 illustrated in clause 4.3.4.2.1.

Scenario 2: A calls B (target non-local ID)

A (originating network) calls B (terminating network). This is an inter-CSP IMS session. B is a target non-local ID in the originating network.

The IBCF provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

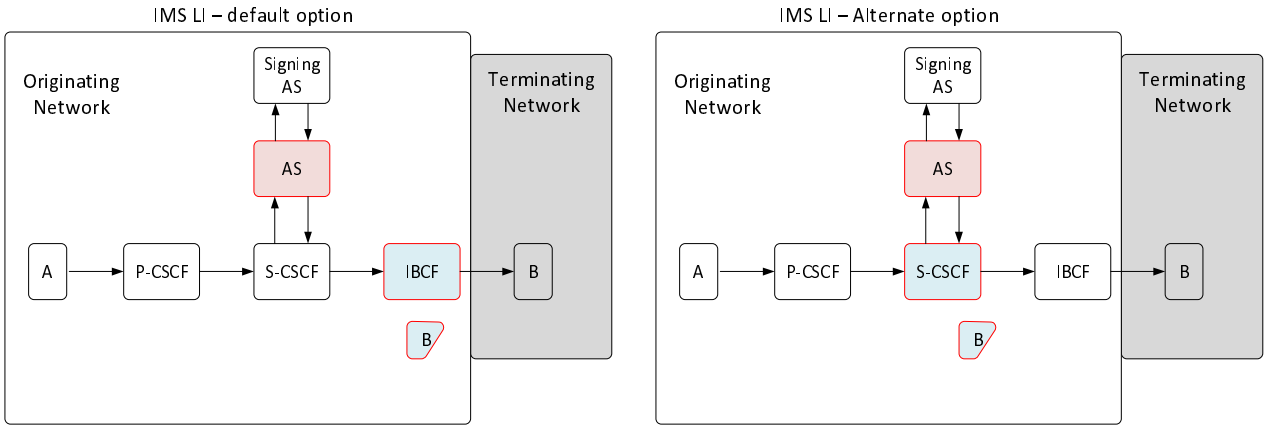


Figure 4.3.4.3.2-2: Scenario 2 – A calls B (target non-local ID)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS that interacts with the Signing AS to authenticate/sign A's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting. Note that B is the target non-local ID here.

Note that this scenario same as the scenario 3 illustrated in clause 4.3.4.2.1.

4.3.4.3.3 Complex scenarios

Basic calls

Scenario 1: Intra-CSP IMS session. A (target) calls B

The S-CSCF (A) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (A) provides the IRI-POI functions for IMS LI with the alternate option.

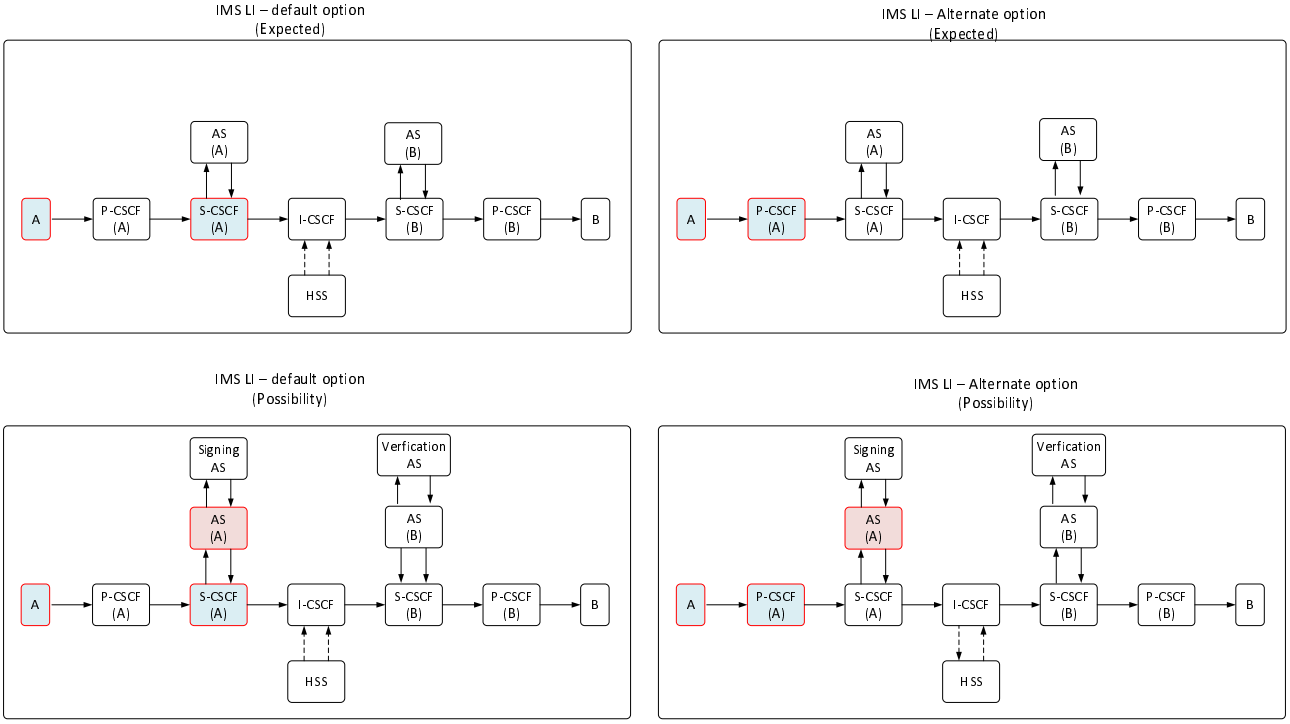


Figure 4.3.4.3.3-1: Scenario 1 – A (target) calls B

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option. From STIR/SHAKEN perspective, the top part of the diagram shows the expected system

behaviour since this is an intra-CSP IMS session and the signing for an intra-CSP IMS session is not required. The bottom part of the diagram shows what would possibly happen with the CSP choice of AS for interacting with the Signing AS.

When the CSP choice is AS to interact with the Signing AS, the AS (A) may authenticate/sign A's identity if it does not know that the current session is an intra-CSP IMS session.

If the AS (A) does interact with the Signing AS to authenticate/signa A's identity (as shown), then the IRI-POI in AS (A) would provide the IRI-POI functions for the STIR/SHAKEN related LI reporting. If the AS (A) does not interact with the Signing AS, then the STIR/SHAKEN related LI reporting does not apply.

Redirected sessions

Scenario 2: A (target) calls B, redirected to C

A (originating network) calls B (originating network) redirected to C (terminating network). A is the target.

The S-CSCF (A) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (A) provides the IRI-POI functions for IMS LI with the alternate option.

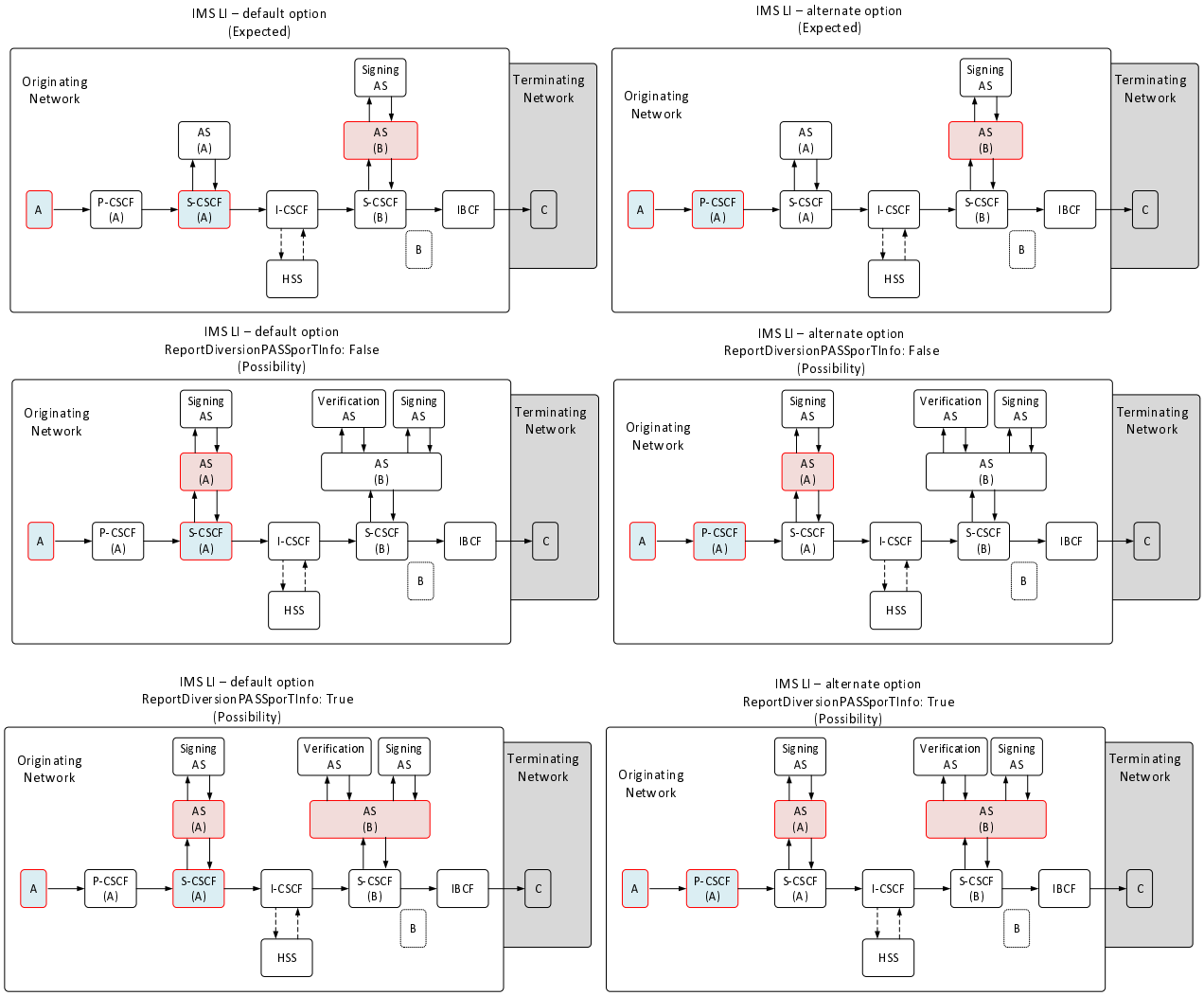


Figure 4.3.4.3.3-2: Scenario 2 – A (target) calls B, redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

From STIR/SHAKEN perspective, the top part of the diagram shows the expected system behaviour since the signing is required only for the inter-CSP IMS session. The AS (B) is expected to interact with the Signing AS to authenticate/sign A's identity and B's identity. In this case, AS (B) that interacts with the Signing AS to

authenticate/sign A's identity and B's identity would provide the IRI-POI functions for STIR/SHAKEN related LI reporting. Whether the "div" PASSporT related information is reported, or not, depends on ReportDiversionPASSporTInfo value.

The middle part and bottom part of the diagram show what could possibly happen with the CSP choice of AS for interacting with the Signing AS when the signing of intra-CSP session is not required. The AS (A) may authenticate/sign A's identity if it does not know that B is in the same network. If the AS (A) without knowing the nature of the call interacts with the Signing AS to authenticate/sign A's identity, then the AS (B) would interact with the Signing AS to authenticate/sign just B's identity. When that happens, the middle part of the diagram applies when the ReportDiversionPASSporTInfo is set to False inferring the scope of STIR/SHAKEN related LI reporting is limited to the reporting of target identities whereas the bottom part of the diagram applies when the ReportDiversionPASSporTInfo is set to True inferring the scope of STIR/SHAKEN related LI reporting includes all identities (i.e. including the diversion). When the ReportDiversionPASSporTInfo is set to True, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity will also provide the IRI-POI functions for the STIR/SHAKEN related LI reporting for target A. Note that here B is not a target.

Scenario 3: Intra-CSP IMS session. A (target) calls, B redirected to C

The S-CSCF (A) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (A) provides the IRI-POI functions for IMS LI with the alternate option.

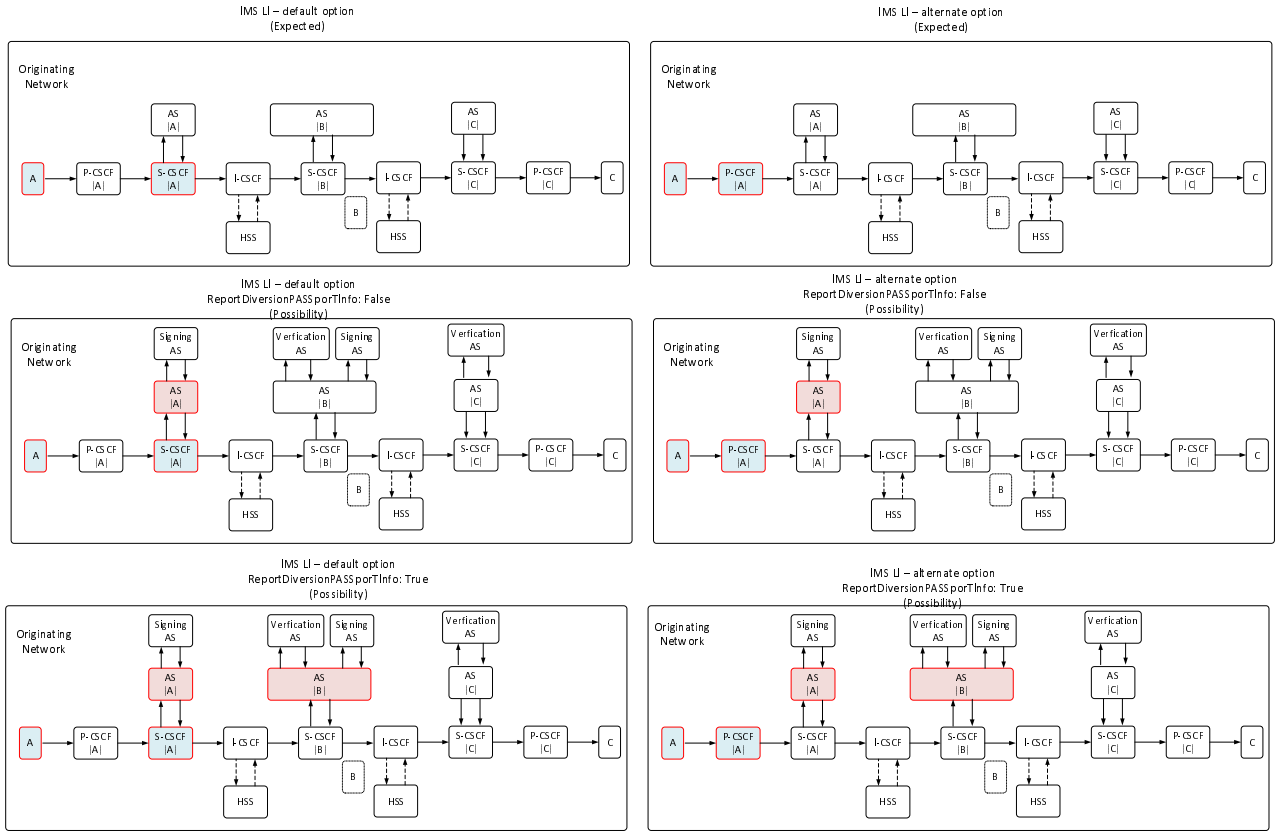


Figure 4.3.4.3.3-3: Scenario 3 – A (target) calls B, redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

From STIR/SHAKEN perspective, the top part of the diagram shows the expected system behaviour since the signing is required only for the inter-CSP IMS session. As expected, if there is no interaction with the Signing AS, then the STIR/SHAKEN related LI reporting does not apply.

The middle part and bottom part of the diagram show what could possibly happen with the CSP choice of AS for interacting with the Signing AS when the signing of intra-CSP session is not required.

The AS (A) and AS (B) may not know that the current session is an intra-CSP IMS session. Assuming neither of them know that the current session is an intra-CSP IMS session, the AS (A) would interact with the Signing AS to

authenticate/sign A's identity and AS (B) would interact with the Signing AS to authenticate/sign B's identity. When that happens, the middle part of the diagram applies when the ReportDiversionPASSporTInfo is set to False inferring the scope of STIR/SHAKEN related LI reporting is limited to the reporting of signatures of calling party (target) whereas the bottom part of the diagram applies when the ReportDiversionPASSporTInfo is set to True inferring the scope of STIR/SHAKEN related LI reporting includes all identities (i.e. including the diversion). The AS (A) that interacts with the Signing AS to authenticate/sign A's identity would provide the IRI-POI functions for the STIR/SHAKEN related LI reporting for A's identity. When the ReportDiversionPASSporTInfo is set to True, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity will also provide the IRI-POI functions for the STIR/SHAKEN related LI reporting for target A. Note that here B is not a target.

With respect to this scenario, what is shown in figure 4.3.4.3.3-4 can also happen.

When A is the target, the S-CSCF (A) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (A) provides the IRI-POI functions for IMS LI with the alternate option.

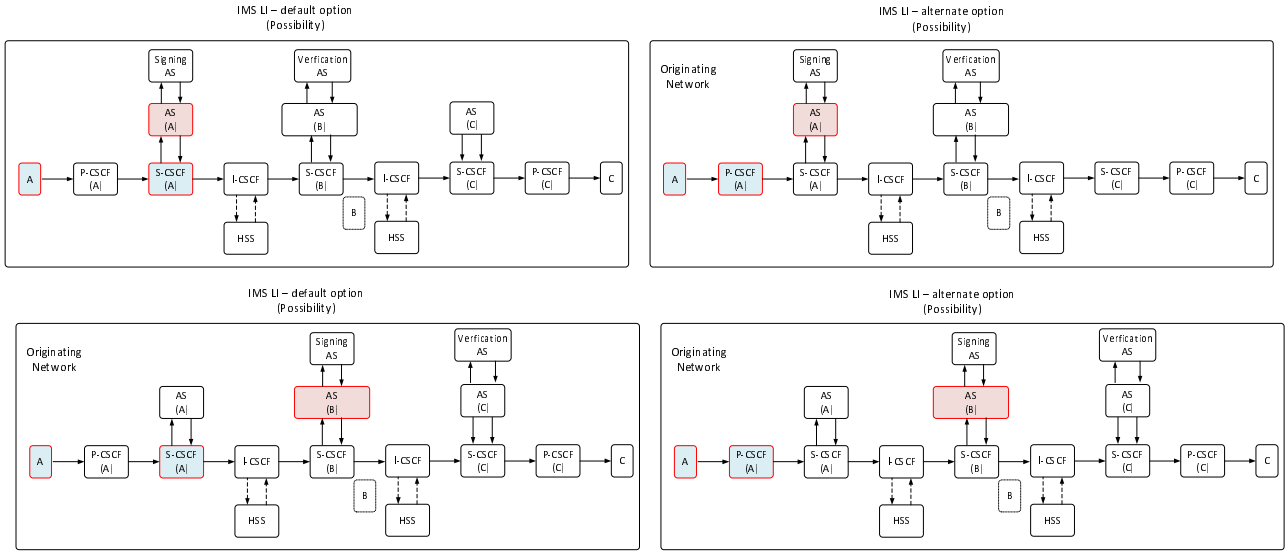


Figure 4.3.4.3.3-4: Scenario 3 (extended) – A (target) calls B, redirected to C, other possibilities

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

It is possible that either AS (A) or AS (B) may know that the current session is an intra-CSP IMS session. In the top part of figure 410.3.4.3.3-4, the AS (A) does not know it is an intra-CSP IMS session, but the AS (B) does. Accordingly, the AS (A) may authenticate/sign A's identity as it does not know that B is in the same network. When that happens, the AS (A) that interacts with the Signing AS to authenticate/sign A's identity would provide the IRI-POI functions for the STIR/SHAKEN related LI reporting for target A.

In the bottom part of the diagram in figure 4.3.4.3.3-4, the AS (A) knows that it is an intra-CSP IMS session, but AS (B) does not. Accordingly, AS (A) does not interact with the Signing AS to authenticate/sign A's identity since signing of intra-CSP IMS session is not required. However, the AS (B) would interact with the Signing AS to authenticate/sign A's identity and B's identity as it does not know that about the redirected leg of the IMS session is an intra-CSP IMS session. When this happens, the IRI-POI present in the AS (B) would provide the IRI-POI functions for STIR/SHAKEN related LI reporting for target A. The scope of STIR/SHAKEN related LI reporting for the target A depends on the provisioned ReportDiversionPASSporTInfo value.

Scenario 4: Intra-CSP IMS session. A calls B (target), redirected to C

A calls B in the originating network, and B has call forwarding to C in the terminating network. B is the target.

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

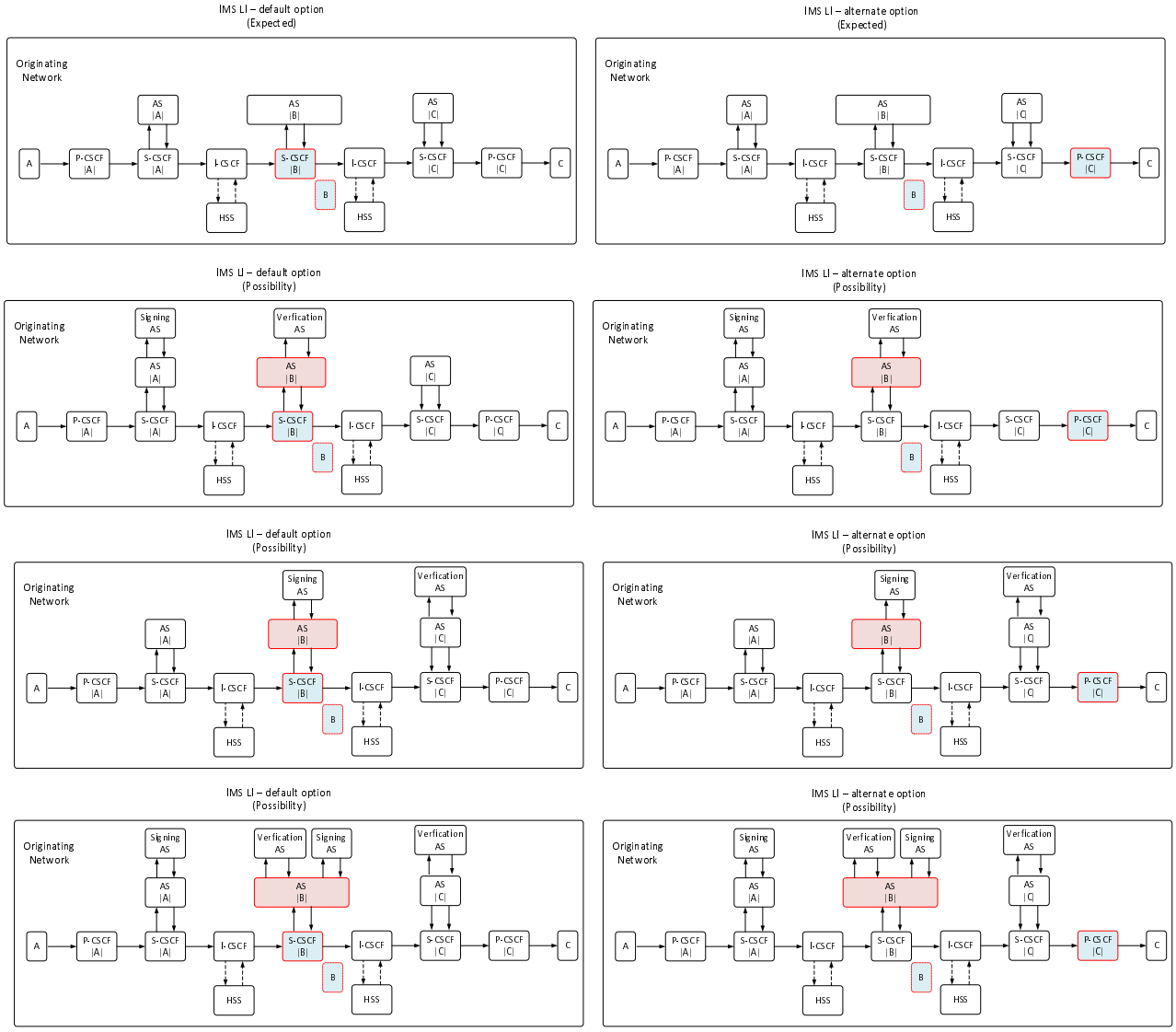


Figure 4.3.4.3.3-5: Scenario 4 –A calls B (target) redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

From STIR/SHAKEN perspective, the top part of the diagram shows the expected system behaviour since the signing is required only for the inter-CSP IMS session. As expected, if there is no interaction with the Signing AS, then the STIR/SHAKEN related LI reporting does not apply.

Any of other three parts of the diagram could possibly happen with the CSP choice of AS for interacting with the Signing AS when the signing of intra-CSP session is not required. The AS (A) may authenticate/sign A's identity if it does not know that B is in the same network. Likewise, The AS (B) may authenticate/sign B's identity if it does not know that C is in the same network.

If what is shown bottom three rows happen, then AS (B) provide the IRI-POI functions for STIR/SHAKEN related LI reporting for target B. Such LI reporting could be just the Validation Result (second row) or Signature (third row) or both (last row).

Scenario 5: A calls B, redirected to C (target non-local ID in the originating network)

A calls B in the originating network, and B has call forwarding to C in the terminating network. C is the target non-local ID in the originating network.

The IBCF provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

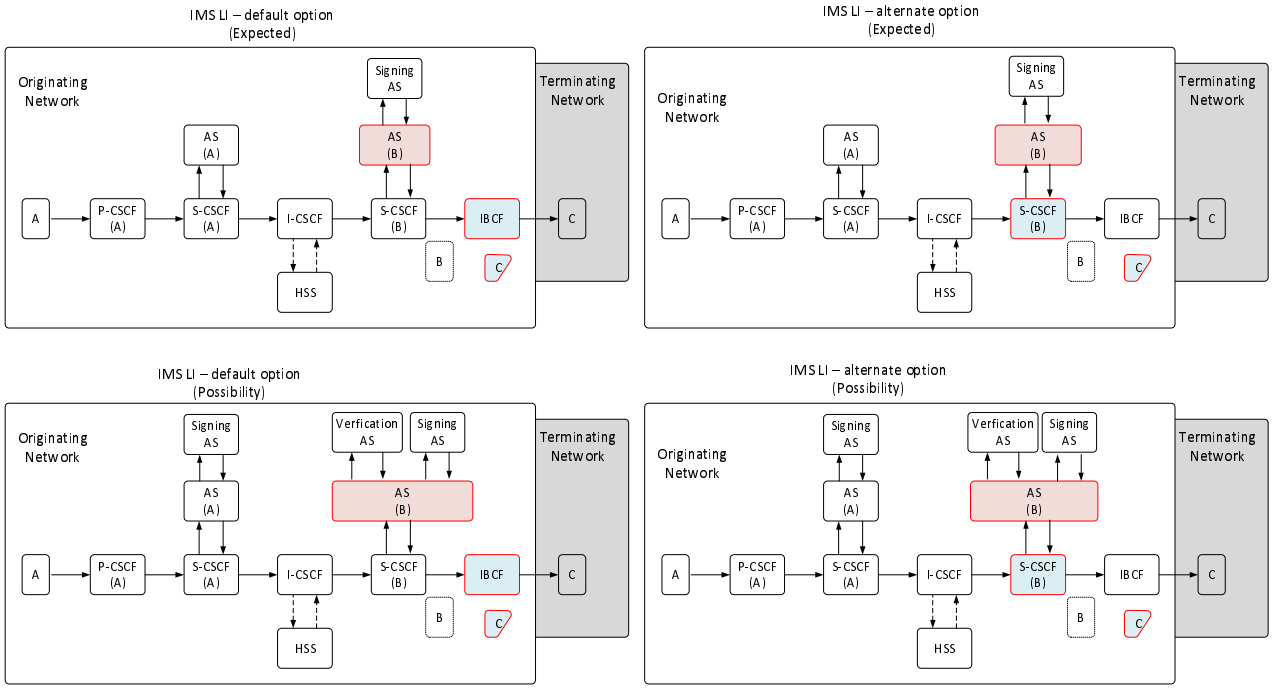


Figure 4.3.4.3.3-6: Scenario 5 –A calls B redirected to C (target non-local ID)

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

From STIR/SHAKEN perspective, the top part of the diagram shows the expected system behaviour since the signing is required only for the inter-CSP IMS session. The AS (B) is expected to interact with the Signing AS to authenticate/sign A's identity and B's identity.

The bottom part of the diagram shows what would possibly happen with the CSP choice of AS for interacting with the Signing AS when the signing of intra-CSP session is not required. The AS (A) may authenticate/sign A's identity if it does not know that B is in the same network. If the AS (A) without knowing the nature of the call interacts with the Signing AS to authenticate/sign A's identity, then the AS (B) would interact with the Signing AS to authenticate/sign just B's identity.

In either case, AS (B) would provide the IRI-POI functions for STIR/SHAKEN related LI reporting A's and B's identity for the target non-local ID C. Note that neither A nor B is a target in this illustration.

4.3.5 Terminating IMS network – Telephony AS interacts with the Verification AS

4.3.5.1 General

In the illustrations shown in this clause, the Telephony AS interacts with the Verification AS. Clause 4.2.3.3 illustrates the conditions under which the Telephony AS would interact with the Verification AS. In brief, Telephony AS would interact with the Verification AS when a Validation Result is not present in the received SIP INVITE and one or more PASSporTs are present in the SIP INVITE. That would be the case even for inter-CSP IMS sessions with CSP choice for verification is Telephony AS.

As a general concept, the IMS NF that interacts with the Verification AS provides the STIR/SHAKEN related LI reporting of the Validation Results (i.e. for verification). However, there are scenarios where an IMS NF may provide the STIR/SHAKEN related LI reporting of the Validation Result even if that IMS NF does not interact with the Verification AS (e.g. redirecting scenarios, inbound roaming).

Within the drawings, AS, AS (A), AS (B), AS (C) represent the Telephony AS.

4.3.5.2 Intra-CSP IMS session verification is required

4.3.5.2.1 Simple scenarios

Scenario 1: Intra-CSP IMS session - A calls B (target).

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

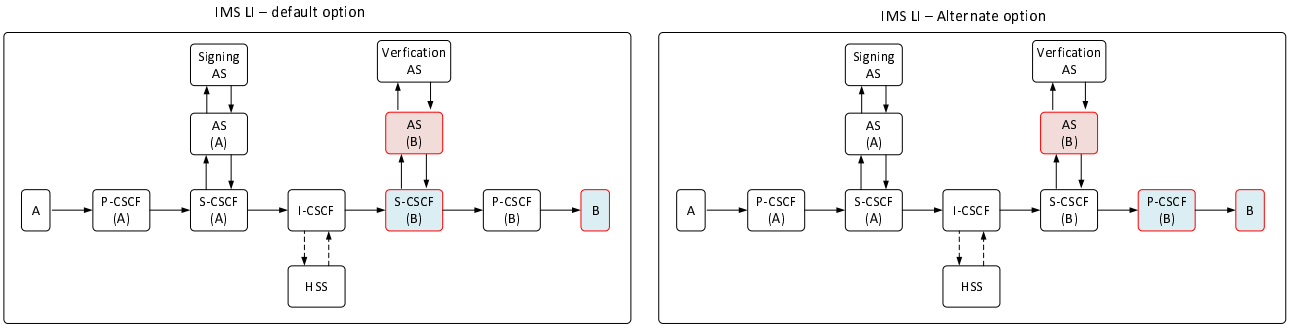


Figure 4.3.5.2.1-1: Scenario 1 – Intra-CSP IMS session, A (target) calls B

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B.

Scenario 2:. Inter-CSP IMS session - A calls B (target)

A in the originating network calls B in the terminating network. B is the target.

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

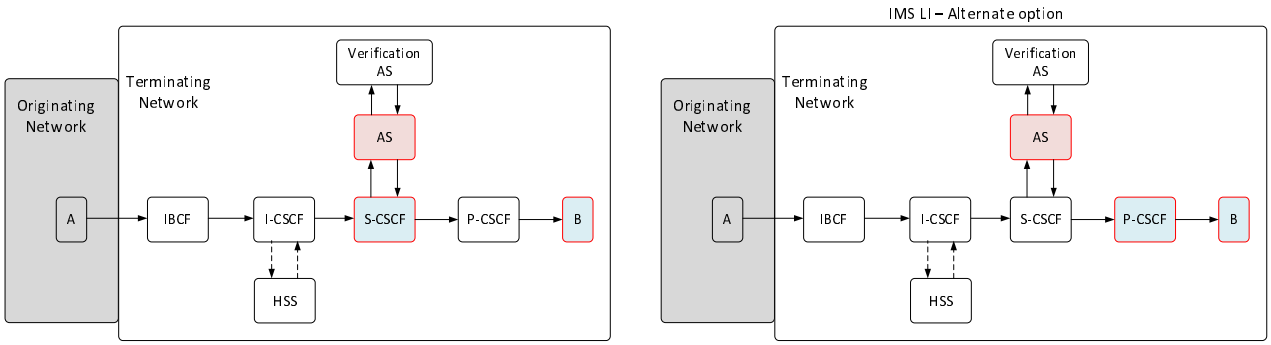


Figure 4.3.5.2.1-2 Scenario 2 – Inter-CSP IMS session, A calls B (target)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B.

Scenario 3: A (target non-local ID) calls B

A in the originating network calls B in the terminating network. A is the target non-local ID in the terminating network.

The IBCF provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

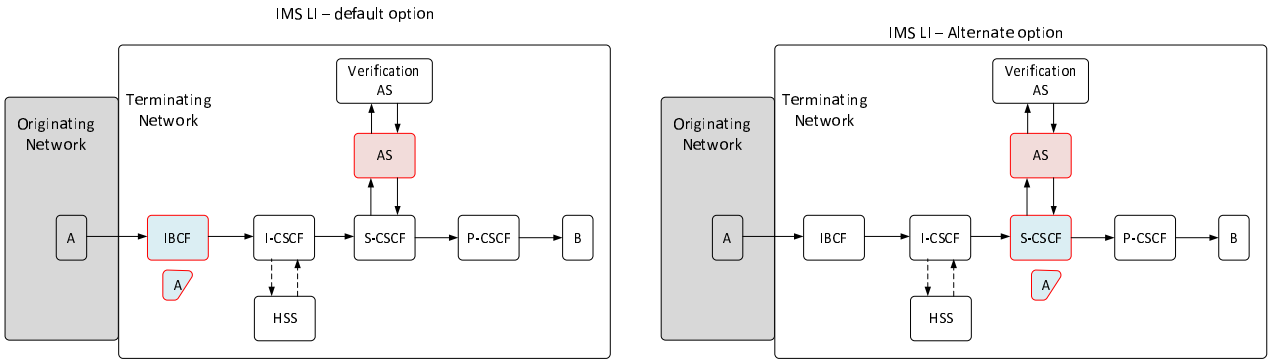


Figure 4.3.5.2.1-3: Scenario 3 – Inter-CSP IMS session, A (target non-local ID) calls B

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B.

4.3.5.2.2 Complex scenarios

Redirected calls, intra-CSP IMS sessions

Scenario 1: A calls B, redirected to C (target)

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

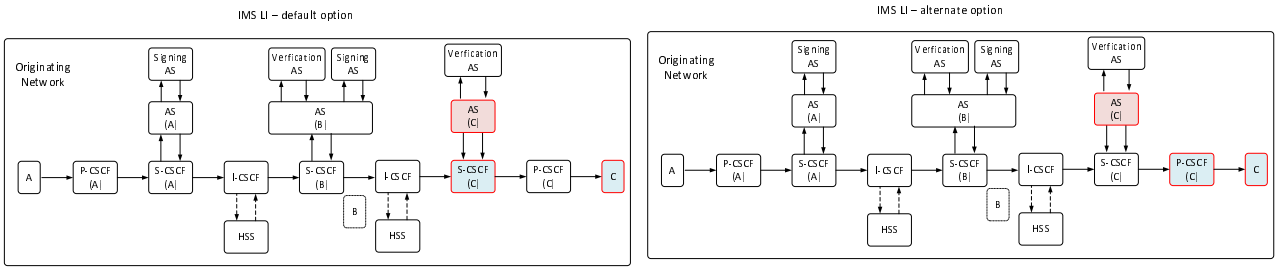


Figure 4.3.5.2.2-1: Scenario 1 – A calls B redirected to C (target)

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (C) that interacts with the Verification AS to validate the received signatures in the PASSporTs provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target C.

Scenario 2: A calls B (target), redirected to C

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

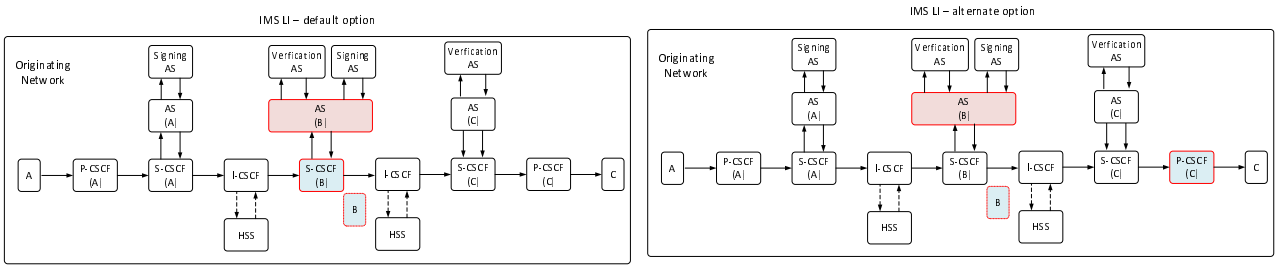


Figure 4.3.5.2.2-2: Scenario 2 – A calls B (target) redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B.

In addition, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target B. Signing part of STIR/SHAKEN related LI reporting when AS interacts with the Signing AS is further illustrated clause 4.3.4.

Scenario 3: A calls B (target,) redirected to C (target)

For target B, the S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

For target C, the S-CSCF (C) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

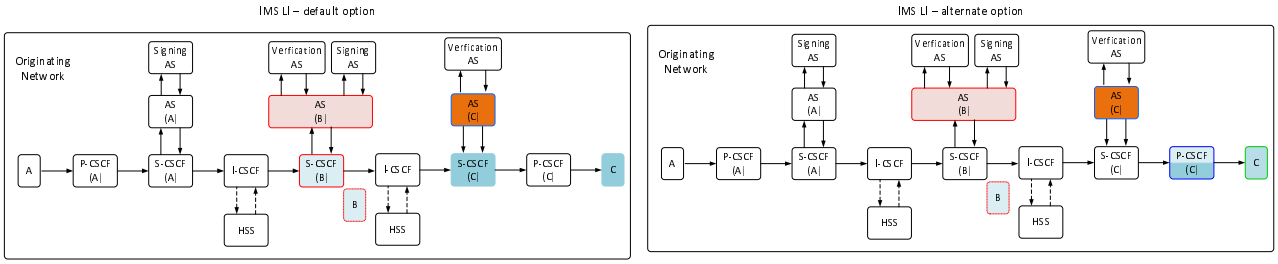


Figure 4.3.5.2.2-3: Scenario 3 – A calls B (target) redirected to C (target)

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B and AS (C) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target C.

In addition, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target B. Signing part of STIR/SHAKEN related LI reporting when AS interacts with the Signing AS is further illustrated clause 4.3.4.

Redirected calls, inter-CSP IMS sessions

Scenario 4: A calls B, redirected to C (target)

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

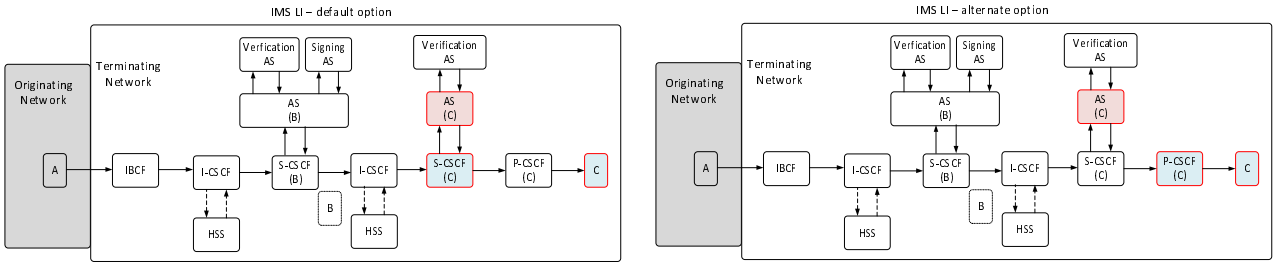


Figure 4.3.5.2.2-4: Scenario 4 – A calls B redirected to C (target)

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (C) that interacts with the Verification AS to validate the received signatures in the PASSporTs provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target C.

Scenario 5: A calls B (target), redirected to C

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

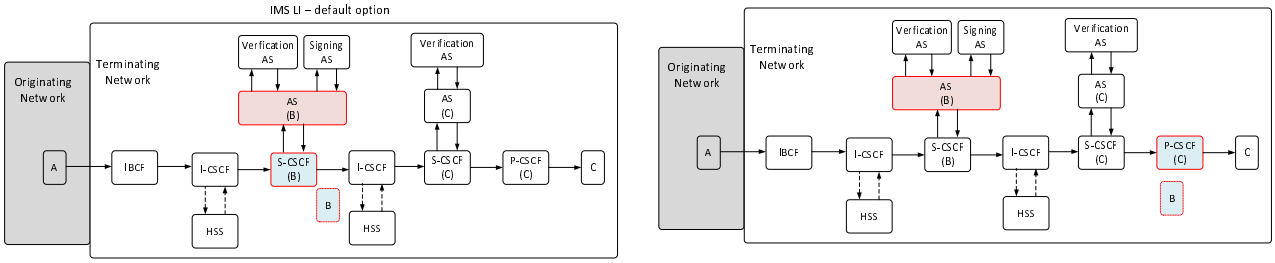


Figure 4.3.5.2.2-5: Scenario 5 – A calls B (target) redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B.

In addition, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target B. Signing part of STIR/SHAKEN related LI reporting when AS interacts with the Signing AS is further illustrated clause 4.3.4.

Scenario 6: A (target non-local ID) calls B, redirected to C

A (originating network) calls B, and B has call forwarding to C. A is the target non-local ID in the terminating network.

The IBCF provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

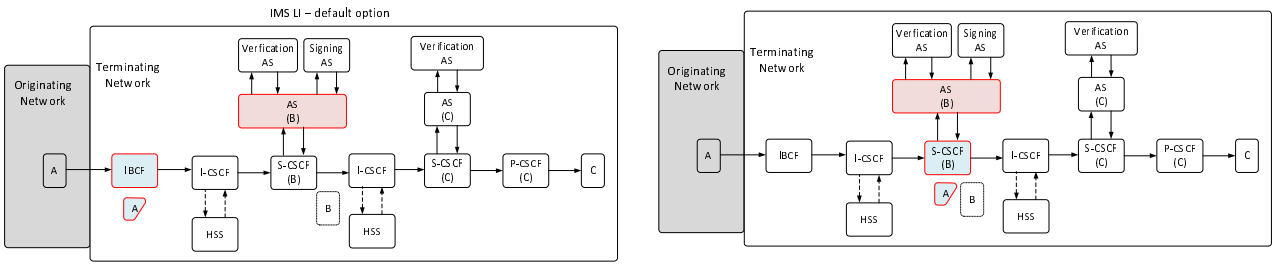


Figure 4.3.5.2.2-6: Scenario 6 – A (target non-local ID) calls B redirected to C

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target non-local ID A.

Scenario 7: A (target non-local ID) calls B (target) redirected to C (target)

A (originating network) calls B and B has call forwarding to C in the terminating network. A is the target non-local ID in the terminating network. B and C are also targets in the terminating network.

For target non-local ID A: The IBCF provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF(B) provides the IRI-POI functions for IMS LI with the alternate option.

For target B: The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

For target C: The S-CSCF (C) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

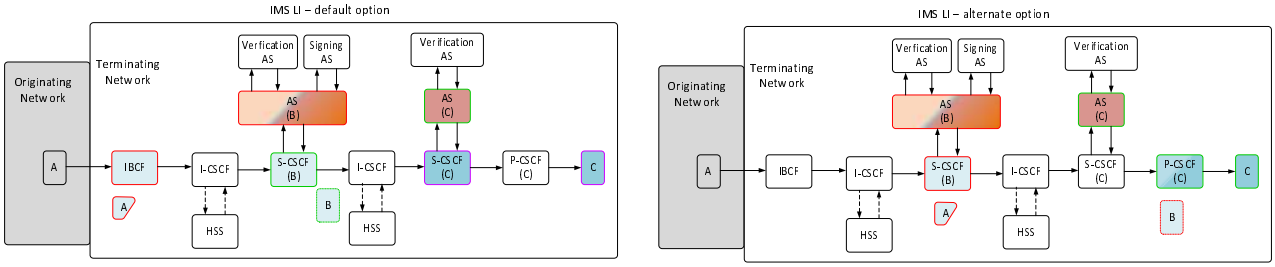


Figure 4.3.4.2.2-7: Scenario 7 – A (target non-local ID) calls B (target) redirected to C (target)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target non-local ID A and target B.

In both cases, the AS (C) that interacts with the Verification AS to validate the received signatures in the PASSporTs provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target C.

4.3.5.3 Intra-CSP IMS session signing is not required

4.3.5.3.1 Simple scenarios

Scenario 1: A calls B (target)

A (originating network) calls B (terminating network). B is the target in the terminating network.

The S-CSCF provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

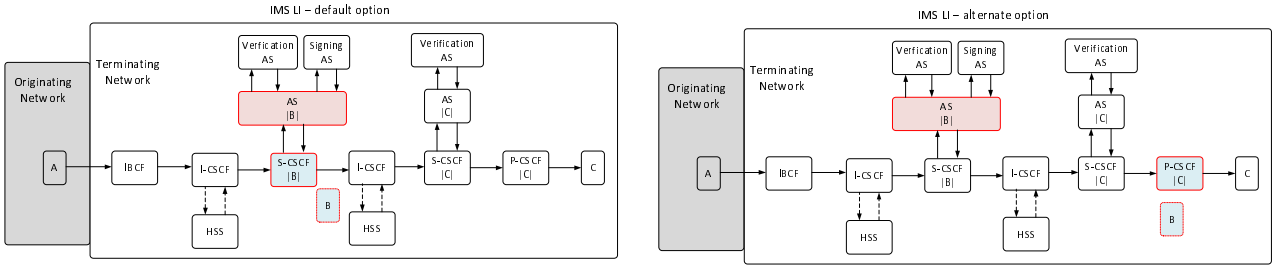


Figure 4.3.5.3.1-1 Scenario 1 - A calls B (target)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B.

Note that this scenario same as the scenario 2 illustrated in clause 4.3.5.2.1.

Scenario 2: A (target non-local ID) calls B

A (originating network) calls B (terminating network). This is an inter-CSP IMS session. A is a target non-local ID in the originating network.

The IBCF provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

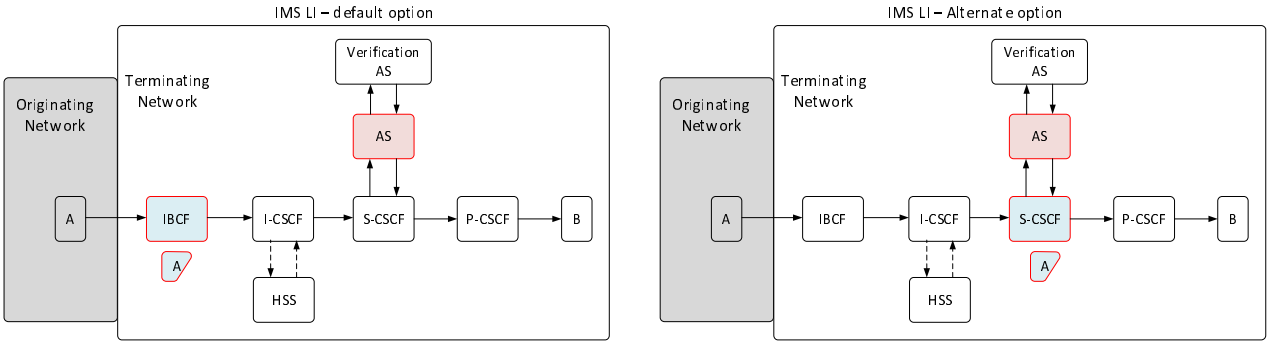


Figure 4.3.5.3.1-2: Scenario 2 – A (target non-local ID) calls B

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target non-local ID A.

Note that this scenario same as the scenario 3 illustrated in clause 4.3.5.2.1.

4.3.5.3.2 Complex scenarios

Scenario 1: Intra-CSP IMS session. A calls B (target)

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

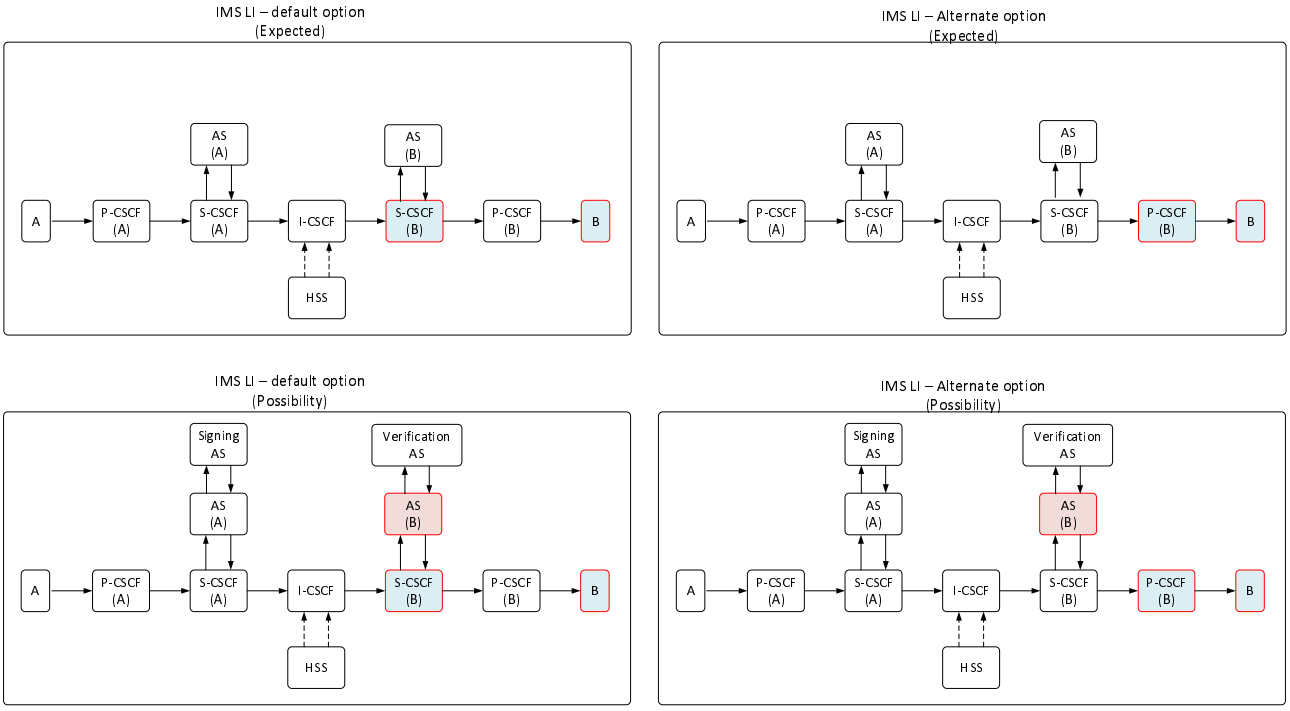


Figure 4.3.5.3.2-1: Scenario 1 – A calls B (target)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option. From STIR/SHAKEN perspective, the top part of the diagram shows the expected system behaviour since this is an intra-CSP IMS session and the signing/verification for an intra-CSP IMS session is not required. The bottom part of the diagram shows what could possibly happen with the CSP choice of AS for interacting with the Signing AS and Verification AS.

When the CSP choice is AS to interact with the Signing AS, the AS (A) may authenticate/sign A's identity if it does not know that the current session is an intra-CSP IMS session. When that happens, AS (B) would interact with the Verification AS to validate the received signature in the PASSporT. In this case, the AS (B) would provide the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for target B.

Redirected sessions

Scenario 2: A calls B, redirected to C (target)

A (originating network) calls B, redirected to C. B and C are in the terminating network. C is the target.

The S-CSCF (C) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

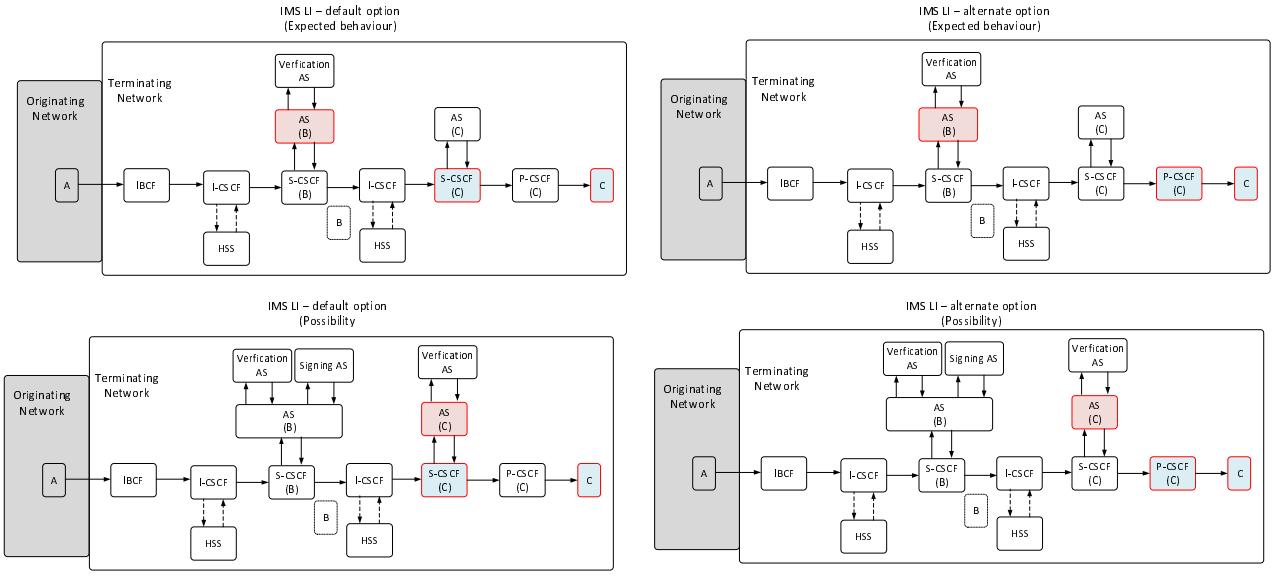


Figure 4.3.5.3.2-2: Scenario 2 – A calls B redirected to C (target)

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

From STIR/SHAKEN perspective, the top part of the diagram shows the expected system behaviour since the redirected leg of the session is an intra-CSP IMS session and the signing/verification for an intra-CSP IMS session is not required. In this case, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target C.

However, the bottom part of the diagram shows what could possibly happen with the CSP choice of AS for interacting with the Signing AS and Verification AS. When the CSP choice is AS to interact with the Signing AS, the AS (B) may authenticate/sign B's identity if it does not know that the redirected leg of the IMS session is an intra-CSP IMS session. When that happens, AS (C) would interact with the Verification AS to validate the received signature in the PASSporT. In this case, the AS (C) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target C.

Scenario 3: A calls B (target), redirected to C

A (originating network) calls B, redirected to C. B and C are in the terminating network. B is the target.

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

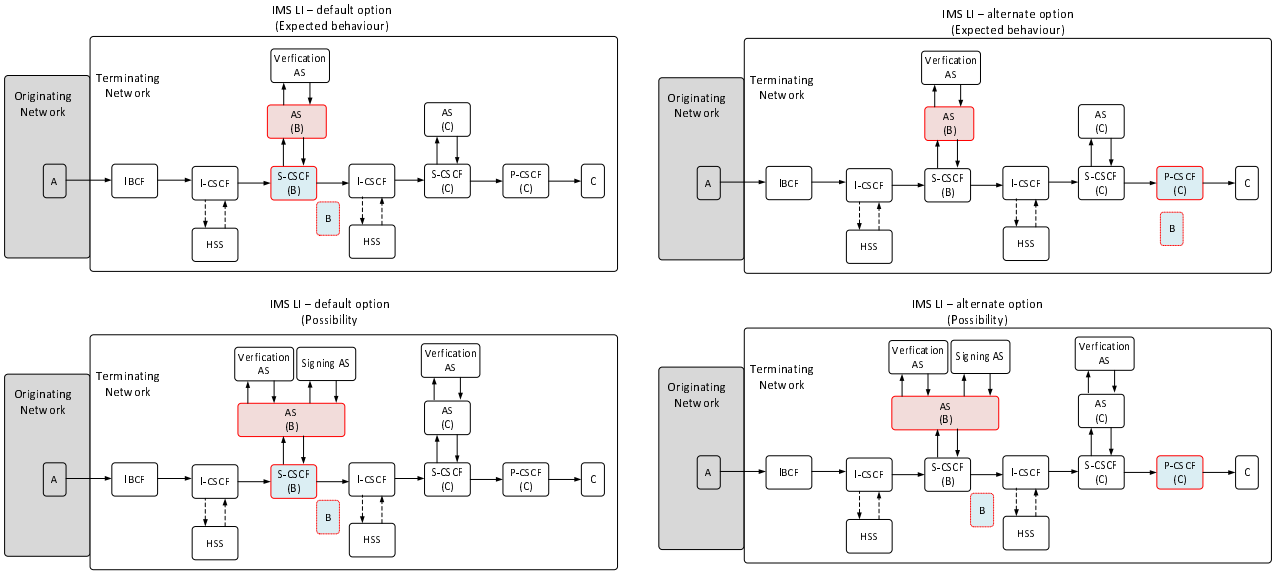


Figure 4.3.5.3.2-3: Scenario 3 – A calls B (target) redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

From STIR/SHAKEN perspective, the top part of the diagram shows the expected system behaviour since the redirected leg of the session is an intra-CSP IMS session and the signing/verification for an intra-CSP IMS session is not required. The bottom part of the diagram shows what could possibly happen with the CSP choice of AS for interacting with the Signing AS and Verification AS. When the CSP choice is AS to interact with the Signing AS, the AS (B) may authenticate/sign B's identity if it does not know that the redirected leg of the IMS session is an intra-CSP IMS session. When that happens, AS (C) would interact with the Verification AS to validate the received signature in the PASSporT.

In all cases, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target B.

In addition, the AS (B) that interacts with the Signing AS to authenticate/sign B's identity provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for the target B. Signing part of STIR/SHAKEN related LI reporting when AS interacts with the Signing AS is further illustrated in clause 4.3.4.

Scenario 4: A (target non-local ID) calls B, redirected to C

A (originating network) calls B, redirected to C. B and C are in the terminating network. A is the target non-local ID in the terminating network.

The IBCF provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

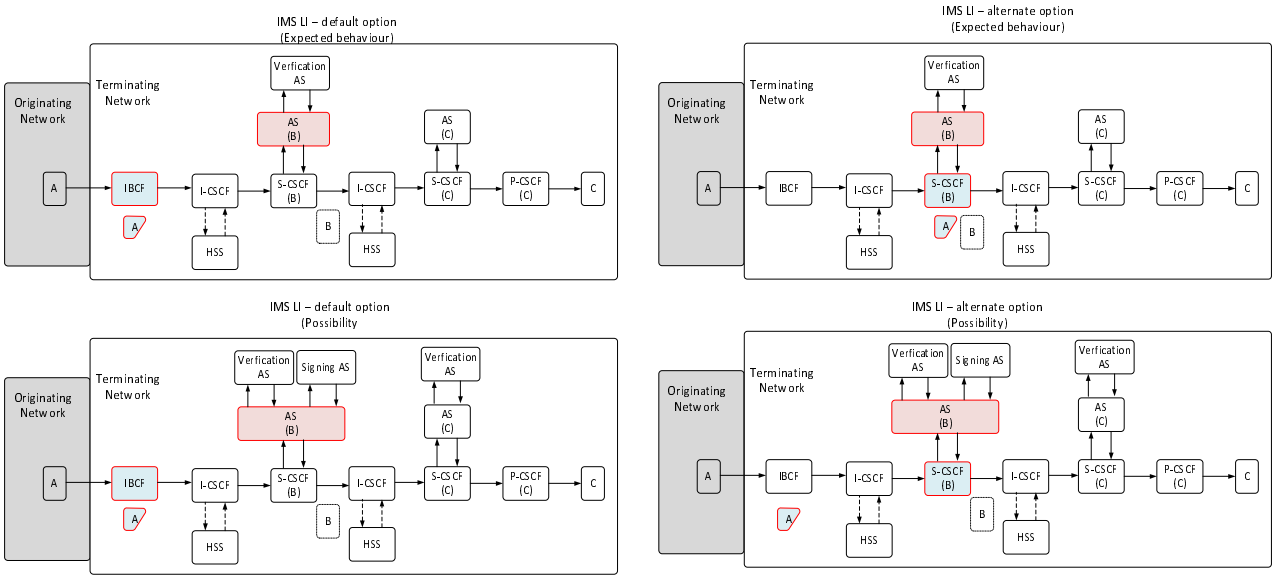


Figure 4.3.5.3.2-4: Scenario 4 – A (target non-local ID) calls B, redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

From STIR/SHAKEN perspective, the top part of the diagram shows the expected system behaviour since the redirected leg of the session is an intra-CSP IMS session and the signing/verification for an intra-CSP IMS session is not required. The bottom part of the diagram shows what could possibly happen with the CSP choice of AS for interacting with the Signing AS and Verification AS. When the CSP choice is AS to interact with the Signing AS, the AS (B) may authenticate/sign B's identity if it does not know that the redirected leg of the session is an intra-CSP IMS session. When that happens, AS (C) would interact with the Verification AS to validate the received signature in the PASSporT.

In all cases, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target non-local ID A.

Scenario 5: A (target non-local ID) calls B (target), redirected to C (target)

A (originating network) calls B, redirected to C. B and C are in the terminating network. A is the target non-local ID in the terminating network. B and C are also targets in the terminating network

For target non-local ID A: The IBCF provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

For target B: The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

For target C: The S-CSCF (C) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

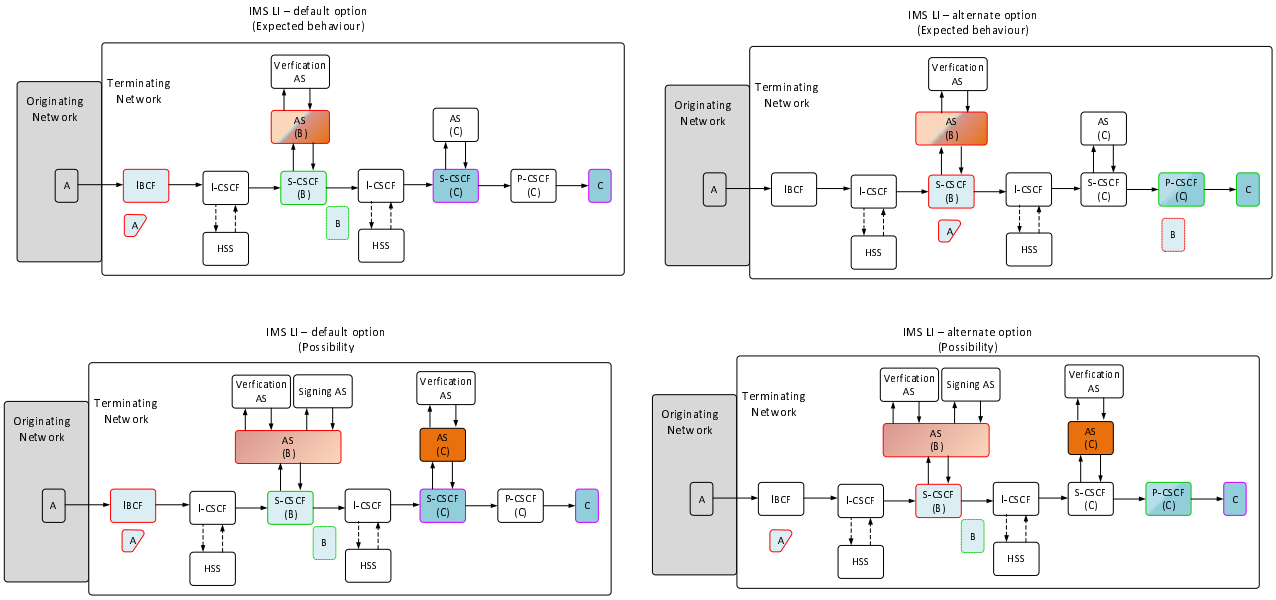


Figure 4.3.5.3.2-5: Scenario 5 – A (target non-local ID) calls B (target), redirected to C (target)

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

From STIR/SHAKEN perspective, the top part of the diagram shows the expected system behaviour since the redirected leg of the session is an intra-CSP IMS session and the signing/verification for an intra-CSP IMS session is not required. In this case, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target non-local ID A, target B and target C.

The bottom part of the diagram shows what could possibly happen with the CSP choice of AS for interacting with the Signing AS and Verification AS. When the CSP choice is AS to interact with the Signing AS, the AS (B) may authenticate/sign B's identity if it does not know that the redirected leg of the session is an intra-CSP IMS session. When that happens, AS (C) would interact with the Verification AS to validate the received signature in the PASSporT. In this case, the AS (B) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target non-local ID A and target B. The AS (B) that interacts with the Signing AS to authenticate/sign B's identity provides the IRI-POI functions for STIR/SHAKEN related LI reporting for target B. The AS (C) that interacts with the Verification AS to validate the received signature in the PASSporT provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for the target C.

Scenario 6: Intra-CSP IMS session. A calls B, redirected to C (target)

The S-CSCF (C) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

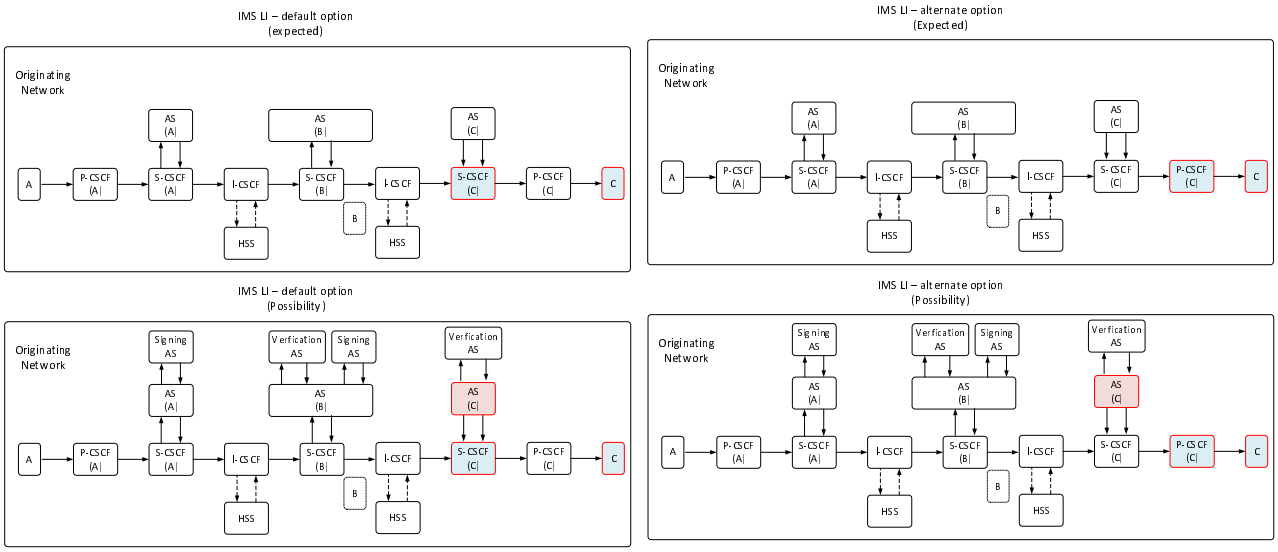


Figure 4.3.5.3.2-6: Scenario 6 – A calls B redirected to C (target)

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

From STIR/SHAKEN perspective, the top part of the diagram shows the expected system behaviour since the signing/verification is required only for the inter-CSP IMS session. As expected, if there is no interaction with the Verification AS, then the STIR/SHAKEN related LI reporting does not apply.

The bottom part and bottom part of the diagram show what could possibly happen with the CSP choice of AS for interacting with the Signing AS and Verification when the signing/verification of intra-CSP session is not required.

The AS (A) and AS (B) may not know that the current session is an intra-CSP IMS session. Assuming neither of them know that the current session is an intra-CSP IMS session, the AS (A) would interact with the Signing AS to authenticate/sign A's identity and in that case, AS (B) would interact with the Verification AS to validate the received PASSporT. If the AS (B) also does not know that the redirected leg of the session is an intra-CSP IMS session, it would also interact with the Signing AS to authenticate/sign B's identity and in that case, the AS (C) would interact with the Verification AS to validate the received PASSporT. Therefore, the AS (C) that interacts with the Verification AS to validate the received PASSporT provides the IRI-POI functions for STIR/SHAKEN related LI reporting with the validation result for target C.

With respect to this scenario, what is shown in figure 4.3.5.3.2-7 can also happen.

When C is the target, the S-CSCF (C) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

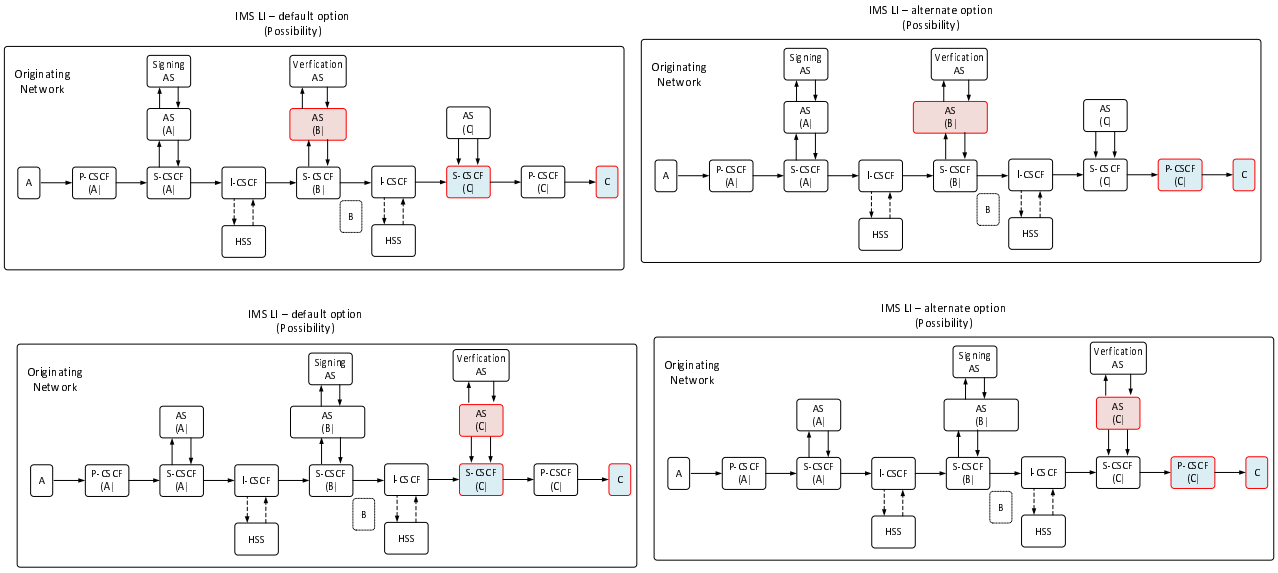


Figure 4.3.5.3.2-7: Scenario 6 (extended) – A calls B redirected to C (target), other possibilities

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

It is possible that either AS (A) or AS (B) may know that the current session is an intra-CSP IMS session. In the top part of figure 410.3.5.3.2-7 the AS (A) does not know it is an intra-CSP IMS session, but the AS (B) does. Accordingly, the AS (A) may authenticate/sign A's identity as it does not know that B is in the same network. When that happens, AS (B) would interact with the Verification AS to validate the received PASSporT. The AS (B) would provide IRI-POI functions for the STIR/SHAKEN related LI reporting with Validation Result for target C.

In the bottom part of the diagram in figure 4.3.5.3.2-7, the AS (A) knows that it is an intra-CSP IMS session, but AS (B) does not. Accordingly, AS (A) does not interact with the Signing AS to authenticate/sign A's identity since signing of intra-CSP IMS session is not required. However, the AS (B) would interact with the Signing AS to sign A's identity and B's identity as it does not know about the redirected leg of the IMS session is an intra-CSP IMS session. When that happens, AS (C) would interact with the Verification AS to validate the received PASSporT and would provide the IRI-POI functions for STIR/SHAKEN related LI reporting with the Validation Result for target C.

Scenario 7: Intra-CSP IMS session. A calls B (target), redirected to C

A calls B. B has a call forwarding to C. B is the target.

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

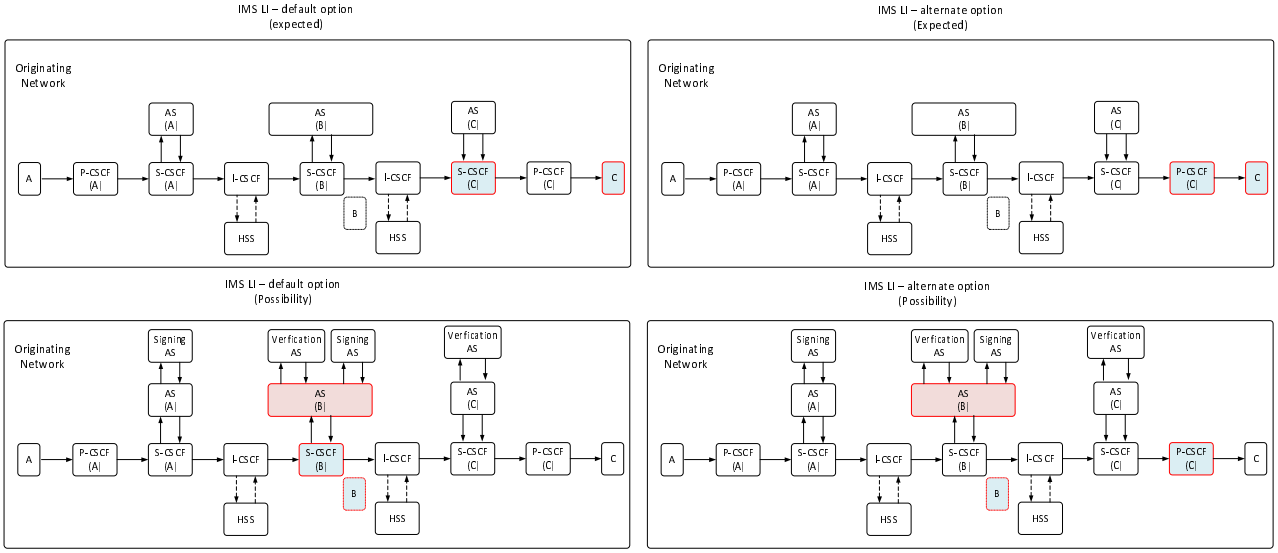


Figure 4.3.5.3.2-8: Scenario 7 –A calls B (target) redirected to C

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

From STIR/SHAKEN perspective, the top part of the diagram shows the expected system behaviour since the signing/verification is required only for the inter-CSP IMS session. As expected, if there is no interaction with the Verification AS, then the STIR/SHAKEN related LI reporting does not apply.

The bottom part of the diagram shows what could possibly happen with the CSP choice of AS for interacting with the Signing AS and Verification when the signing/verification of intra-CSP session is not required.

The AS (A) and AS (B) may not know that the current session is an intra-CSP IMS session. Assuming neither of them know that the current session is an intra-CSP IMS session, the AS (A) would interact with the Signing AS to authenticate/sign A's identity and in that case, AS (B) would interact with the Verification AS to validate the received PASSporT. The AS (B) provides the IRI-POI functions for STIR/SHAKEN related LI reporting with the Validation Result for target B.

If the AS (B) also does not know that the redirected leg of the session is an intra-CSP IMS session, it would also interact with the Signing AS to authenticate/sign B's identity. In this case, the IRI-POI in AS (B) would provide STIR/SHAKEN related LI reporting for target B. The AS (C) would interact with the Verification AS to validate the received PASSporT.

With respect to this scenario, what is shown in figure 4.3.5.3.2-9 can also happen.

When B is the target, the S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF (C) provides the IRI-POI functions for IMS LI with the alternate option.

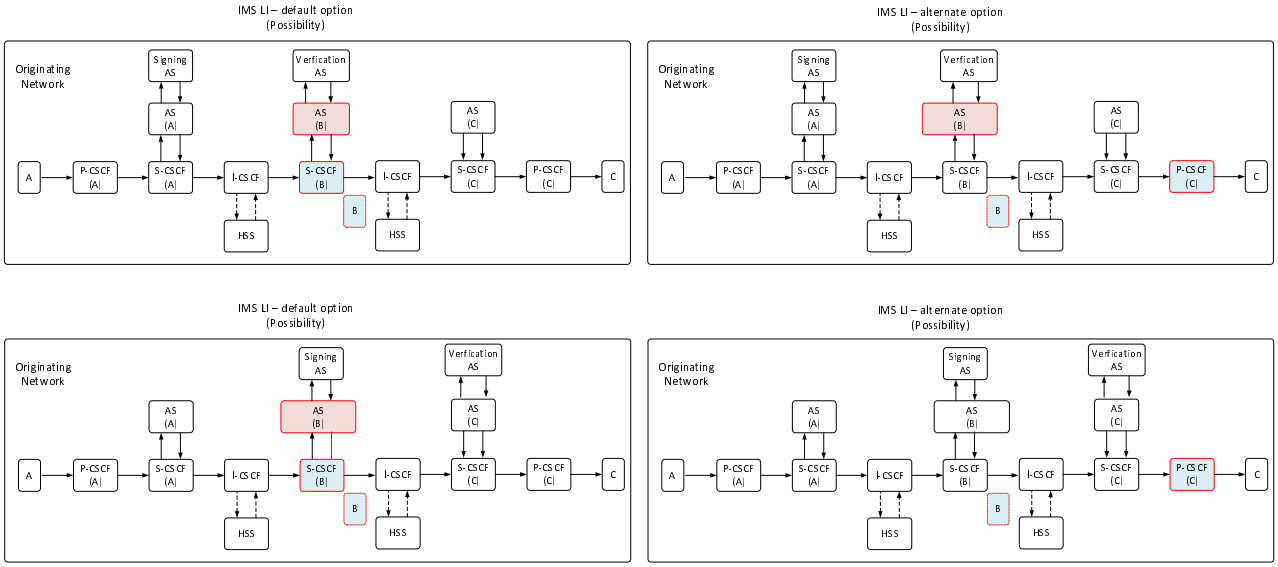


Figure 4.3.5.3.2-9: Scenario 7 (extended) – A calls B (target) redirected to C, other possibilities

The left part of diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

It is possible that either AS (A) or AS (B) may know that the current session is an intra-CSP IMS session. In the top part of figure 410.3.5.3.2-9 the AS (A) does not know it is an intra-CSP IMS session, but the AS (B) does. Accordingly, the AS (A) may authenticate/sign A's identity as it does not know that B is in the same network. When that happens, AS (B) would interact with the Verification AS to validate the received PASSporT. The AS (B) would provide IRI-POI functions for the STIR/SHAKEN related LI reporting with Validation Result for target B.

In the bottom part of the diagram in figure 4.3.5.3.2-9, the AS (A) knows that it is an intra-CSP IMS session, but AS (B) does not. Accordingly, AS (A) does not interact with the Signing AS to authenticate/sign A's identity since signing of intra-CSP IMS session is not required. However, the AS (B) would interact with the Signing AS to sign A's identity and B's identity as it does not know that the redirected leg of the IMS session is an intra-CSP IMS session. In this case, the AS (B) would provide the IRI-POI functions for STIR/SHAKEN related LI reporting for target B. The AS (C) would interact with the Verification AS to validate the received PASSporT.

4.3.6 Intermediate IMS network

4.3.6.1 General

The illustrations shown in this clause apply to an intermediate IMS network in the sense an incoming call from an originating network is redirected as an outgoing call to the terminating network.

The STIR/SHAKEN related LI reporting for the incoming part of the IMS session follow the terminating IMS network procedures as illustrated in clauses 4.3.3 and 4.3.5.

The STIR/SHAKEN related LI reporting for the outgoing part of the IMS session follow the originating IMS network procedures as illustrated in clauses 4.3.2 and 4.3.4.

4.3.6.2 Telephony AS interacts with the Signing/Verification AS

Within the drawings, AS (B) represents the Telephony AS.

Scenario 1: A calls B (target), redirected to C

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the IBCF (Egress) provides the IRI-POI functions for IMS LI with the alternate option.

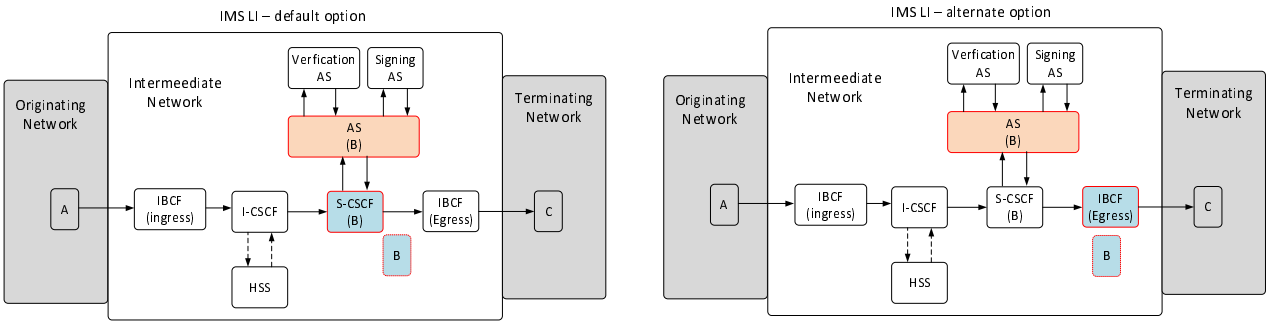


Figure 4.3.6.2-1: Scenario 1 – A calls B (target), redirected to C

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Signing AS and the Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for target B.

Scenario 2: A (target non-local ID) calls B, redirected to C

The IBCF (Ingress) provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

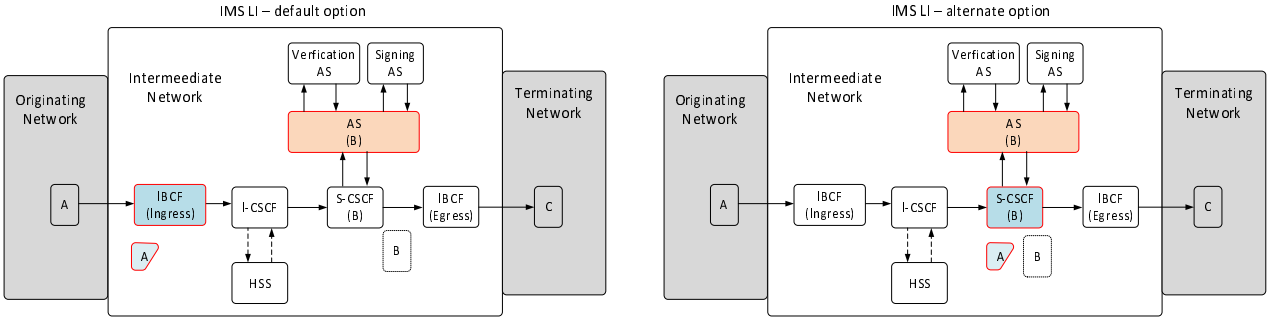


Figure 4.3.6.2-2: Scenario 2 – A (target non-local ID) calls B, redirected to C

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

In both cases, the AS (B) that interacts with the Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for target non-local ID A.

When the ReportDivPASSporTInfo provisioned for target non-local ID A is true, AS (B) that interacts with the Signing AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for target non-local ID A.

Scenario 3: A calls B, redirected to C (target non-local ID)

The IBCF (Egress) provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

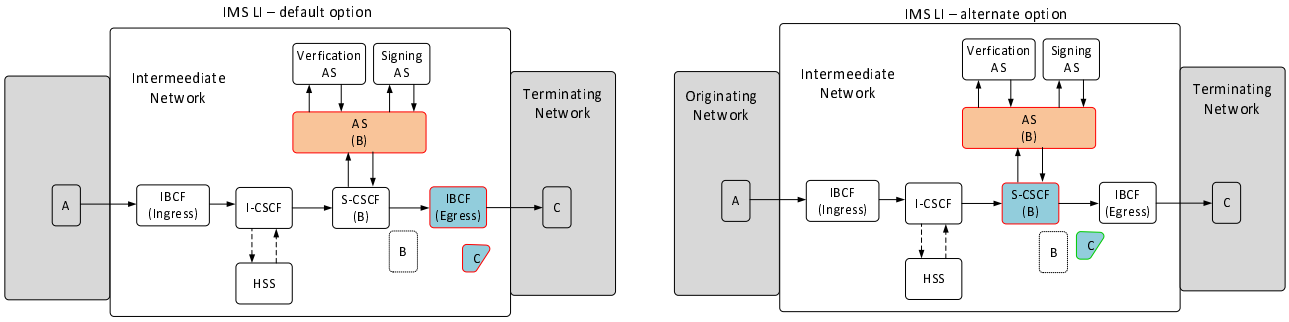


Figure 4.3.6.2-3: Scenario 3 – A calls B, redirected to C (target non-local ID)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

The AS (B) that interacts with the Signing AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for target non-local ID C.

Scenario 4: A (target non-local ID) calls B, redirected to C (target non-local ID)

For target non-local ID A: The IBCF (Ingress) provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

For target non-local ID C: The IBCF (Egress) provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

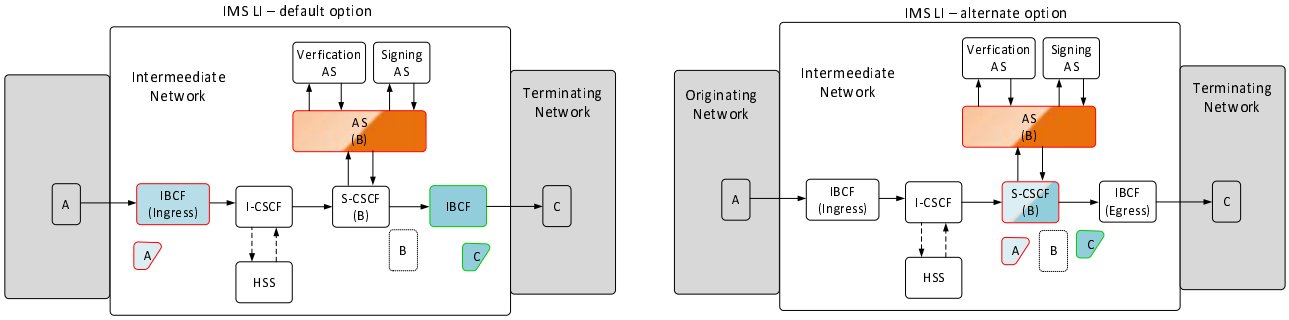


Figure 4.3.6.2-4: Scenario 4 – A (target non-local ID) calls B, redirected to C (target non-local ID)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

The AS (B) that interacts with the Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for target non-local ID A.

The AS (B) that interacts with the Signing AS provides the STIR/SHAKEN related LI reporting for target non-local ID C and for target non-local ID A (when the ReportDivPASSporTInfo provisioned for target non-local ID A is true).

Scenario 5: A (target non-local ID) calls B (target), redirected to C (target non-local ID)

For target non-local ID A: The IBCF (Ingress) provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

For target B: The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the IBCF (Egress) provides the IRI-POI functions for IMS LI with the alternate option.

For target non-local ID C: The IBCF (Egress) provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

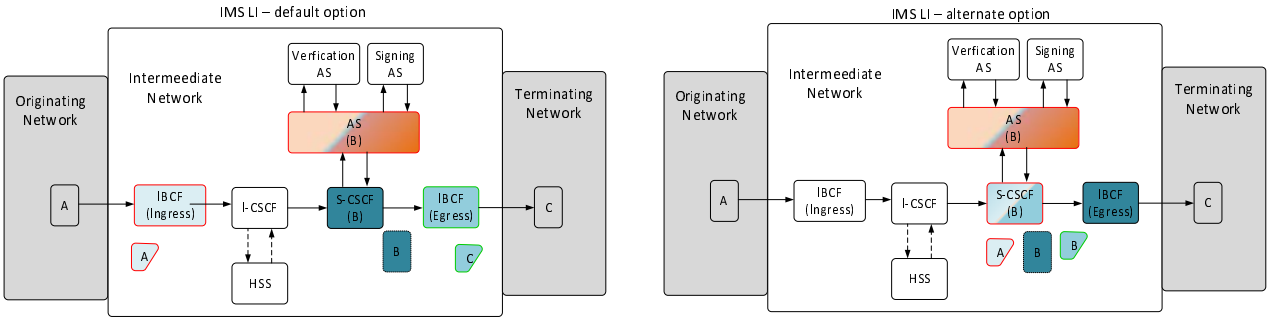


Figure 4.3.6.2-5: Scenario 5 – A (target non-local ID calls B (target), redirected to C (target non-local ID)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

The AS (B) that interacts with Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for target non-local ID A and target B.

The AS (B) that interacts with the Signing AS provides the STIR/SHAKEN related LI reporting for target non-local ID C, target B and target non-local ID A (when the ReportDivPASSporTInfo provisioned for target non-local ID A is true).

4.3.6.3 IBCF interacts with the Signing/Verification AS

Within the drawings, AS (B represents the Telephony AS.

Scenario 1: A calls B (target), redirected to C

The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the IBCF (Egress) provides the IRI-POI functions for IMS LI with the alternate option.

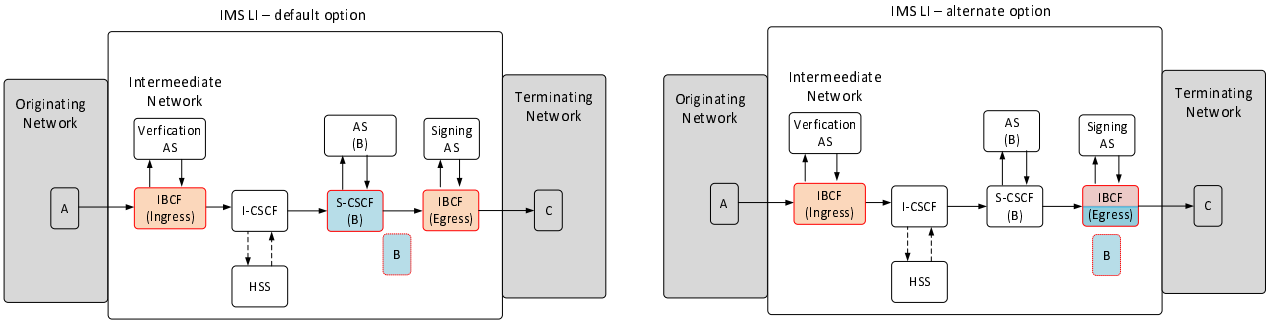


Figure 4.3.6.3-1: Scenario 1 – A calls B (target), redirected to C

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

The IBCF (Ingress) that interacts with the Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for target B.

The IBCF (Egress) that interacts with the Signing AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for target B.

Scenario 2: A (target non-local ID) calls B, redirected to C

The IBCF (Ingress) provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

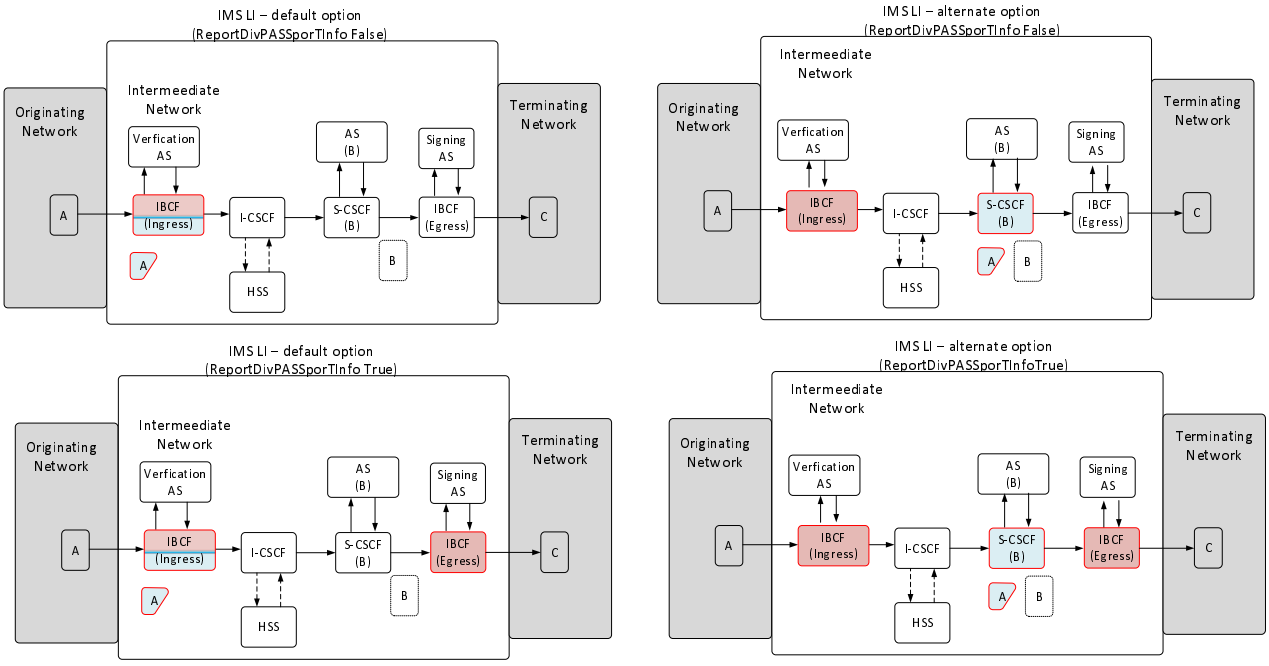


Figure 4.3.6.3-2: Scenario 2 – A (target non-local ID) calls B, redirected to C

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

The IBCF (Ingress) that interacts with the Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for target non-local ID A.

When the ReportDivPASSporTInfo provisioned for target non-local ID A is true, IBCF (Egress) that interacts with the Signing AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for target non-local ID A.

Scenario 3: A calls B, redirected to C (target non-local ID)

The IBCF (Egress) provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

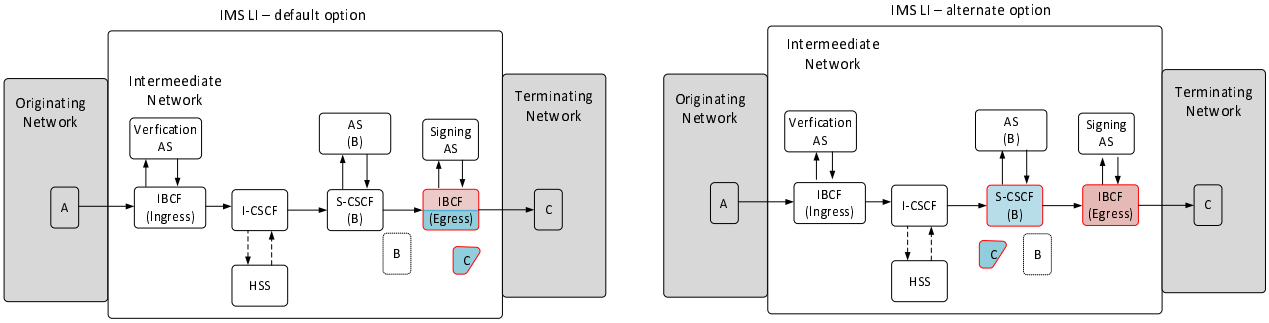


Figure 4.3.6.3-3: Scenario 3 – A calls B, redirected to C (target non-local ID)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

The IBCF (B) that interacts with the Signing AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for target non-local ID C.

Scenario 4: A (target non-local ID) calls B, redirected to C (target non-local ID)

For target non-local ID A: The IBCF (Ingress) provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

For target non-local ID C: The IBCF (Egress) provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

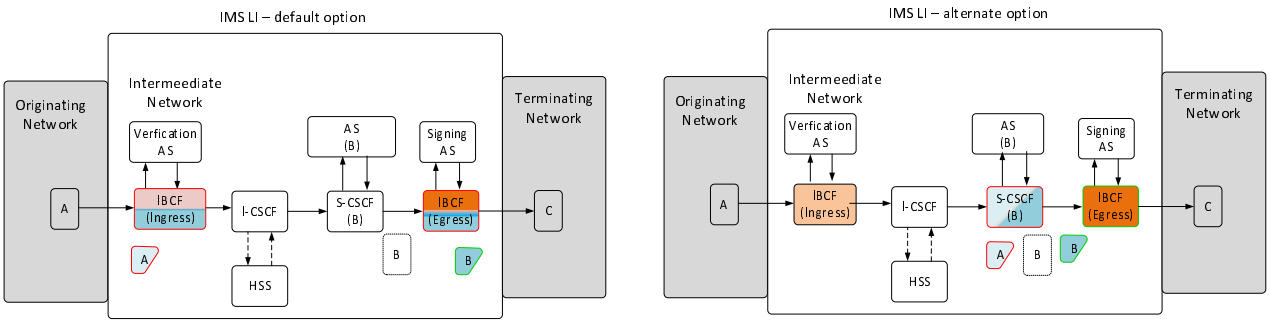


Figure 4.3.6.3-4: Scenario 4 – A (target local ID calls B, redirected to C (target local ID)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

The IBCF (Ingress) that interacts with the Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for target non-local ID A.

The IBCF (Egress) that interacts with the Signing AS provides the STIR/SHAKEN related LI reporting for target non-local ID C and for target non-local ID A (when the ReportDivPASSporTInfo provisioned for target non-local ID A is true).

Scenario 5: A (target non-local ID) calls B (target), redirected to C (target non-local ID)

For target non-local ID A: The IBCF (Ingress) provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

For target B: The S-CSCF (B) provides the IRI-POI functions for IMS LI with the default option, and the IBCF (Egress) provides the IRI-POI functions for IMS LI with the alternate option.

For target non-local ID C: The IBCF (Egress) provides the IRI-POI functions for IMS LI with the default option, and the S-CSCF (B) provides the IRI-POI functions for IMS LI with the alternate option.

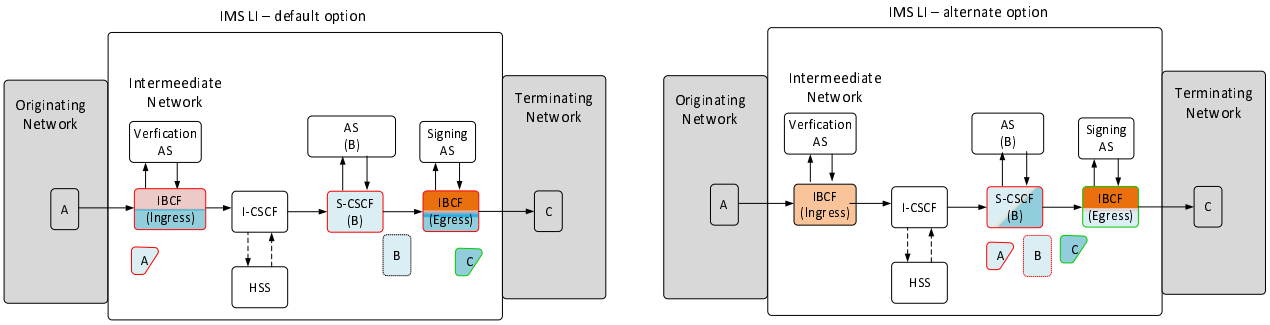


Figure 4.3.6.3-5: Scenario 5 – A (target non-local ID) calls B (target) redirected to C (target non-local ID)

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

The IBCF (Ingress) that interacts with Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for target non-local ID A and target B.

The IBCF (Egress) that interacts with the Signing AS provides the STIR/SHAKEN related LI reporting for target non-local ID C, target B and target non-local ID A (when the ReportDivPASSporTInfo provisioned for target non-local ID A is true).

4.3.7 Emergency sessions

The illustrations shown in this clause apply to emergency sessions made by a target.

Scenario 1: A (target) makes an emergency call

The E-CSCF provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

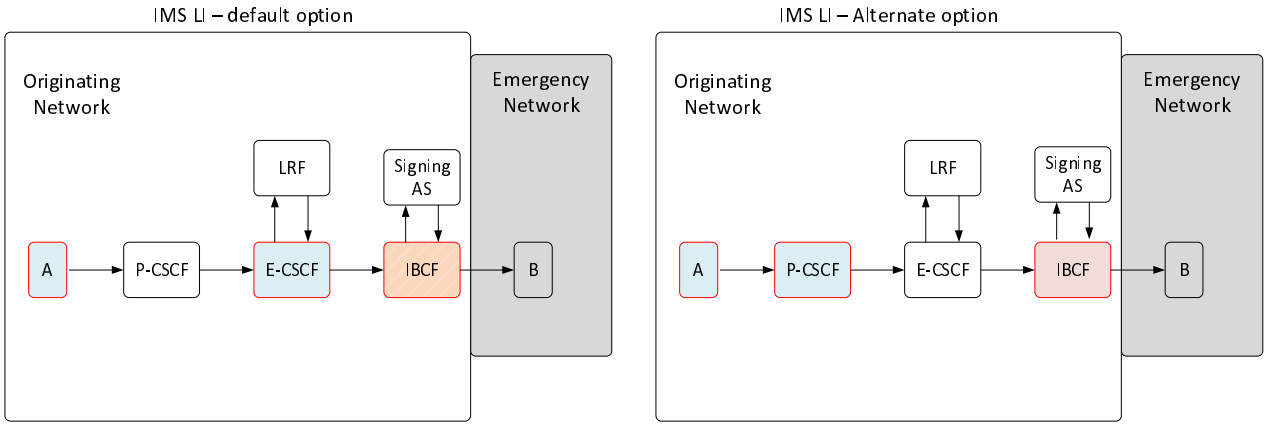


Figure 4.3.7-1: Scenario 1 – A (target) makes an emergency call

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

The Egress IBCF that interacts with the Signing AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting for target A.

4.3.8 Emergency callback

The illustrations shown in this clause apply to emergency call backs to the target.

Scenario 1: A (target) receives an emergency call back, Telephony AS interacts with the Verification AS

The target A receives an emergency callback from the PSAP.

The S-CSCF provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

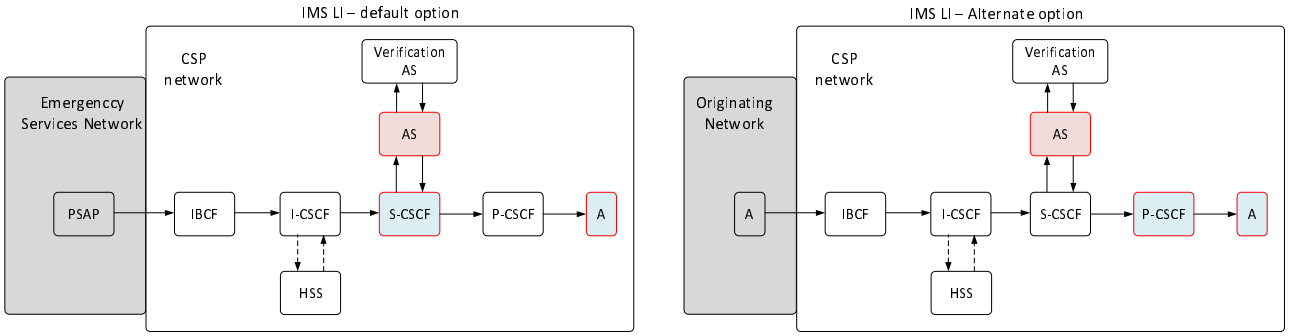


Figure 4.3.8-1: Scenario 1 – A (target) receives an emergency callback

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

The AS that interacts with the Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for target A.

Scenario 2: A (target) receives an emergency call back, IBCF interacts with the Verification AS

The target A receives an emergency callback from the PSAP.

The S-CSCF provides the IRI-POI functions for IMS LI with the default option, and the P-CSCF provides the IRI-POI functions for IMS LI with the alternate option.

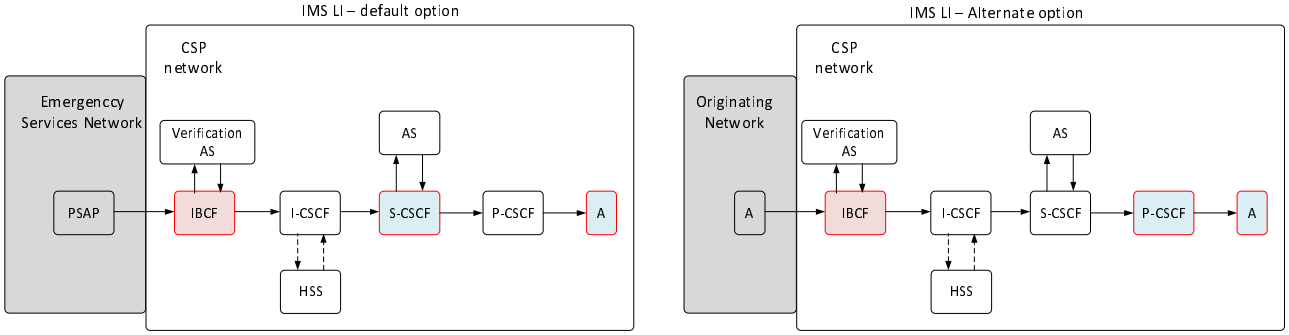


Figure 4.3.8-2: Scenario 2 – A (target) receives an emergency callback

The left part of the diagram shows for IMS LI with default option and the right part of the diagram shows for IMS LI with alternate option.

The Ingress IBCF that interacts with the Verification AS provides the IRI-POI functions for the STIR/SHAKEN related LI reporting with the Validation Result for target A.

4.4 LI reporting details

4.4.1 General

In the illustrations shown in the subsequent clauses, the IRI-POIs present in the Telephony AS or the IBCF that interacts with the Signing AS or the Verification AS would generate the xIRIs for STIR/SHAKEN related reporting. In a roaming scenario, the IRI-POIs present in the P-CSCF or the LMISF-IRI in the terminating end of the VPLMN would generate the xIRIs for the STIR/SHAKEN verification related reporting.

In the illustrations, the Telephony AS is shown as AS, Party A is shown as A, Party B is shown as B, Party C is shown as C, Party D is shown as D.

Some of the other guidelines that are used within the diagrams are listed in the following notes:

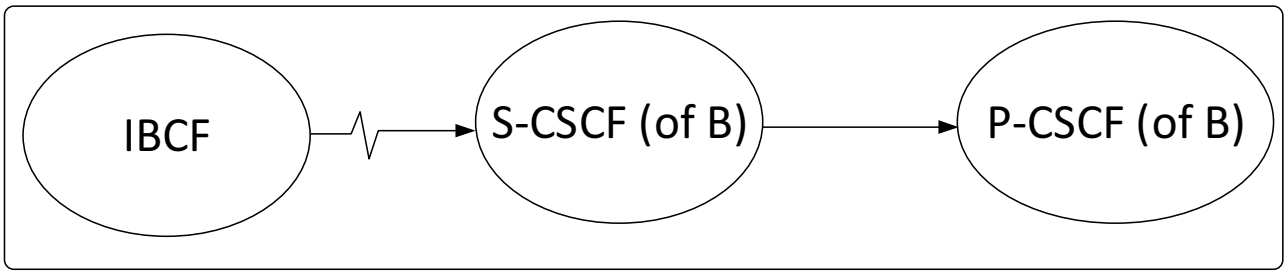
NOTE 1: The illustrations show STIR/SHAKEN related LI reporting along with the IMS LI related reporting for various scenarios, at a conceptual level. In the illustrations, IMSMessage represents the xIRI/IRI for the IMS LI and xIRISTIRSHAKENSignatureGeneration and xIRISTIRSHAKENSignatureValidation represent the xIRIs and STIRSHAKENSignatureGeneration and STIRSHAKENSignatureValidation represent the IRIs for the STIR/SHAKEN related LI reporting.

NOTE 2: Only the cases that may be useful to understand the STIR/SHAKEN related LI reporting and its acquaintance with the IMS LI are illustrated. In other words, not every scenario illustrated in clause 4.3 has an equivalent illustration here.

NOTE 3: The illustrations do not show all the IMS NFs involved in the signaling path. The illustrations do not show the media interceptions.

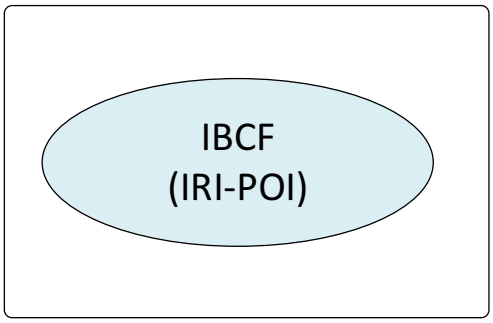
NOTE 4: The diagrams shown in the subsequent clauses are presumed to imply the transfer of SIP INVITEs. The diagrams deliberately skip some of the IMS NFs that are part of an IMS session setup. To infer that when the two IMS NFs do not interact directly, a curly line is used between them. When the two IMS NFs interact directly, a straight line is used.

Accordingly, in the below presentation, the IBCF does not interact with the S-CSCF (of B) directly (e.g. SIP INVITE goes through the I-CSCF before hitting the S-CSCF (of B)) whereas the S-CSCF (of B) interacts with the P-CSCF (of B) directly.



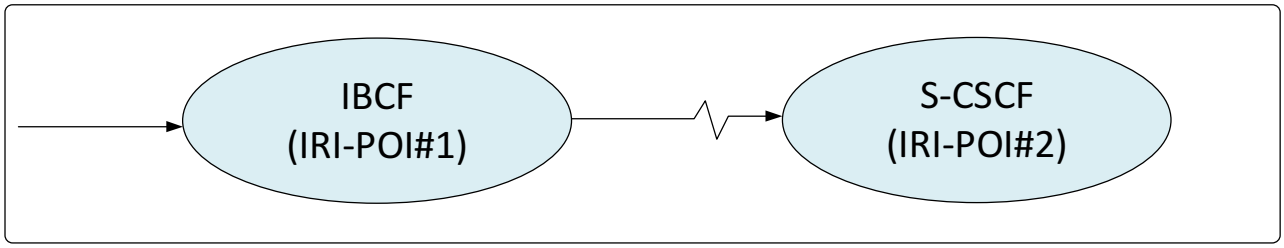
NOTE 5: When an IMS NF has the LI functions, that IMS NF is shown with blue coloured shade.

In the below presentation, the IRI-POI is present in the IBCF.



NOTE 6: Within the illustrations, #1, #2, etc. are used as in (IRI-POI#1), (IRI-POI#2) to differentiate the two distinct IRI-POIs within the same illustrations (for the same IMS session).

In the below presentation, IRI-POI#1 in IBCF and IRI-POI#2 in S-CSCF provide the LI functions for the same IMS session.

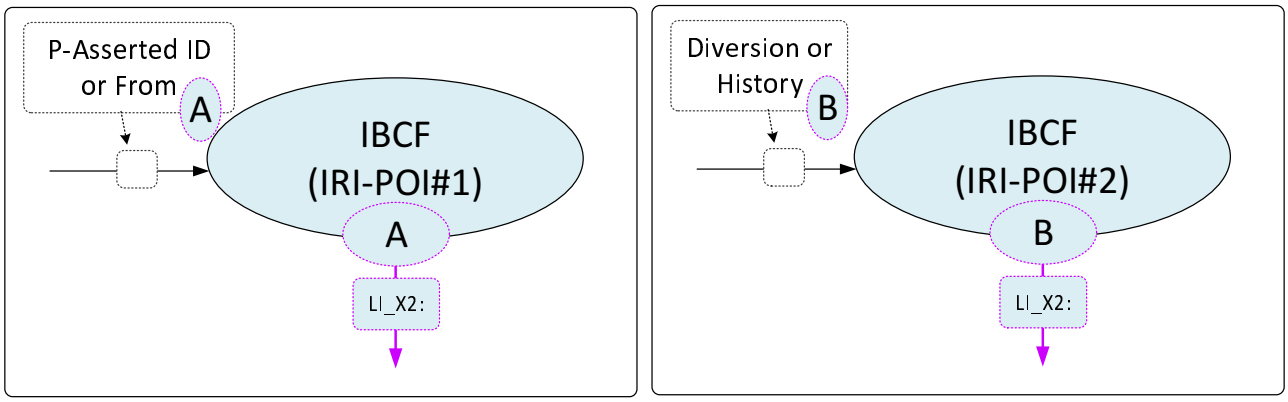


NOTE 7: The diagrams show the SIP headers from the SIP INVITE used in performing a target match. The xIRIs generated from the IRI-POIs are delivered over the LI_X2 interface for that target.

In the below presentation, the IRI-POI#1 in IBCF uses the P-Asserted ID (A) or the From (A) headers for a target match and delivers the generated xIRIs over LI_X2 for the target A.

Likewise, in the below presentation, the IRI-POI #2 uses the Diversion (B) or History Info (B) headers for a target match and delivers the xIRIs over LI_X2 for target B.

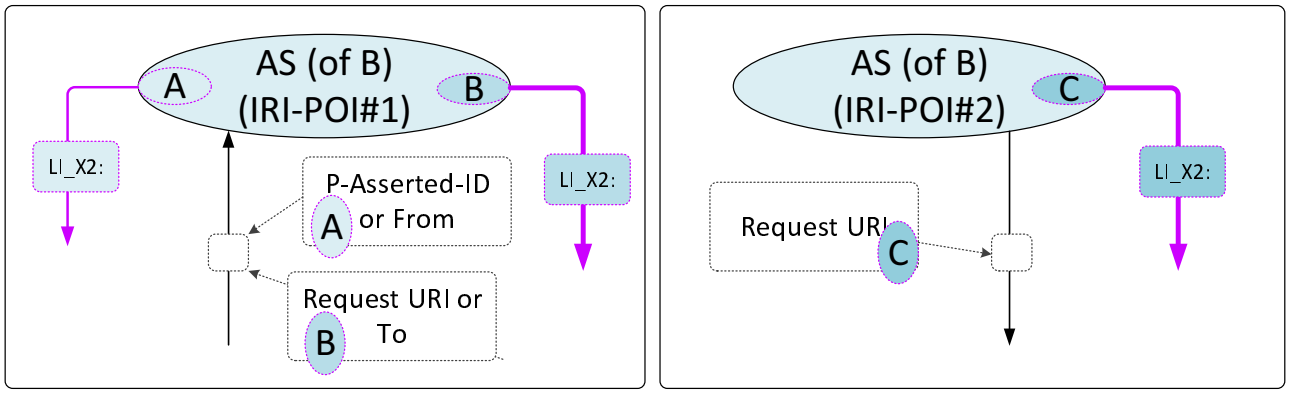
A purple colour indicates that it is specific to LI.



NOTE 8: The NF may provide the IRI-POI functions for more than one targets and the identities used for a target match is be taken from the incoming SIP INVITE unless specifically indicated to say the identities are taken from the outgoing SIP INVITE.

In the below illustration, the IRI-POI#1 in AS (of B) uses P-Asserted ID (A) or From (A) and Request URI (B) or To (B) from the incoming SIP INVITE. The xIRIs are delivered over LI_X2 for target A and target B. The LI line has differing thickness.

Also, in the below presentation, the IRI-POI#2 in AS (of B) uses the SIP header Request URI (C) from the outgoing SIP INVITE for a target match. The xIRIs are delivered over LI_X2 for target C. The IRI-POI#1 uses the SIP headers from the incoming SIP INVITE. For redirection cases to perform a target match on the new destination, the To header is not used since it would be pointing to the original destination.



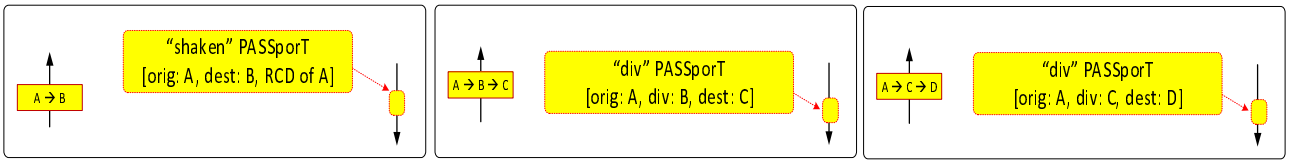
NOTE 9: The diagrams show the parties involved in the session and the type and scope of PASSporTs.

In the below presentations, A → B represents that A has called B, A → B → C represents that A is the calling party, B is the redirecting party and C is the new destination, and A → C → D represents that A is the calling party, C is the redirecting party and D is the new destination. In the A → C → D case, there is a prior redirection.

For A→ B case, in the illustration, the "shaken" PASSporT includes A in the origination claim, B in the destination claim is, and RCD of A. The descriptions may just say "shaken" PASSporT for Party A.

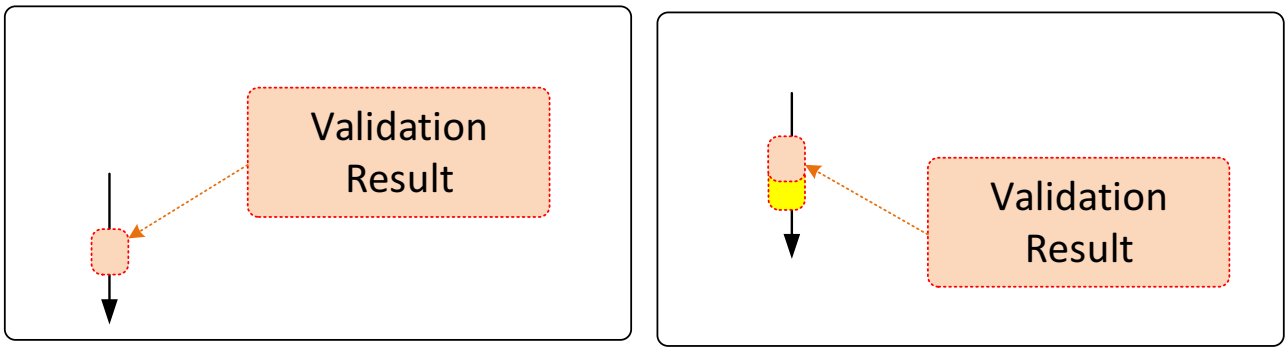
For A → B → C case, in the illustration, the "div" PASSporT includes A in the origination claim, B in the diversion claim and C in the destination claim. The descriptions may just say "div" PASSporT for Party B.

For A → C → D case, in the illustration, the "div" PASSporT includes A in the origination claim, C in the diversion claim and D in the destination claim. The descriptions may just say "div" PASSporT for Party C.



NOTE 10: The diagrams show two types of Validation Results: 1) that carries the PASSporT(s) and the one that does not carry the PASSporTs. The Validation Result without the PASSporT(s) are received from the Verification AS and delivered to the terminating UE whereas the Validation Result with the PASSporT (s) are sent in the SIP INVITEs between the two IMS NFs.

In the below illustration, the left diagram shows the Validation Result without the PASSporT (s) and the right diagram shows the Validation Result with PASSporT (s).



NOTE 11: All the illustrations show only three LI messages. The xIRIs generated/delivered from the IRI-POIs over LI_X2 and the IRIs forwarded from the MDF2 to LEMF over LI_HI2.

LI_X2: xIRIIMSMessage → LI_HI2: IMSMessage.

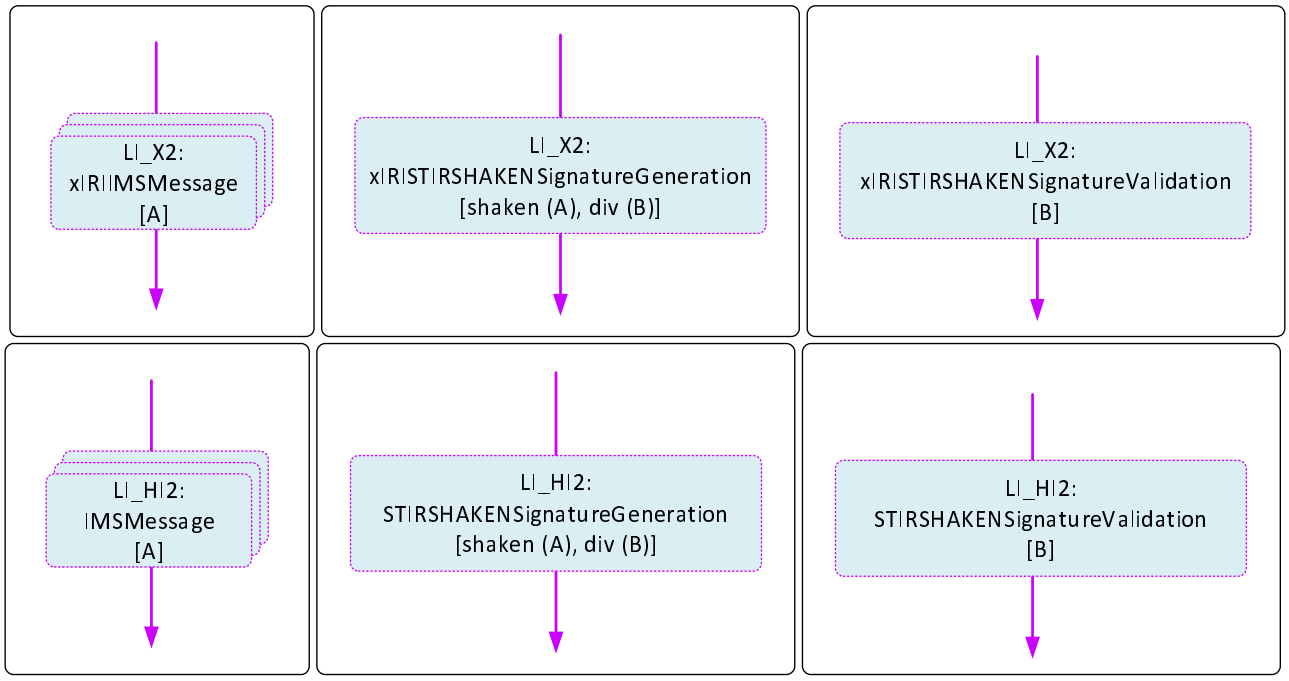
LI_X2: xIRISTIRSHAKENGeneration → LI_HI2: STIRSHAKENSignatureGeneration.

LI_X2: STIRSHAKENValidation → LI_HI2: STIRSHAKENSignatureValidation.

In the below illustration, [A] in IRIMessage [A] indicates that the LI message is for target A.

In the below illustration, the shaken (A) and div (B) within the STIRSHAKENSignatureGeneration indicate that they are the "shaken" PASSporT for A and "div" PASSporT for B, respectively. Neither of them (A, B) needs to be the targets.

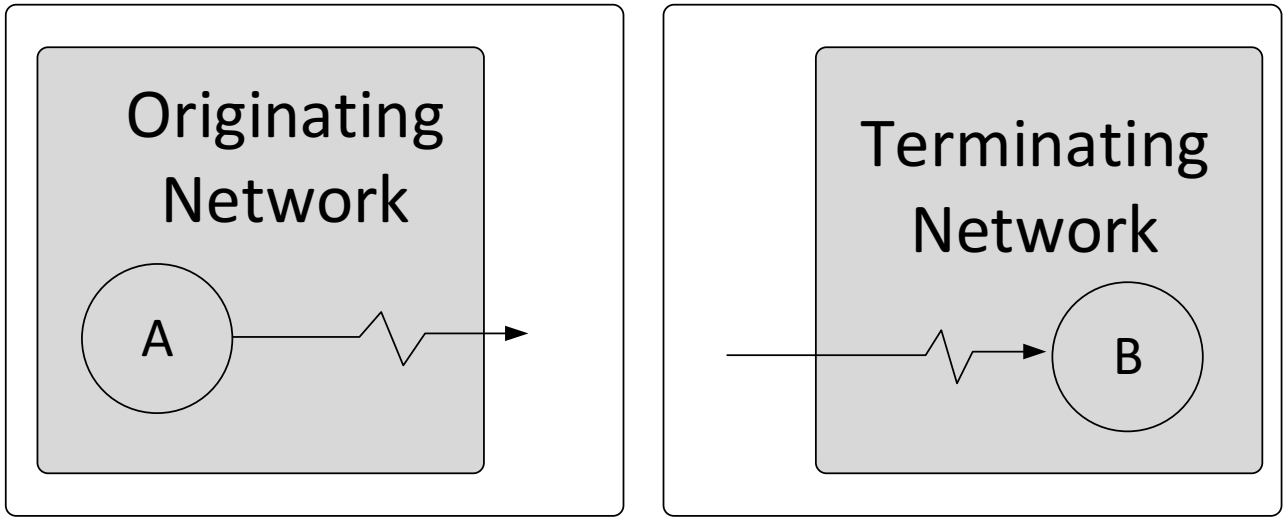
In the below illustration, [B] in STIRSHAKENSigatureValidation indicates that it is the signature validation is done for incoming call to target B.



NOTE 12: For inter-network sessions, the other networks are shown as Originating Network or Terminating Network.

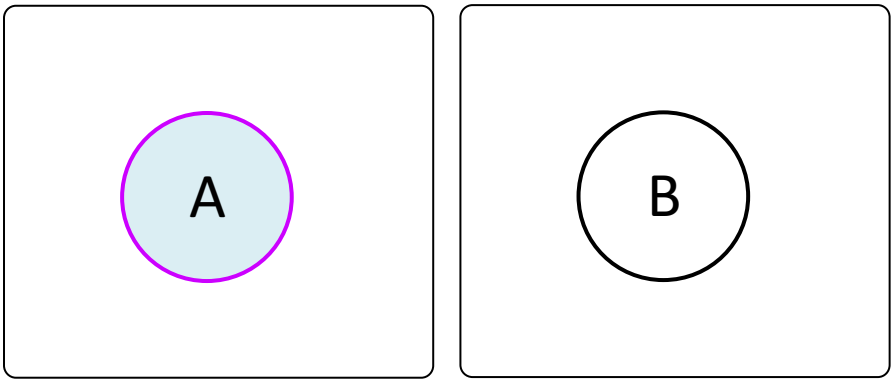
In the below illustration, Party A has originated a session from the Originating Network.

In the below illustration, Party B is the destination party in the Terminating Network.

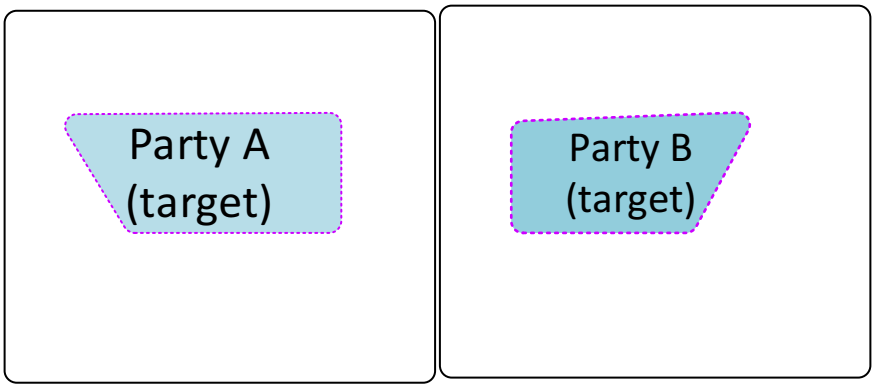


NOTE 13: A target is indicated with a blue shaded circle.

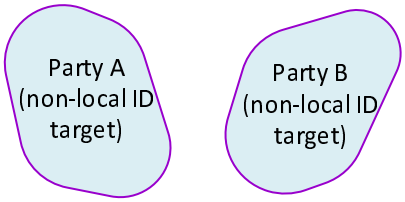
In the below illustration, Party A is a target whereas Party B is not.



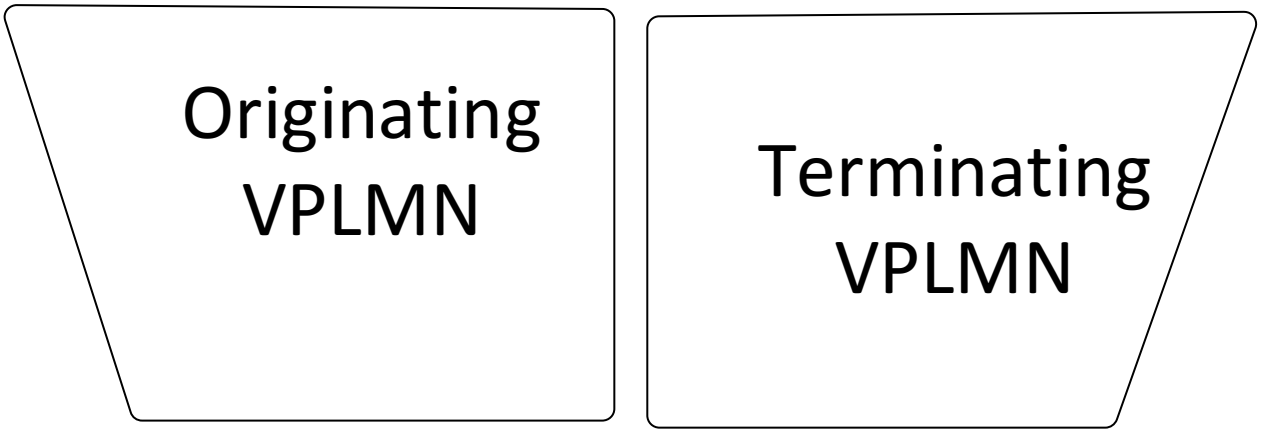
NOTE 14: An outbound roaming UE is shown with a dotted/slanted boundary. In the following Party A (originating) and Party B (terminating) and both happen to be the outbound roaming targets.



NOTE 15: A non-local ID target is shown with slanted solid boundary as shown below. The direction of the slant indicates whether a session is incoming from a non-local ID target (left) or whether the session is to the non-local ID target (right).



NOTE 16: The VPLMNs are shown with a slanted solid boundary with direction of the slant showing whether it is an originating VPLMN or terminating VPLMN.



4.4.2 General concept overview

In the illustration shown in figure 4.4.2-1, the Party A is inbound roaming in the Originating VPLMN and outbound roaming from the Originating HPLMN. Party B is outbound roaming from Terminating HPLMN and inbound roaming in the Terminating VPLMN.

Party A makes a call to Party B.

In this illustration, Party A is a target in the Originating VPLMN (LEMF-1) and Originating HPLMN (LEMF-2). Part B is a target in the Terminating HPLMN (LEMF-3) and Terminating VPLMN (LEMF-4). The illustration focusses only on the STIR/SHAKEN related reporting (in other words, the IMS related LI is outside the scope of this illustration).

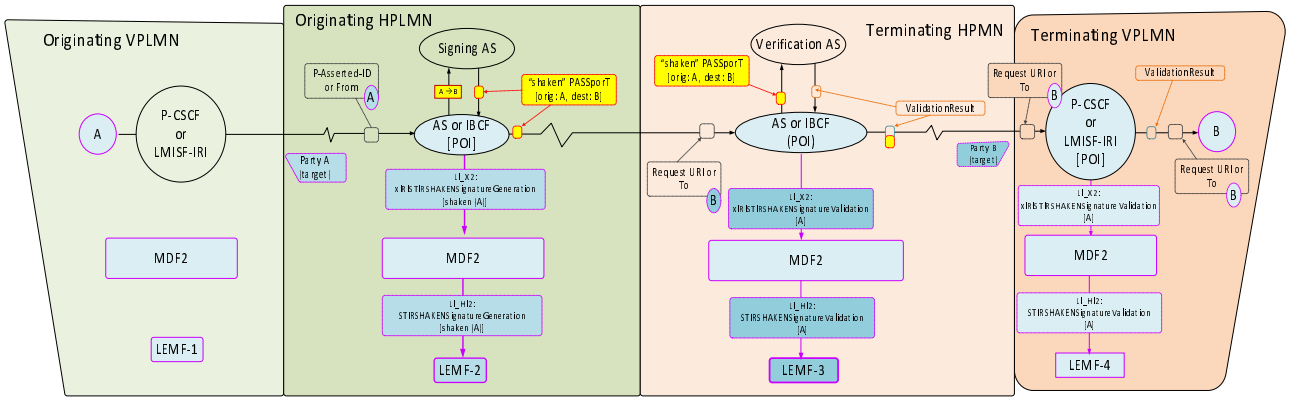


Figure 4.4.2-1: STIR/SHAKEN related reporting for signature (originating network)

Originating VPLMN

There is no STIR/SHAKEN related LI reporting happens in the Originating VPLMN since the calling party number is not signed in the Originating VPLMN. The IMS session would be intercepted, but that aspect is outside scope of this illustration.

Originating HPLMN

Telephony AS or the IBCF that interacts with the Signing AS in the Originating HPLMN includes the "shaken" PASSporT in the outgoing SIP INVITE.

Since Party A is a target, the IRI-POI present in the Telephony AS or the IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record to MDF2 over the LI_X2 interface.

The MDF2 forwards the IRI report STIRSHAKENSignatureGeneration over the LI_HI2 to the LEMF-2.

Terminating HPLMN

Telephony AS or the IBCF that interacts with the Verification AS in the Terminating HPLMN includes the received Validation Result in the outgoing SIP INVITE.

Since Party B is a target, the IRI-POI present in the Telephony AS or the IBCF generates and delivers the xIRISTIRSHAKENSignatureValidation record to MDF2 over the LI_X2 interface.

The MDF2 forwards the IRI report STIRSHAKENSignatureValidation over the LI_HI2 to the LEMF-3.

Terminating VPLMN

The LMISF-IRI (Home-routed roaming) or the IRI-POI present in the P-CSCF (Local Break-out), upon seeing the Validation Result in the SIP INVITE sent to the target would generate and deliver the xIRISTIRSHAKENSignatureValidation record to MDF2 over the LI_X2 interface.

The MDF2 forwards the IRI report STIRSHAKENSignatureValidation over the LI_HI2 to the LEMF-4.

4.4.3 General principles of STIR/SHAKEN related LI reporting

4.4.3.1 General

In general, the IMS NF that interacts with the Signing AS or Verification AS provide the IRI-POI functions for STIR/SHAKEN related LI reporting. However, there are exceptions made to this generality, mainly, to avoid any double interceptions, e.g. the case of redirections. The illustrations shown in the subsequent clauses cover these exceptions as well.

4.4.3.2 Scope of ReportDiversionPASSporTInfo

For the STIR/SHAKEN related LI reporting a Boolean flag is provisioned into the IRI-POIs to indicate whether or not, to report the subsequent "div" PASSporTs generated by the same network for the IMS session. The flag is referred to as ReportDiversionPASSporTInfo.

For example, when Party A calls Party B redirected to Party C, redirected to Party D, redirected to party E, the ReportDiversionPASSporTInfo comes into play as shown next three figures shown below:

NOTE: Party A, Party B, Party C and Party D are target in the originating network. Party E is a target non-local ID in the originating network.

Three cases shown below:

- Intra-network signing is required.
- Intra-network signing is not required, RCD is not present.
- Intra-network signing is not required, RCD is present.

Each an identity is signed/authenticated, a PASSporT ('shaken" for the calling party, "div" for the redirecting party) is generated.

The illustrations in the next three figures are based on the following three basic principles:

- When the intra-network signing is required, the Telephony AS of each party signs/authenticates the party's identities and the Telephony AS would always verify the received PASSporTs.
- When the intra-network signing is not required, either egress Telephony AS or the egress IBCF signs/authenticates identities of all parties.
- When the intra-network signing is not required but the RCD is present for the calling party, the Telephony AS of calling party signs/authenticates calling party's identity, the Telephony AS of first called party verifies the received PASSporT and after that the egress Telephony AS or the egress IBCF signs/authenticates the identities of the other redirecting parties present on the call.

When the received PASSporTs are verified, the Validation Result is sent to the LEAs of the terminating target along with the received PASSporTs. The ReportDiversionPASSporTInfo, is however, to determine the scope of PASSporTs reported when subsequent redirections happen.

The illustrations are to be viewed as the totality of PASSporTs reported to the LEMFs of each targets.

Intra-network signing is required

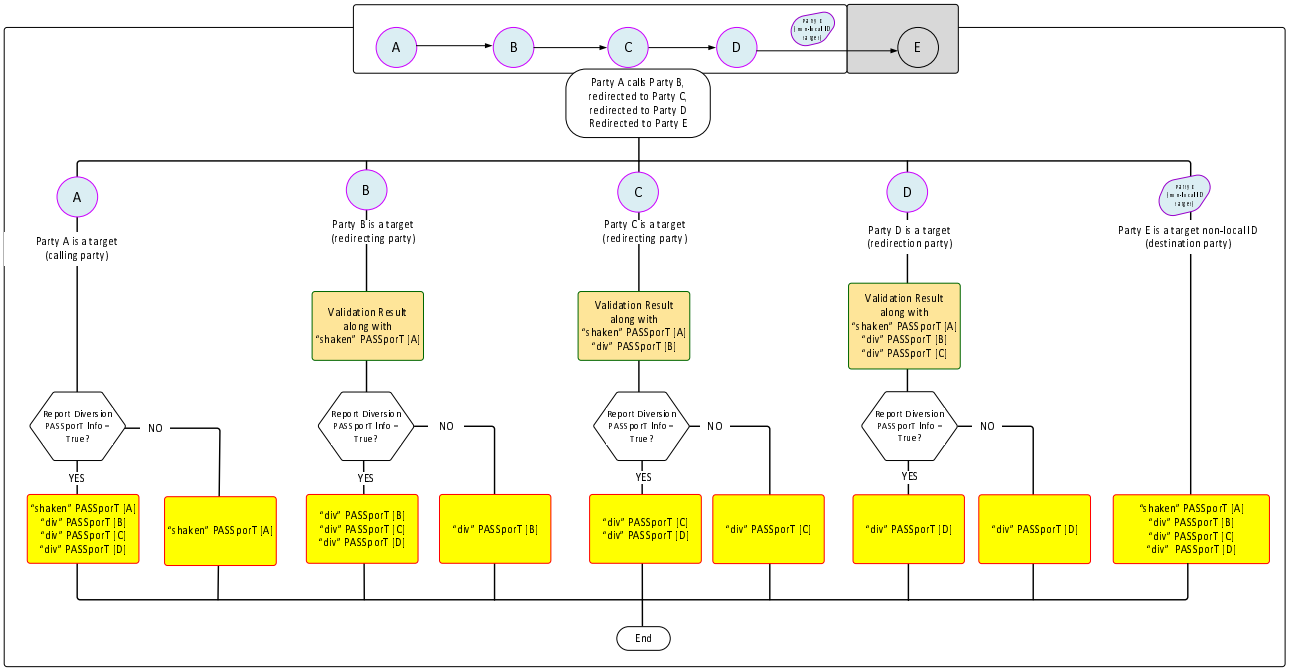


Figure 4.4.3.2-1: Scope of ReportDiversionPASSporTInfo

Intra-network signing is not required, RCD is not present

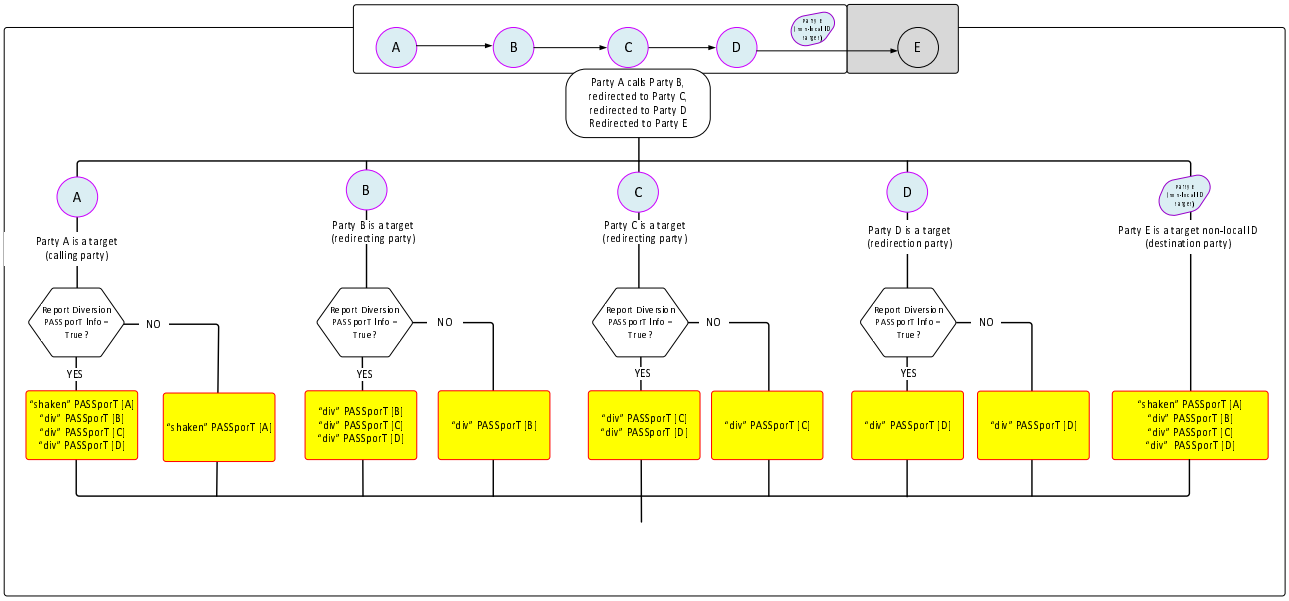


Figure 4.4.3.2-2: Scope of ReportDiversionPASSporTInfo (intra-network signing not required, no RCD)

Intra-network signing is not required, RCD is present

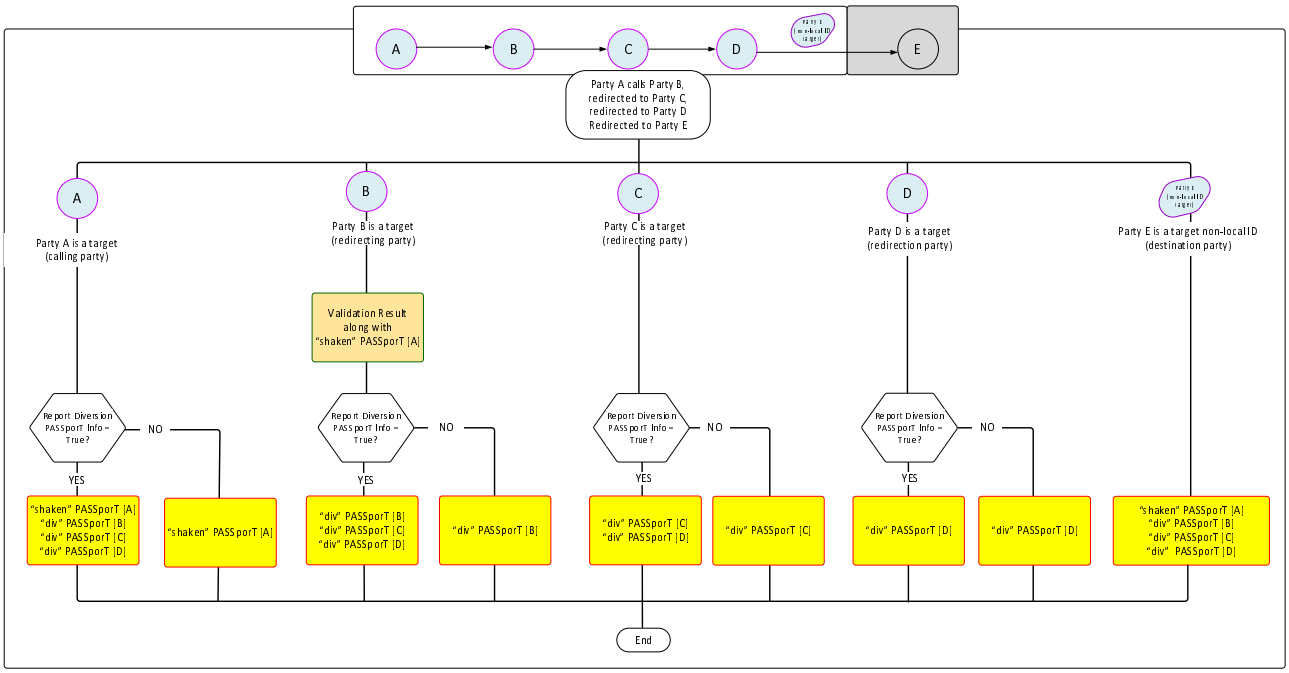


Figure 4.4.3.2-3: Scope of ReportDiversionPASSporTInfo (intra-network signing not required, no RCD)

The illustrations that involve redirections shown in the subsequent clauses adhere to the above basic concepts.

4.4.3.3 Validation result reporting

It is assumed that all the PASSporTs that are validated are reported in the STIRSHAKENSignatureValdiation message to the LEMFs.

For reporting of the STIRSHAKENSignatureValidation covering the redirection scenarios to avoid the double reporting, the logic shown in figure 4.4.3.3-1 is used.

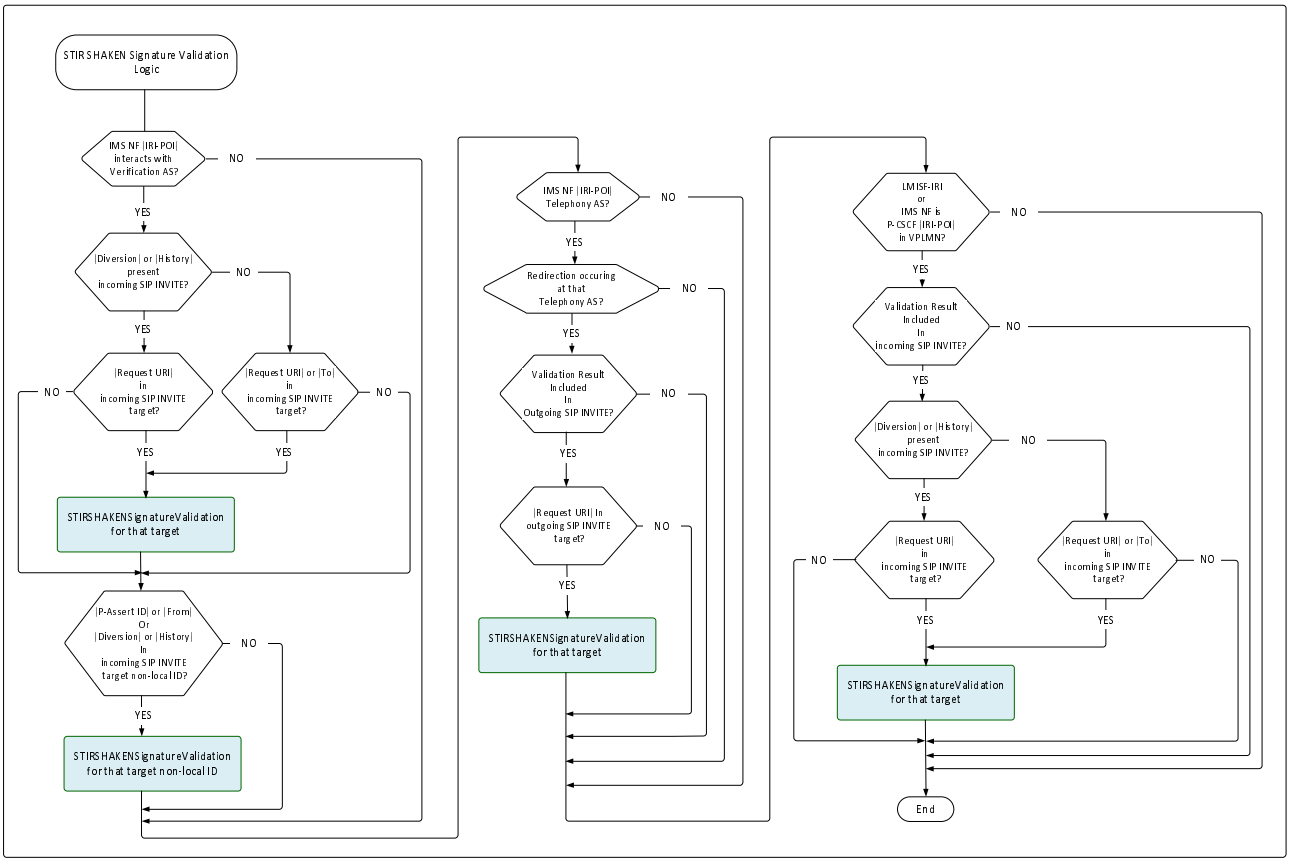


Figure 4.4.3.3-1: Logic used for STIR SHAKEN Signature Validation

When the egress IBCF interacts with the Verification AS, the IRI-POI in that IBCF would report the Validation Result for the target pointed by the Request URI. The Telephony AS that sees the Validation Result for the same target should not report the same again. When a redirection happens, the Validation Result for the target pointed by the redirecting party will have to be reported by the Telephony AS where the redirection occurs.

In addition, for a target match, if a prior redirection has occurred, the To header (which points to the original called party) has not be used for the redirected to parties.

4.4.4 Signing required for intra-network IMS sessions

4.4.4.1 General

The subsequent sub-clauses illustrate the scenarios when signing is required for intra-network IMS sessions.

When the intra-network signing is required, it is always the Telephony AS that interacts with the Signing AS and with Verification AS unless it is an emergency call (see clause 4.4.7).

The Telephony AS at the originating end of the call (i.e. Telephony AS associated with the calling party) interacts with the Signing AS and the Telephony AS at the terminating end (i.e. Telephony AS associated with the called party) of the call interacts with the Verification AS. The validation result is forwarded to the called party.

The IMS NFs that provide the STIR/SHAKEN related LI reporting are dependent on call scenario and the target.

The illustrations show the IMS LI reporting and STIR/SHAKEN related LI reporting as defined in TS 33.127 [3], TS 33.128 [4]. The TR 33.929-1 illustrates various scenarios for IMS LI with the default option.

Clause 4.3 of the present document indicates the conditions for an IMS NF to have the IRI-POI for the STIR/SHAKEN related reporting.

The Telephony AS on both ends of the call provide the LI functions for STIR/SHAKEN related reporting. It is assumed that all the PASSporTs that are validated are reported in the STIRSHAKENSignatureValdiation message to the LEMFs.

4.4.4.2 Intra-network IMS sessions, basic scenarios

4.4.4.2.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI for basic call scenarios (i.e. no redirections).

The calling party's identity is signed and included in the "shaken" PASSporT with calling party's identity in the origination claim and the called party's identity in the destination claim.

4.4.4.2.2 Scenario 1: Party A (target) calls Party B

4.4.4.2.2.1 The scenario

In this scenario, Party A calls Party B. Party A is the target.



Figure 4.4.4.2.2.1-1: Party A (target) calls Party B.

Clauses 4.4.4.2.2.2 and 4.4.4.2.2.3 illustrate the LI aspects for this scenario.

4.4.4.2.2.2 With default option for IMS LI

The illustration shown in figure 4.4.4.2.2.2-1 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of A provides the IRI-POI functions for IMS LI.

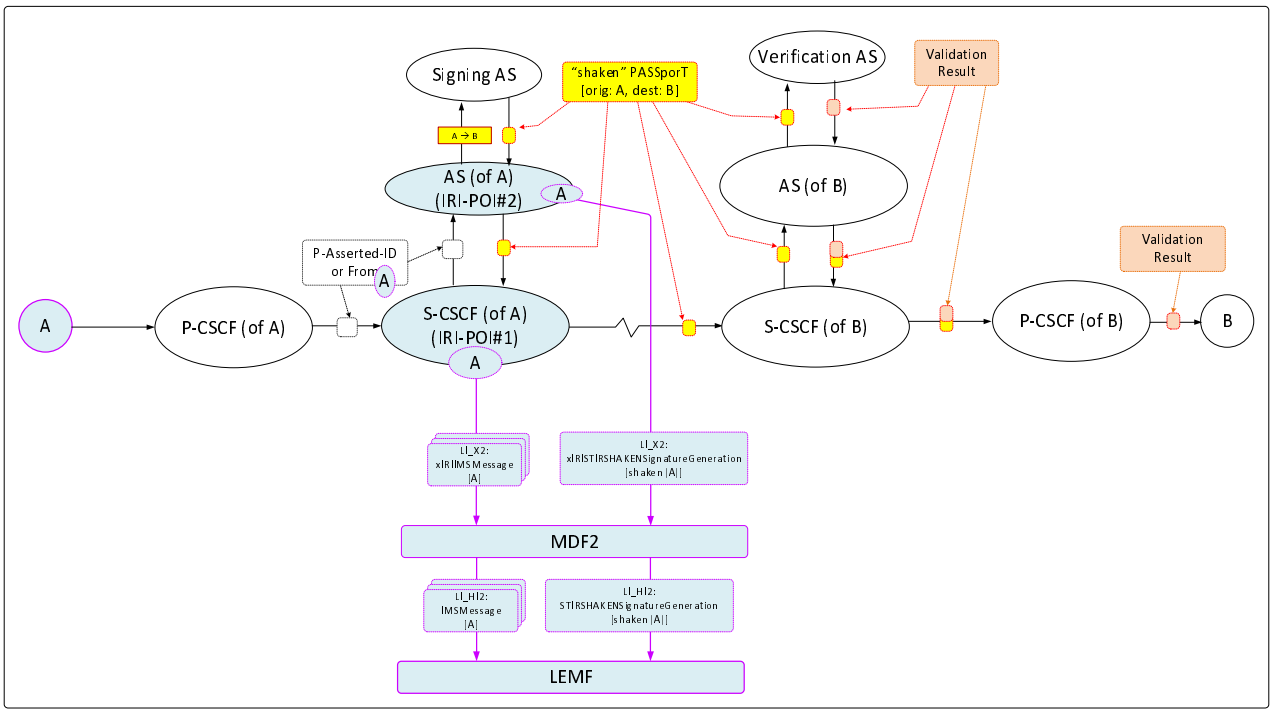


Figure 4.4.4.2.2.2-1: The LI aspects of scenario 1 (default option for IMS LI)

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record with the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

4.4.4.2.2.3 With alternate option for IMS LI

The illustration shown in 4.4.4.2.2.3-1 assumes that the alternate option (see TS 33.127 [3]) for IMS LI and therefore, the P-CSCF of A provides the IRI-POI functions for IMS LI.

NOTE: The alternate option for IMS LI is not shown in every illustration. It is shown here so as to illustrate influence it may have on the STIR/SHAKEN related LI reporting.

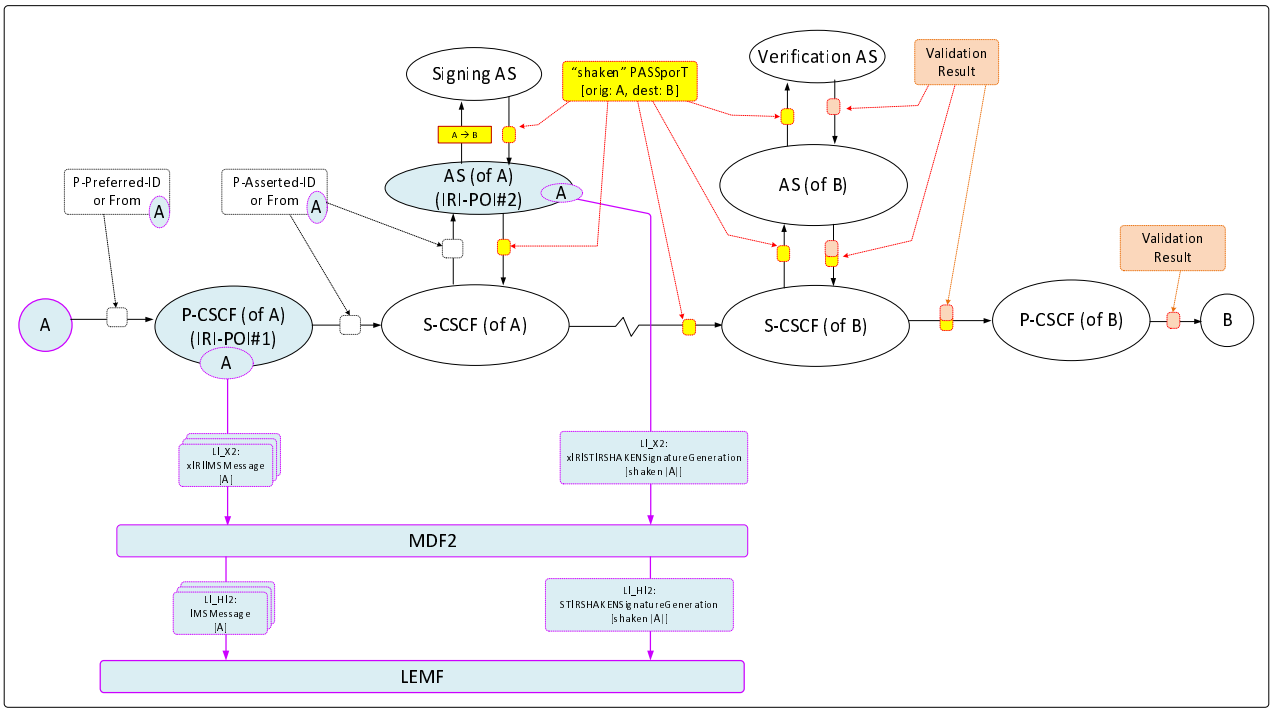


Figure 4.4.4.2.2.3-1: The LI aspects for scenario 1 (alternate option for IMS LI)

For target (Party A)

The IRI-POI present in the P-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record with the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the P-CSCF (of A) uses the P-Asserted ID or the From headers (pointing to Party A) present in the outgoing SIP INVITE to perform a target match. Alternatively, the P-Preferred ID may also be used from the incoming SIP INVITE in some implementations.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

4.4.4.2.3 Scenario 2: Party A calls Party B (target)

4.4.4.2.3.1 The scenario

In this scenario, Party A calls Party B. Party B is the target.



Figure 4.4.4.2.3.1-1: Party A calls Party B (target)

Figures 4.4.4.2.3.2-1 and 4.4.4.2.3.3-1 illustrate the LI aspects for this scenario.

4.4.4.2.3.2 With default option for IMS LI

The illustration shown in figure 4.4.4.2.3.2-1 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI.

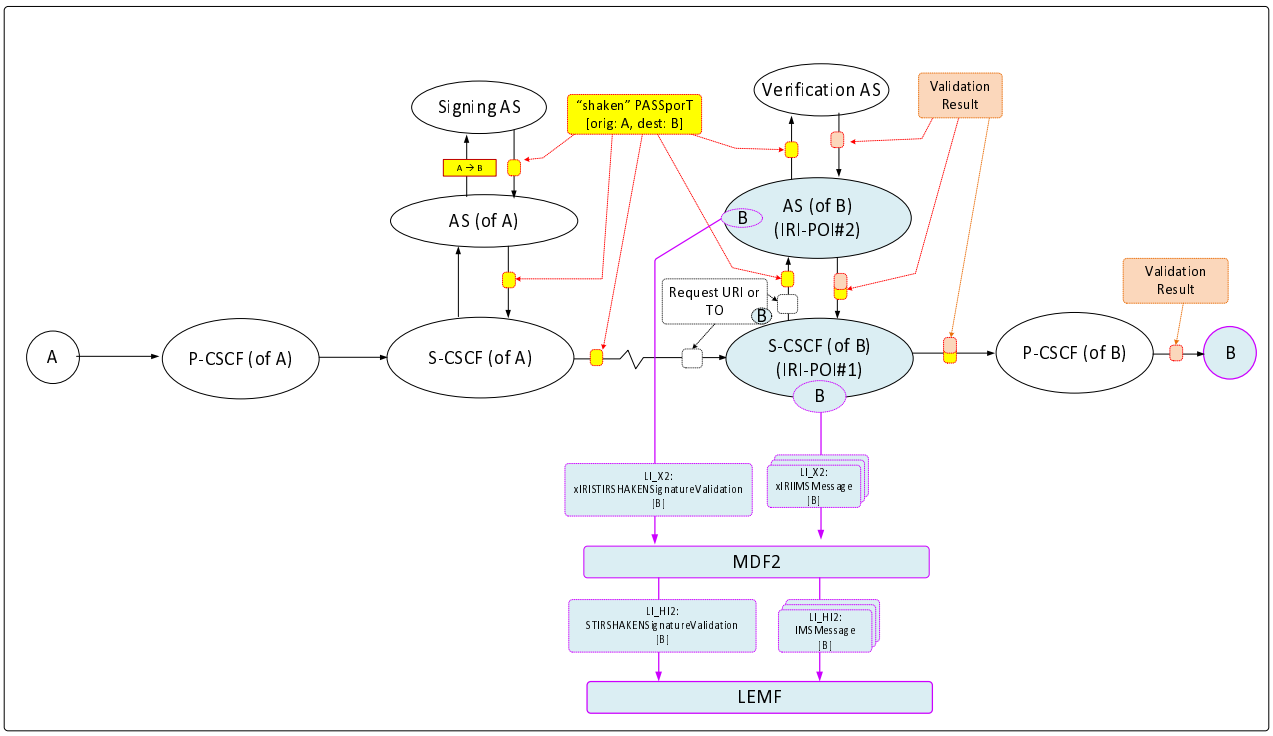


Figure 4.4.4.2.3.2-1: The LI aspects for scenario 2 (default option for IMS LI)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing to Party B) headers present in the incoming SIP INVITE to perform a target match.

4.4.4.2.3.3 With alternate option for IMS LI

The illustration shown in 4.4.4.2.3.3-1 assumes that the alternate option (see TS 33.127 [3]) for IMS LI and therefore, the P-CSCF of B provides the IRI-POI functions.

NOTE: The alternate option for IMS LI is not shown in every illustration. It is shown here so as to illustrate influence it may have on the STIR/SHAKEN related LI reporting.

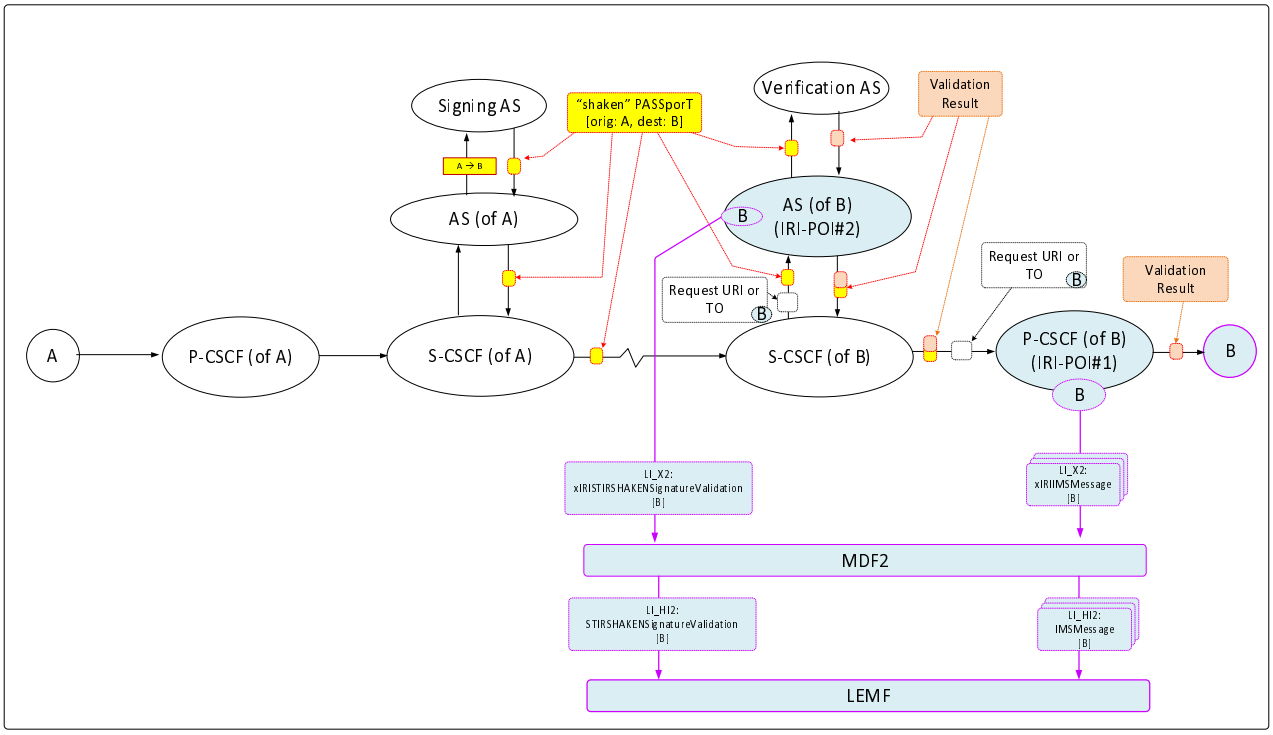


Figure 4.4.4.2.3.3-1: The LI aspects for scenario 2 (alternate option for IMS LI

For target (Party B)

The IRI-POI present in the P-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the P-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing to Party B) headers present in the incoming SIP INVITE to perform a target match.

4.4.4.3 Intra-network IMS sessions, redirecting scenarios

4.4.4.3.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI when an IMS session setup encounters a redirection.

The calling party's identity is signed and included in the "shaken" PASSporT with calling party's identity in the origination claim and the called party's identity in the destination claim.

The redirecting party's identity is signed and included in the "div" PASSporT with calling party's identity in the origination claim, redirecting party's identity in the diversion claim and the redirected-to party's identity in the destination claim. As illustrated in clause 4.4.3, the ReportDiversionPASSporTInfo has a play in determining the scope of PASSporTs reported to the LEMFs.

4.4.4.3.2 Scenario 1: Party A (target) calls Party B redirected to Party C

4.4.4.3.2.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party A is the target.

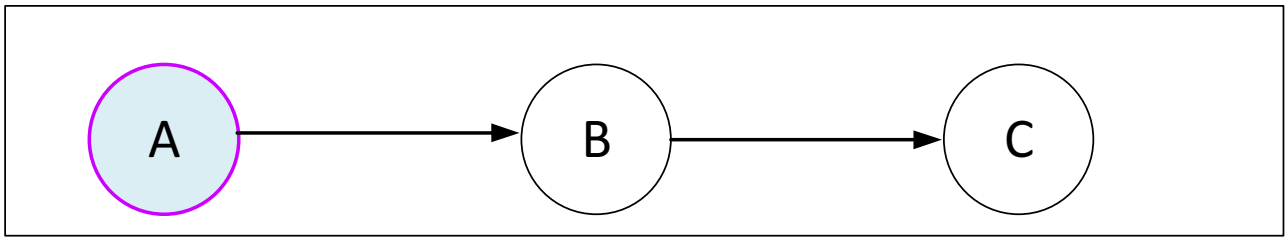


Figure 4.4.4.3.2.1-1: Party A (target) calls Party B redirected to Party C

Figures 4.4.4.3.2.2-1 and 4.4.4.3.2.3-1 illustrate the LI aspects for this scenario. Both illustrations assume the default option (see TS 33.127 [3]) for IMS LI.

4.4.4.3.2.2 When ReportDiversionPASSporTInfo is set to "False"

The illustration assumes that the ReportDiversionPASSporTInfo is set to "False".

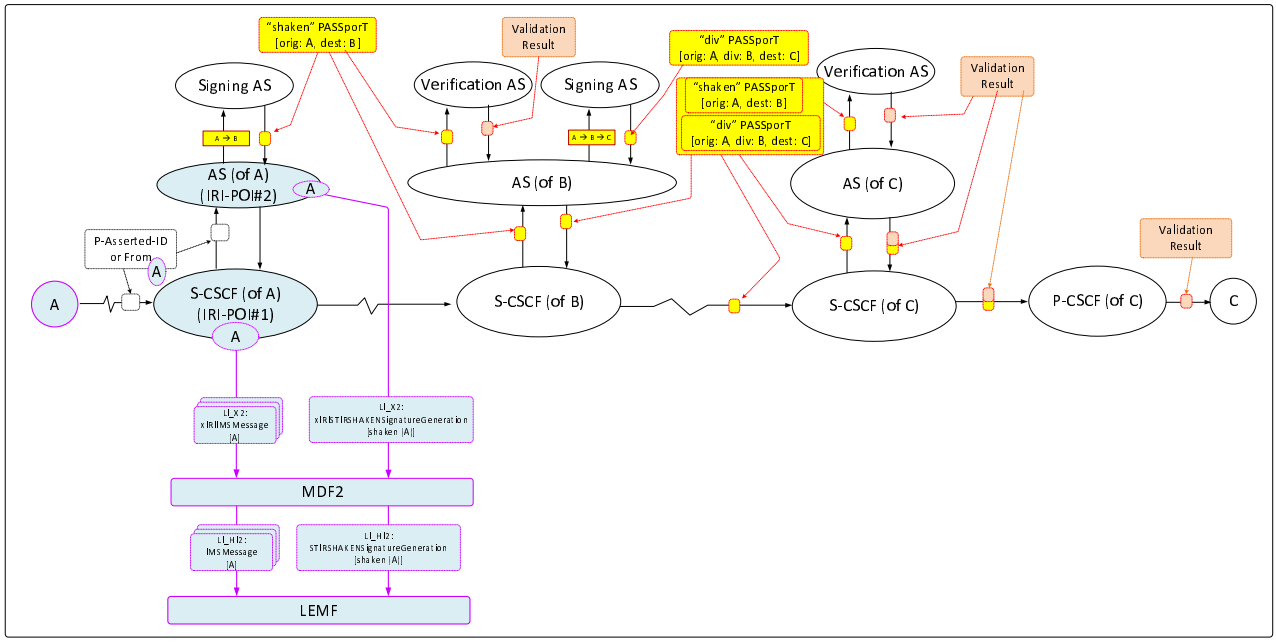


Figure 4.4.4.3.2.2-1: The LI aspects for scenario 1 (ReportDiversionPASSporTInfo is "False")

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record with the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

4.4.4.3.2.3 When ReportDiversionPASSporTInfo is set to "True"

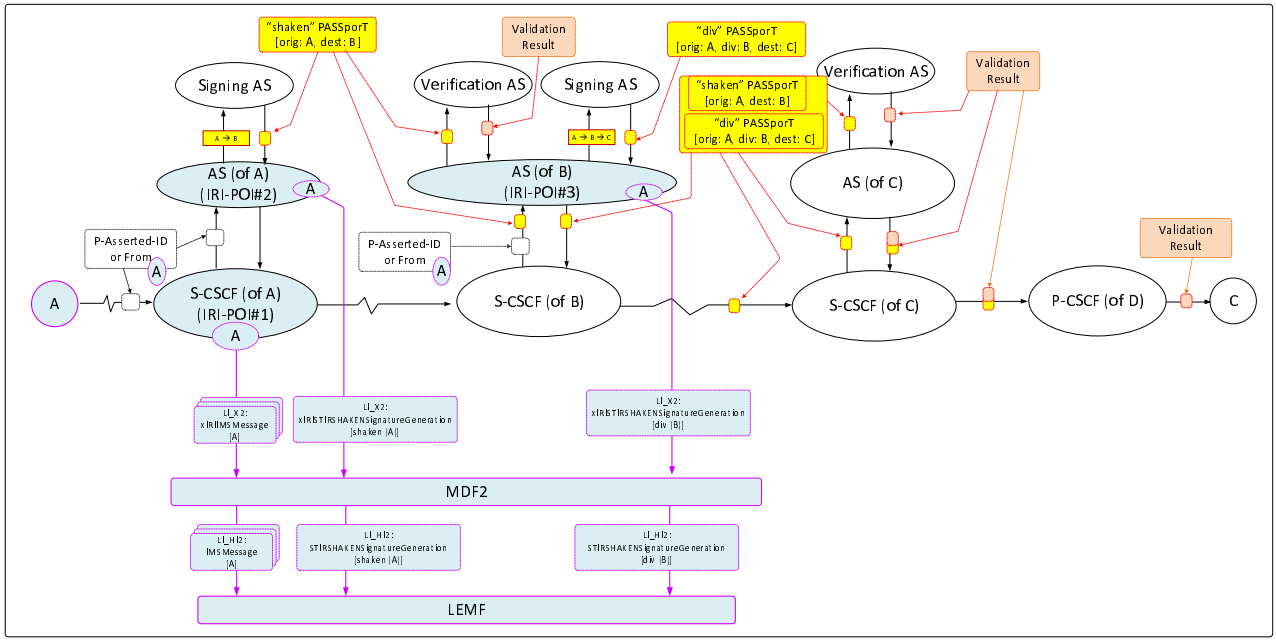


Figure 4.4.4.3.2.3-1: The LI aspects for scenario 1 (ReportDiversionPASSporTInfo is "True")

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record with "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The Telephony AS (of B) interacts with the Signing AS to authenticate/sign B's identity and with the ReportDiversionPASSporTInfo set to "True", the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POIs present in the Telephony AS (of A) and Telephony AS (of B) use the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

4.4.4.3.3 Scenario 2: Party A calls Party B (target) redirected to Party C

4.4.4.3.3.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party B is the target.

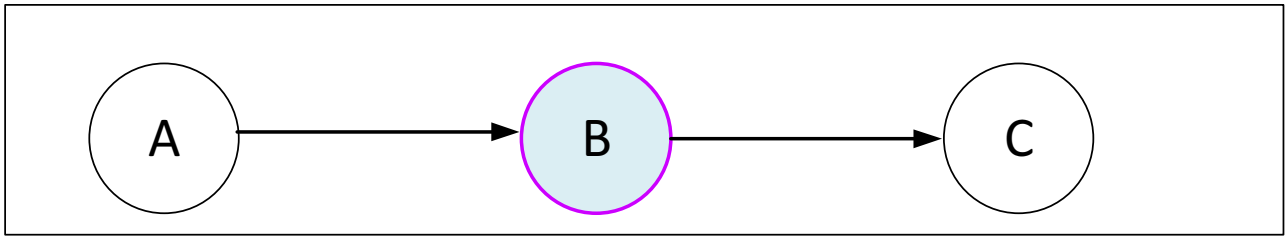


Figure 4.4.4.3.3.1-1: Party A calls Party B (target) redirected to Party C

Figures 4.4.4.3.3.2-1 and 4.4.4.3.3.3-1 illustrate the LI aspects for this scenario.

4.4.4.3.3.2 With default option for IMS LI

The illustration shown in figure 4.4.4.3.3.2-1 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI.

NOTE: The alternate option for IMS LI is not shown in every illustration. It is shown here so as to illustrate influence, it may have on the STIR/SHAKEN related LI reporting.

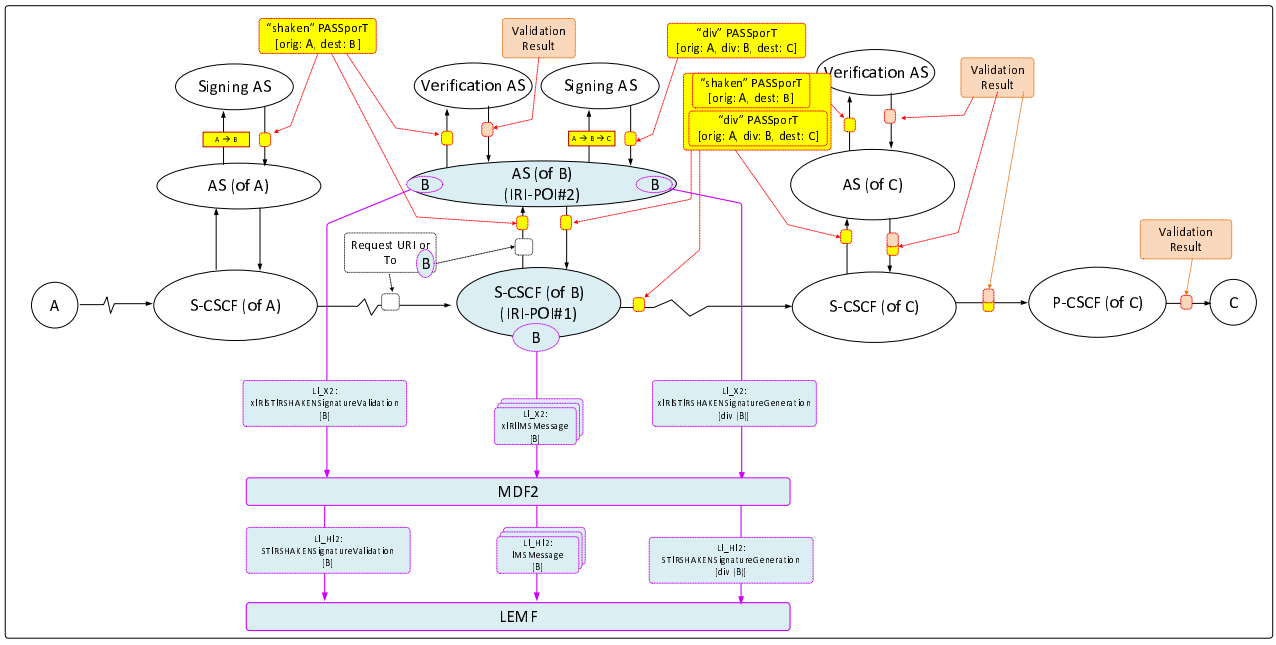


Figure 4.4.4.3.3.2-1: The LI aspects of scenario 2 (default option for IMS LI)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The Telephony AS (of B) interacts with the Signing AS to authenticate/sign B's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

NOTE: The provisioned ReportDiversionPASSporTInfo will have an affect (similar to the illustrations shown in clause 4.4.4.3.2.3), if further redirections happen.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or To headers (pointing to Party B) present in the incoming SIP INVIE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

4.4.4.3.3.3 With alternate option for IMS LI

The illustration shown in figure 4.4.4.3.3.3-1 assumes the alternate option (see TS 33.127 [3]) for IMS LI and therefore, the P-CSCF of C provides the IRI-POI functions for IMS LI.

NOTE: The alternate option for IMS LI is not shown in every illustration. It is shown here to illustrate influence it may have on the STIR/SHAKEN related LI reporting.

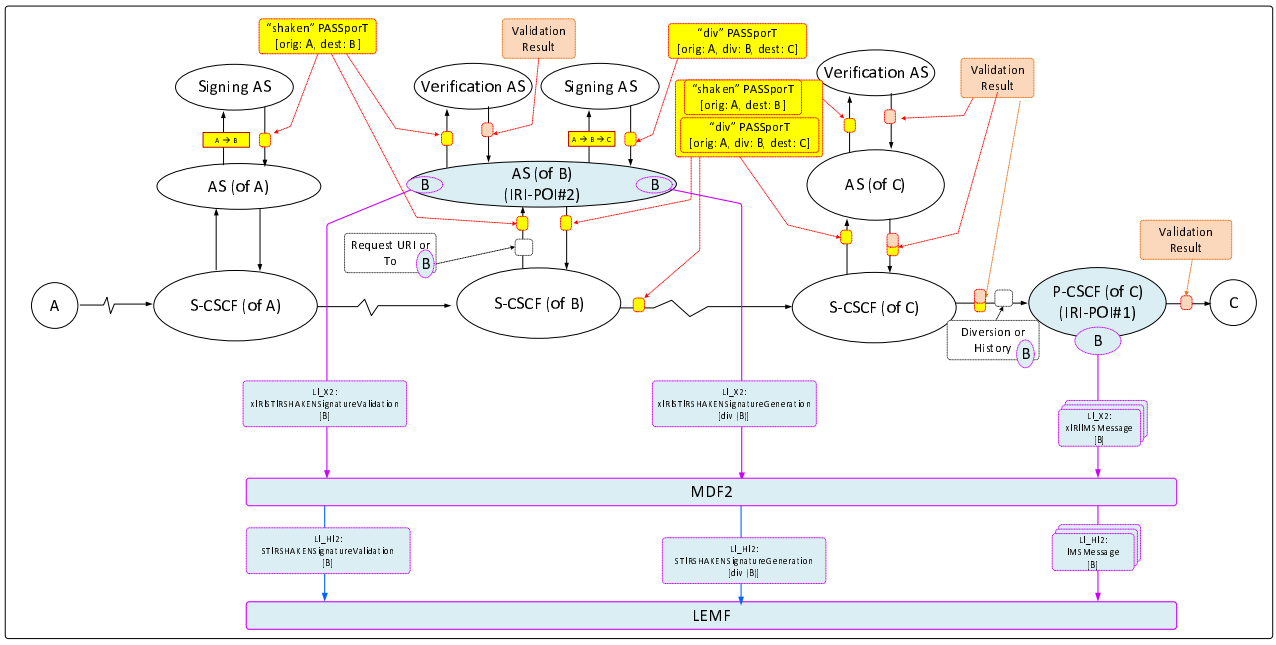


Figure 4.4.4.3.3.3-1: Party A (target) calls Party B (alternate option for IMS LI)

For target (Party B)

The IRI-POI present in the P-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The Telephony AS (of B) interacts with the Signing AS to authenticate/sign B's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

NOTE: The provisioned ReportDiversionPASSporTInfo will have an affect (similar to the illustrations shown in clause 4.4.4.3.2.3), if further redirections happen.

The IRI-POI present in the P-CSCF (of C) uses the Diversion or the History Info (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

4.4.4.3.4 Scenario 3: Party A calls Party B redirected to Party C (target)

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party C is the target.

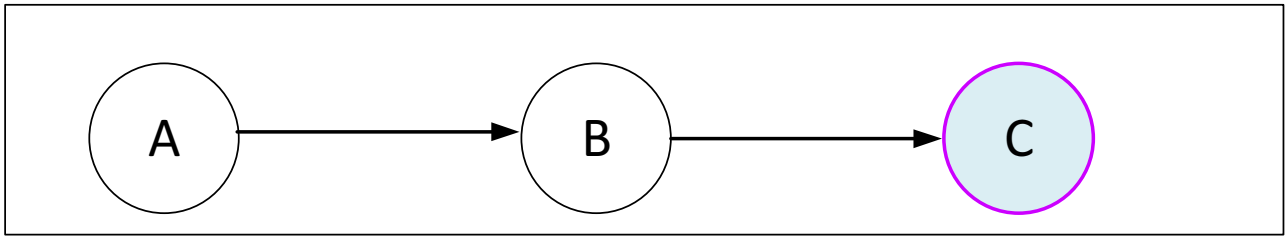


Figure 4.4.4.3.4-1: Party A calls Party B redirected to Party C (target)

Figure 4.4.4.3.4-2 illustrates the LI aspects for this scenario.

The illustration assumes default option (TS 33.127 [3]) for IMS LI.

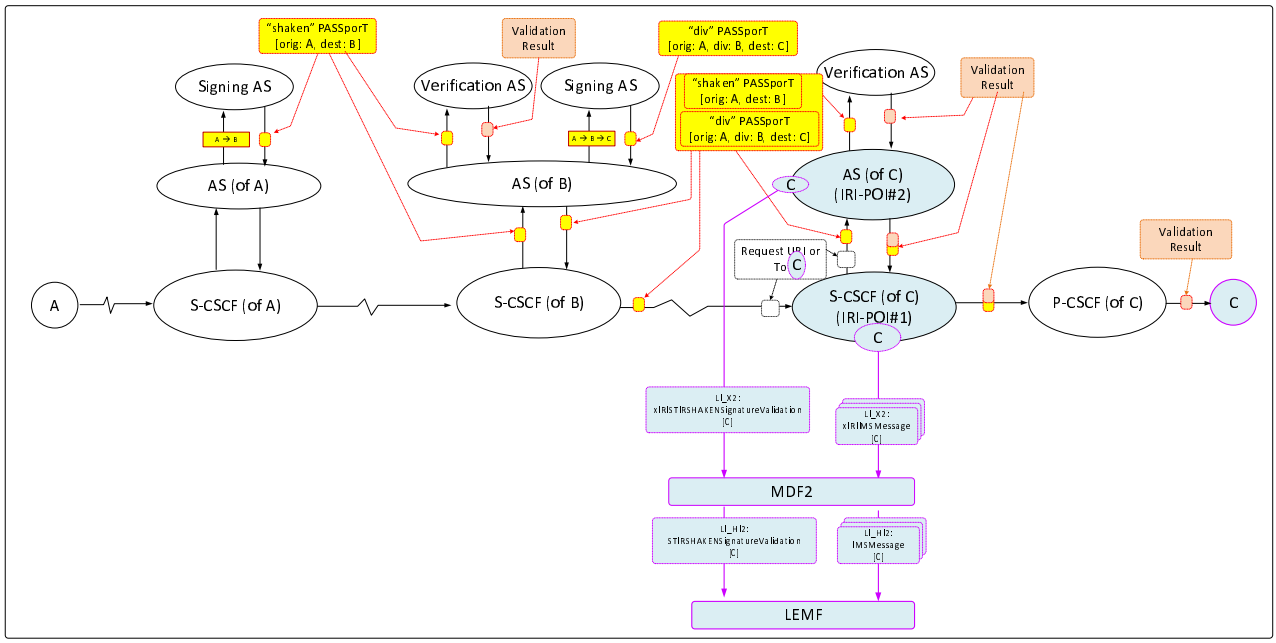


Figure 4.4.4.3.4-2: The LI aspects for scenario 3

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of C) interacts with the Verification AS to verify the received PASSporTs ("shaken" and "div") and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the S-CSCF (of C) uses the Request URI header (pointing to Party C) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Request URI header (pointing to Party C) of the incoming SIP INVITE to perform a target match.

4.4.4.4 Inter-network IMS sessions (originating network)

4.4.4.4.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI for inter-network IMS sessions in the originating network when signing is required for intra-network IMS sessions.

The calling party's identity is signed and included in the "shaken" PASSporT with calling party's identity in the origination claim and the called party's identity in the destination claim.

The Telephony AS at the originating end of the call (i.e. Telephony AS associated with the calling party) interacts with the Signing AS. In the case of redirections, the Telephony AS associated with the redirecting party would interact with the Verification AS to validate the PASSporT (s) and then would interact with the Signing AS to sign the redirecting party's identity. One (for basic call) or more (redirected calls), PASSporTs are sent to the Terminating Network in the SIP INVITE.

The Telephony AS provides the LI functions for STIR/SHAKEN related reporting. For destination target non-local ID, two implementation options are considered to provide the LI functions for STIR/SHAKEN related reporting (see clause 4.4.4.4.3.1).

4.4.4.4.2 Scenario 1: Party A (target) calls Party B

In this scenario, Party A (in originating network) calls Party B (in the terminating network). Party A is the target in the originating network.

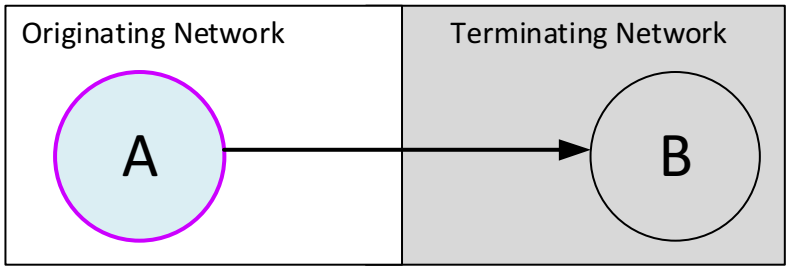


Figure 4.4.4.4.2-1: Scenario 1 - Party A (target) calls Party B.

Figure 4.4.4.4.2-2 illustrates the LI aspects for this scenario.

The illustration shown in figure 4.4.4.4.2-2 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of A provides the IRI-POI functions for IMS LI.

Since the signing of intra-network session is required, the Telephony AS of A interacts with the Signing AS to authenticate/sign A's identity and the signature is sent to the terminating network in the "shaken" PASSporT.

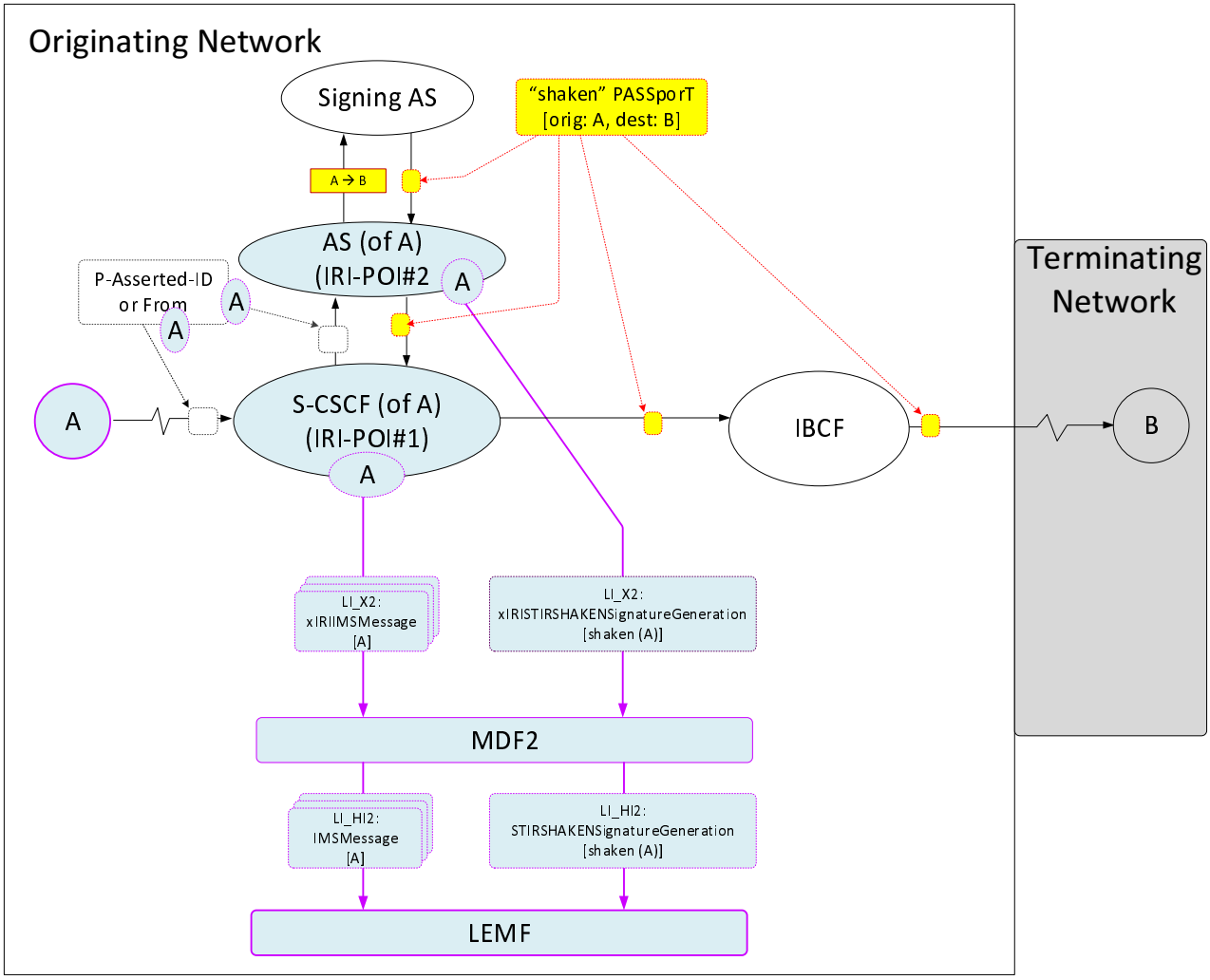


Figure 4.4.4.4.2-2: The LI aspects for scenario 1

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that has the "shaken" PASSporT for Party A) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

4.4.4.4.3 Scenario 2: Party A calls Party B (target non-local ID)

4.4.4.4.3.1 The scenario

In this scenario, Party A (in originating network) calls Party B (in the terminating network). Party B is a target non-local ID in the originating network.

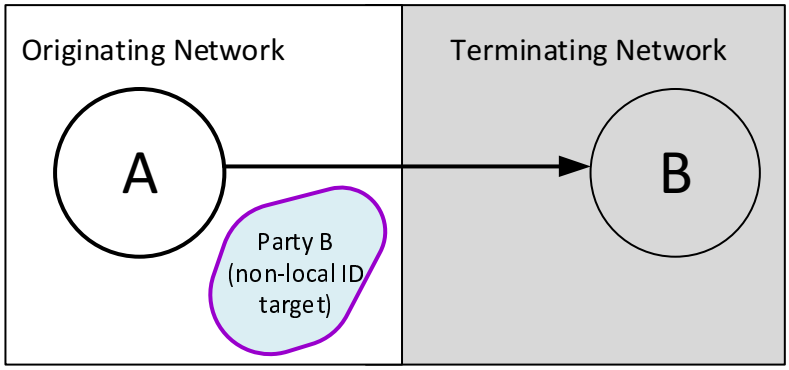


Figure 4.4.4.4.3-1: Scenario 2 - Party A calls Party B (target non-local ID)

Figures in the subsequent clauses illustrate the LI aspects for this scenario. The illustrations present two different implementation options for the STIR/SHAKEN related LI reporting when the destination party is a target non-local ID:

- Option 1: Telephony AS that interacts with Signing AS provides the IRI-POI functions.
- Option 2: IBCF provides the IRI-POI functions.

The TS 33.128 [4] specifies the stage 3 requirements for option 1 in the above. However, an implementation that has default option (TS 33.127 [3]) for IMS LI (target non-local ID) where the egress IBCF would provide the IRI-POI functions for the destination target non-local ID could have some benefit from the option 2.

4.4.4.4.3.2 Option 1

4.4.4.4.3.2.1 General

Since the signing of intra-network session is required, the Telephony AS of A interacts with the Signing AS to authenticate/sign A's identity and the signature is sent to the terminating network in the "shaken" PASSporT.

The IRI-POI present in the Telephony AS provides the STIR/SHAKEN related LI reporting.

4.4.4.4.3.2.2 Default option for IMS LI (target non-local ID)

The illustration shown in figure 4.4.4.4.3.2.2-1 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the egress IBCF provides the IRI-POI functions for IMS LI.

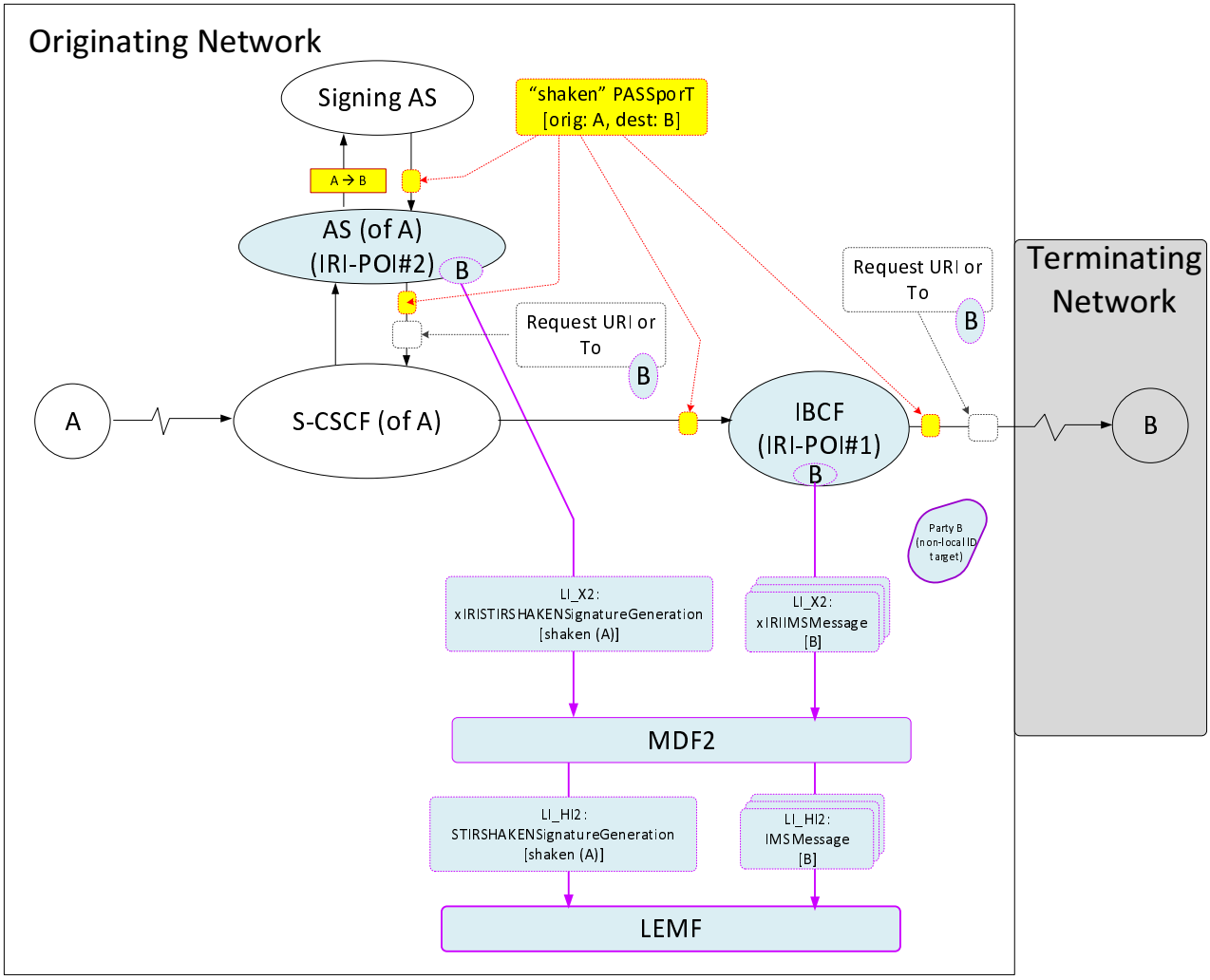


Figure 4.4.4.4.3.2.2-1: The LI aspects for scenario 2, option 1 (default option for IMS LI)

For target non-local ID (Party B)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that has the "shaken" PASSporT for Party A) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the egress IBCF uses the Request URI or the To headers (pointing to Party B) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the Request URI or the To headers (pointing to the target non-local ID Party B) present in the outgoing SIP INVITE to perform a target match.

4.4.4.4.3.2.3 Alternate option for IMS LI (target non-local ID)

The illustration shown in figure 4.4.4.4.3.2.3-1 assumes the alternate option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF (of A) provides the IRI-POI functions for IMS LI.

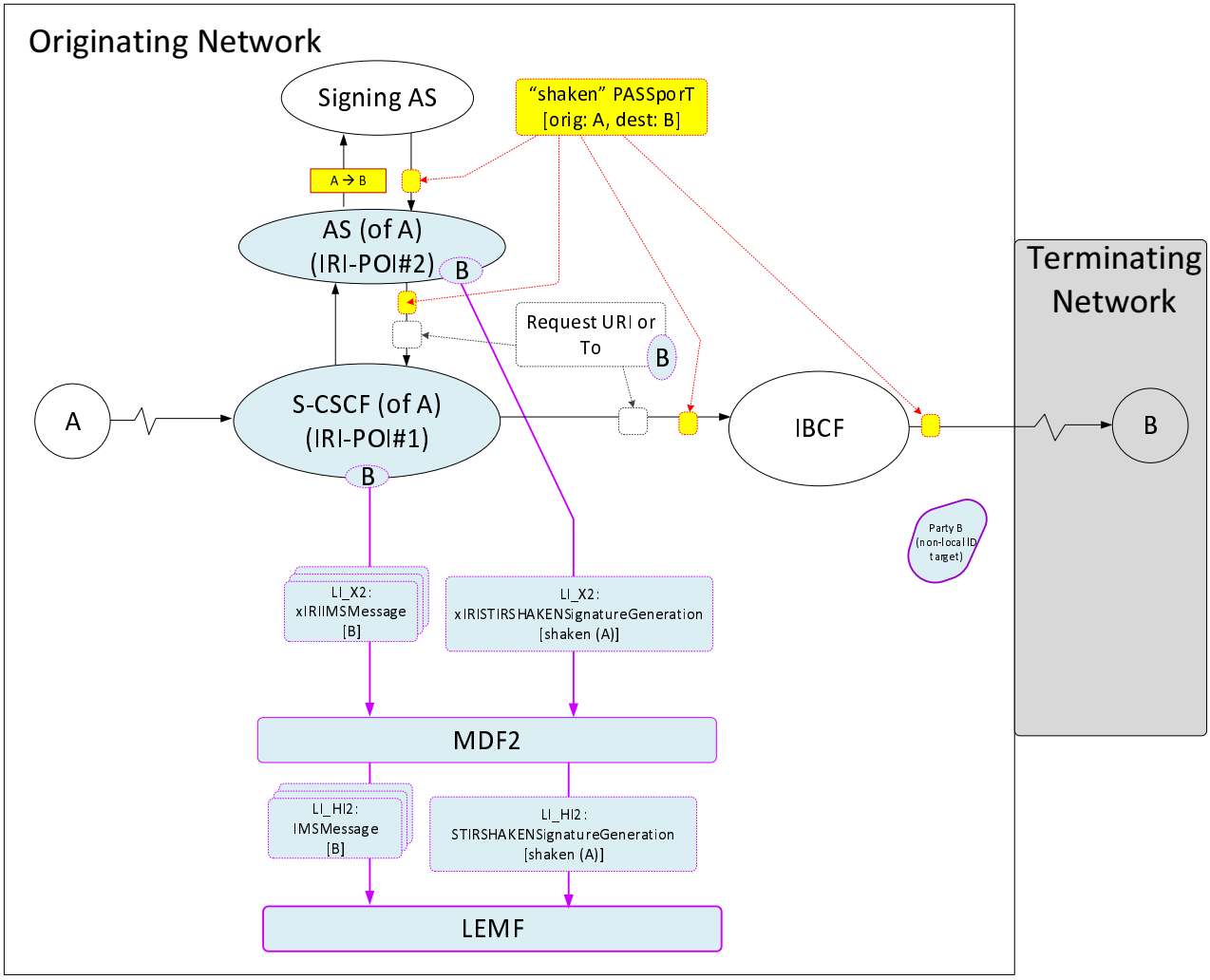


Figure 4.4.4.4.3.2.3-1: The LI aspects for scenario 2, option 1 (alternate option for IMS LI)

For target non-local ID (Party B)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that has the "shaken" PASSporT for Party A) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the Request URI or the To headers (pointing to Party B) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the Request URI or the To headers (pointing to the target non-local ID Party B) present in the outgoing SIP INVITE to perform a target match.

4.4.4.4.3.3 Option 2

4.4.4.4.3.3.1 General

Since the signing of intra-network session is required, the Telephony AS of A interacts with the Signing AS to authenticate/sign A's identity and the signature is sent to the terminating network in the "shaken" PASSporT.

In this option, the IRI-POI present in the egress IBCF provides the STIR/SHAKEN related LI reporting.

4.4.4.4.3.3.2 Default option for IMS LI (target non-local ID)

The illustration shown in figure 4.4.4.4.3.3.2-1 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the egress IBCF provides the IRI-POI functions for IMS LI.

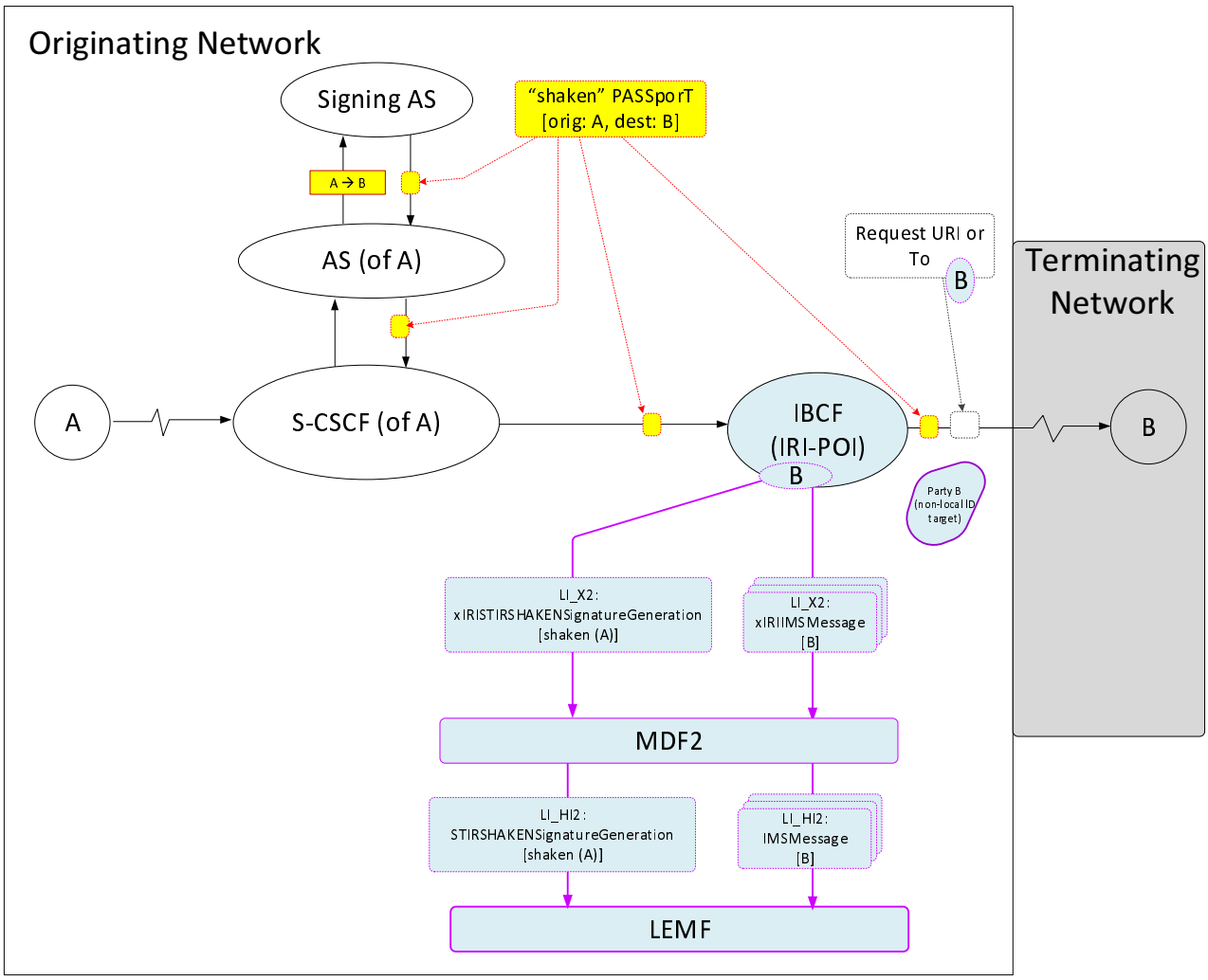


Figure 4.4.4.4.3.3.2-1: LI aspects for scenario 2, option 2 (default option for IMS LI)

For target non-local ID (Party B)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The egress IBCF that sees the PASSporT in SIP INVITE generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the egress IBCF uses the Request URI or the To headers (pointing to Party B) present in the outgoing SIP INVITE to perform a target match.

4.4.4.4.3.3.3 Alternate option for IMS LI (target non-local ID)

The illustration shown in figure 4.4.4.4.3.3.3-1 assumes the alternate option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF (of A) provides the IRI-POI functions for IMS LI.

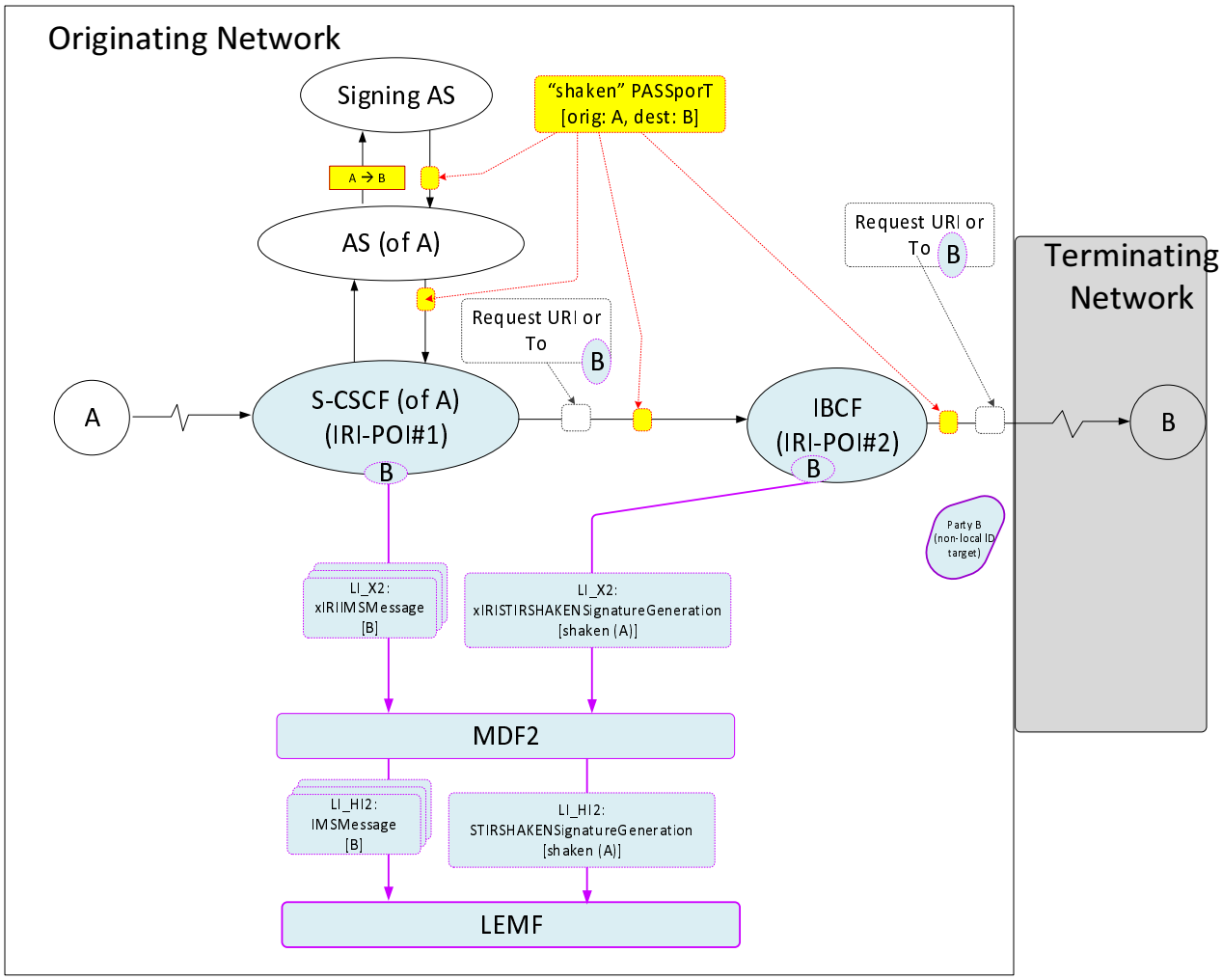


Figure 4.4.4.4.3.3.3-1: LI aspects for scenario 2, option 2 (alternate option for IMS LI)

For target non-local ID (Party B)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The egress IBCF that sees the PASSporT in the SIP INVITE generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the Request URI or the To headers (pointing to Party B) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the Request URI or the To headers (pointing to Party B) present in the outgoing SIP INVITE to perform a target match.

4.4.4.4.4 Scenario 3: Party A (target) calls Party B (target) redirected to Party C (target non-local ID)

4.4.4.4.4.1 The scenario

Multiple targets are considered here.

In the scenario, Party A calls Party B who has a call forwarding to Party C. Party A and Part B are the targets in the originating network. Party C is a target non-local ID in the originating network.

LEMF-1 is intercepting the target Party A, LEMF-2 is intercepting the target Party B and LEMF-3 is intercepting the target non-local ID Party C.

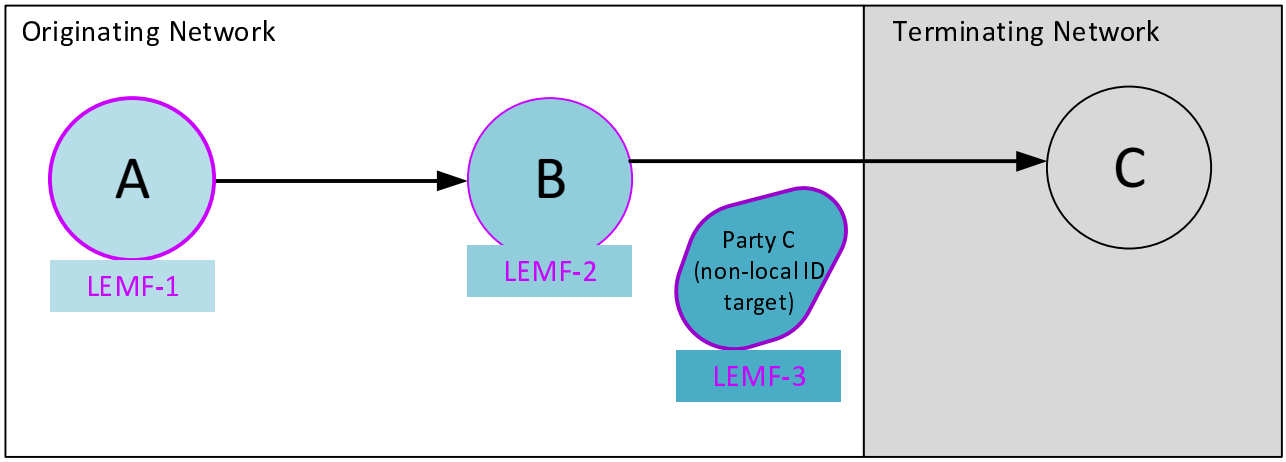


Figure 4.4.4.4.4.1-1: Party A (target) calls Party B (target) redirected to Party C (target non-local ID)

The descriptions given in clause 4.4.3 apply to this scenario as well.

Figures 4.4.4.4.4.2-1 and 4.4.4.4.4.3-1 illustrate the LI aspects for this scenario.

Both illustrations assume the default option (see TS 33.127 [3]) for IMS LI for targets Party A and Party B and target non-local ID (Party C). Therefore, the S-CSCF of A provides the IRI-POI functions for IMS LI for target Party A, S-CSCF of B provides the IRI-POI functions for IMS LI for target Party B and egress IBCF provides the IRI-POI functions for target non-local ID (Party C).

4.4.4.4.4.2 When ReportDiversionPASSporTInfo is set to "False"

For target non-local ID (Party C), the illustration assumes option 1 (see clause 4.4.4.4.3.1) for the STIR/SHAKEN related LI reporting. Therefore, the Telephony AS (of B) provides the IRI-POI functions for STIR/SHAKEN related LI reporting for target non-local ID (Party C).

In this illustration, the ReportDiversionPASSporTInfo is set to "False".

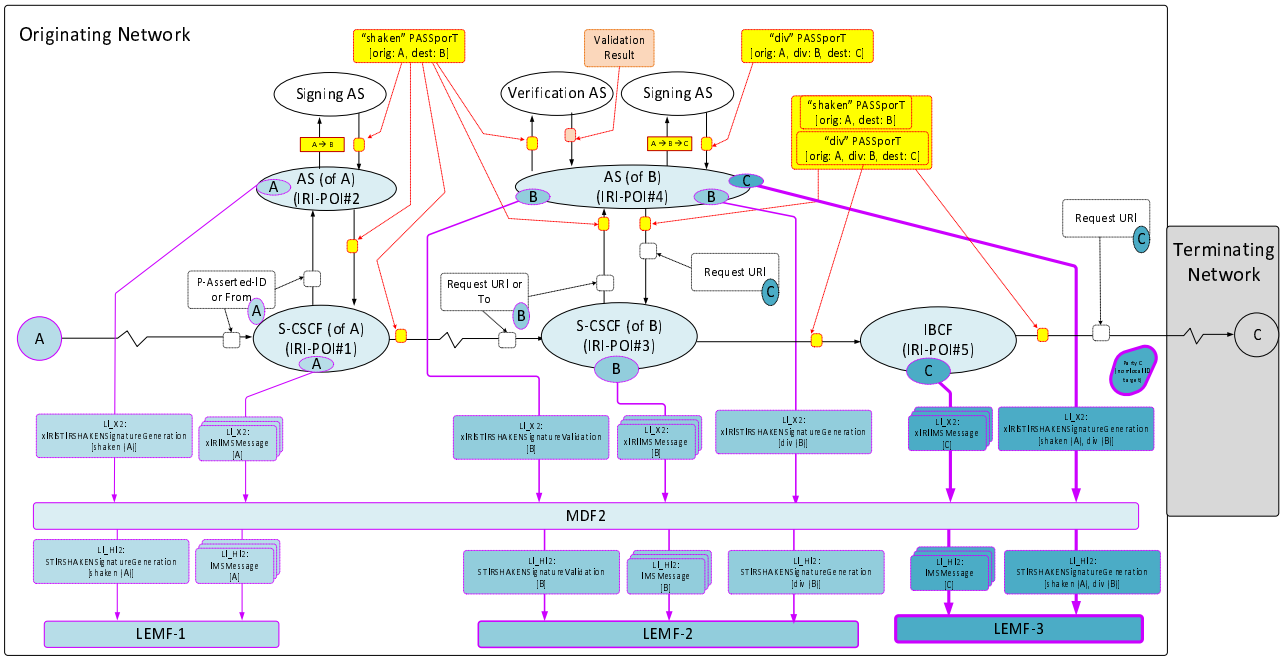


Figure 4.4.4.4.4.2-1: The LI aspects of scenario 3 (ReportDiversionPASSporTInfo is "False")

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-2.

The Telephony AS (of B) interacts with the Signing AS to authenticate/sign B's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that has the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POIs present in the Telephony AS (of B) use the Request URI or the To headers (pointing to Party B) of the incoming SIP INVITE to perform a target match.

For target non-local ID (Party C)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-3.

The IRI-POI present in the egress Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A and the "div" PASSporT for Party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-3.

The IRI-POI present in the egress IBCF uses the Request URI (pointing to Party C) in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) use the Request URI header (pointing to Party C) of the outgoing SIP INVITE to perform a target match.

4.4.4.4.4.3 When ReportDiversionPASSporTInfo is set to "True"

The illustration shown in figure 4.4.4.4.4.3-1 assumes that the ReportDiversionPASSporTInfo is set to "True".

For target non-local ID (Party C), the illustration assumes option 1 (see clause 4.4.4.4.3.1) for the STIR/SHAKEN related LI reporting. Therefore, the Telephony AS (of B) provides the IRI-POI functions for STIR/SHAKEN related LI reporting for target non-local ID (Party C).

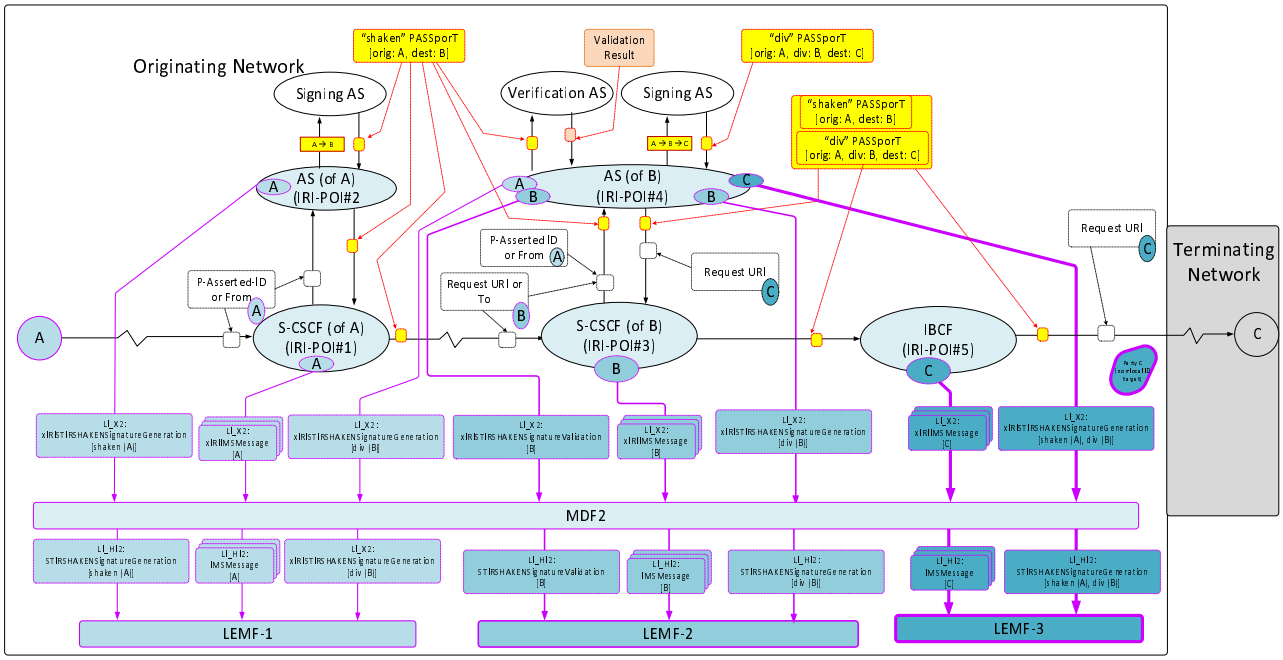


Figure 4.4.4.4.4.3-1: The LI aspects of scenario 3 (ReportDiversionPASSporTInfo is "True")

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

With ReportDiversionPASSporTInfo being set to "True", the IRI-POI present in the Telephony AS (of B) (which interacts with the Signing AS to authenticate/sign B's identity) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-2.

The Telephony AS (of B) interacts with the Signing AS to authenticate/sign B's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POIs present in the Telephony AS (of B) use the Request URI or the To headers (pointing to Party B) of the incoming SIP INVITE to perform a target match.

For target non-local ID (Party C)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-3.

The IRI-POI present in the Telephony AS (of B), upon detecting the destination is a target non-local ID, generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A and the "div" PASSporT for Party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the egress IBCF uses the Request URI header (pointing Party C) of the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) use the Request URI header (pointing to Party C) in the outgoing SIP INVITE to perform a target match.

4.4.4.4.5 Scenario 4: Party A (target) calls Party B (target) redirected to Party C, redirected to Party D (target non-local ID)

Multiple targets are considered here.

In the scenario, Party A calls Party B who has a call forwarding to Party C. who in turn has a call forwarding to Party D. Party A and Part B are the targets in the originating network. Party C and Party D are in the terminating network and Party D is a target non-local ID in the originating network.

LEMF-1 is intercepting the target Party A, LEMF-2 is intercepting the target Party B and LEMF-3 is intercepting the target non-local ID Party D.

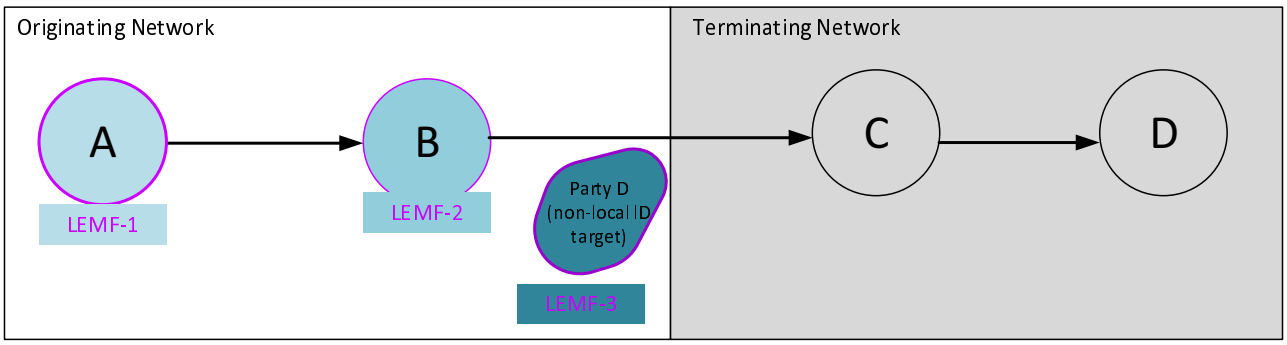


Figure 4.4.4.4.5-1: Party A (target) calls Party B (target) redirected to Party C redirected to Party D (target non-local ID)

The descriptions given in clause 4.4.3 apply to this scenario as well.

Figure 4.4.4.4.5-2 illustrates the LI aspects for this scenario where the ReportDiversionPASSporTInfo is set to "True" for all targets.

The illustration figure 4.4.4.4.5-2 assumes the default option (see TS 33.127 [3]) for IMS LI for targets Party A and Party B and target non-local ID (Party D). Therefore, the S-CSCF of A provides the IRI-POI functions for IMS LI for target Party A, S-CSCF of B provides the IRI-POI functions for IMS LI for target Party B.

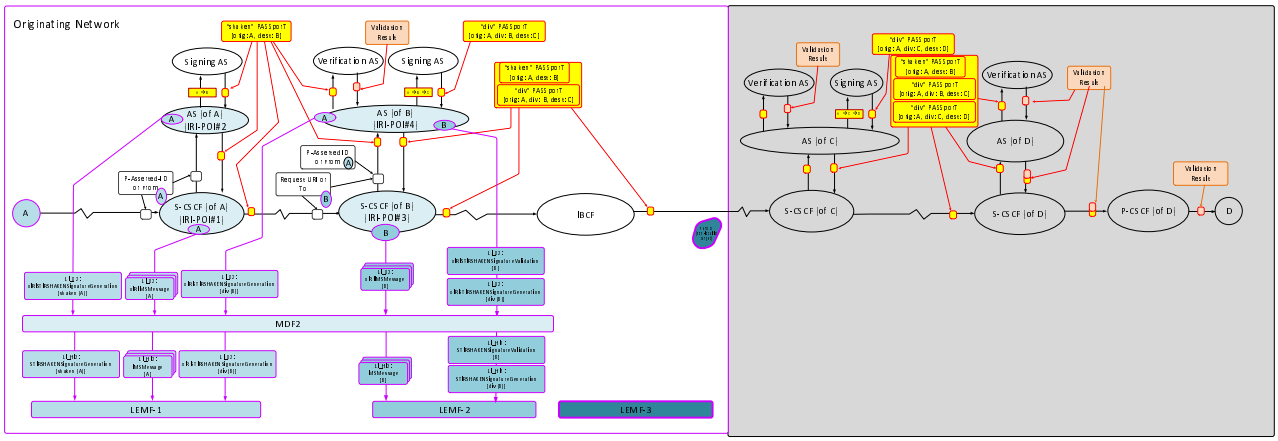


Figure 4.4.4.4.5-2: The LI aspects of scenario 3 (ReportDiversionPASSporTInfo is "True")

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

With ReportDiversionPASSporTInfo being set to "True", the IRI-POI present in the Telephony AS (of B) that interacts with the Signing AS to authenticate/sign B's identity generates and delivers the xIRISTIRSHAKENSignatureGeneration record with the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-2.

The Telephony AS (of B) interacts with the Signing AS to authenticate/sign B's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record with "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POIs present in the Telephony AS (of B) use the Request URI or the To headers (pointing to Party B) of the incoming SIP INVITE to perform a target match.

For target non-local ID (Party D)

Since the call forwarding from Party C to Party D happens in the terminating network and since the originating network is unaware of Party D's presence on the call, there is no LI in the originating network.

Point to note in this scenario

As illustrated, in the scenarios for the terminating network (see clause 4.4.4.5), the Telephony AS (of C) would interact with the Signing AS to sign Party C's identity and forwards the "div" PASSporT for Party C to the next hop on the signaling path.

The originating network does not see that redirection from Party C to Party D and therefore, even though ReportDiversionPASSporTInfo is set to "True" for the target (Party A) and the target (Party B), the "div" PASSPORT for Party C is not reported to either of the two LEMFs (LEMF-1 or LEMF-2).

In contrast, in the terminating network, if Party A and Party B were target non-local IDs, then STIR/SHAKEN related LI reporting for verification and further diversions would have applied to both the target non-local IDs (see the illustration in clause 4.4.4.5.4.3).

4.4.4.5 Inter-network IMS sessions (terminating network)

4.4.4.5.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI for inter-network IMS sessions for the terminating network.

The calling party's identity is signed and included in the "shaken" PASSporT with calling party's identity in the origination claim and the called party's identity in the destination claim. The terminating network that receives the "shaken" and one or more "div" PASSporTs (if redirection had happened in the originating network) would validate the received PASSporT (s) before offering the call to the terminating party.

The Telephony AS in the terminating network interacts with the Verification AS to validate the received PASSporTs. In the case of further redirections in the terminating network, the Telephony AS associated with the redirecting party would interact with the Signing AS to sign the redirecting party identity. The Telephony AS associated with the terminating party interacts with the Verification AS to validate the received PASSporTs.

The Telephony AS provides the LI functions for STIR/SHAKEN related reporting.

4.4.4.5.2 Scenario 1: Party A calls Party B (target)

In this scenario, Party A (in originating network) calls Party B (in the terminating network). Party A is the target in the originating network.

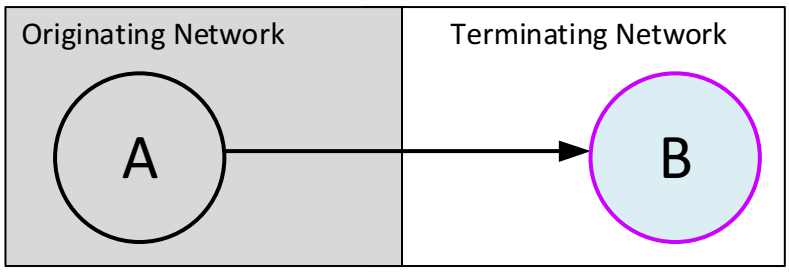


Figure 4.4.4.5.2-1: Party A calls Party B (target)

Figure 4.4.4.5.2-2 illustrates the LI aspects for this scenario.

The illustration shown in figure 4.4.4.5.2-2 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI. Note that alternate option as defined in TS 33.127 [3] for IMS LI is also possible. In that alternate option, the P-CSCF (of B) provides the IMS LI for target B (see clause 4.4.4.2.3.3).

Since the signing of intra-network session is required, the Telephony AS of B interacts with the Verification AS to validate the received "shaken" PASSporT.

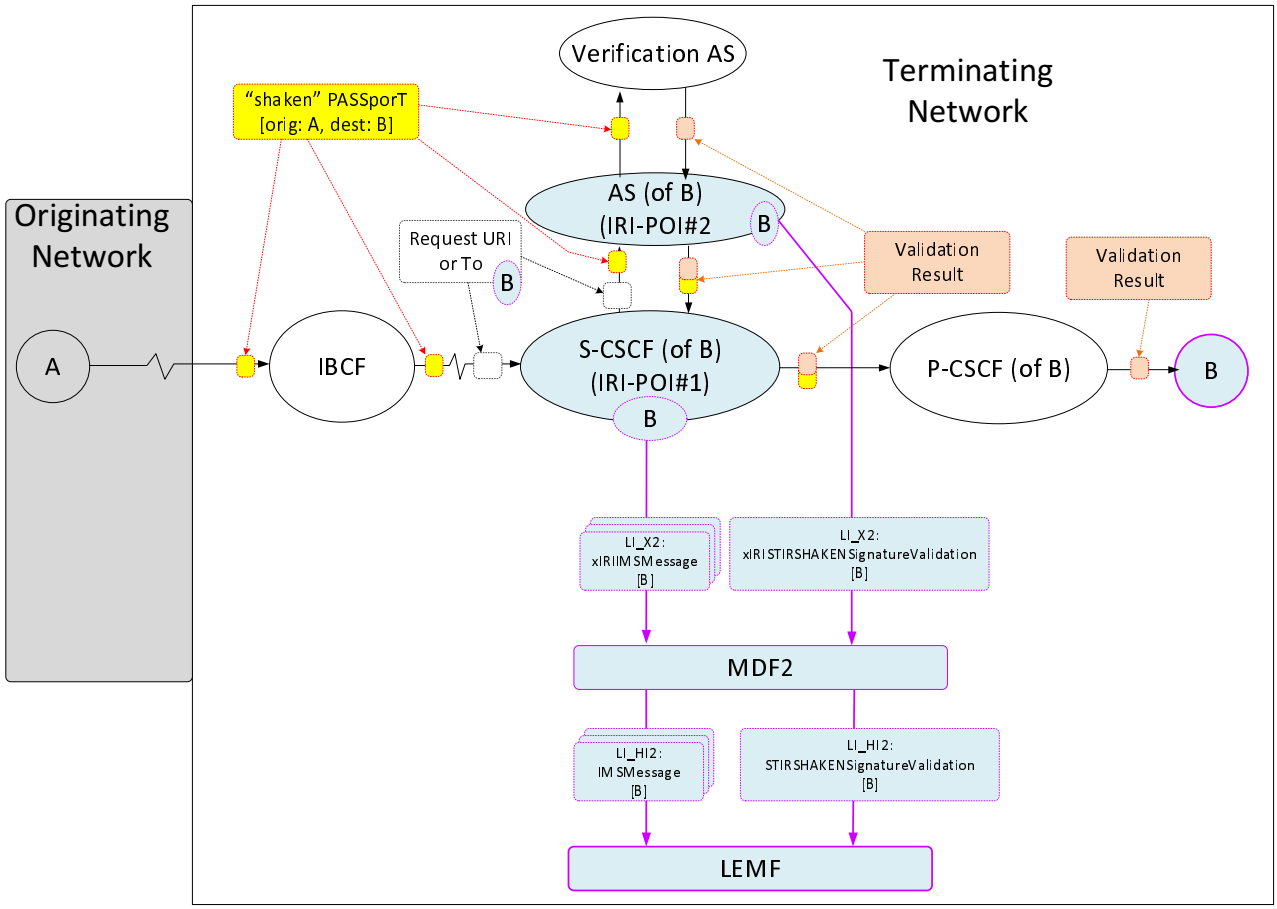


Figure 4.4.4.5.2-2 The LI aspects for scenario 1

For the target Party (B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing to Party B) headers present in the incoming SIP INVITE to perform a target match.

4.4.4.5.3 Scenario 2: Party A (target non-local ID) calls Party B

In this scenario, Party A (from the originating network) calls Party B (in the terminating network). Party A is a target non-local ID in the terminating network.

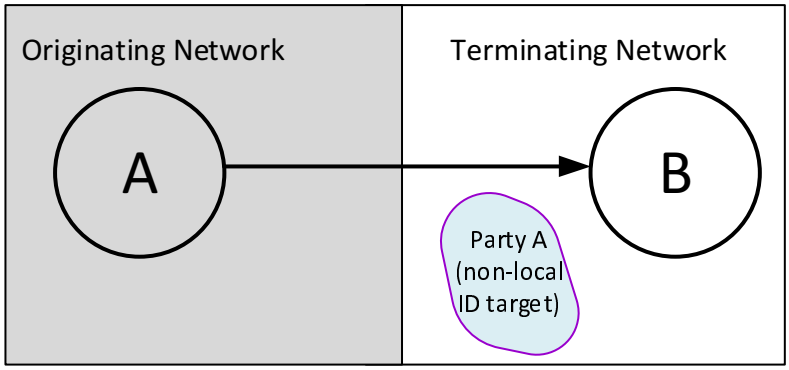


Figure 4.4.4.5.3-1: Party A calls Party B. Party A is the target non-local ID in the terminating network

Figure 4.4.4.5.3-2 illustrates the LI aspects for this scenario.

The illustration shown in figure 4.4.4.5.3-2 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the Ingress IBCF provides the IRI-POI functions for IMS LI.

Since the signing of intra-network session is required, the Telephony AS of B interacts with the Verification AS to validate the received "shaken" PASSporT.

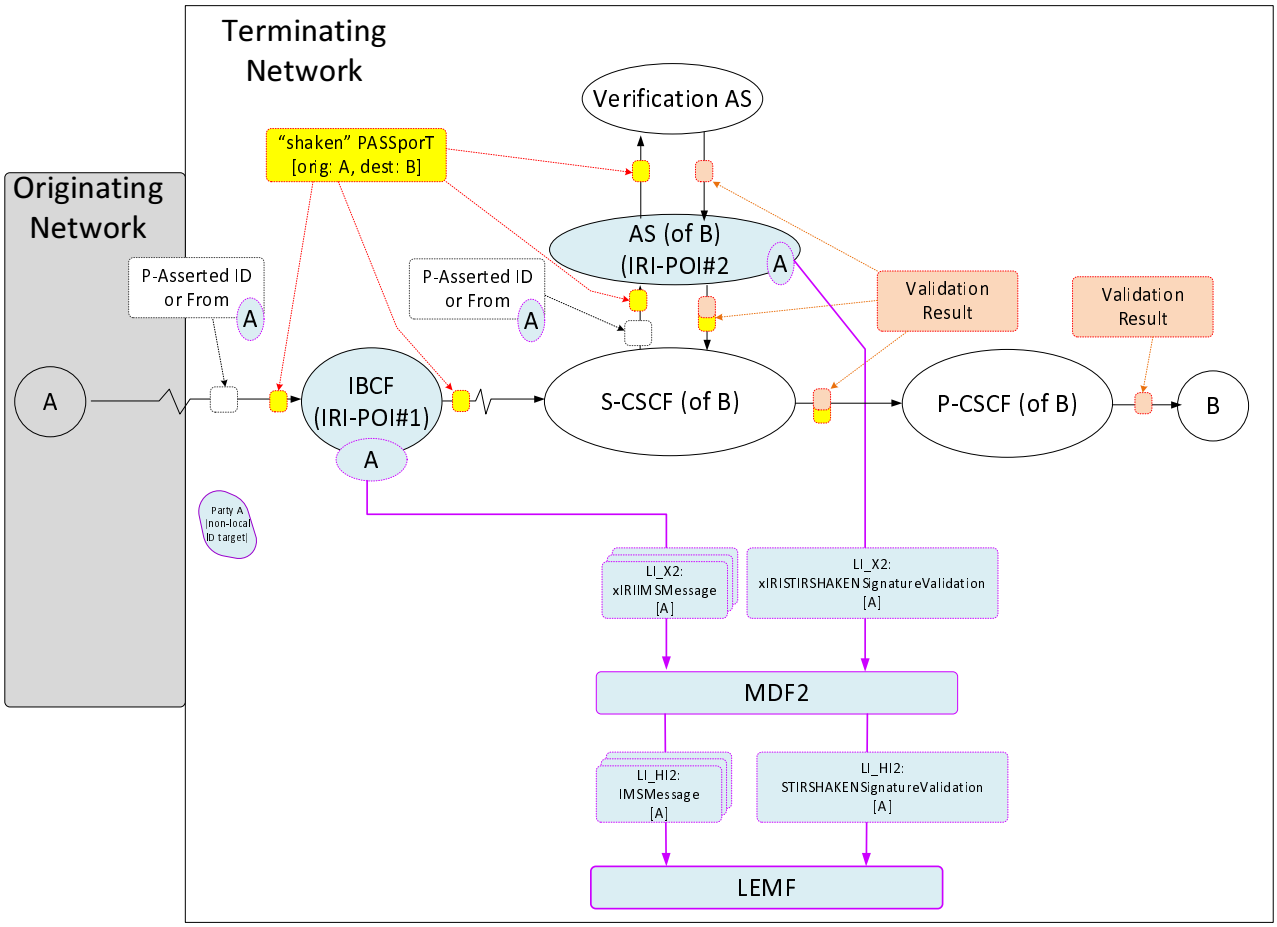


Figure 4.4.4.5.3-2: The LI aspects for scenario 2

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (which points to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the P-Asserted ID or the From headers (which points to Party A) headers present in the incoming SIP INVITE to perform a target match.

4.4.4.5.4 Scenario 3: Party A (target non-local ID) calls Party B (target non-local ID) redirected to Party C (target) redirected to Party D (target)

4.4.4.5.4.1 The scenario

Multiple targets are considered here.

In the scenario, Party A calls Party B who has a call forwarding to Party C. Party A and Part B are the target non-local IDs in the terminating network. Party C has a call forwarding to Party D in the terminating network. Both Party C and Party D are targets in the terminating network.

LEMF-1 is intercepting the target non-local Party A and target non-local ID Party B. The LEMF-2 is intercepting the target Party C and LEMF-3 is intercepting the target Party D.

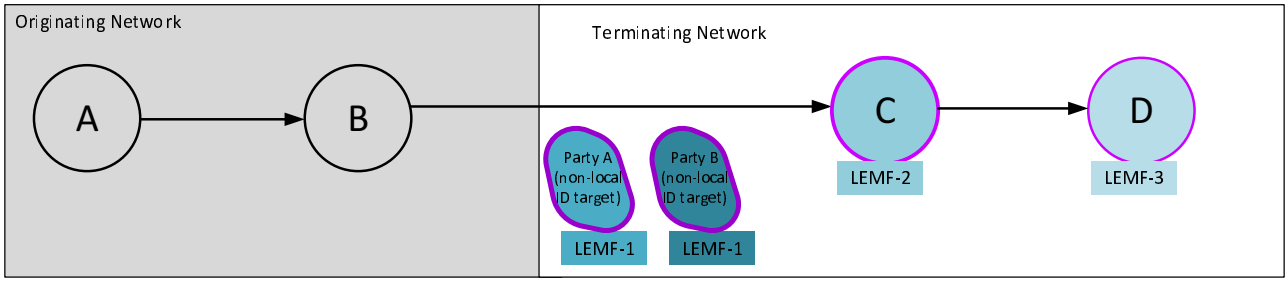


Figure 4.4.4.5.4.1-1: Redirection scenario, multiple targets

The descriptions given in clause 4.4.3 apply to this scenario as well.

Figures 4.4.4.5.4.2-1 and 4.4.4.5.4.3-1 illustrate the LI aspects for this scenario.

4.4.4.5.4.2 ReportDiversionPASSporTInfo is False

The illustration shown in figure 4.4.4.5.4.2-1 assumes that the ReportDiversionPASSporTInfo is False.

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI for all targets. Accordingly, the S-CSCF of C provides the IRI-POI functions for IMS LI for target Party C, S-CSCF of D provides the IRI-POI functions for IMS LI for target Party C and Ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID Party A and target non-local ID B.

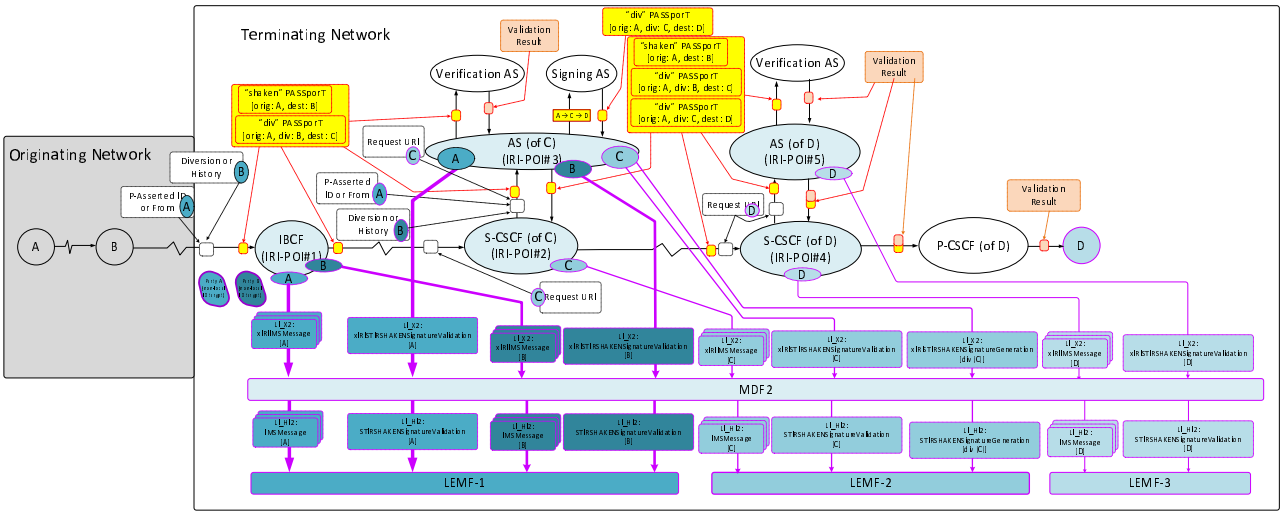


Figure 4.4.4.5.4.2-1: The LI aspects for scenario 3 (ReportDiversionPASSporTInfo is False)

For target non-local ID (Party A)

The ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of C) interacts with the Verification AS to verify the received "shaken" and "div" PASSporTs and the IRI-POI present in the same Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) headers present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party B)

The ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of C) interacts with the Verification AS to verify the received "shaken" and "div" PASSporTs and the IRI-POI present in the same Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the Diversion or the History Info headers (pointing to Party B) headers present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Diversion or the History Info headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The Telephony AS (of C) interacts with the Verification AS to verify the received "shaken" and "div" PASSporTs and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-2.

The Telephony AS (of C) interacts with the Signing AS to authenticate/sign C's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the S-CSCF (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

For target (Party D)

The IRI-POI present in the S-CSCF (of D) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-3.

The Telephony AS (of D) interacts with the Verification AS to verify the received "shaken" and "div" PASSporTs and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-3.

The IRI-POI present in the S-CSCF (of D) uses the Request URI header (pointing to Party D) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of D) uses the Request URI header (pointing to Party D) present in the incoming SIP INVITE to perform a target match.

4.4.4.5.4.3 ReportDiversionPASSporTInfo is set to "True"

The illustration shown in figure 4.4.4.5.4.3-1 assumes that the ReportDiversionPASSporT info is True.

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI for all targets. Accordingly, the S-CSCF of C provides the IRI-POI functions for IMS LI for target Party C, S-CSCF of D provides the IRI-POI functions for IMS LI for target Party C and Ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID (Party A) and target non-local ID (Party B).

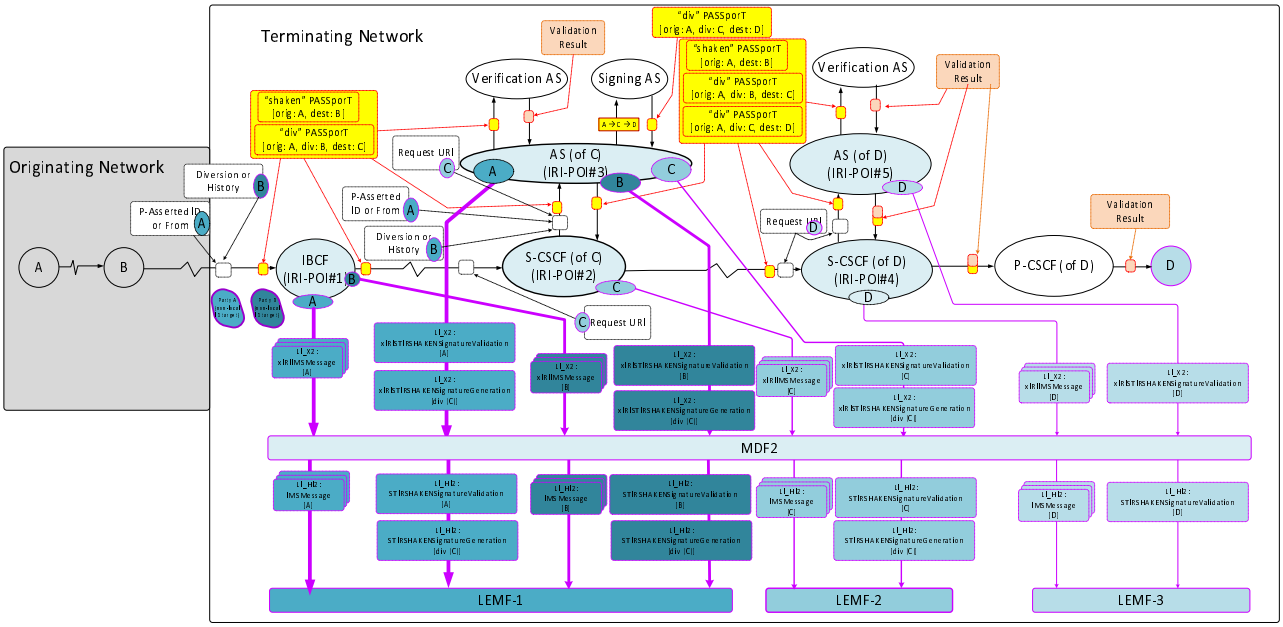


Figure 4.4.4.5.4.3-1: The LI aspects for scenario 3 (ReportDiversionPASSporTInfo is True)

For target non-local ID (Party A)

The ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of C) interacts with the Verification AS to verify the received "shaken" and "div" PASSporTs and the IRI-POI present in the same Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The Telephony AS (of C) interacts with the Signing AS to authenticate/sign C's identity and with ReportDiversionPASSporTInfo set to "True" for target non-local ID (Party A), the IRI-POI present in the Telephony AS (Party C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) headers present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the P-Asserted ID or the From headers (pointing to Party A) headers present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party B)

The ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of C) interacts with the Verification AS to verify the received "shaken" and "div" PASSporTs and the IRI-POI present in the same Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The Telephony AS (of C) interacts with the Signing AS to authenticate/sign C's identity and with ReportDiversionPASSporTInfo set to "True" for target non-local ID (Party B), the IRI-POI present in the Telephony AS (Party C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the Diversion or the History Info headers (pointing to Party B) headers present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Diversion or the History Info headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The Telephony AS (of C) interacts with the Verification AS to verify the received "shaken" and "div" PASSporTs and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-2.

The Telephony AS (of C) interacts with the Signing AS to authenticate/sign C's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the S-CSCF (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

For target (Party D)

The IRI-POI present in the S-CSCF (of D) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-3.

The Telephony AS (of D) interacts with the Verification AS to verify the received "shaken" and "div" PASSporTs and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-3.

The IRI-POI present in the S-CSCF (of D) uses the Request URI header (pointing to Party D) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of D) uses the Request URI header (which points to Party D) present in the incoming SIP INVITE to perform a target match.

4.4.4.6 Inter-network IMS sessions (intermediate network)

4.4.4.6.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI for inter-network IMS sessions for the intermediate network.

The originating network signs and authenticates the calling party's identity and any redirection party identities (if redirection happens in the originating network) and includes the PASSporT(s) in the SIP INVITE message.

The Telephony AS in the intermediate network interacts with the Verification AS to validate the received PASSporTs. In the case of further redirections in the intermediate network, the Telephony AS associated with the redirecting party would interact with the Verification AS to verify the received PASSporT(s) and then interacts with the Signing AS to sign the redirecting party identity.

The Telephony AS provides the LI functions for STIR/SHAKEN related reporting. For the destination target non-local ID, with the option 1 (see clause 4.4.4.4.3.1), the egress IBCF provides the LI functions for STIR/SHAKEN related reporting.

4.4.4.6.2 Scenario 1: Party A calls Party B (target) redirected to Party C

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party A is in the originating network and Part C is in the terminating network. Party B is in the intermediate network and is a target in the intermediate network.

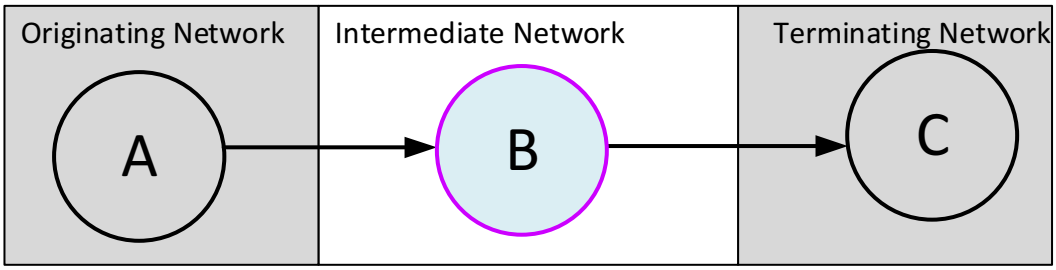


Figure 4.4.4.6.2-1: Party (originating network) calls Party B (target) redirected to Party C (terminating network)

Figure 4.4.4.6.2-2 illustrates the LI aspects for this scenario.

The illustration shown in figure 4.4.4.6.2-2 assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI for target Party B.

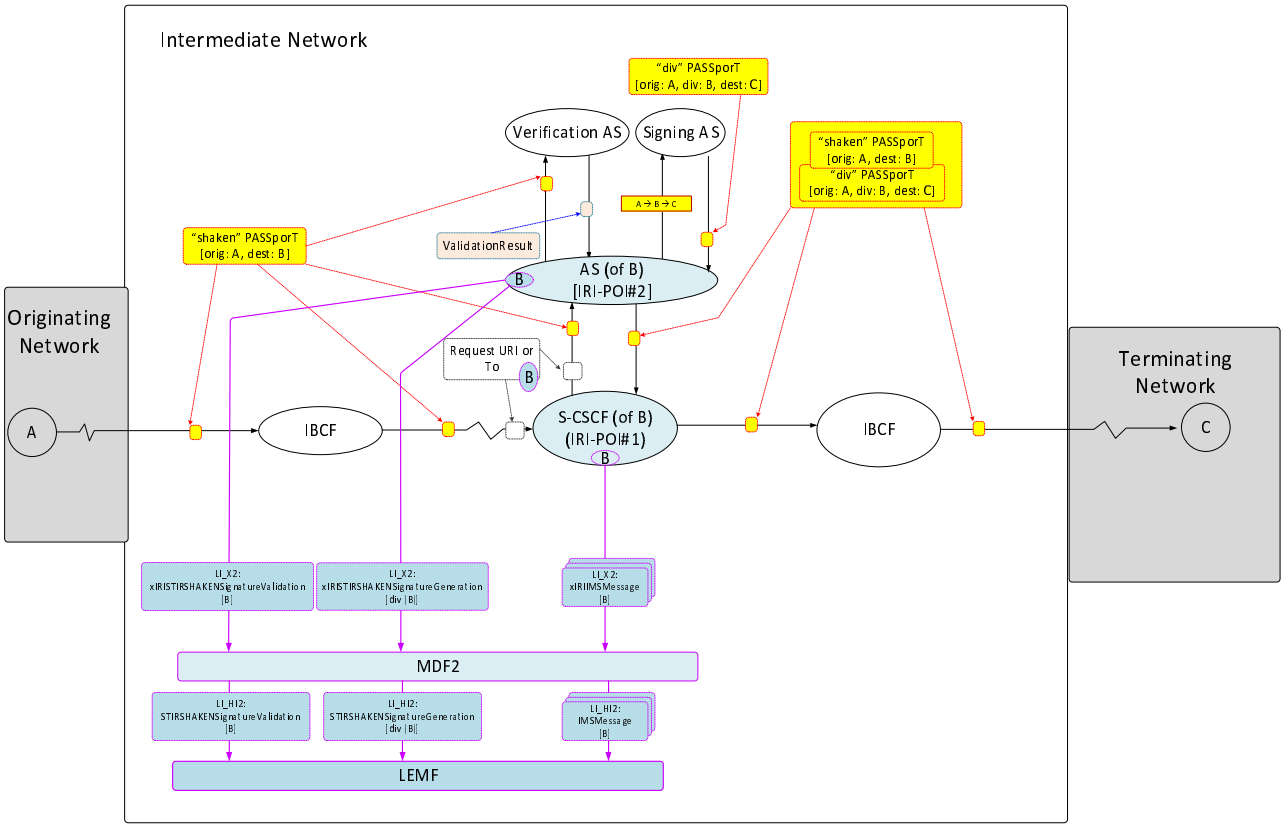


Figure 4.4.4.6.2-2: The LI aspects for scenario 1

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The Telephony AS (of B) interacts with the Signing AS to authenticate/sign B's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

4.4.4.6.3 Scenario 2: Party A (target non-local ID) calls Party B redirected Party C (target non-local ID)

4.4.4.6.3.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party A is in the originating network and is target non-local IDs in the intermediate network. Party C is in the terminating network and is a target non-local ID in the intermediate network.

LEMF-1 is intercepting the target non-local ID Party A. The LEMF-2 is intercepting the target non-local ID (Party C).

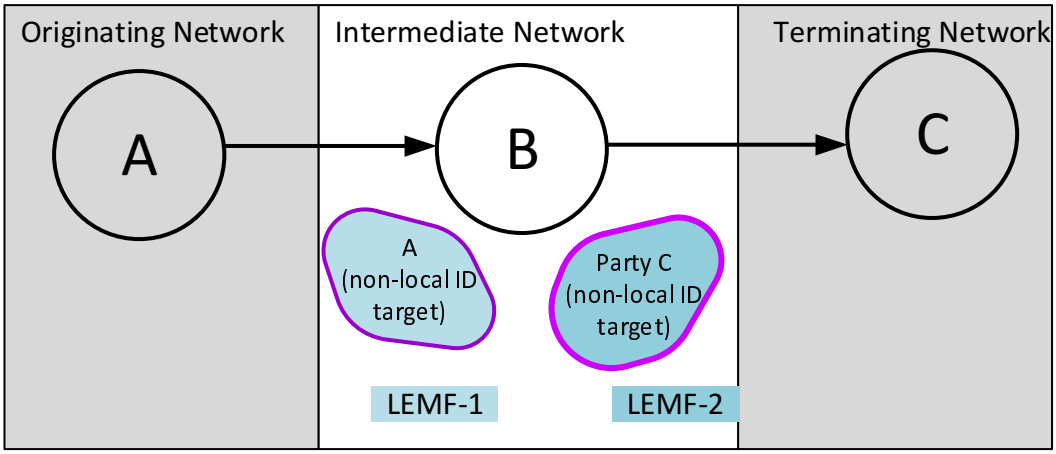


Figure 4.4.4.6.3.1-1: Party A (target non-local ID) calls Party B redirected to Party C (target non-local ID)

Figures 4.4.4.6.3.2-1 and 4.4.4.6.3.3-1 illustrate the LI aspects for this scenario.

4.4.4.6.3.2 ReportDiversionPASSporTInfo is set to "False"

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID Party A and the egress IBCF provides the IRI-POI for IMS LI for the target non-local ID Party C.

The illustration assumes option 1 for the STIR/SHAKEN related LI reporting when the destination is a target non-local ID. Therefore, the Telephony AS (of B) provides the IRI-POI for STIR/SHAKEN related LI reporting for target non-local ID (Party C).

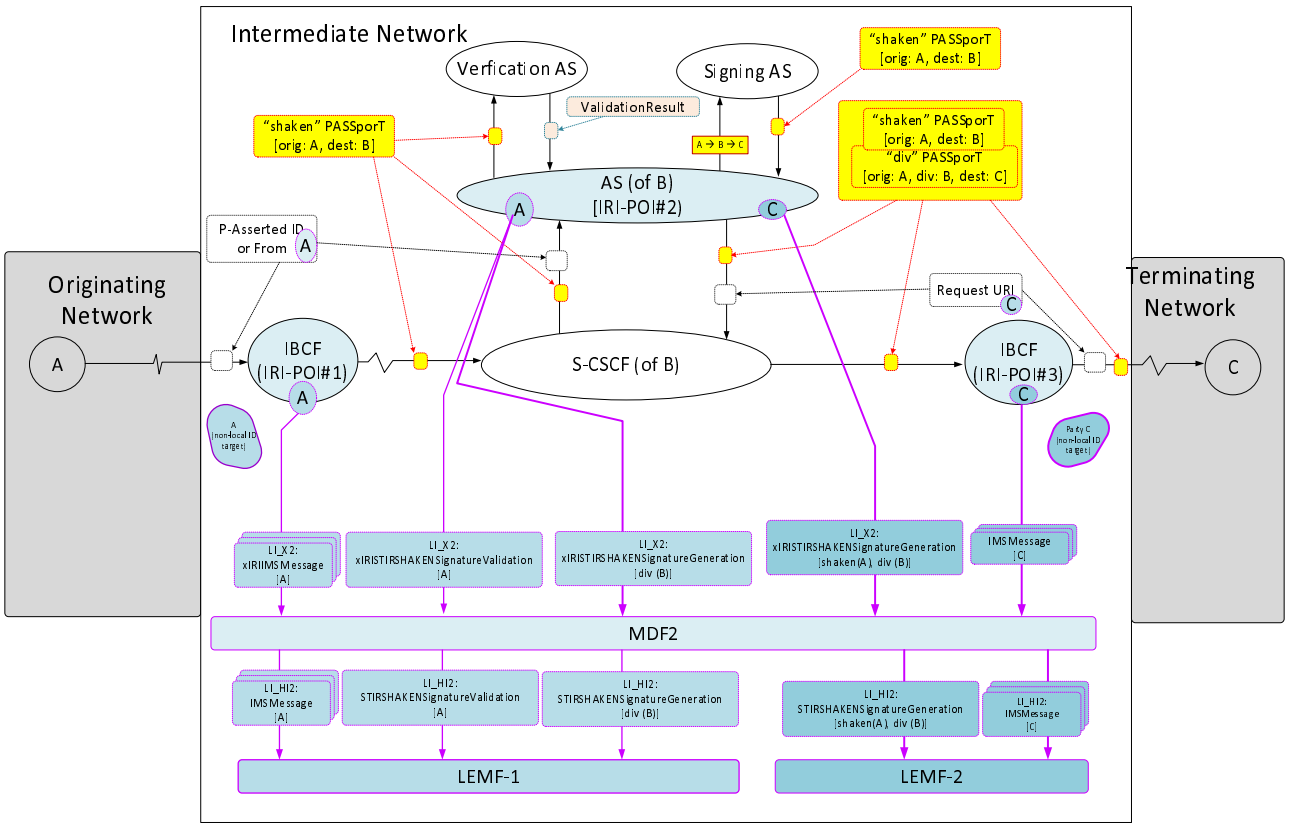


Figure 4.4.4.6.3.2-1: The LI aspects for scenario 2 (ReportDiversionPASSporT is "False")

For target non-local ID (Party A)

The IRI-POI present in the Ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the P-Asserted ID or the From headers (pointing to Party A) headers present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party C)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in the Telephony AS (of B) that signs/authenticates B's identity, upon seeing that the destination is a target non-local ID, generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A and the "div" PASSporT for party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the egress IBCF uses the Request URI (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

4.4.4.6.3.3 ReportDiversionPASSporTInfo is set "True"

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID Party A and the Egress IBCF provides the IRI-POI for IMS LI for the target non-local ID Party C.

The illustration assumes option 1 for the STIR/SHAKEN related LI reporting when the destination is a target non-local ID. Therefore, the Telephony AS (of B) provides the IRI-POI for STIR/SHAKEN related LI reporting for target non-local ID (Party C).

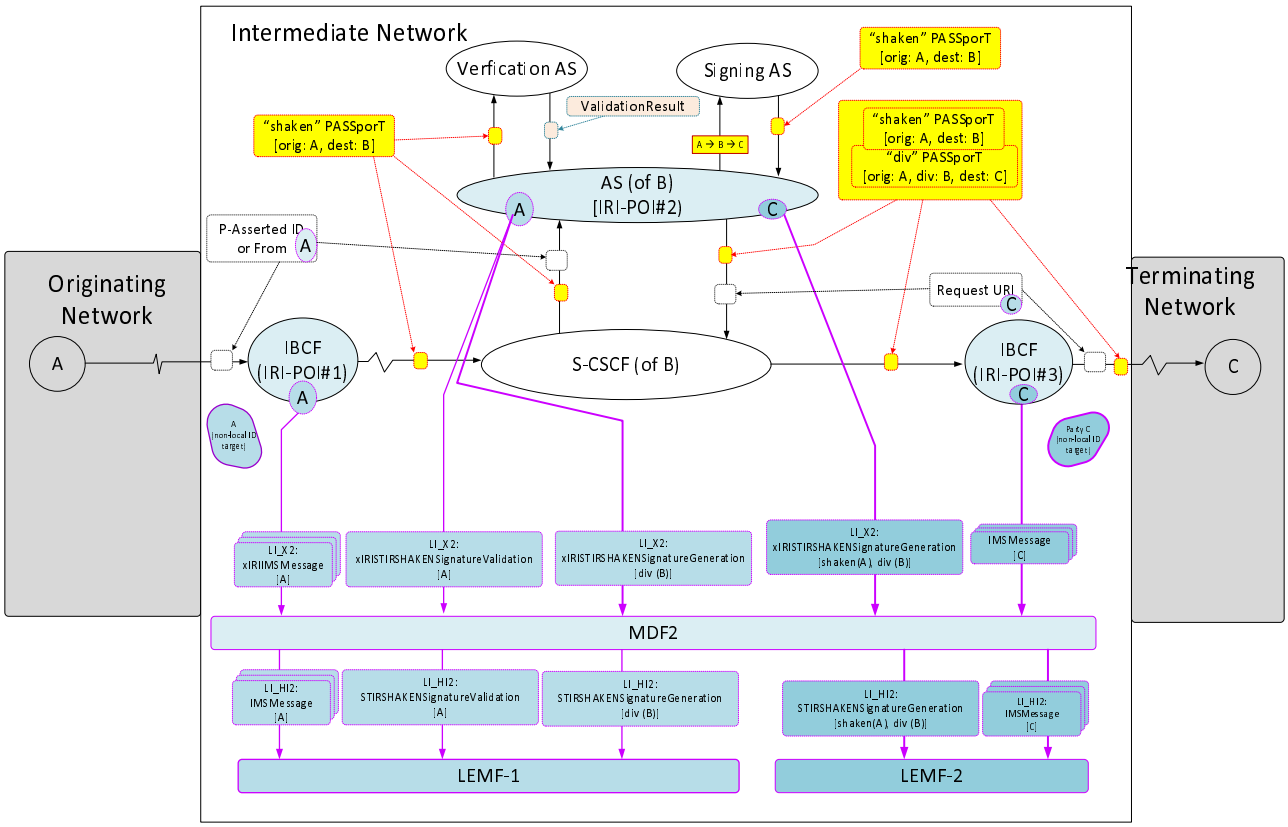


Figure 4.4.4.6.3.3-1: The LI aspects for scenario 2 (ReportDiversionPASSporT is "True")

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The Telephony AS (of B) interacts with the Signing AS to authenticate/sign B's identity and with ReportDiversionPASSporTInfo set to "True" for target non-local ID (Party A), the IRI-POI present Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the P-Asserted ID or the From headers (which points to Party A) headers present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party C)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in the Telephony AS (of B) that signs/authenticates B's identity, upon detecting that the destination is a target non-local ID generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A and the "div" PASSporT for party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2. The xIRI includes all the PASSporTs.

The IRI-POI present in the egress IBCF uses the Request URI (pointing to Party D) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

4.4.4.6.4 Scenario 3: Party A calls Party B (target) redirected to Party C (target) redirected to Party D

4.4.4.6.4.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C who in turn has call forwarding to Party D. Party A is in the originating network and Part D is in the terminating network. Party B and Party C are in the intermediate network and both Party B and Party C are the targets in the terminating network.

LEMF-1 is intercepting the target Party B and the LEMF-2 is intercepting the target Party C.

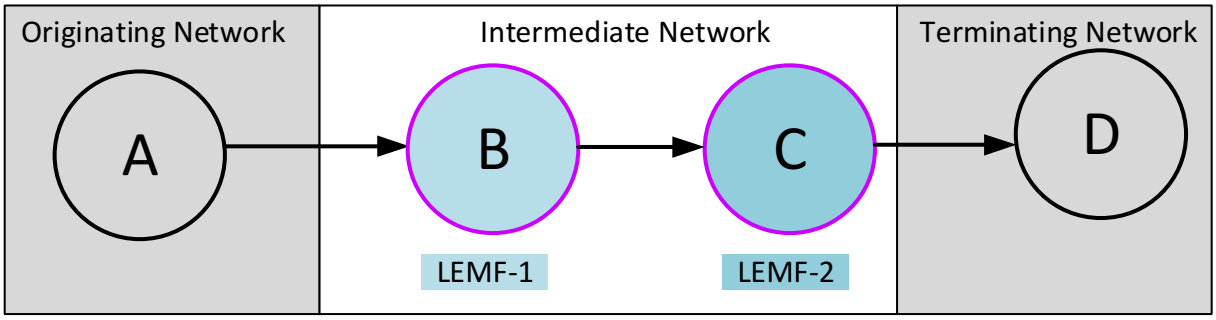


Figure 4.4.4.6.4.1-1: Party (originating network) calls Party B (target), redirected to Party C (target) redirected to Party D (terminating network)

The descriptions given in clause 4.4.3 apply to this scenario as well.

Figures 4.4.4.6.4.2-1 and 4.4.4.6.4.3-1 illustrate the LI aspects for this scenario.

4.4.4.6.4.2 With ReportDiversionPASSporTInfo is set to "False"

The illustration shown in figure 4.4.4.6.4.2-1 assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI for target Party B. S-CSCF of C provides the IRI-POI functions for IMS LI for target Party C.

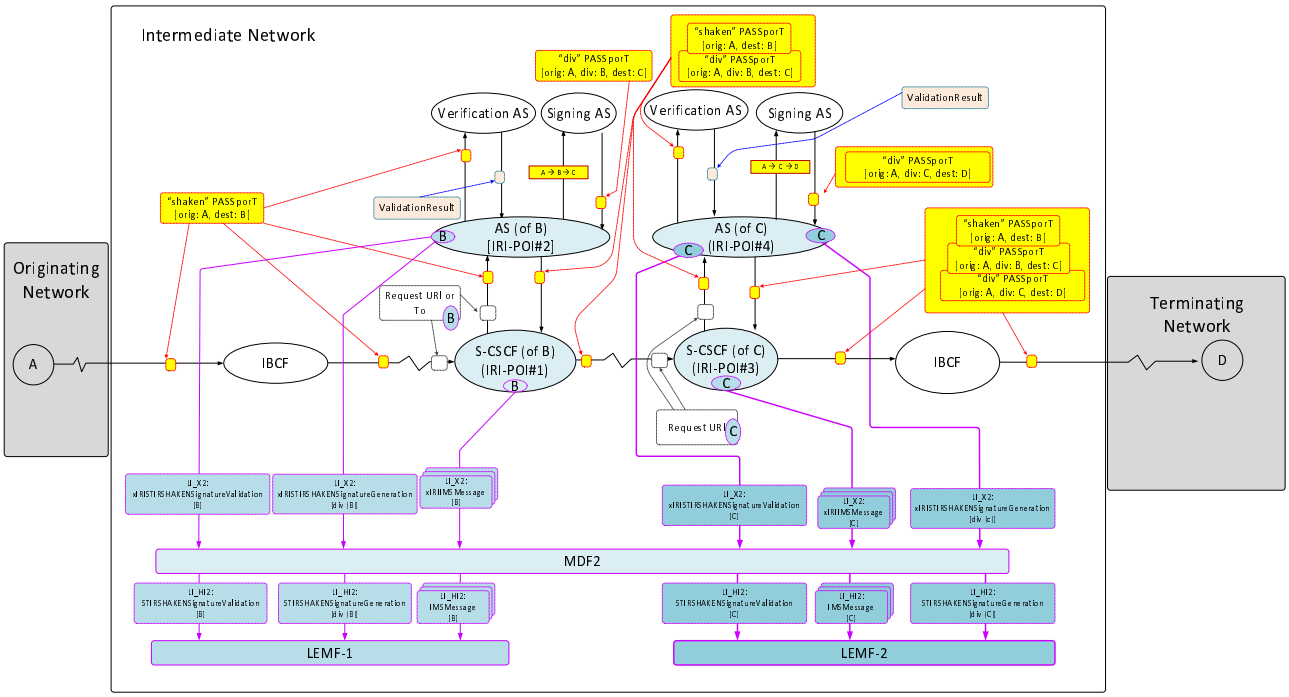


Figure 4.4.4.6.4.2-1: The LI aspects for scenario 3 (ReportDiversionPASSporTInfo is set to "False")

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The Telephony AS (of B) interacts with the Signing AS to authenticate/sign B's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The Telephony AS (of C) interacts with the Verification AS to verify the received "shaken" and "div" PASSporTs and the IRI-POI present in the same Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-2.

The Telephony AS (of C) interacts with the Signing AS to authenticate/sign C's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the S-CSCF (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

4.4.4.6.4.3 With ReportDiversionPASSporTInfo is set to "True"

The illustration shown in figure 4.4.4.6.4.3-1 assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI for target Party B. S-CSCF of C provides the IRI-POI functions for IMS LI for target Party C.

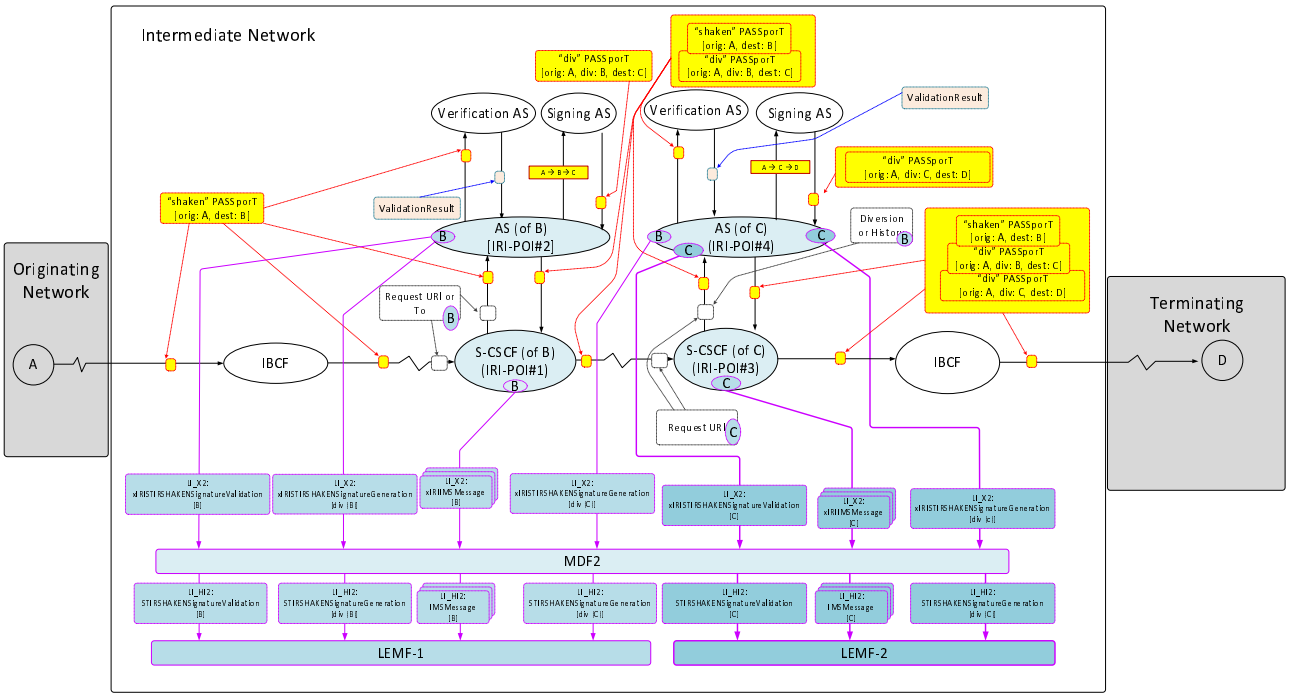


Figure 4.4.4.6.4.3-1: The LI aspects for scenario 3 (ReportDiversionPASSporTInfo is set to "True")

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The Telephony AS (of B) interacts with the Signing AS to authenticate/sign B's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The Telephony AS (of C) interacts with the Signing AS to authenticate/sign C's identity and with ReportDiversionPASSporTInfo set to True for target (Party B), the IRI-POI present in the Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or To headers (pointing to Party B) headers present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Diversion or History Info headers (pointing to Party B) headers present in the incoming SIP INVITE to perform a target match.

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The Telephony AS (of C) interacts with the Verification AS to verify the received "shaken" and "div" PASSporTs and the IRI-POI present in the same Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-2.

The Telephony AS (of C) interacts with the Signing AS to authenticate/sign C's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the S-CSCF (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

4.4.4.6.5 Scenario 4: Party A (target non-local ID) calls Party B redirected Party C (target non-local ID).

4.4.4.6.5.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C who in turn has a call forwarding to Party D. Party A is in the originating network and is target non-local ID in the intermediate network. Party D is in the terminating network and is a target non-local ID in the intermediate network.

LEMF-1 is intercepting the target non-local Party A. The LEMF-2 is intercepting the target non-local ID (Party D).

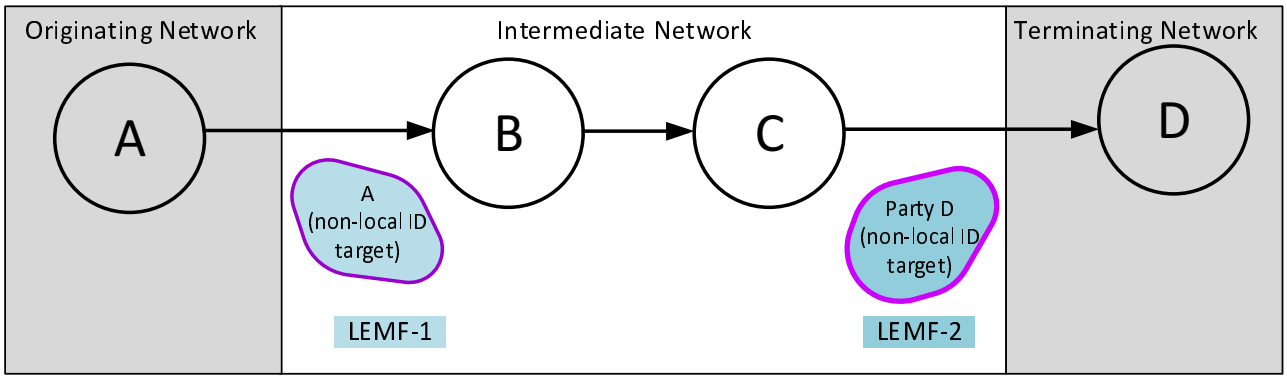


Figure 4.4.4.6.5.1-1: Party A (target non-local ID) calls Party B redirected to Party C redirected to Party D (target non-local ID)

The descriptions given in clause 4.4.3 apply to this scenario as well.

4.4.4.6.5.2 ReportDiversionPASSporTInfo is set to "False"

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID Party A and the Egress IBCF provides the IRI-POI for IMS LI for the target non-local ID Party D.

The illustration assumes option 1 for the STIR/SHAKEN related LI reporting when the destination is a target non-local ID. Therefore, the Telephony AS (of C) provides the IRI-POI for STIR/SHAKEN related LI reporting for the target non-local ID Party D.

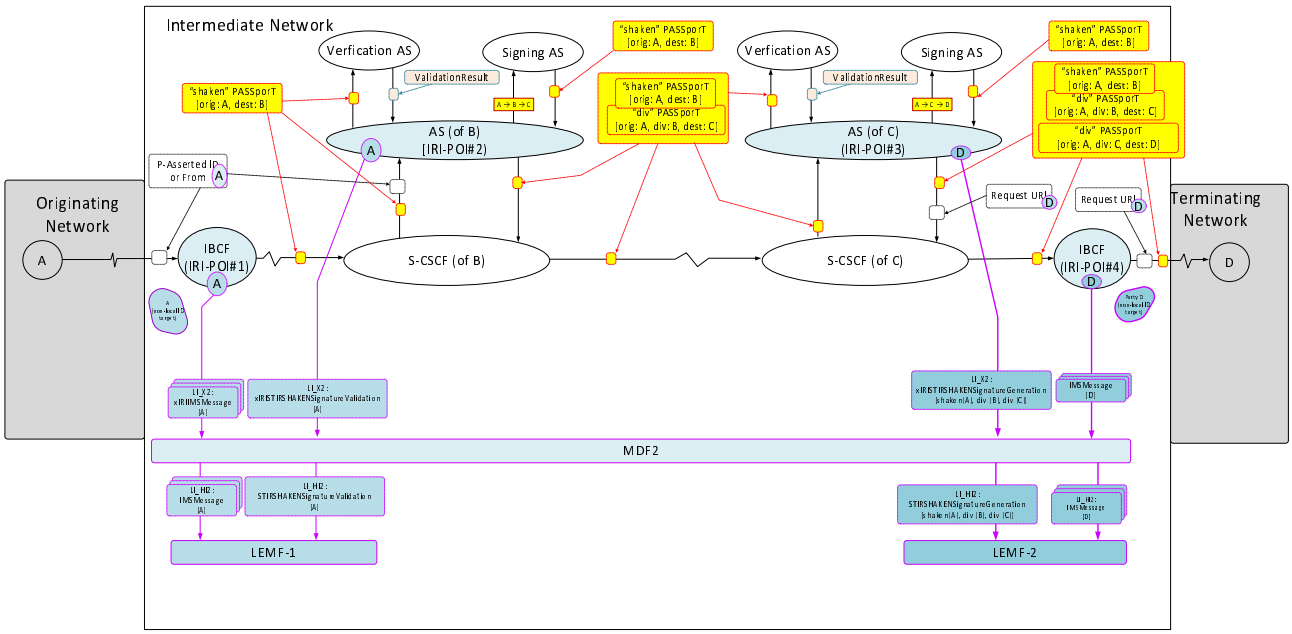


Figure 4.4.4.6.5.2-1: The LI aspects for scenario 4 (ReportDiversionPASSporT is set to "False")

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the P-Asserted ID or the From headers (pointing to Party A) headers present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party D)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in Telephony AS (of C) that signs/authenticates C's identity, upon seeing that the destination is a target non-local ID, generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A and the two "div" PASSporTs (for Party B and Party C)) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the egress IBCF uses the Request URI (pointing to Party D) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Request URI (pointing to Party D) present in the outgoing SIP INVITE to perform a target match.

4.4.4.6.5.3 ReportDiversionPASSporTInfo is set to "True"

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID Party A and the egress IBCF provides the IRI-POI for IMS LI for the target non-local ID Party D.

The illustration assumes option 1 for the STIR/SHAKEN related LI reporting when the destination is a target non-local ID. Therefore, the Telephony AS (of C) provides the IRI-POI for STIR/SHAKEN related LI reporting for the target non-local ID Party D.

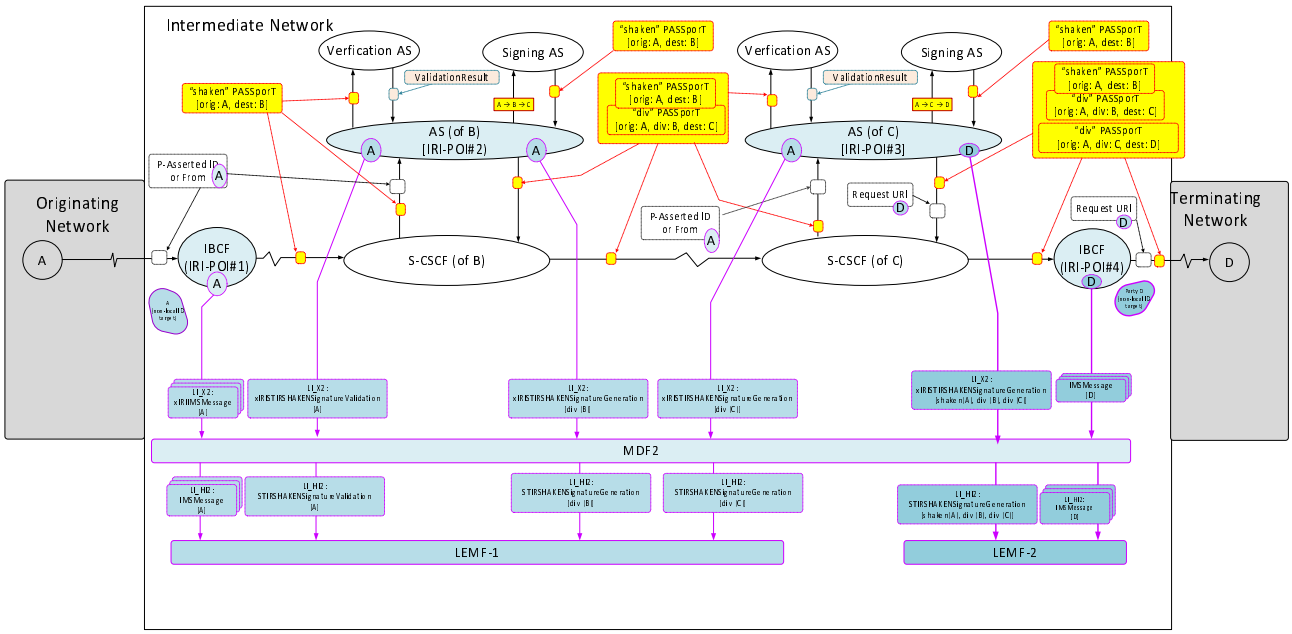


Figure 4.4.4.6.5.3-1: The LI aspects for scenario 4 (ReportDiversionPASSporT is set to "True")

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The Telephony AS (of B) interacts with the Signing AS to authenticate/sign B's identity and with ReportDiversionPASSporTInfo set to "True" for target non-local ID (Party A), the IRI-POI present Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The Telephony AS (of C) interacts with the Signing AS to authenticate/sign C's identity and with ReportDiversionPASSporTInfo set to True for target non-local ID (Party A), the IRI-POI present Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the Telephony AS (of B) and Telephony AS (of C) use the P-Asserted ID or the From headers (which points to Party A) headers present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party D)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in Telephony AS (of C) that signs/authenticates C's identity, upon seeing that the destination is a target non-local ID, generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A and the two "div" PASSporTs (for Party B and Party C)) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the egress IBCF uses the Request URI (pointing to Party D) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Request URI (pointing to Party D) present in the outgoing SIP INVITE to perform a target match.

4.4.5 Signing is not required for intra-network IMS sessions

4.4.5.1 General

The subsequent sub-clauses illustrate the scenarios when signing is not required for intra-network IMS sessions. The IMS NFs that provide the STIR/SHAKEN related LI reporting are dependent on the call scenario and the target.

When the intra-network signing is not required, based on CSP choice either the egress IBCF or the last Telephony AS (in the signalling path within the network) would interact with the Signing AS in the originating network (or intermediate network) and either the ingress IBCF or the first Telephony AS (in the signalling path within the network) would interact with the Verification AS in the terminating network (or the intermediate network).

There are a few exceptions to the above facts:

- The Telephony AS interacts with the Signing AS if RCD is supported for the call.

- The Telephony AS that is not able to determine the whereabouts of terminating user may interact with the Signing AS even if it is not the last Telephony AS (in the signalling path within the CSP network) when the CSP choice is Telephony AS.

- The Telephony AS always interacts with the Verification AS if it sees PASSporT (s) within the incoming SIP INVITE without any Validation Results.

- The IBCF does not interact with the Signing AS if it sees PASSporT (s) without any Validation Results in the incoming SIP INVITE message, even if the CSP choice is IBCF.

The illustrations show the IMS LI reporting and STIR/SHAKEN related LI reporting as defined in TS 33.127 [3], TS 33.128 [4]. The TR 33.929-1 illustrates various scenarios for IMS LI with the default option.

Clause 4.3 of the present document indicates the conditions for an IMS NF to have the IRI-POI for the STIR/SHAKEN related reporting.

4.4.5.2 Intra-network IMS sessions

4.4.5.2.1 General

When the signing of intra-network IMS sessions is not required, STIR/SHAKEN related LI reporting applies to an intra-network IMS sessions only when the Telephony AS is not able to determine the whereabouts of the destination party or when RCD is supported for the call. The scenarios presented in the subsequent sub-clauses focus on the RCD aspects.

The Telephony AS that interacts with the Signing AS/Verification AS provides the LI functions for STIR/SHAKEN related reporting. It is assumed that all the PASSporTs that are validated are reported in the STIRSHAKENSignatureValdiation message to the LEMFs.

The calling party's identity, when signed, is included in the "shaken" PASSporT (with calling party's identity in the origination claim and the called party's identity in the destination claim). The redirecting party identity, when signed, is included in the "div" PASSporT (with calling party's identity in the origination claim, redirecting party's identity in the diversion claim and the redirected-to party's identity in the destination claim).

4.4.5.2.2 Scenario 1: Party A (target) calls Party B

4.4.5.2.2.1 The scenario

In this scenario, Party A calls Party B. Party A is the target.



Figure 4.4.5.2.2.1-1: Party A calls Party B. Party A is the target

Clauses 4.4.5.2.2.2 and 4.4.5.2.2.3 illustrate the LI aspects for this scenario.

4.4.5.2.2.2 RCD is not supported

The illustration shown in figure 4.4.5.2.2.2-1 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of A provides the IRI-POI functions for IMS LI.

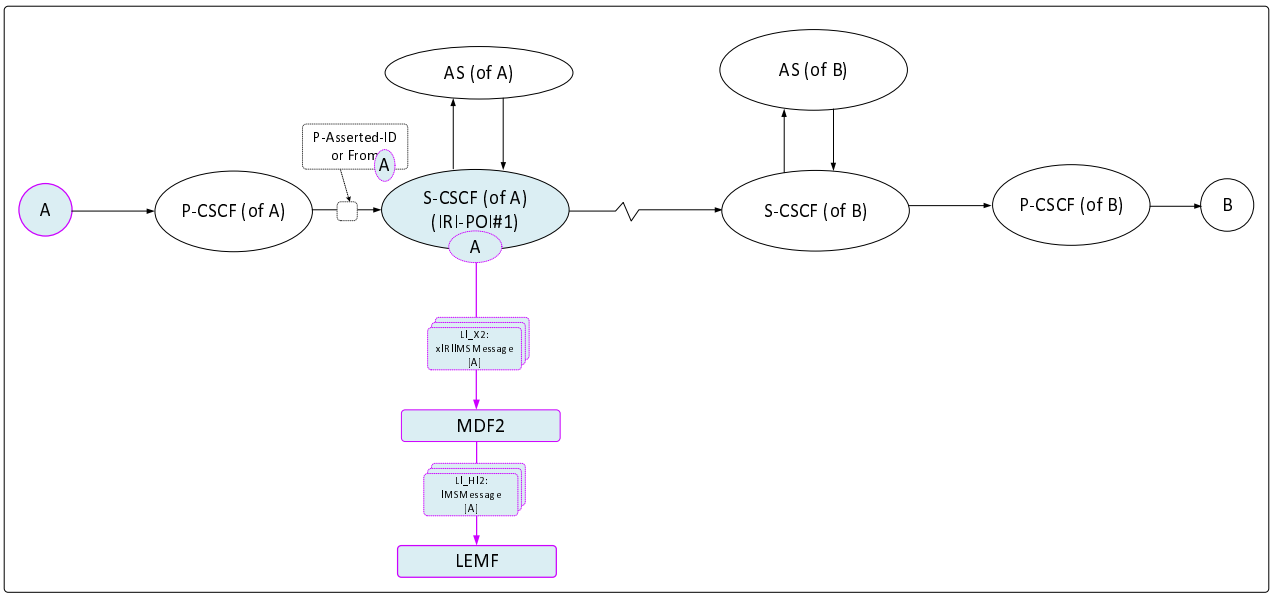


Figure 4.4.5.2.2.2-1: Party A (target) calls Party B (RCD is not supported for the call)

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

Since signing is not required for intra-network sessions, and the Telephony AS knows that the Party B served by the same CSP as that of Party A, the Telephony AS of A does not interact with the Signing AS. Therefore, STIR/SHAKEN related LI reporting does not apply to this case.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

4.4.5.2.2.3 RCD is supported

The illustration shown in figure 4.4.5.2.2.3-1 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of A provides the IRI-POI functions for IMS LI.

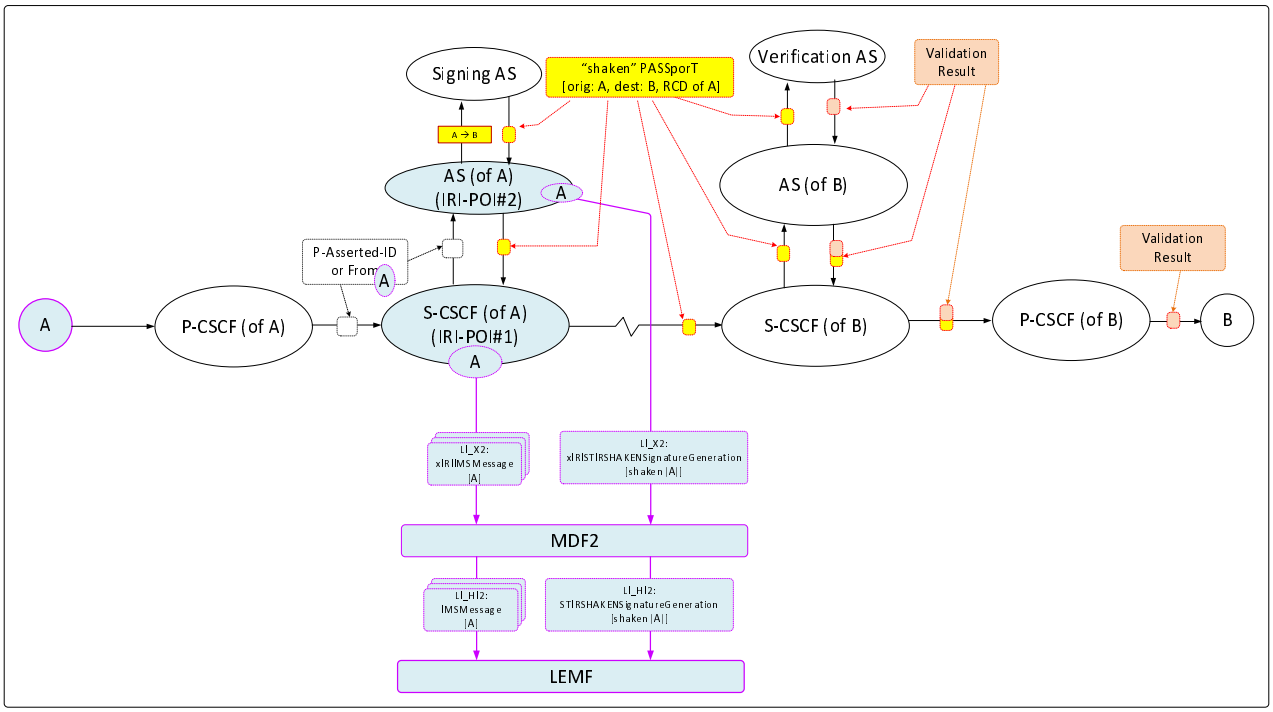


Figure 4.4.5.2.2.3-1: Party A (target) calls Party B (RCD is supported)

For target (Party A)

The IRI-POI present in the P-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

Since RCD is supported for the call from Party A to Party B, the Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the shaken PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

4.4.5.2.3 Scenario 2: Party A calls Party B (target)

4.4.5.2.3.1 The scenario

In this scenario, Party A calls Party B. Party B is the target.



Figure 4.4.5.2.3.1-1: Party A calls Party B. Party B is the target

Clauses 4.4.5.2.3.2 and 4.4.5.2.3.3 illustrate the LI aspects for this scenario.

4.4.5.2.3.2 RCD is not supported

The illustration shown in figure 4.4.5.2.3.2-1 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI.

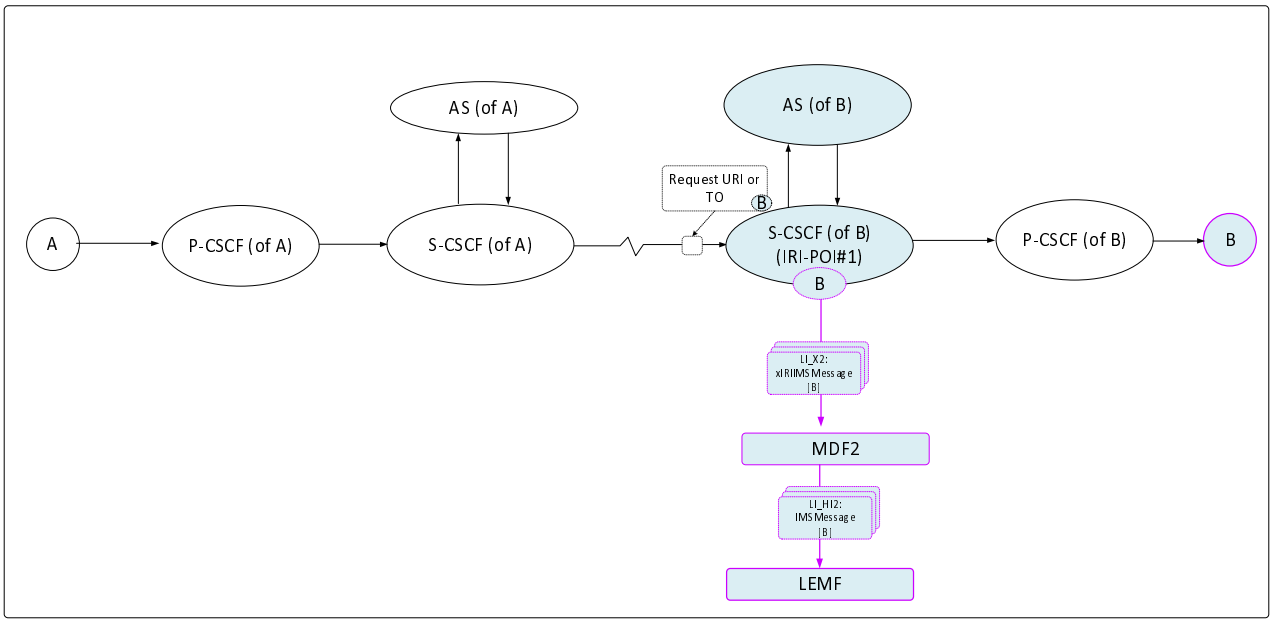


Figure 4.4.5.2.3.2-1: Party A calls Party B (target) - RCD is not supported

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

Since signing is not required for intra-network sessions, and the Telephony AS of A knows that the Party B served by the same CSP as that of Party A, the Telephony AS of A does not interact with the Signing AS. Accordingly, the Telephony AS of B does not see any PASSporTs and hence, does not interact with the Verification AS. Therefore, STIR/SHAKEN related LI reporting does not apply to this case.

For the target (Party B), the IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

4.4.5.2.3.3 RCD is supported

The illustration shown in figure 4.4.5.2.3.3-1 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI.

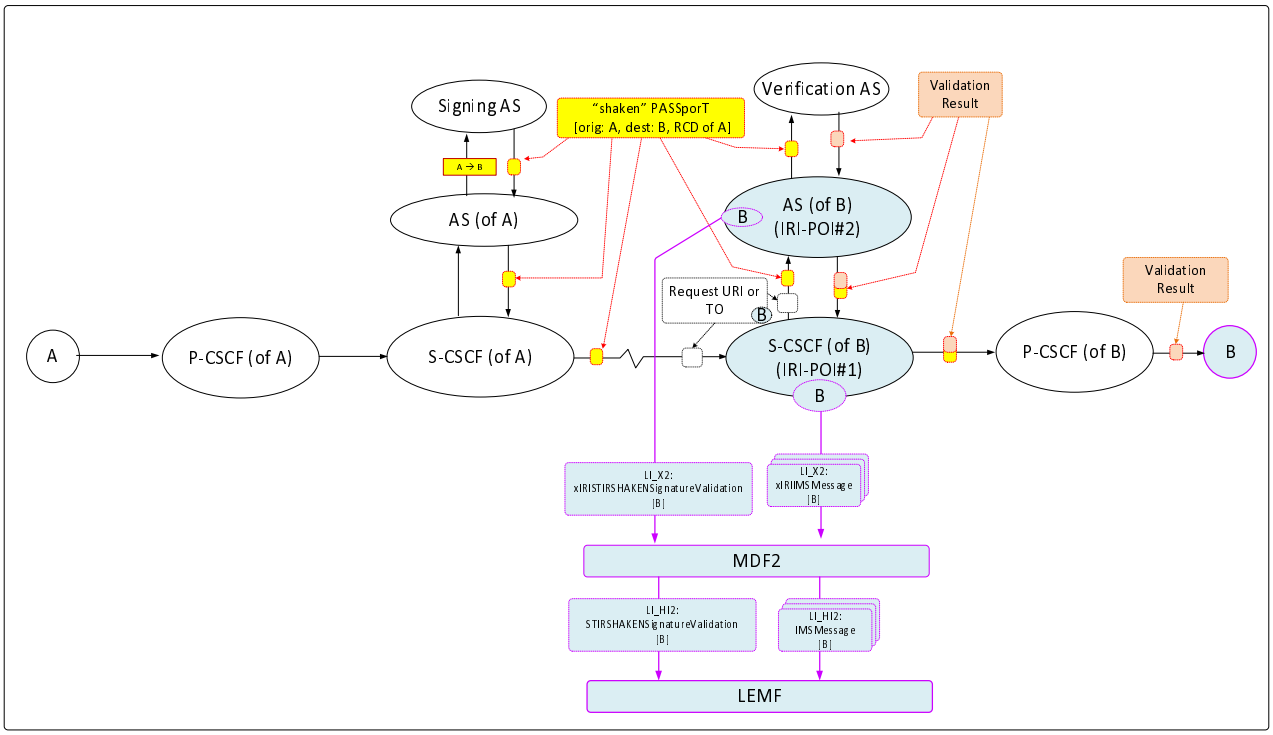


Figure 4.4.5.2.3.3-1: Party A calls Party B (target) - RCD is supported

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record that includes the RCD to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation that includes the RCD to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing to Party B) headers present in the incoming SIP INVITE to perform a target match.

4.4.5.2.4 Scenario 3: Party A (target) calls Party B redirected to Party C

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party A is the target.

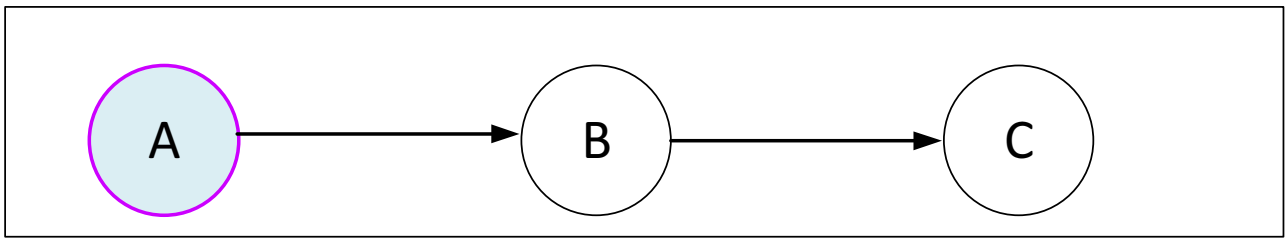


Figure 4.4.5.2.4-1: Party A (target) calls Party B redirected to Party C

Figure 4.4.5.2.4-2 illustrates the LI aspects for this scenario.

In this illustration, RCD is supported with Party A as the target. The illustration assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of A provides the IRI-POI functions for IMS LI.

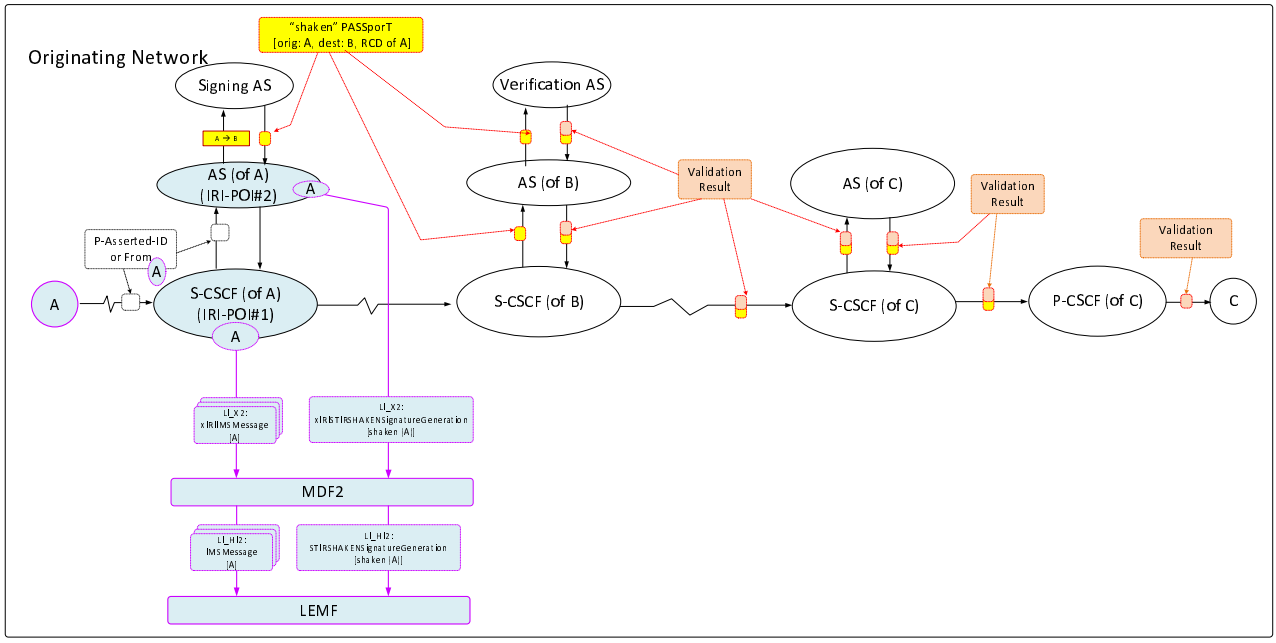


Figure 4.4.5.2.4-2: The LI aspects for scenario 3 (RCD supported)

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

Since RCD is supported for the call, the Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the shaken PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

4.4.5.2.5 Scenario 4: Party A calls Party B (target) redirected to Party C

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party B is the target.

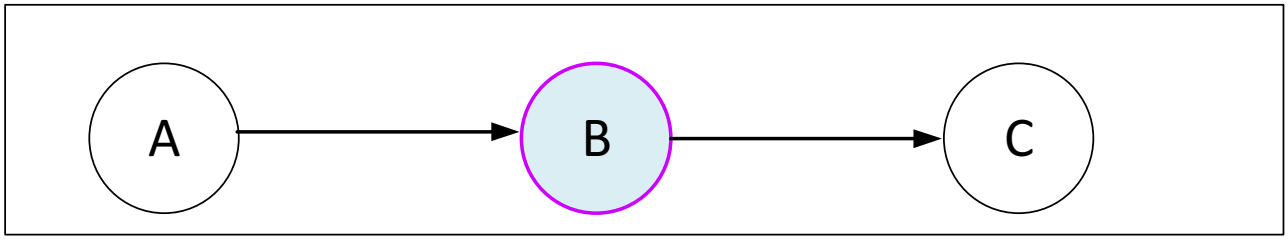


Figure 4.4.5.2.5-1: Party A calls Party B (target) redirected to Party C

Figure 4.4.5.2.5-2 illustrates the LI aspects for this scenario.

In this illustration, RCD is supported with Party B as the target. The illustration assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI.

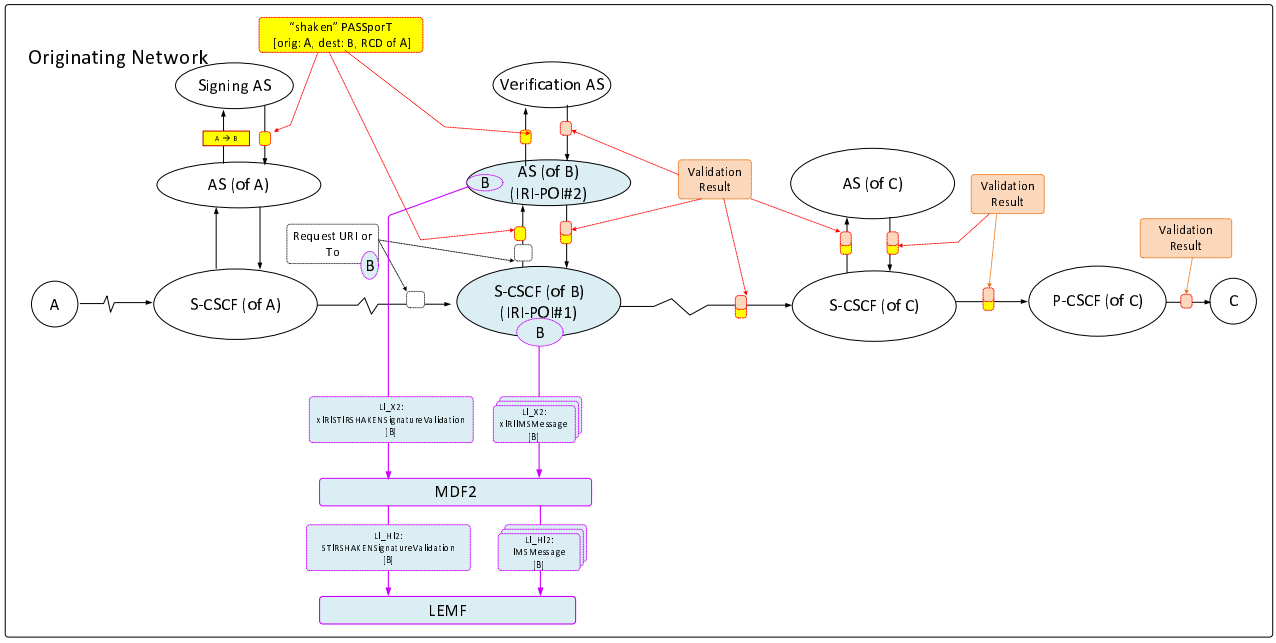


Figure 4.4.5.2.5-2: The LI aspects for scenario 4 (RCD supported)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

Since RCD is supported for the call, the Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity includes the "shaken" PASSporT that includes Party A's identity in the origination claim and Party B's identity in the destination claim in the outgoing SIP INVITE.

The Telephony AS (of B) upon seeing the "shaken" PASSporT in the incoming SIP INVITE interacts with the Verification AS to validate the received PASSporT. The IRI-POI present in the Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

4.4.5.2.6 Scenario 5: Party A calls Party B redirected to Party C (target)

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party C is the target.

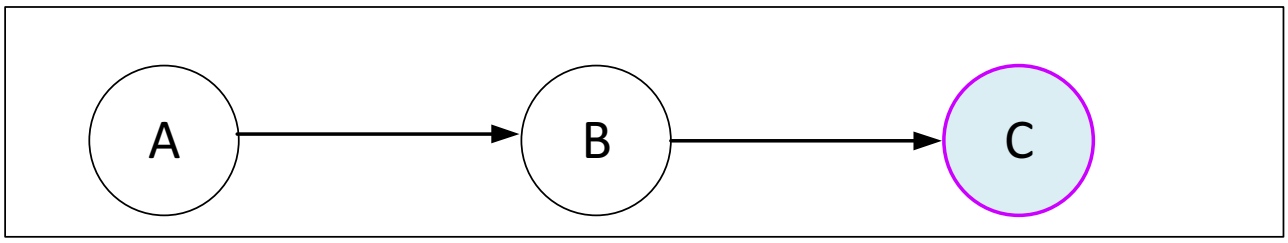


Figure 4.4.5.2.6-1: Party A calls Party B redirected to Party C (target)

Figure 4.4.5.2.6-2 illustrate the LI aspects for this scenario.

In this illustration, RCD is supported on the call with Party C as the target. The illustration assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of C provides the IRI-POI functions for IMS LI.

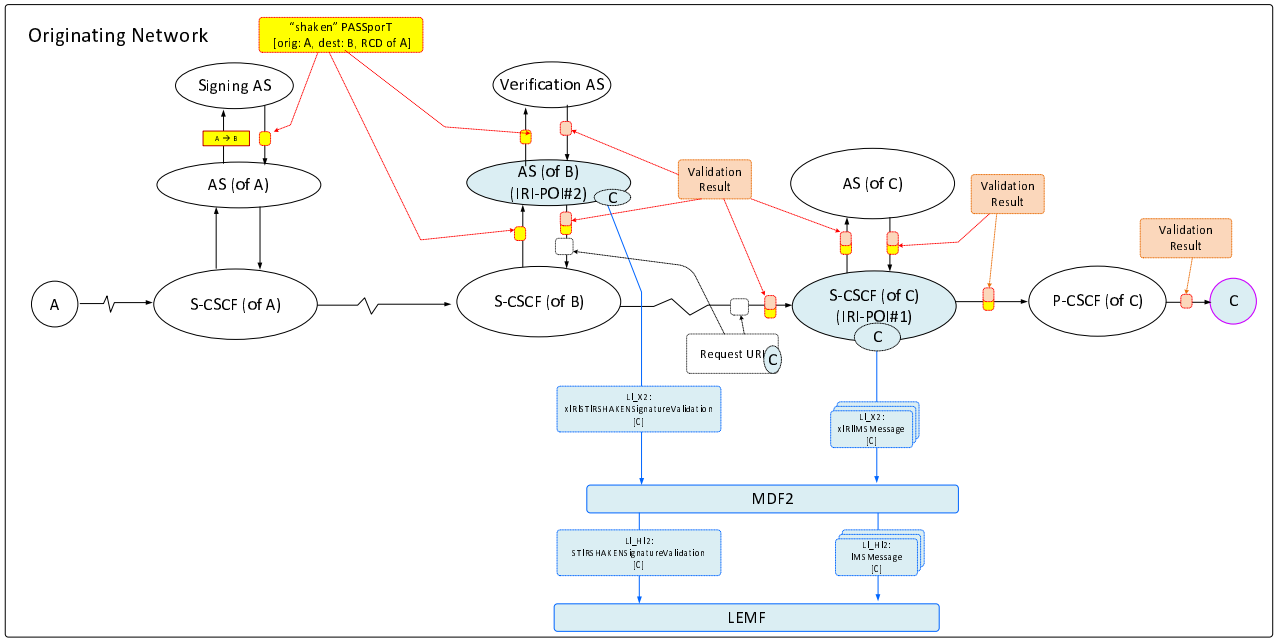


Figure 4.4.5.2.6-2: The LI aspects for scenario 5 (RCD supported)

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

Since RCD is supported for the call, the Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and includes the "shaken" PASSporT that includes Party A's identity in the origination claim and Party B's identity in the destination claim in the outgoing SIP INVITE.

The Telephony AS (of B) upon seeing the "shaken" PASSporT in the incoming SIP INVITE interacts with the Verification AS to validate the received PASSporT. The IRI-POI present in the Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the S-CSCF (of C) uses the Request URI header (pointing to Party C) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI header (pointing to Party C) of the outgoing SIP INVITE to perform a target match.

4.4.5.3 Inter-network IMS sessions (originating network, basic scenarios)

4.4.5.3.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI for inter-network IMS sessions in the originating network.

The calling party's identity, when signed, is included in the "shaken" PASSporT (with calling party's identity in the origination claim and the called party's identity in the destination claim).

4.4.5.3.2 Scenario 1: Party A (target) calls Party B

In this scenario, Party A (in originating network) calls Party B (in the terminating network). Party A is the target in the originating network.

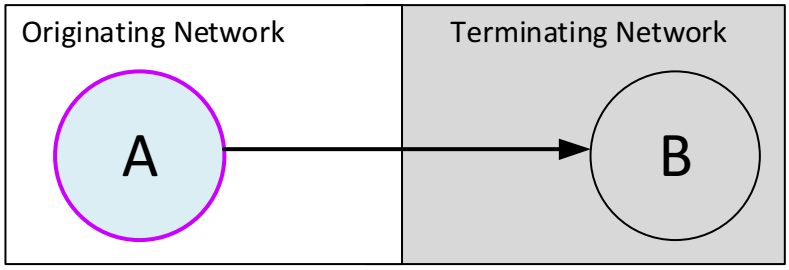


Figure 4.4.5.3.2-1: Scenario 1 - Party A (target) calls Party B.

Figure 4.4.5.4.2-2 illustrates the LI aspects for this scenario when the CSP choice is IBCF.

NOTE: When the CSP choice is Telephony AS, for this call scenario, the LI aspects are the same as in the case when intra-network signing is required since the Telephony AS (of A) is the only one and the last one on the signalling path. Accordingly, the figure 4.4.4.4.2-2 can be referenced for the illustration. The same is true when the RCD is supported for the call.

The illustration shown in figure 4.4.5.3.2-2 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of A provides the IRI-POI functions for IMS LI.

As shown, in the illustration, the egress IBCF interacts with the Signing AS.

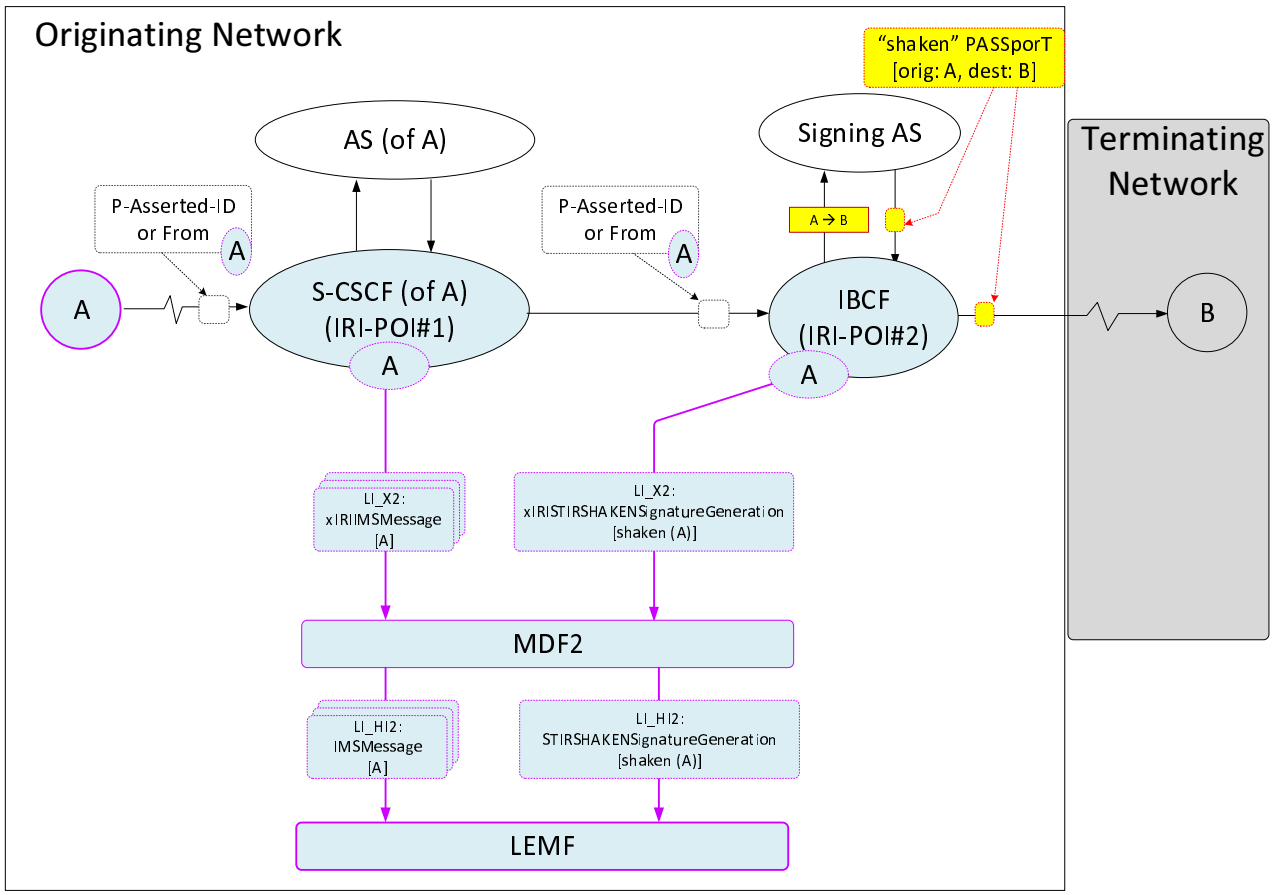


Figure 4.4.5.3.2-2 The LI aspects for scenario 1 (CSP choice IBCF)

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The egress IBCF interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken"

PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

4.4.5.3.3 Scenario 2: Party A calls Party B (target non-local ID)

4.4.5.3.3.1 The scenario

In this scenario, Party A (in originating network) calls Party B (in the terminating network). Party B is a target non-local ID in the originating network.

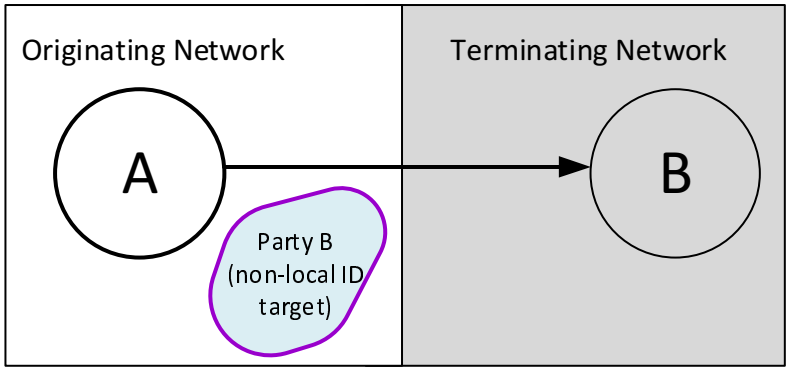


Figure 4.4.5.3.3.1-1: Scenario 2 - Party A calls Party B (target non-local ID)

Figures 4.4.5.3.3.2-1 and 4.4.5.3.3.3-1 illustrate the LI aspects for this scenario when the CSP choice is IBCF.

NOTE: When the CSP choice is Telephony AS, the LI aspects are the same as in the case when intra-network signing is required since the Telephony AS (of A) is the only one and the last one on the signalling path. Accordingly, the figure 4.4.4.4.3.2-1 can be referenced for the illustration with the option 1 and the figure 4.4.4.4.3.3-1 can be referenced for the illustration with the option 2.

Even when the CSP choice is IBCF, the Telephony AS would interact with the Signing AS when RCD is supported for the call.

4.4.5.3.3.2 CSP choice is IBCF (no RCD)

The illustration shown in figure 4.4.5.3.3.2-1 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of A provides the IRI-POI functions for IMS LI.

As shown, in the illustration, the egress IBCF interacts with the Signing AS.

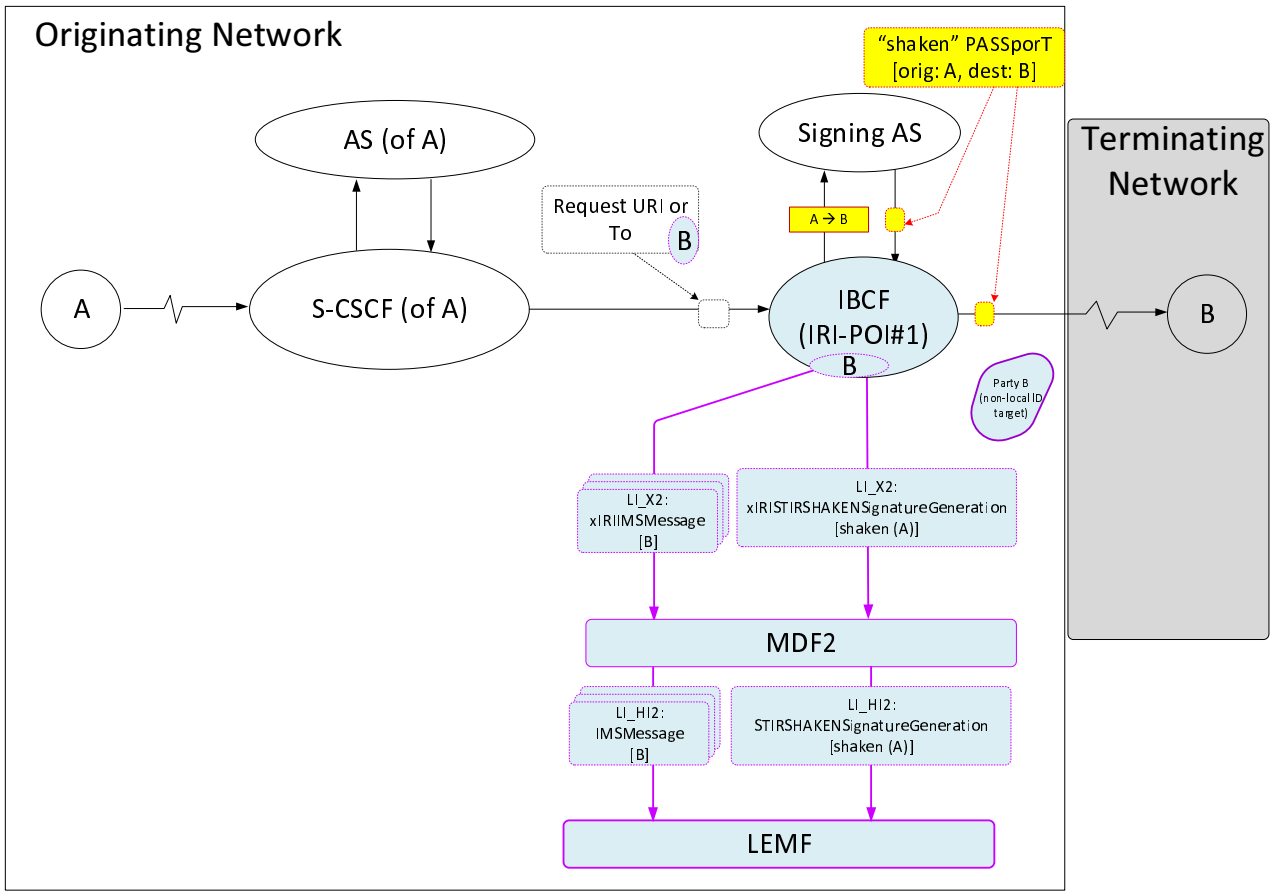


Figure 4.4.5.3.3.2-1: LI aspects for scenario 2 (CSP choice is IBCF, no RCD)

For target non-local ID (Party B)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The egress IBCF interacts with the Signing AS to authenticate/sign A's identity and the same egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the egress IBCF uses the Request URI or the To headers (pointing to Party B) present in the outgoing SIP INVITE to perform a target match.

4.4.5.3.3.3 CSP choice is IBCF (with RCD)

The illustration shown in figure 4.4.5.3.3.3-1 assumes the default option (see TS 33.127 [3]) for IMS LI and therefore, the S-CSCF of A provides the IRI-POI functions for IMS LI.

As shown, in the illustration, the Telephony AS of A interacts with the Signing AS. For STIR/SHAKEN related LI reporting option 1 (see clause 4.4.4.4.3.1) is presumed.

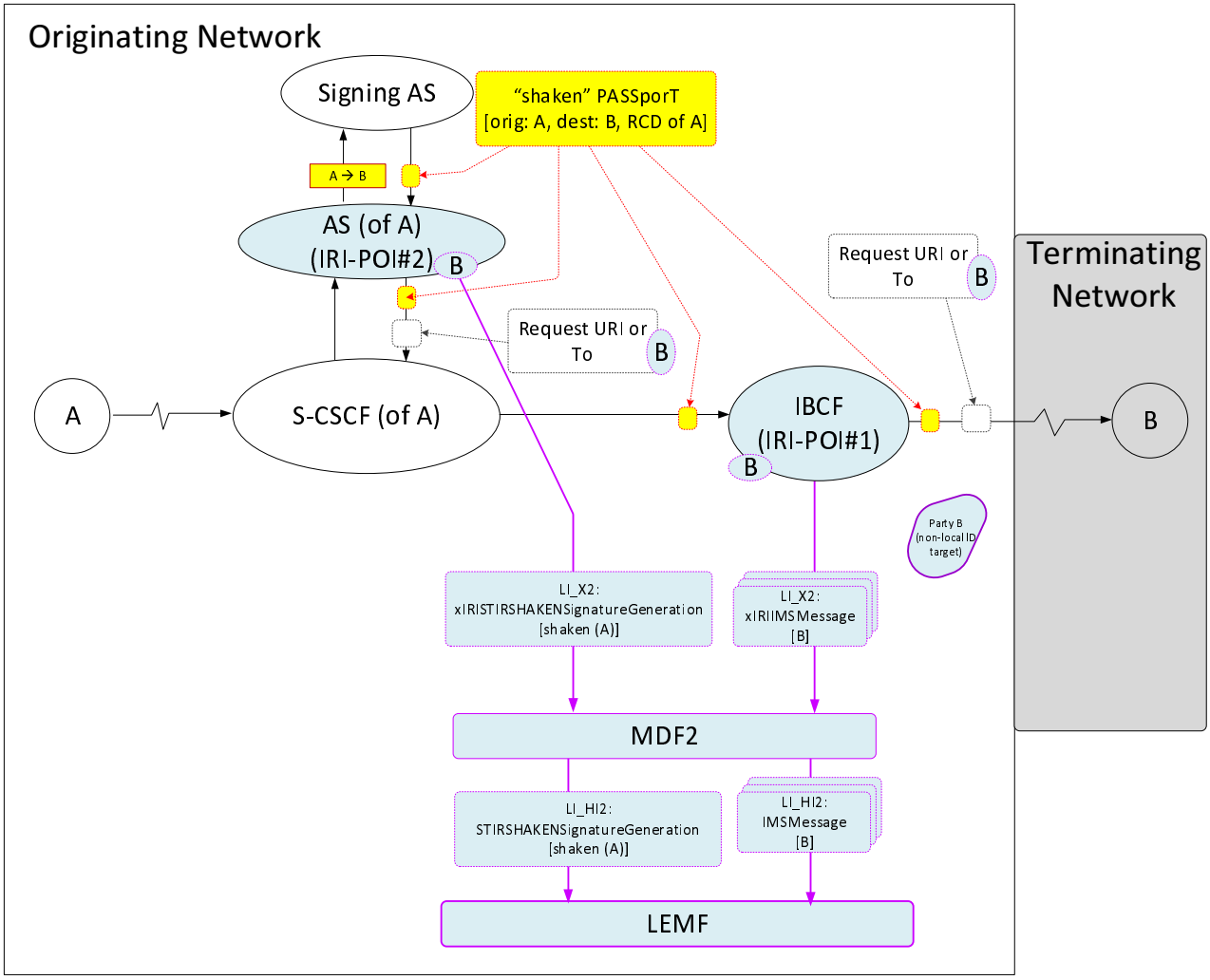


Figure 4.4.5.3.3.3-1: LI aspects for scenario 2 (CSP choice is IBCF, with RCD)

For target non-local ID (Party B)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

Since the RCD is supported for the call, the Telephony AS (A) interacts with the Signing AS to authenticate/sign A's identity and includes the "shaken" PASSporT that has A's identity in the origination claim and B's identity in the destination claim and the RCD for A in.

The IRI-POI present in the Telephony AS (of A), upon detecting that the destination (Party B) is a target non-local generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A along with the RCD for A) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the egress IBCF uses the Request URI or the To headers (pointing to Party B) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the Request URI or the To headers (pointing to Party B) present in the outgoing SIP INVITE to perform a target match.

4.4.5.4 Inter-network IMS sessions (originating network, redirecting scenarios without RCD)

4.4.5.4.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI when an IMS session setup encounters a redirection in the originating network.

The calling party's identity is signed and included in the "shaken" PASSporT with calling party's identity in the origination claim and the called party's identity in the destination claim.

The redirecting party's identity is signed and included in the "div" PASSporT with calling party's identity in the origination claim, redirecting party's identity in the diversion claim and the redirected-to party's identity in the destination claim. As illustrated in clause 4.4.3, the ReportDiversionPASSporTInfo has a role to play in determining the scope of PASSporTs reported to the LEMFs.

4.4.5.4.2 Scenario 1: Party A (target) calls Party B redirected to Party C

4.4.5.4.2.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party A and Party B are in the originating network. Party A is the target.

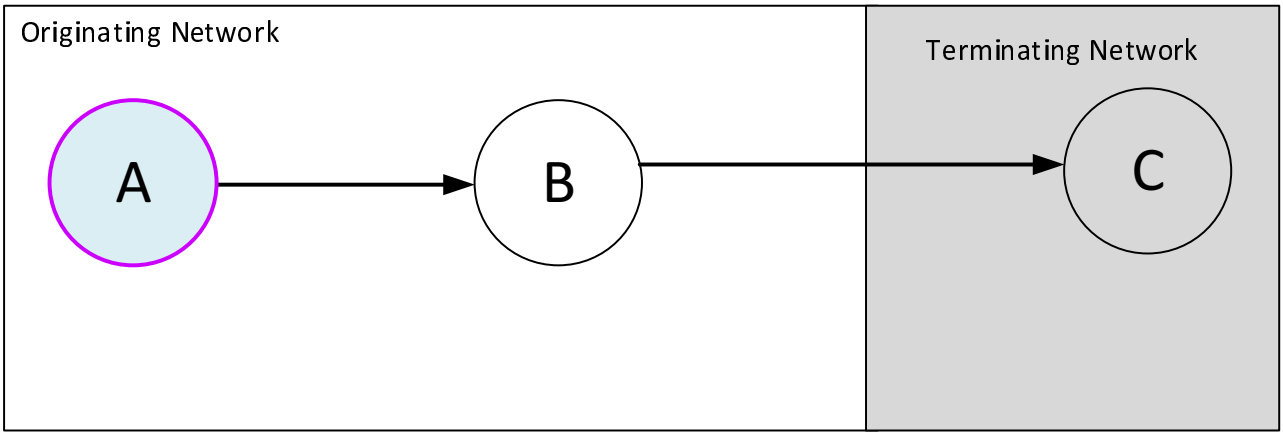


Figure 4.4.5.4.2.1-1: Party A (target) calls Party B redirected to Party C (terminating network)

The subsequent sub-clauses illustrate the LI aspects for these scenarios with the two CSP choices: IBCF Vs Telephony AS. In these illustrations, it is presumed that RCD is not present for the call.

All the illustrations assume the default option (see TS 33.127 [3]) for IMS LI.

4.4.5.4.2.2 CSP choice is IBCF: ReportDiversionPASSporTInfo is set to "False"

In the illustration, the S-CSCF (of A) provides the IMS-LI for target (Party A). The egress IBCF interacts with the Signing AS to authenticate/sign Party A's identity and Party B's identity.

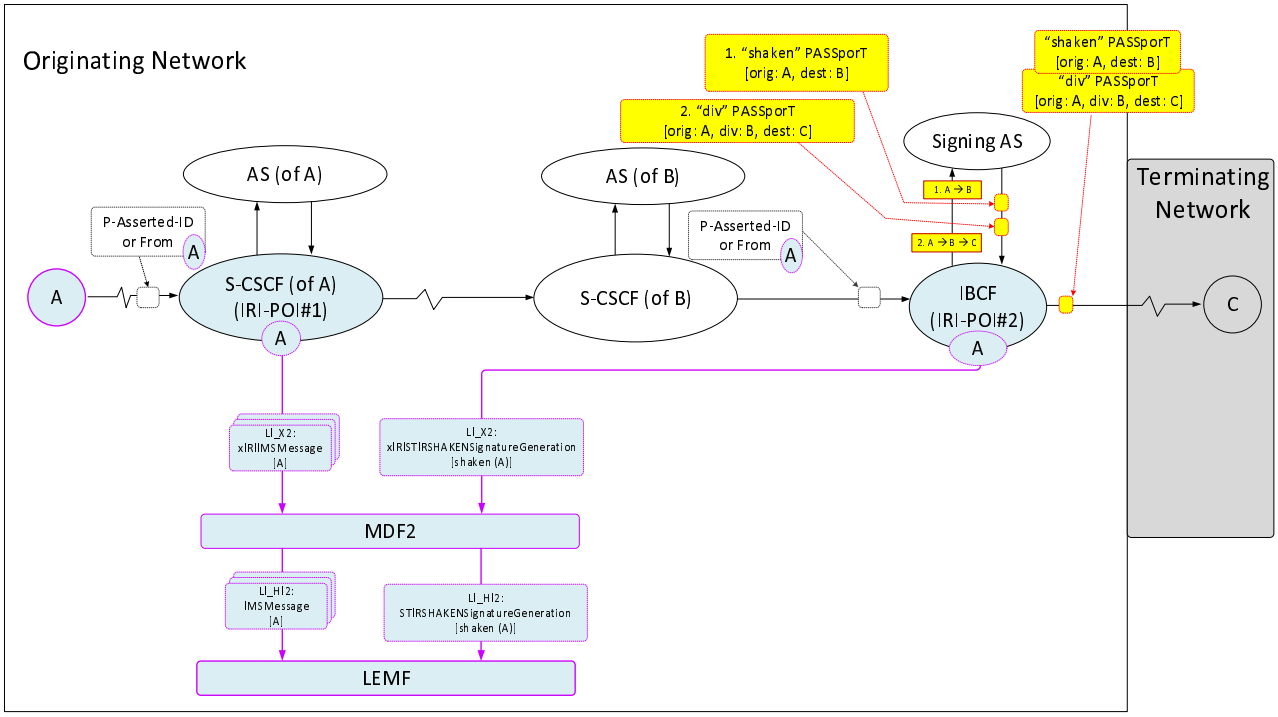


Figure 4.4.5.4.2.2-1: The LI aspects of scenario 1 with CSP choice IBCF (Party A is target) (ReportDiversionPASSporTInfo is set to "False")

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The egress IBCF interacts 1) with the Signing AS to authenticate/sign A's identity and, again interacts (2) with the Signing AS, to authenticate/sign B's identity.

The IRI-POI present in the egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

4.4.5.4.2.3 CSP choice is IBCF - ReportDiversionPASSporTInfo is set to "True"

In the illustration, the S-CSCF (of A) provides the IMS-LI for target (Party A). The egress IBCF interacts with the Signing AS to authenticate/sign Party A's identity and Party B's identity.

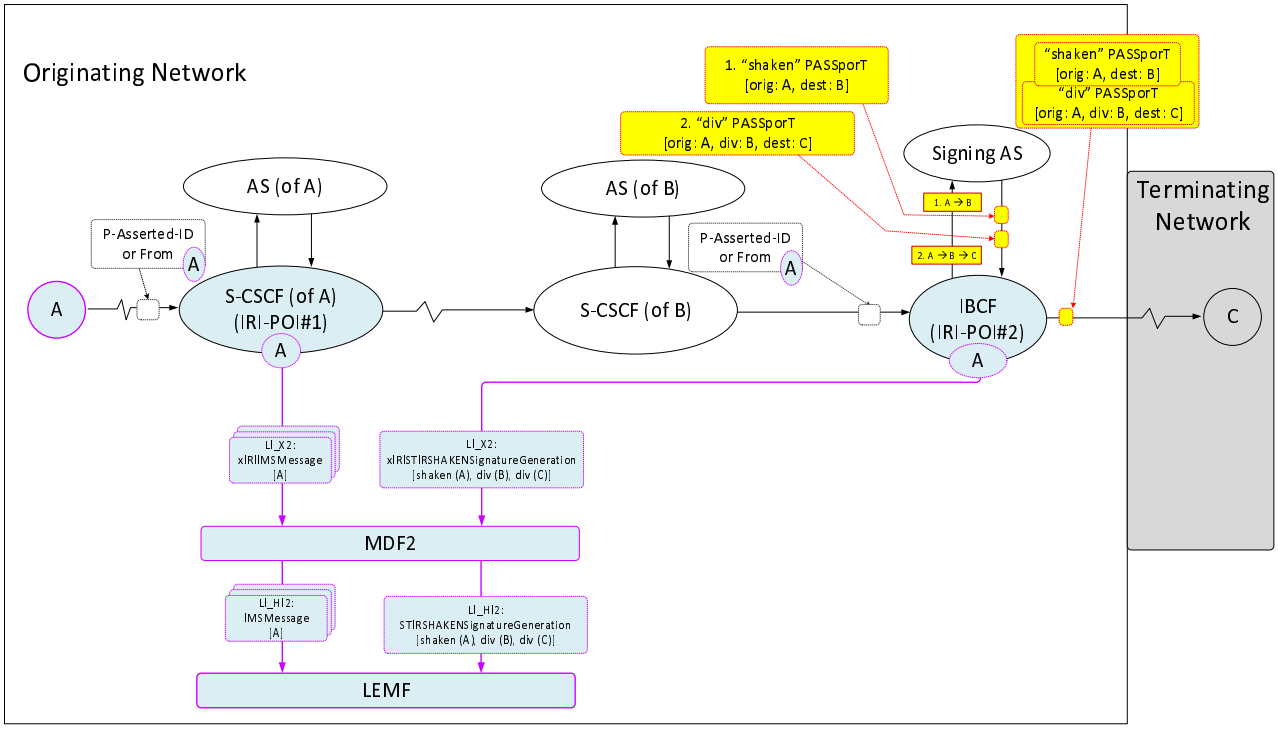


Figure 4.4.5.4.2.3-1: The LI aspects of scenario 1 with CSP choice IBCF (Party A is target) (ReportDiversionPASSporTInfo is "True")

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The egress IBCF interacts 1) with the Signing AS to authenticate/sign A's identity and, again interacts (2) with the Signing AS, to authenticate/sign B's identity.

The IRI-POI present in the egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT (for Party A) and the "div" PASSporT (for Party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

4.4.5.4.2.4 CSP choice is Telephony AS: ReportDiversionPASSporTInfo is set to "False"

In the illustration, the S-CSCF (of A) provides the IMS-LI for target (Party A). The Telephony AS (of B) interacts with the Signing AS to authenticate/sign Party A's identity and Party B's identity.

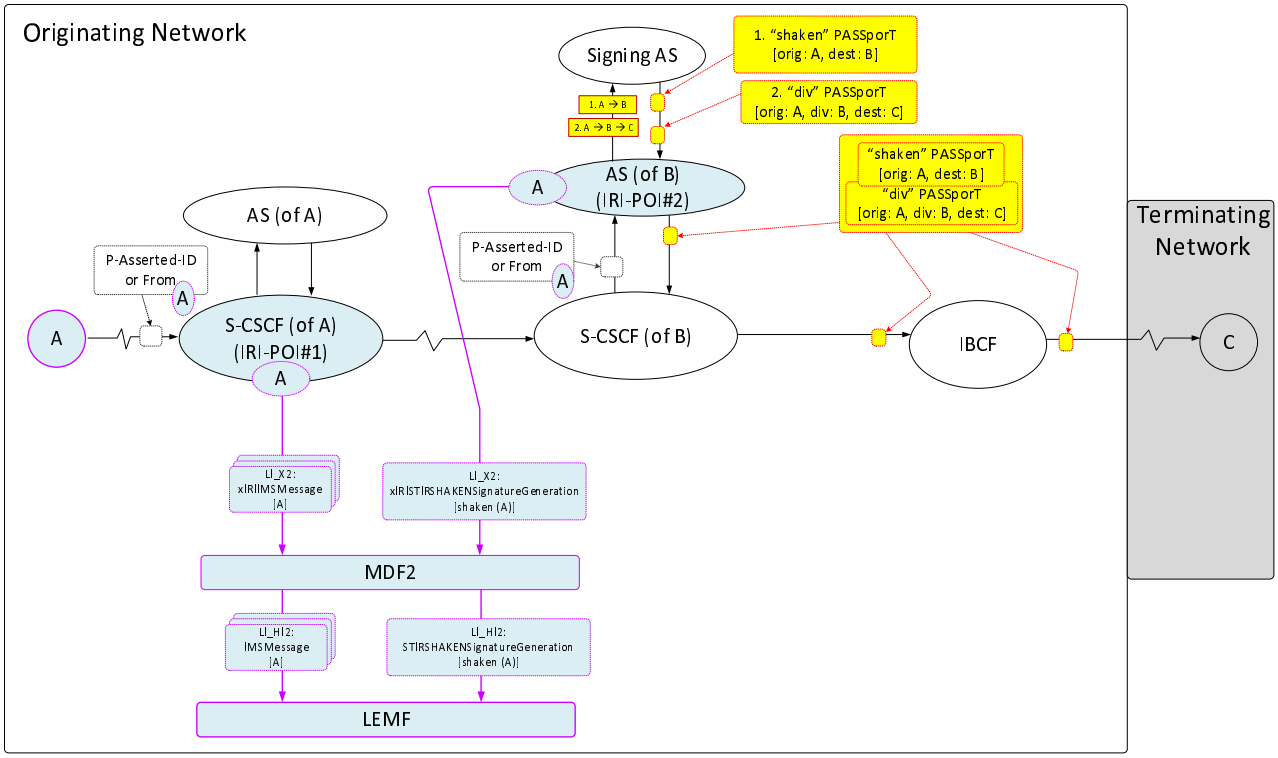


Figure 4.4.5.4.2.4-1: The LI aspects of scenario 1 with CSP choice Telephony AS (Party A is target) (ReportDiversionPASSporTInfo is set to "False")

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts 1) with the Signing AS to authenticate/sign A's identity and, again interacts (2) with the Signing AS, to authenticate/sign B's identity.

The IRI-POI present in the Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

4.4.5.4.2.5 CSP choice is Telephony AS: ReportDiversionPASSporTInfo is set to "True"

In the illustration, the S-CSCF (of A) provides the IMS-LI for target (Party A). The Telephony AS (of B) interacts with the Signing AS to authenticate/sign Party A's identity and Party B's identity.

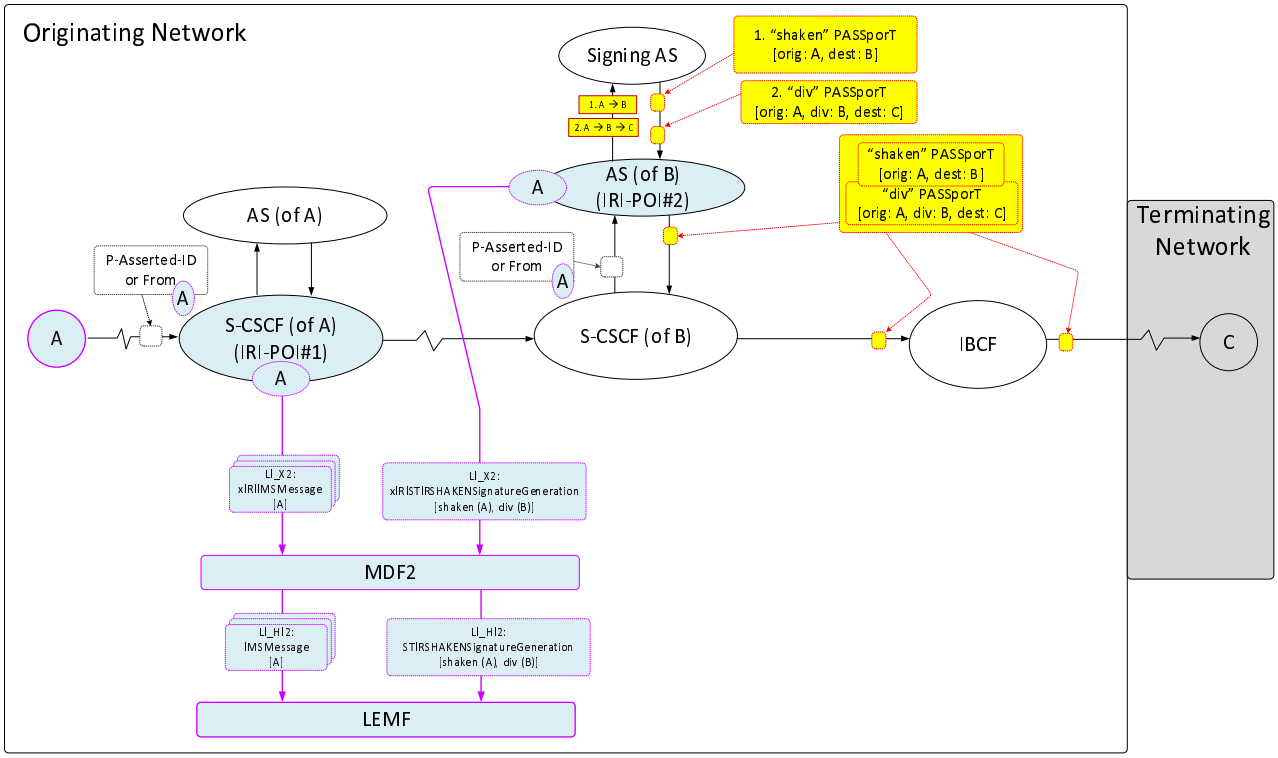


Figure 4.4.5.4.2.5-1: The LI aspects of scenario 1 with CSP choice Telephony AS (Party A is target) (ReportDiversionPASSporTInfo is "True")

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts 1) with the Signing AS to authenticate/sign A's identity and, again interacts (2) with the Signing AS, to authenticate/sign B's identity.

The IRI-POI present in the Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT (for Party A) and the "div" PASSporT (for Party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

4.4.5.4.3 Scenario 2: Party A calls Party B (target) redirected to Party C

4.4.5.4.3.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party A and Party B are in the originating network. Party B is the target.

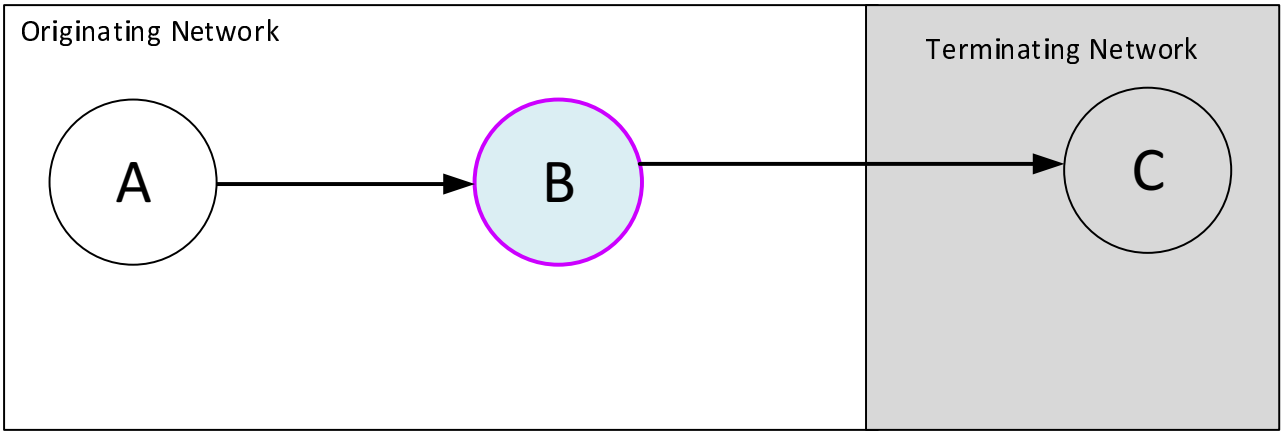


Figure 4.4.5.4.3.1-1: Party A calls Party B (target) redirected to Party C (terminating network)

The subsequent sub-clauses illustrate the LI aspects for these scenarios with the two CSP choices: IBCF Vs Telephony AS. In these illustrations, it is presumed that RCD is not present for the call.

All the illustrations assume the default option (see TS 33.127 [3]) for IMS LI.

In the illustration, since there is no subsequent redirection, the ReportDivesrionPASSporTInfo (even if it is set to "True") for target (Party B) has no impact.

4.4.5.4.3.2 CSP choice is IBCF

In the illustration, the S-CSCF (of B provides the IMS-LI for target (Party B). The egress IBCF interacts with the Signing AS to authenticate/sign Party A's identity and Party B's identity.

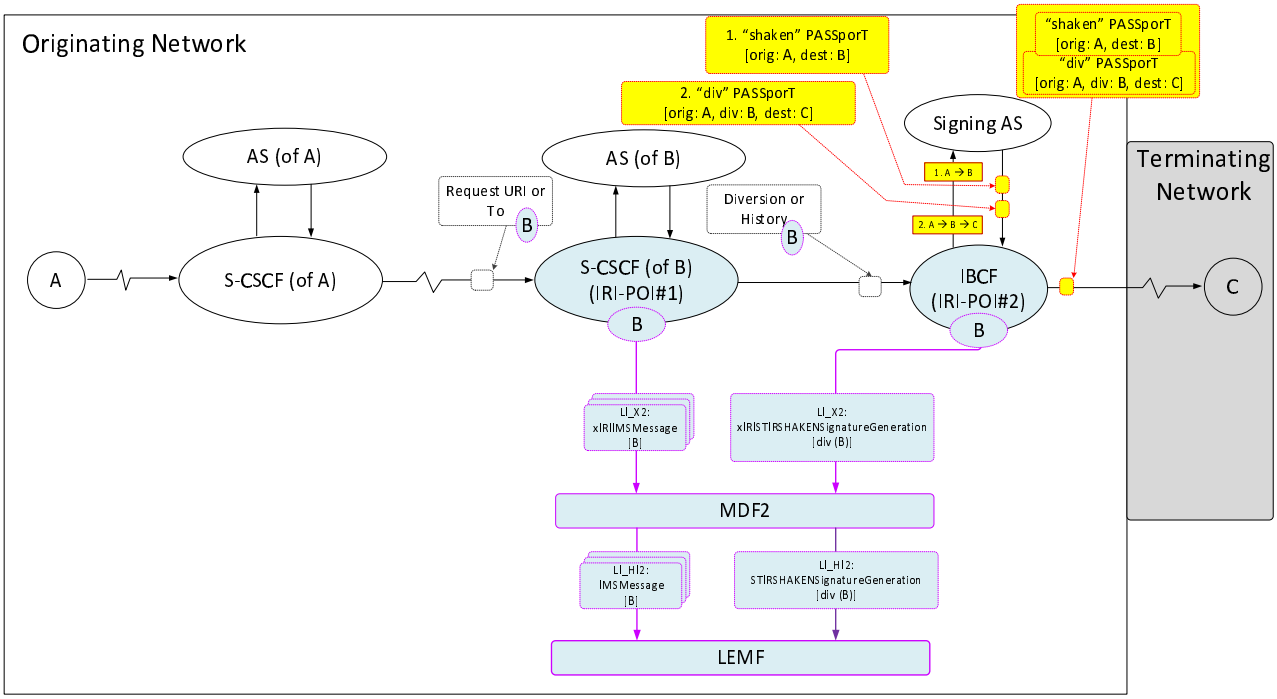


Figure 4.4.5.4.3.2-1: The LI aspects of scenario 1 with CSP choice IBCF (Party B is target)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The egress IBCF interacts 1) with the Signing AS to authenticate/sign A's identity and, again interacts (2) with the Signing AS, to authenticate/sign B's identity.

The IRI-POI present in the egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the Diversion or the History Info headers (pointing to Party B) of the incoming SIP INVITE to perform a target match.

4.4.5.4.3.3 CSP choice is Telephony AS

In the illustration, the S-CSCF (of B) provides the IMS-LI for target (Party B). The Telephony AS (of B) interacts with the Signing AS to authenticate/sign Party A's identity and Party B's identity.

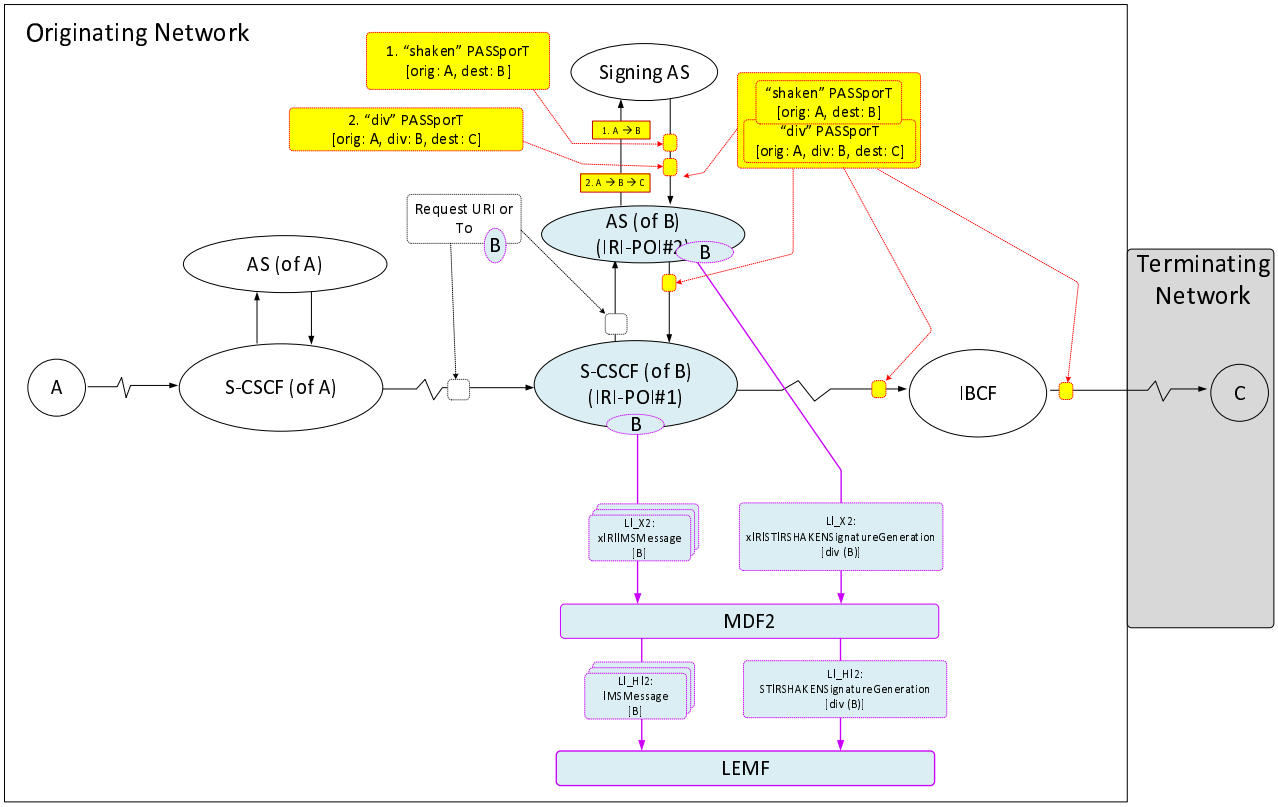


Figure 4.4.5.4.3.3-1: The LI aspects of scenario 1 with CSP choice Telephony AS (Party B is target)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts 1) with the Signing AS to authenticate/sign A's identity and, again interacts (2) with the Signing AS, to authenticate/sign B's identity.

The IRI-POI present in the Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

4.4.5.4.4 Scenario 3: Party A calls Party B redirected to Party C (target non-local ID)

4.4.5.4.4.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party A and Party B are in the originating network. Party B is the target.

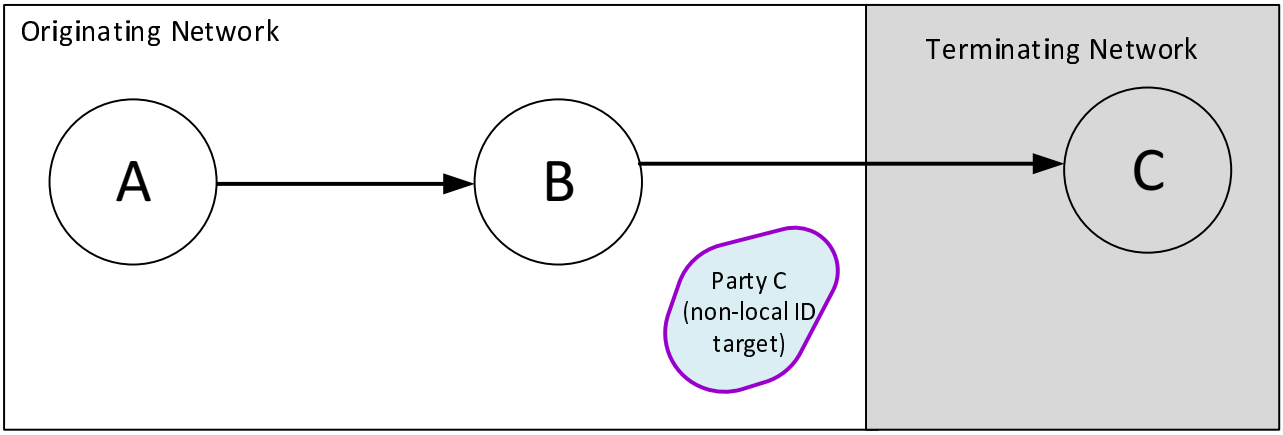


Figure 4.4.5.4.4.1-1: Party A calls Party B redirected to Party C ((target non-local ID)

The subsequent sub-clauses illustrate the LI aspects for these scenarios with the two CSP choices: IBCF Vs Telephony AS. In these illustrations, it is presumed that RCD is not present for the call.

All the illustrations assume the default option (see TS 33.127 [3]) for IMS LI.

4.4.5.4.4.2 CSP choice is IBCF

In the illustration, the egress IBCF provides the IMS-LI for target non-local ID (Party C). The egress IBCF interacts with the Signing AS to authenticate/sign Party A's identity and Party B's identity.

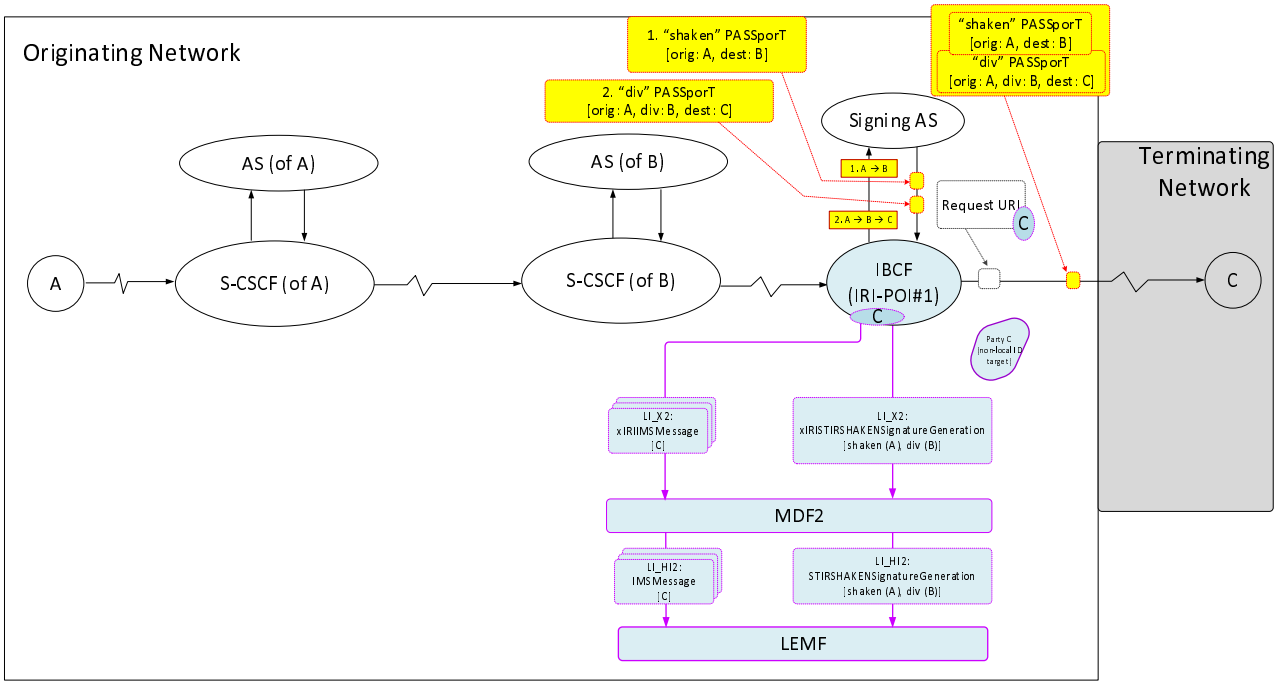


Figure 4.4.5.4.4.2-1: The LI aspects of scenario 3 with CSP choice IBCF (Party C is target non-local ID)

For target non-local ID (Party C)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The egress IBCF interacts 1) with the Signing AS to authenticate/sign A's identity and, again interacts (2) with the Signing AS, to authenticate/sign B's identity.

The IRI-POI present in the egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT (for Party A) and the "div" PASSporT (for Party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the egress IBCF uses the Request URI header (pointing Party C) of the outgoing SIP INVITE to perform a target match.

4.4.5.4.4.3 CSP choice is Telephony AS

In the illustration, the S-CSCF (of B) provides the IMS-LI for target (Party B). The Telephony AS (of B) interacts with the Signing AS to authenticate/sign Party A's identity and Party B's identity.

For STIR/SHAKEN related LI reporting, the option 1 (introduced in clause 4.4.4.4.3.1) is presumed.

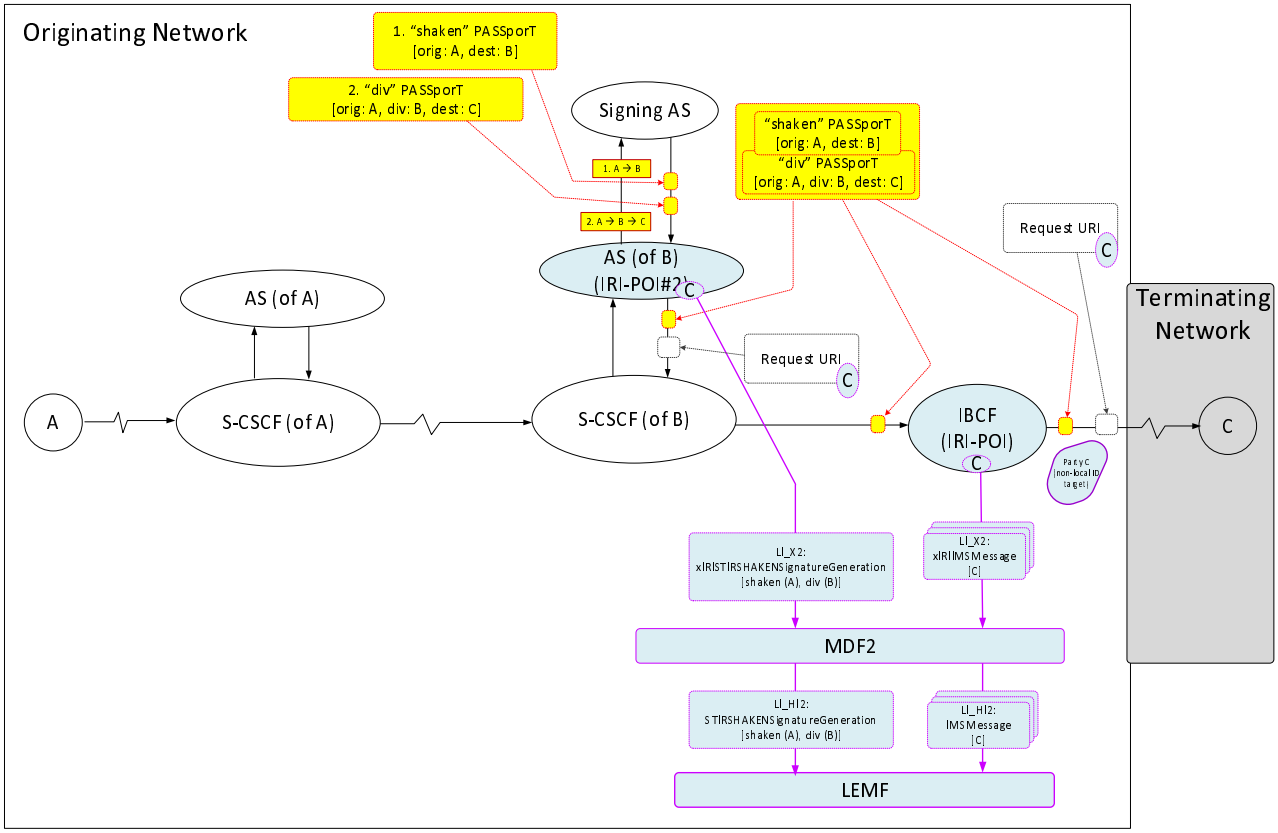


Figure 4.4.5.4.4.3-1: The LI aspects of scenario 3 with CSP choice Telephone AS (Party C is target non-local ID)

For target non-local ID (Party C)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts 1) with the Signing AS to authenticate/sign A's identity and, again interacts (2) with the Signing AS, to authenticate/sign B's identity.

The IRI-POI present in the Telephony AS (of B), upon detecting that the destination (Party C) is a target non-local ID, generates and delivers the xIRISTIRSHAKENSignatureGeneration record(that includes the "shaken" PASSporT for Party A and the "div" PASSporT for Party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the egress IBCF uses the Request URI header (pointing Party C) of the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI header (pointing Party C) of the outgoing SIP INVITE to perform a target match.

4.4.5.5 Inter-network IMS sessions (originating network, redirecting scenarios with RCD)

4.4.5.5.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI when an IMS session setup encounters a redirection in the originating network and CSP network supports RCD for the calling party.

The calling party's identity is signed and included in the "shaken" PASSporT with calling party's identity in the origination claim and the called party's identity in the destination claim. The "shaken" PASSporT also includes the RCD for the caller (i.e. Party A).

The redirecting party's identity is signed and included in the "div" PASSporT with calling party's identity in the origination claim, redirecting party's identity in the diversion claim and the redirected-to party's identity in the destination claim. As illustrated in clause 4.4.3, the ReportDiversionPASSporTInfo has a role to play in determining the scope of PASSporTs reported to the LEMFs.

4.4.5.5.2 Scenario 1: Party A (target) calls Party B redirected to Party C redirected to Party D

4.4.5.5.2.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C who in turn has a call forwarding to Party D. Party A, Party B and Party C are in the originating network. Party A is the target.

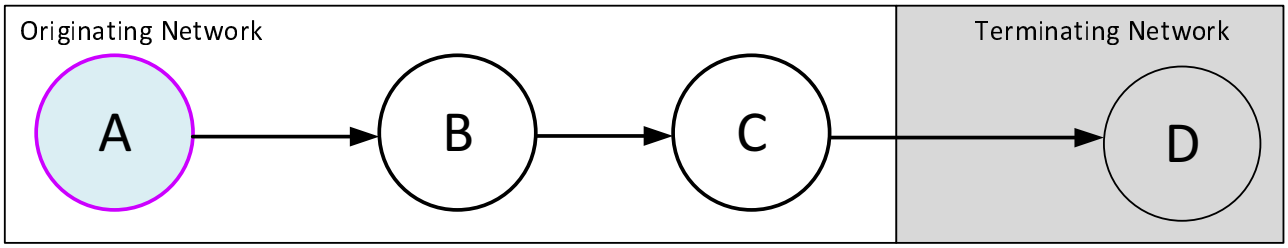


Figure 4.4.5.5.2.1-1: Party A (target) calls Party B redirected to Party C redirected to Party D (Party D in the terminating network)

The scenario is used to illustrate the impacts of RCD support on the STIR/SHAKEN related LI reporting. The present document assumes that the RCD support is applicable only for the calling party (in the scenario, Party A).

The subsequent sub-clauses illustrate the LI aspects for these scenarios for various call situations and the CSP choice used in selecting the IMS NF that interacts with the Signing AS (IBCF Vs Telephony AS).

All the illustrations assume the default option (see TS 33.127 [3]) for IMS LI.

4.4.5.5.2.2 CSP choice is IBCF

4.4.5.5.2.2.1 General

In the illustrations, the S-CSCF (of A) provides the IMS-LI for target (Party A).

Since RCD is supported, the Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the egress IBCF interacts with the Signing AS to authenticate/sign B's and C's identities.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT.

4.4.5.5.2.2.2 ReportDiversionPASSporTInfo is set to "False"

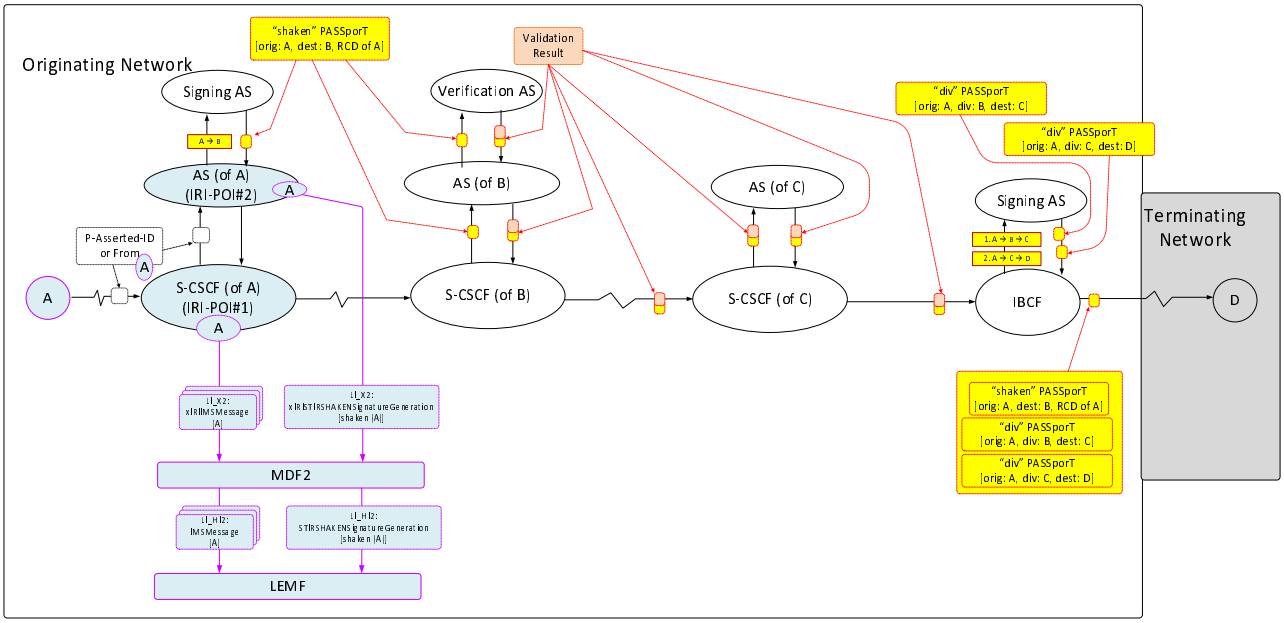


Figure 4.4.5.5.2.2.2-1: The LI aspects of scenario 1 (ReportDiversionPASSporTInfo is "False", RCD is supported)

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for Party to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

4.4.5.5.2.2.3 ReportDiversionPASSporTInfo is set to "True"

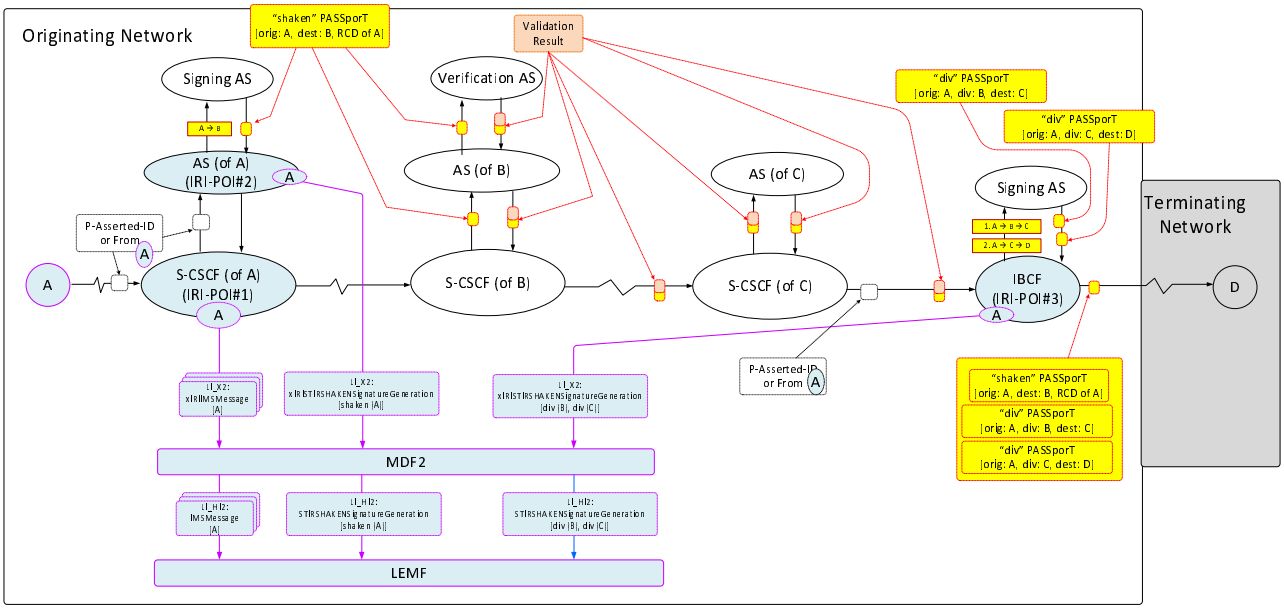


Figure 4.4.5.5.2.2.3-1: The LI aspects of scenario 1 (ReportDiversionPASSporTInfo is "True"", RCD is supported)

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The egress IBCF interacts 1) with the Signing AS to authenticate/sign B's identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

The IRI-POI present in the egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the two "div" PASSporTs (for Party B and for Party C) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) of the incoming SIP INVITE to perform a target match.

4.4.5.5.2.3 CSP choice is Telephony AS

4.4.5.5.2.3.1 General

In the illustrations, the S-CSCF (of A) provides the IMS-LI for target (Party A).

Since RCD is supported, the Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the Telephony AS (of C) interacts with the Signing AS to authenticate/sign B's and C's identities.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT.

4.4.5.5.2.3.2 ReportDiversionPASSporTInfo is set to "False"

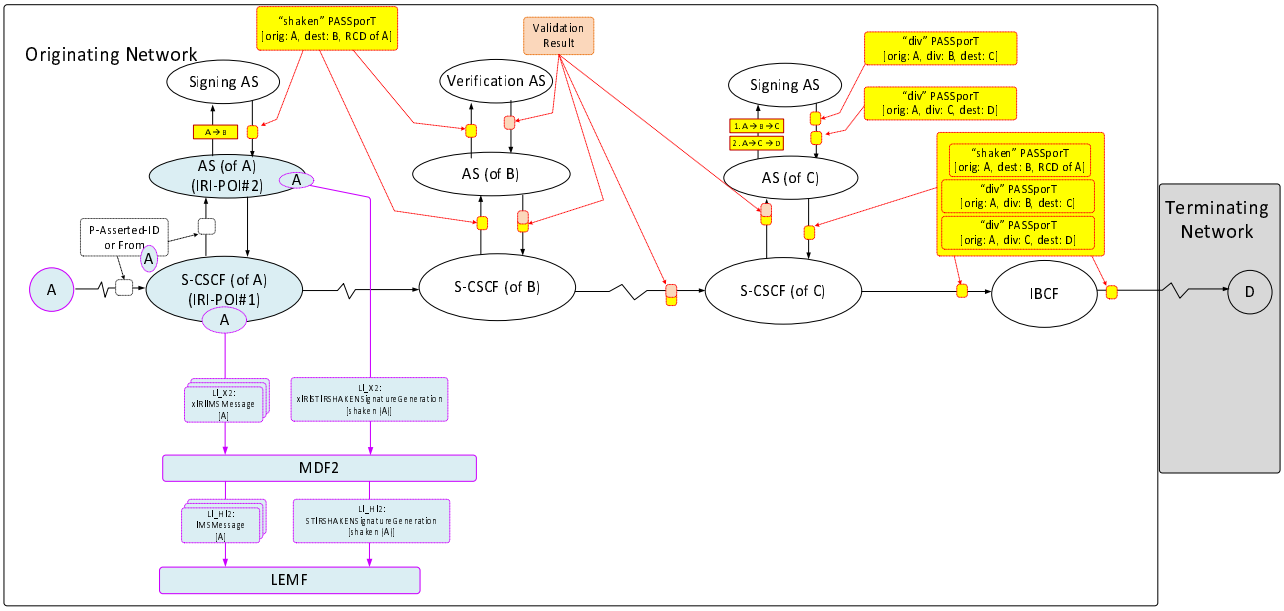


Figure 4.4.5.5.2.3.2-1: The LI aspects of scenario 1 (ReportDiversionPASSporTInfo is "False", RCD is supported)

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

4.4.5.5.2.3.3 ReportDiversionPASSporTInfo is set to "True"

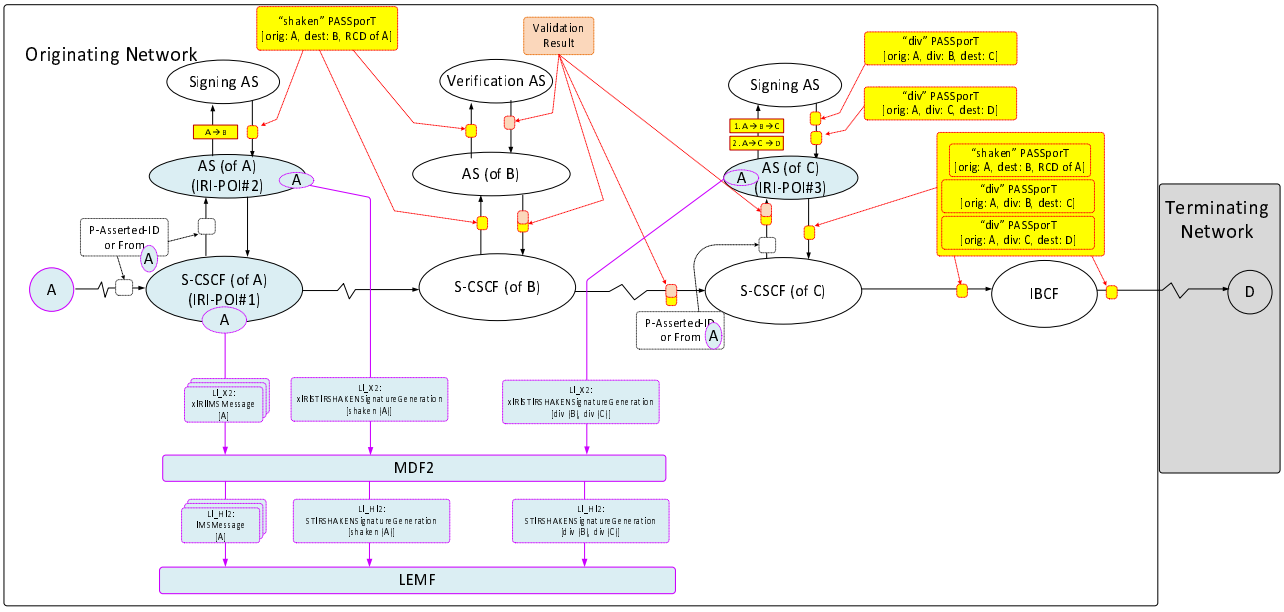


Figure 4.4.5.5.2.3.3-1: The LI aspects of scenario 1 (ReportDiversionPASSporTInfo is "True", RCD is supported)

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The Telephony AS(of C) interacts 1) with the Signing AS to authenticate/sign B's identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

The IRI-POI present in the Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the two "div" PASSporTs (for Party B and for Party C) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of A) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

4.4.5.5.3 Scenario 2: Party A calls Party B (target) redirected to Party C redirected to Party D

4.4.5.5.3.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C who in turn has a call forwarding to Party D. Party A, Party B and Party C are in the originating network. Party B is the target.

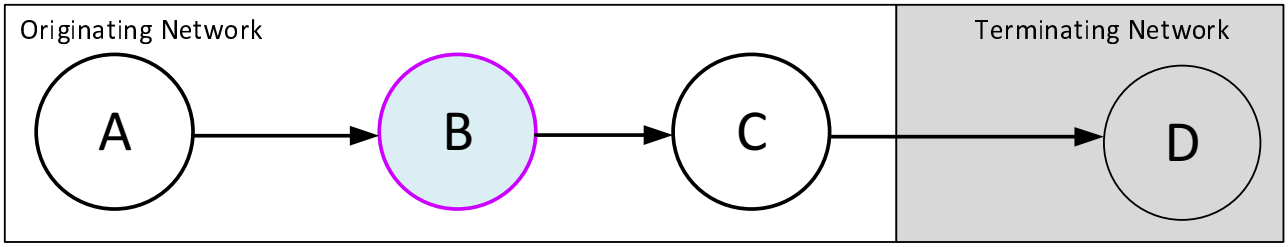


Figure 4.4.5.5.3.1-1: Party calls Party B (target) redirected to Party C redirected to Party D (Party D in the terminating network)

The scenario is used to illustrate the impacts of RCD support on the STIR/SHAKEN related LI reporting. The present document assumes that the RCD support is applicable only for the calling party (in the scenario, Party A).

The subsequent sub-clauses illustrate the LI aspects for these scenarios for various call situations and the CSP choice used in selecting the IMS NF that interacts with the Signing AS (IBCF Vs Telephony AS).

All the illustrations assume the default option (see TS 33.127 [3]) for IMS LI.

4.4.5.5.3.2 CSP choice is IBCF

4.4.5.5.3.2.1 General

In the illustrations, the S-CSCF (of B) provides the IMS-LI for target (Party B).

Since RCD is supported, the Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the egress IBCF interacts with the Signing AS to authenticate/sign B's and C's identities.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT.

4.4.5.5.3.2.2 ReportDiversionPASSporTInfo is set to "False"

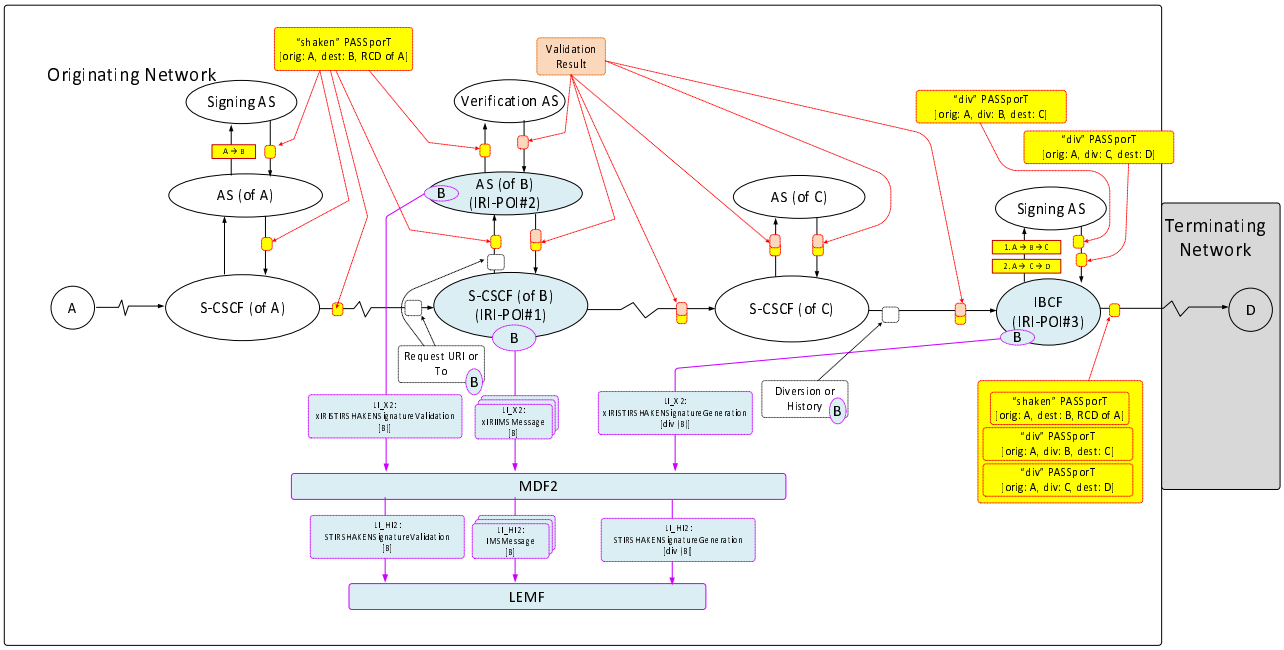


Figure 4.4.5.5.3.2.2-1: The LI aspects of scenario 2 (ReportDiversionPASSporTInfo is "False", RCD is supported)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The egress IBCF interacts 1) with the Signing AS to authenticate/sign B's identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

The IRI-POI present in the egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the Diversion or the History Info headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

4.4.5.5.3.2.3 ReportDiversionPASSporTInfo is set to "True"

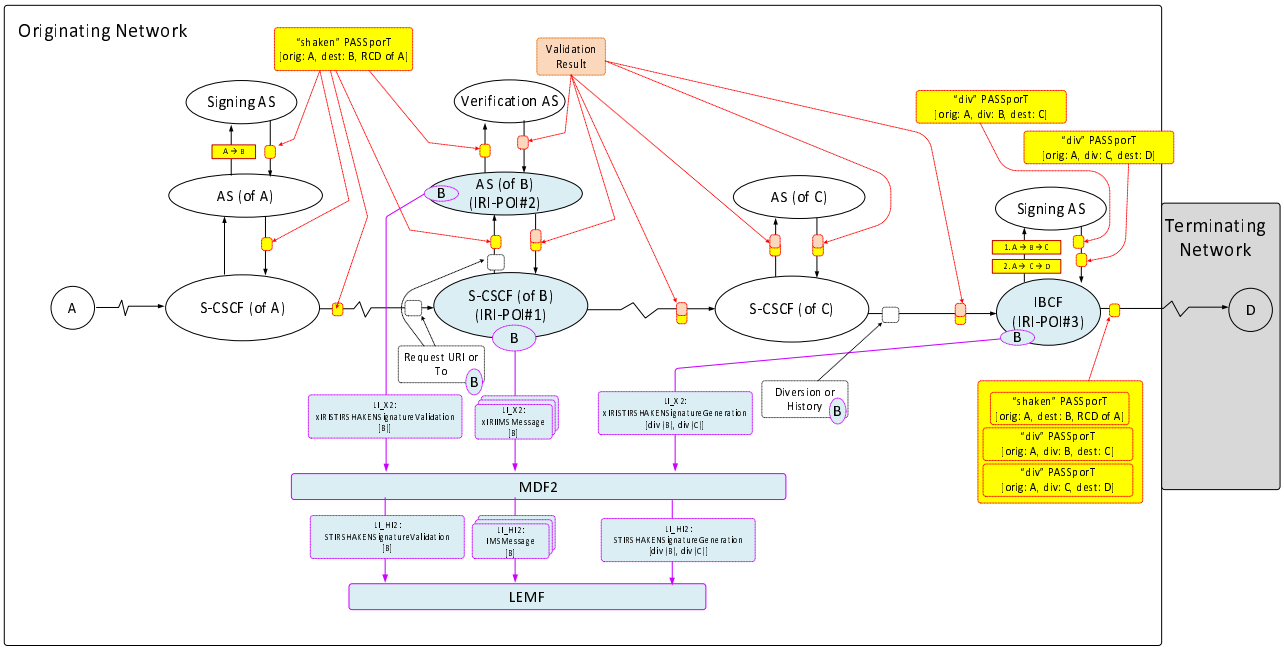


Figure 4.4.5.5.3.2.3-1: The LI aspects of scenario 2 (ReportDiversionPASSporTInfo is "True"", RCD is supported)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The egress IBCF interacts 1) with the Signing AS to authenticate/sign B's identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

The IRI-POI present in the egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the two "div" PASSporTs (for Party B and for Party C) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the Diversion or the History Info headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

4.4.5.5.3.3 CSP choice is Telephony AS

4.4.5.5.3.3.1 General

In the illustrations, the S-CSCF (of B) provides the IMS-LI for target (Party B).

Since RCD is supported, the Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the Telephony AS (of C) interacts with the Signing AS to authenticate/sign B's and C's identities.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT.

4.4.5.5.3.3.2 ReportDiversionPASSporTInfo is set to "False"

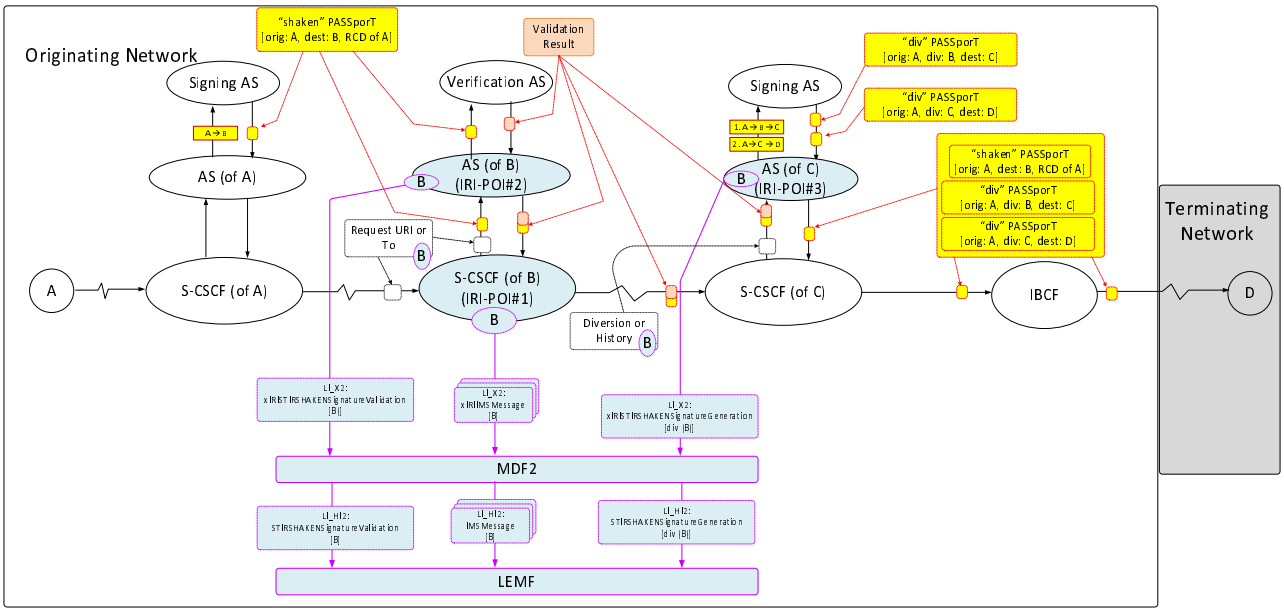


Figure 4.4.5.5.3.3.2-1: The LI aspects of scenario 2 (ReportDiversionPASSporTInfo is "False", RCD is supported)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The Telephony AS (of C) interacts 1) with the Signing AS to authenticate/sign B’s identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

The IRI-POI present in the Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Diversion or the History Info headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

4.4.5.5.3.3.3 ReportDiversionPASSporTInfo is set to "True"

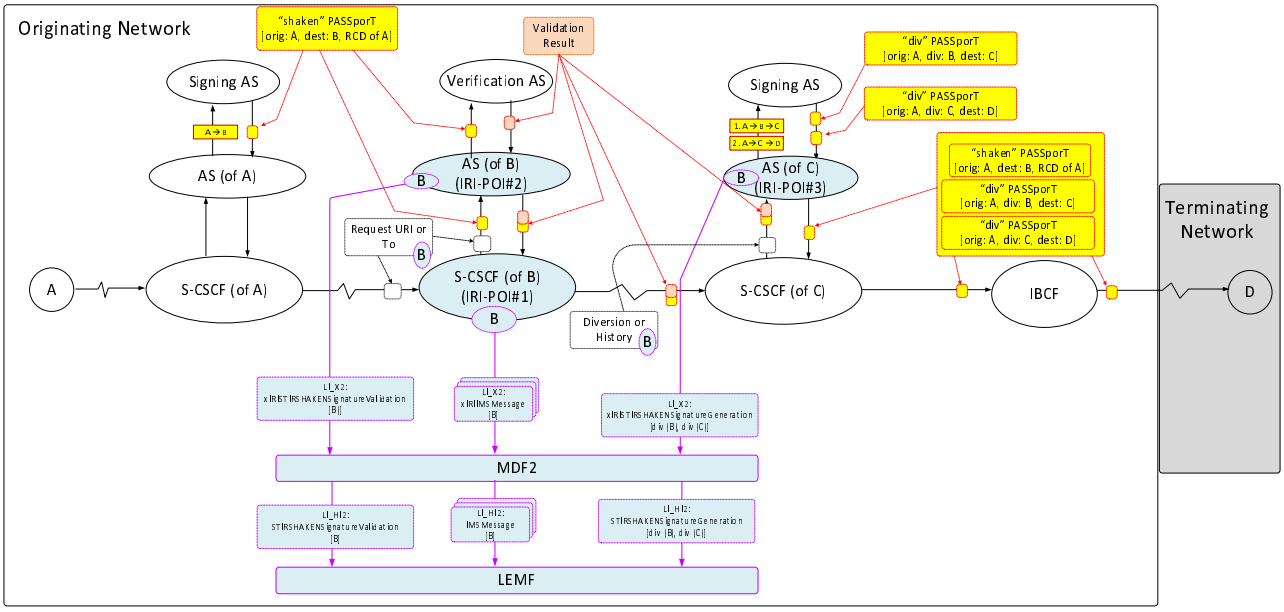


Figure 4.4.5.5.3.3.3-1: The LI aspects of scenario 2 (ReportDiversionPASSporTInfo is "True", RCD is supported)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The Telephony AS (of C) interacts 1) with the Signing AS to authenticate/sign B's identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

The IRI-POI present in the Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the two "div" PASSporTs (for Party B and for Party C) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Diversion or the History Info headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

4.4.5.5.4 Scenario 3: Party A calls Party B redirected to Party C (target) redirected to Party D

4.4.5.5.4.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C who in turn has a call forwarding to Party D. Party A, Party B and Party C are in the originating network. Party C is the target.

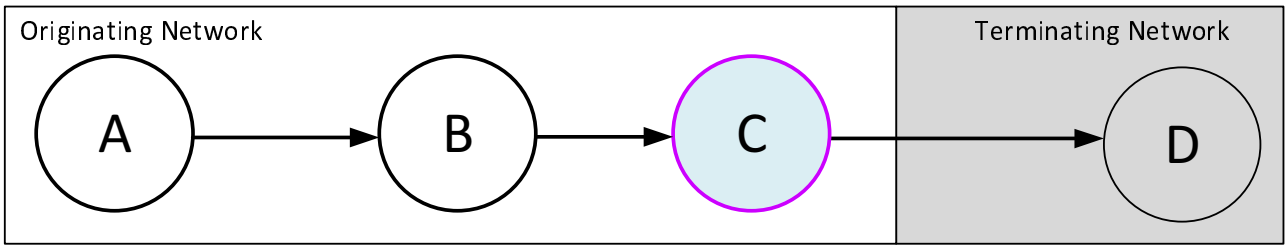


Figure 4.4.5.5.4.1-1: Party calls Party B redirected to Party C (target) redirected to Party D (Party D in the terminating network)

The scenario is used to illustrate the impacts of RCD support on the STIR/SHAKEN related LI reporting. The present document assumes that the RCD support is applicable only for the calling party (in the scenario, Party A).

The subsequent sub-clauses illustrate the LI aspects for based on the CSP choice used in selecting the IMS NF that interacts with the Signing AS (IBCF Vs Telephony AS).

All the illustrations assume the default option (see TS 33.127 [3]) for IMS LI.

4.4.5.5.4.2 CSP choice is IBCF

In the illustration, the S-CSCF (of C) provides the IMS-LI for target (Party C).

Since RCD is supported, the Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the egress IBCF interacts with the Signing AS to authenticate/sign B's and C's identities.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT.

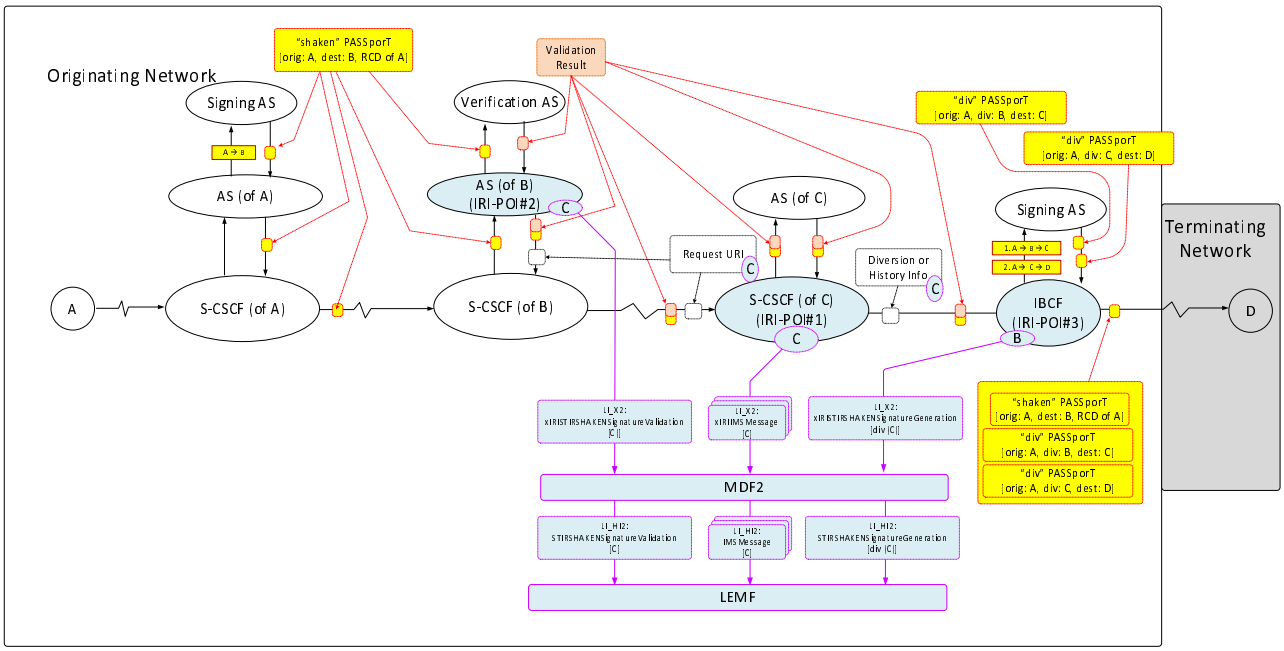


Figure 4.4.5.5.4.2-1: The LI aspects of scenario 3 (RCD is supported)

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The egress IBCF interacts 1) with the Signing AS to authenticate/sign B's identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

The IRI-POI present in the egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of C) uses the Request URI header (pointing Party C) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI header (pointing Party C) of the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the Diversion or the History Info headers (pointing Party C) of the incoming SIP INVITE to perform a target match.

4.4.5.5.4.3 CSP choice is Telephony AS

In the illustrations, the S-CSCF (of B) provides the IMS-LI for target (Party B).

Since RCD is supported, the Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the Telephony AS (of C) interacts with the Signing AS to authenticate/sign B's and C's identities.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT.

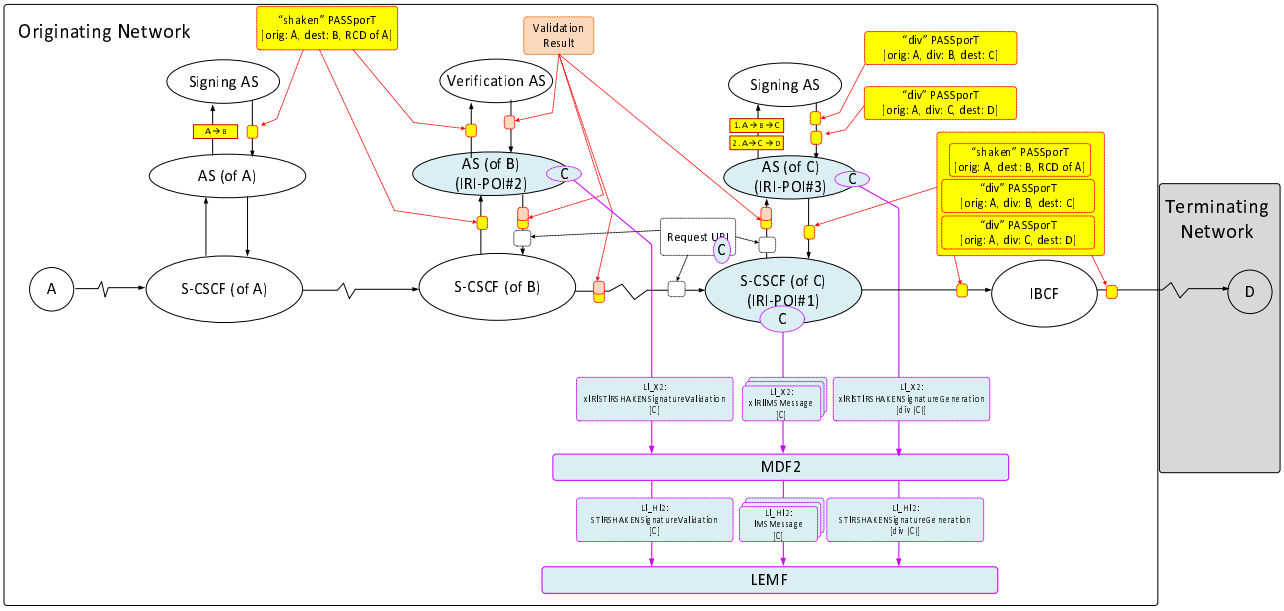


Figure 4.4.5.5.4.3-1: The LI aspects of scenario 3 (RCD is supported)

For target (Party B)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The Telephony AS (of C) interacts 1) with the Signing AS to authenticate/sign B's identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

The IRI-POI present in the Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of C) uses the Request URI header (pointing Party C) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI header (pointing Party C) of the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Request URI header (pointing Party C) of the incoming SIP INVITE to perform a target match.

4.4.5.5.5 Scenario 4: Party A calls Party B redirected to Party C redirected to Party D (target non-local ID)

4.4.5.5.5.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C who in turn has a call forwarding to Party D. Party A, Party B and Party C are in the originating network. Party D is the target non-local ID.

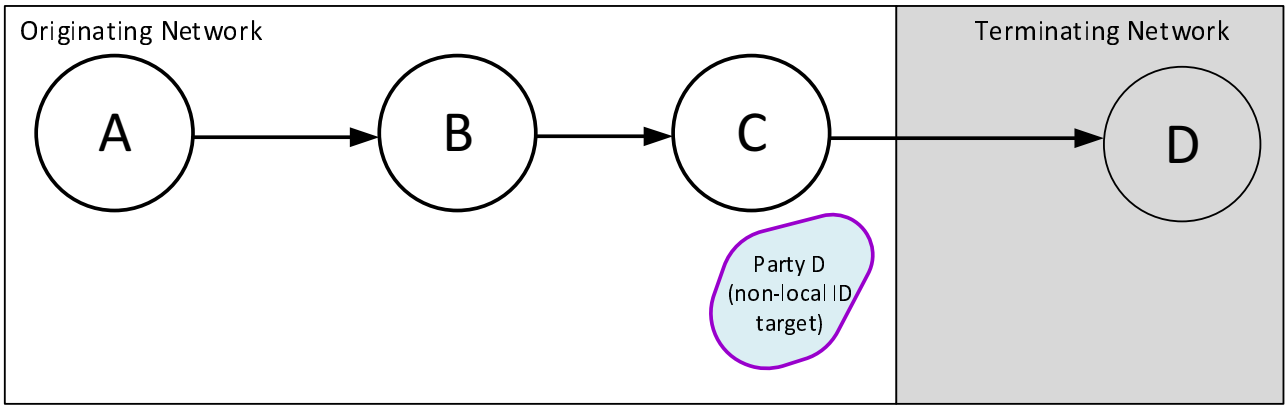


Figure 4.4.5.5.5.1-1: Party calls Party B redirected to Party C (target) redirected to Party D (Party D in the terminating network)

The scenario is used to illustrate the impacts of RCD support on the STIR/SHAKEN related LI reporting. The present document assumes that the RCD support is applicable only for the calling party (in the scenario, Party A).

The subsequent sub-clauses illustrate the LI aspects for based on the CSP choice used in selecting the IMS NF that interacts with the Signing AS (IBCF Vs Telephony AS).

All the illustrations assume the default option (see TS 33.127 [3]) for IMS LI.

4.4.5.5.5.2 CSP choice is IBCF

In the illustration, the egress IBCF provides the IMS-LI for target non-local ID (Party D).

Since RCD is supported, the Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the egress IBCF interacts with the Signing AS to authenticate/sign B's and C's identities.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT.

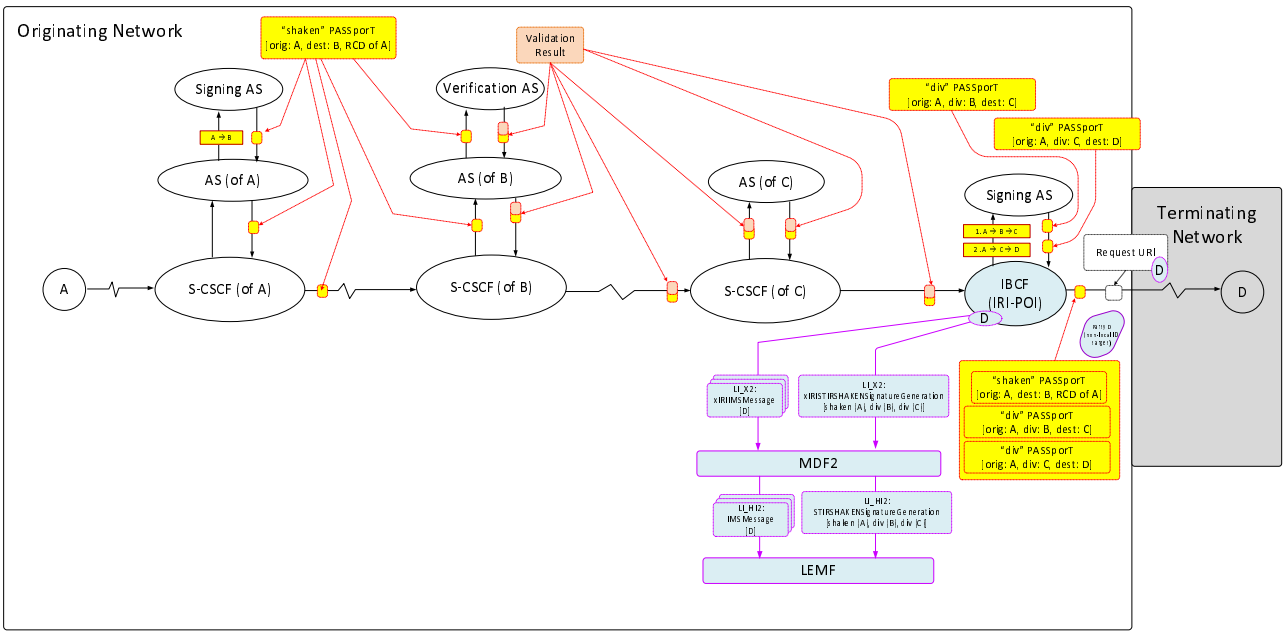


Figure 4.4.5.5.5.2-1: The LI aspects of scenario 4 (RCD is supported)

For target non-local ID (Party D)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The egress IBCF interacts 1) with the Signing AS to authenticate/sign B's identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

The IRI-POI present in the egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT (for Party A) and the two "div" PASSporTs (for Party B and for Party C) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the egress IBCF uses the Request URI (pointing Party D) of the outgoing SIP INVITE to perform a target match.

4.4.5.5.5.3 CSP choice is Telephony AS

In the illustration, the egress IBCF provides the IMS-LI for target non-local ID (Party D).

Since RCD is supported, the Telephony AS (of A) interacts with the Signing AS to authenticate/sign A's identity and the Telephony AS (of C) interacts with the Signing AS to authenticate/sign B's and C's identities.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT.

Fort target non-local ID (Party D), the option 1 (introduced in clause 4.4.4.4.3.1) is presumed.

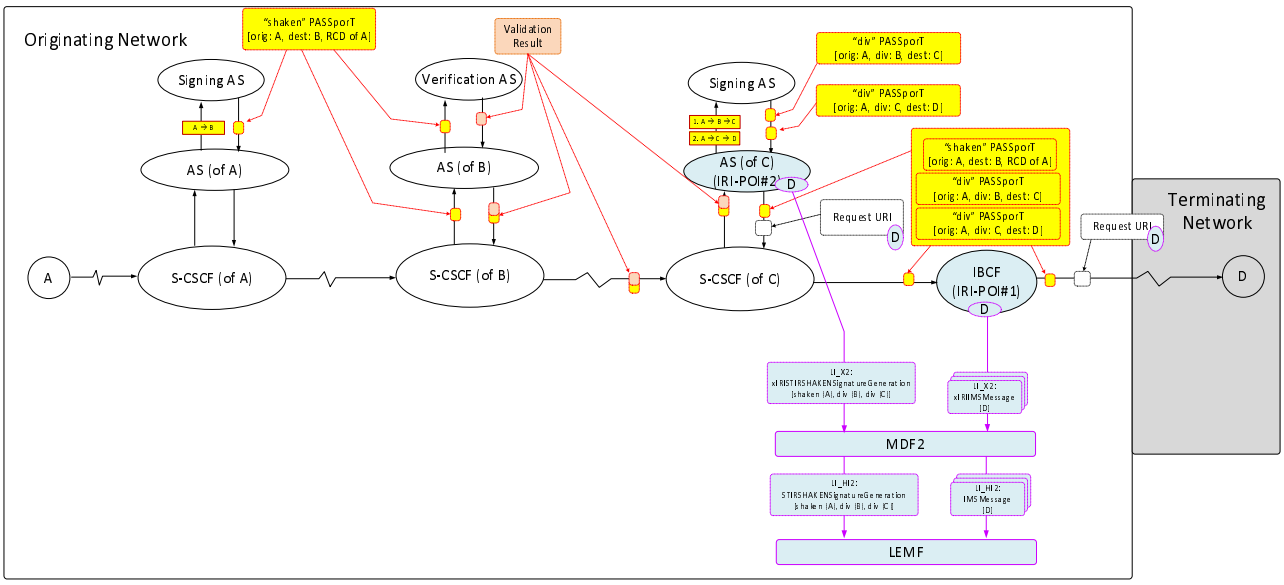


Figure 4.4.5.5.5.3-1: The LI aspects of scenario 4 (RCD is supported)

For target non-local ID (Party D)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of C) interacts 1) with the Signing AS to authenticate/sign B's identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

The IRI-POI present in the Telephony AS (of C) upon detecting that the destination (Party D) is a target non-local ID, generates and delivers the xIRISTIRSHAKENSignatureGeneration record that (includes the "shaken" PASSporT for Party A with RCD and the two "div" PASSporTs (for Party B and for Party C)) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the egress IBCF uses the Request URI (pointing Party D) of the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Request URI (pointing Party D) of the outgoing SIP INVITE to perform a target match.

4.4.5.6 Inter-network IMS sessions (terminating network, basic scenarios)

4.4.5.6.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI for inter-network IMS sessions for the terminating network when signing is not required for intra-network IMS sessions.

The calling party's identity is signed and included in the "shaken" PASSporT with calling party's identity in the origination claim and the called party's identity in the destination claim. The terminating network that receives the "shaken" and one or more "div" PASSporTs (if redirection had happened in the originating network) would validate the received PASSporT (s) before offering the call to the terminating party.

When the CSP choice is IBCF, the ingress IBCF in the terminating network interacts with the Verification AS to validate the received PASSporTs.

When the CSP choice is the Telephony AS, the Telephony AS of the first party in terminating network interacts with the Verification AS to validate the received PASSporTs.

The IMS NFs that provide the LI functions for the STIR/SHAKEN related reporting depends on the call scenario and the CSP choice.

4.4.5.6.2 Scenario 1: Party A calls Party B (target, in the terminating network)

4.4.5.6.2.1 The scenario

In this scenario, Party A calls Party B. Party A is in the originating network and Party B is in the terminating network. Party B is the target.

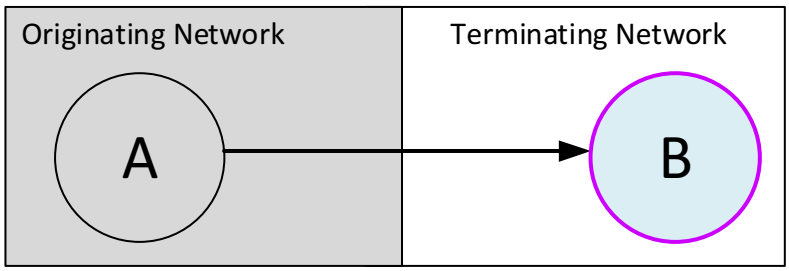


Figure 4.4.5.6.2.1-1: Party A calls Party B (target, the terminating network)

The subsequent sub-clauses illustrate the LI aspects for this scenario for the CSP choice used in selecting the IMS NF that interacts with the Verification AS (IBCF Vs Telephony AS).

All the illustrations assume the default option (see TS 33.127 [3]) for IMS LI.

4.4.5.6.2.2 CSP choice is IBCF

In the illustrations, the S-CSCF (of B) provides the IMS-LI for target (Party B).

The ingress IBCF interacts with the Verification AS to validate the received PASSporT (s).

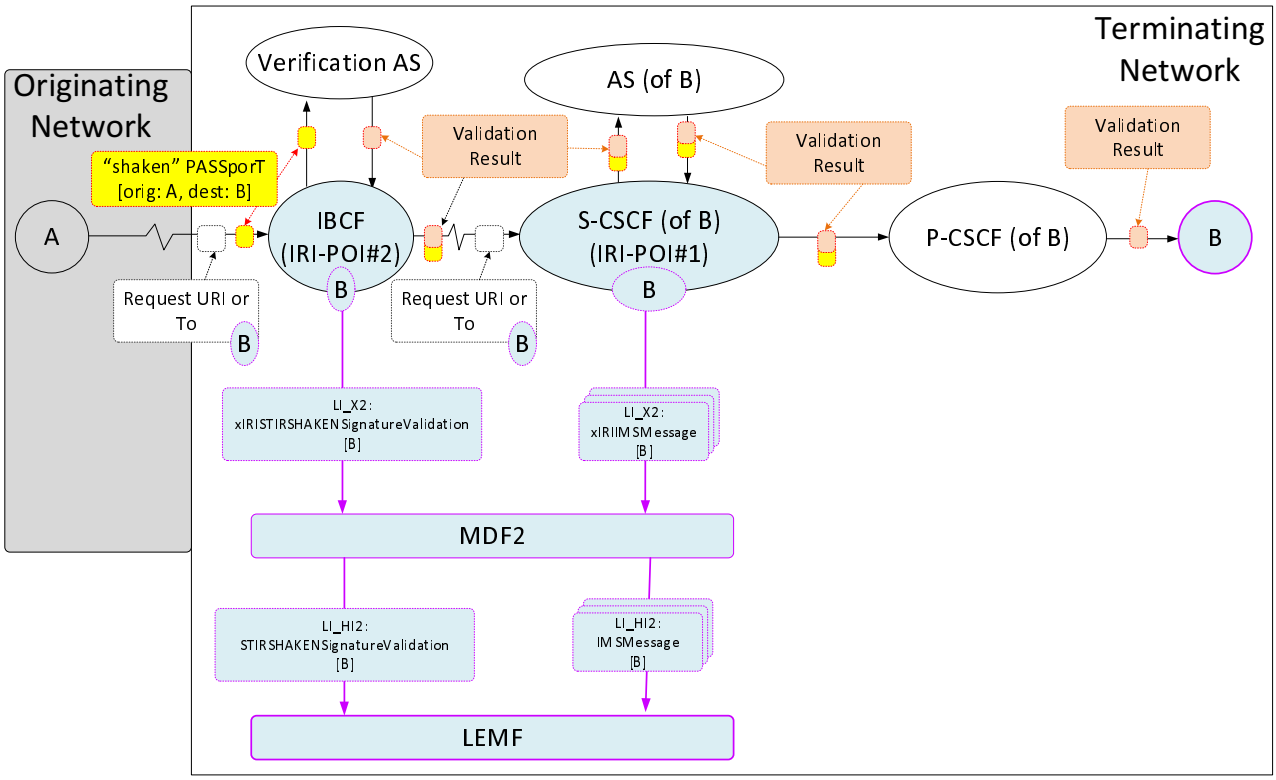


Figure 4.4.5.6.2.2-1: The LI aspects of scenario 1 (IBCF)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The ingress IBCF interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same ingress IBCF generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the ingress IBCF uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

4.4.5.6.2.3 CSP choice is Telephony AS

In the illustrations, the S-CSCF (of B) provides the IMS-LI for target (Party B).

The Telephony AS (of B) interacts with the Verification AS to validate the received PASSporT (s).

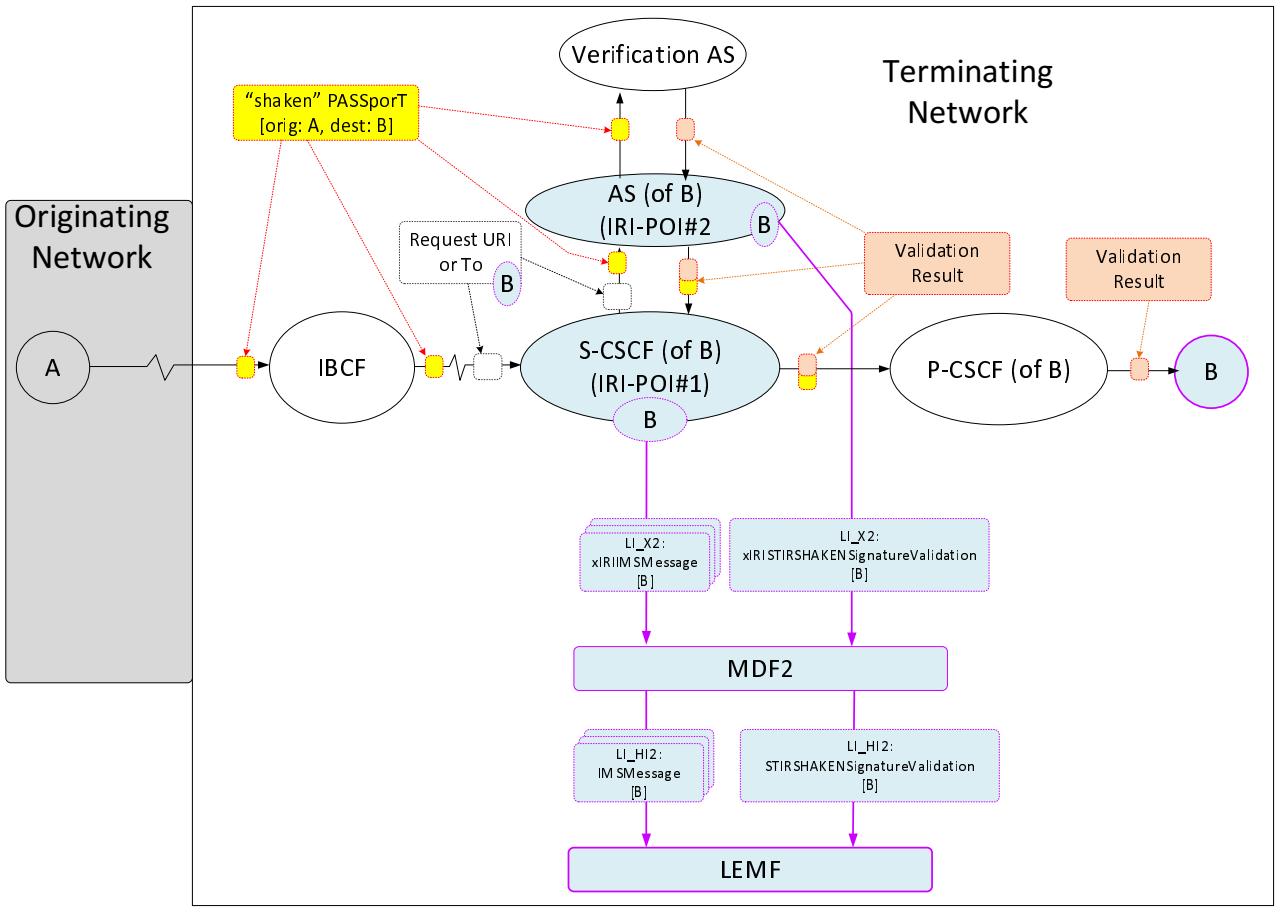


Figure 4.4.5.6.2.3-1: The LI aspects of scenario 1 (Telephony AS)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

4.4.5.6.3 Scenario 2: Party A (target non-local ID) calls Party B

4.4.5.6.3.1 The scenario

In this scenario, Party A calls Party B. Party A is in the originating network and Party B is in the terminating network. Party A is the target non-local ID in the terminating network.

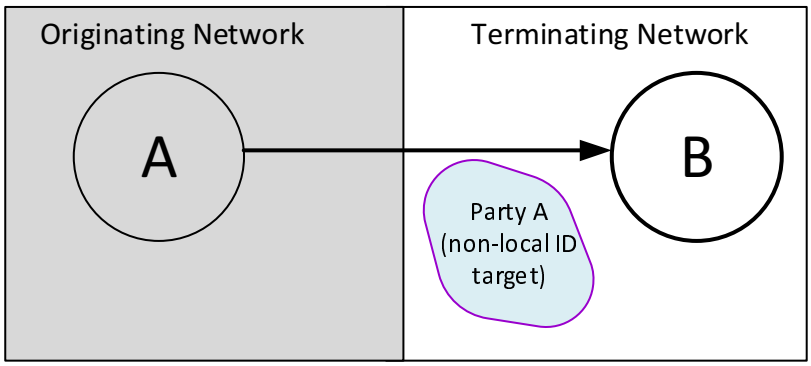


Figure 4.4.5.6.3.1-1: Party A (target non-local ID) calls Party B

The subsequent sub-clauses illustrate the LI aspects for this scenario for the CSP choice used in selecting the IMS NF that interacts with the Verification AS (IBCF Vs Telephony AS).

All the illustrations assume the default option (see TS 33.127 [3]) for IMS LI.

4.4.5.6.3.2 CSP choice is IBCF

In the illustrations, the ingress IBCF provides the IMS-LI for target non-local ID (Party A).

The ingress IBCF interacts with the Verification AS to validate the received PASSporT (s).

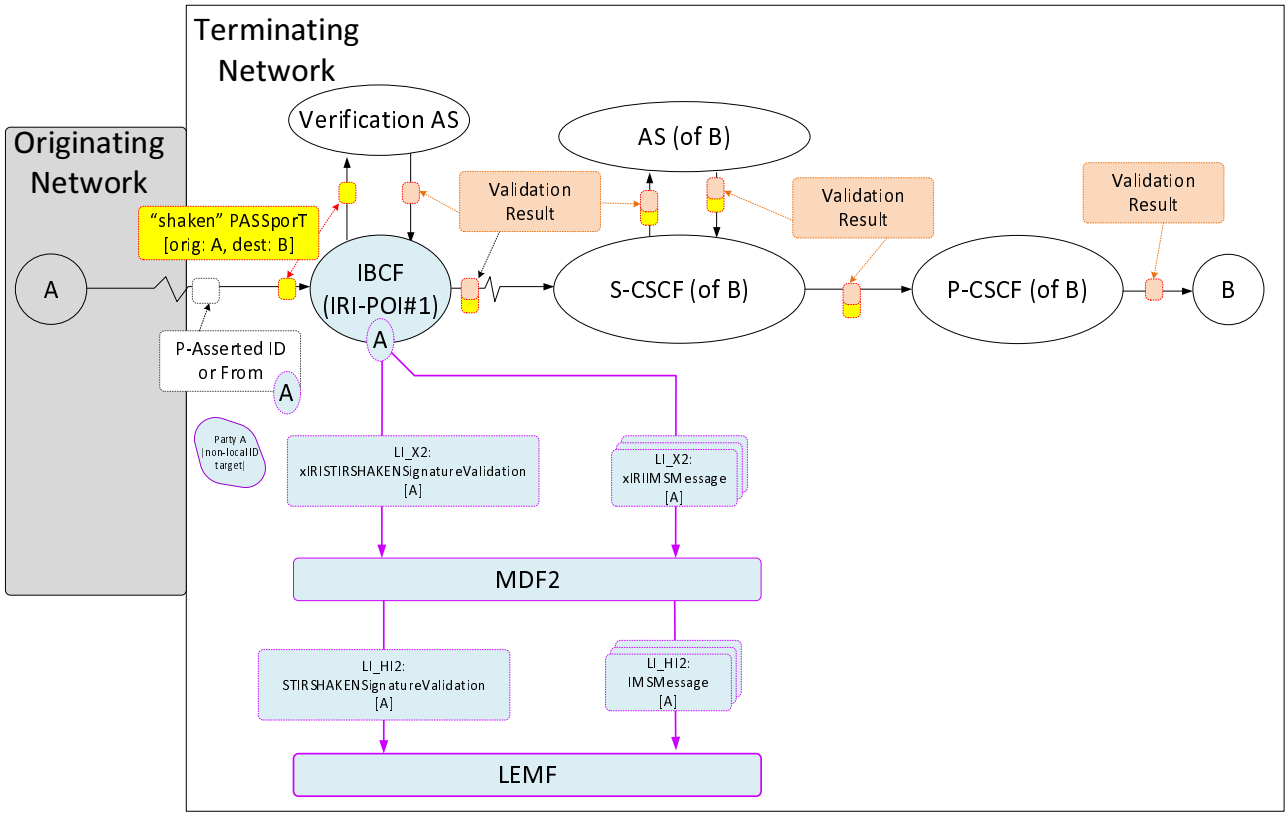


Figure 4.4.5.6.3.2-1: The LI aspects of scenario 2 (IBCF)

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The ingress IBCF interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same ingress IBCF generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the ingress IBCF uses P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

4.4.5.6.3.3 CSP choice is Telephony AS

In the illustrations, the ingress IBCF provides the IMS-LI for target non-local ID (Party A).

The Telephony AS (of B) interacts with the Verification AS to validate the received PASSporT (s).

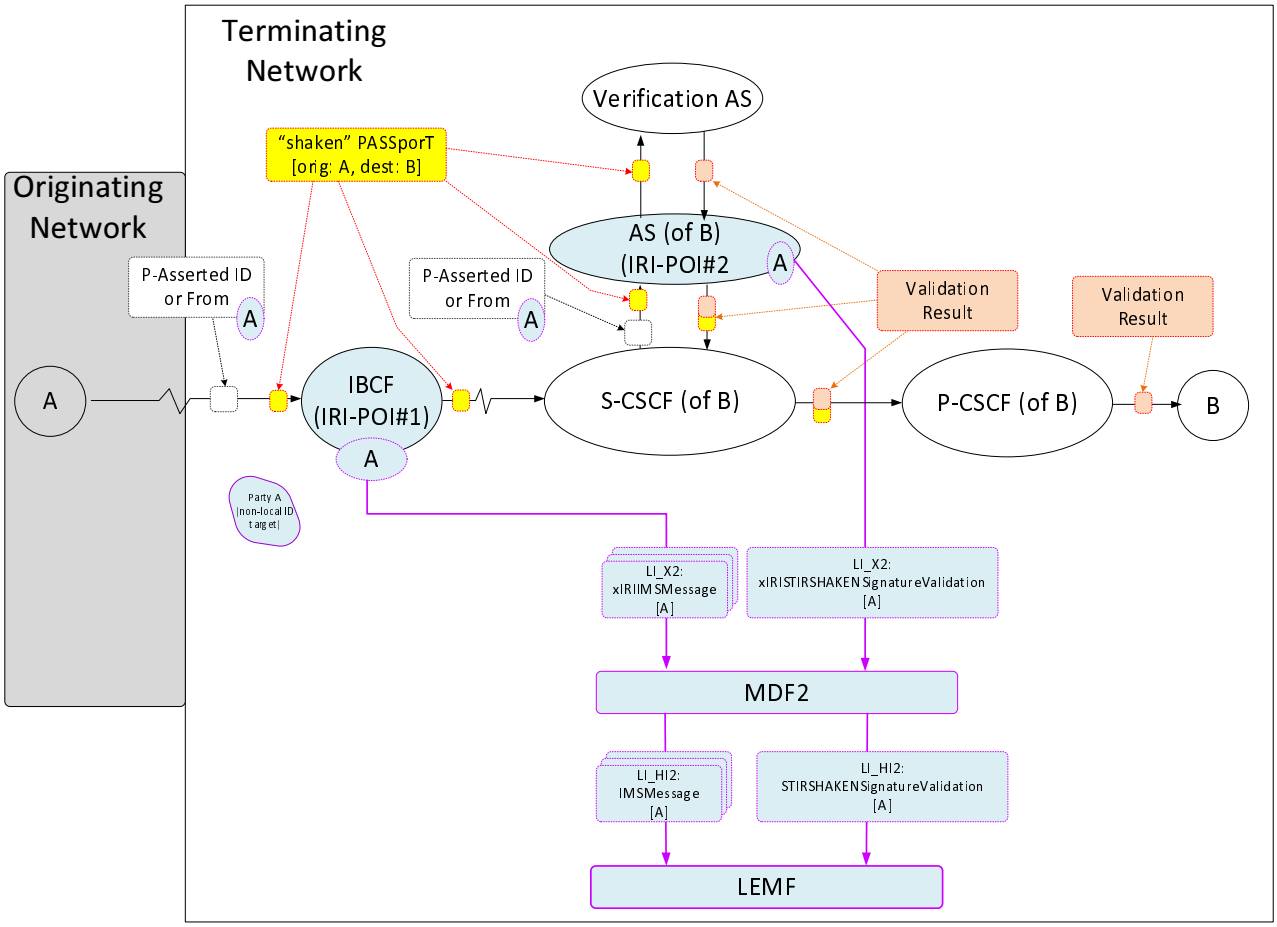


Figure 4.4.5.6.3.3-1: The LI aspects of scenario 2 (Telephony AS)

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the ingress IBCF uses P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses P-Asserted ID or the From headers (pointing Party A) of the incoming SIP INVITE to perform a target match.

4.4.5.7 Inter-network IMS sessions (terminating network, redirecting scenarios)

4.4.5.7.1 Scenario 1: Party A calls Party B (target) redirected to Party C

4.4.5.7.1.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party A is in the originating network and Party B and Party C are in the terminating network. Party B is the target.

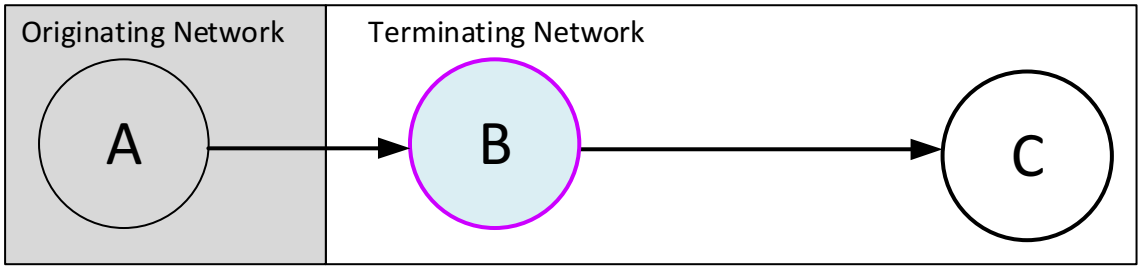


Figure 4.4.5.7.1.1-1: Party A calls Party B (target) redirected to Party C

The subsequent sub-clauses illustrate the LI aspects for this scenario for the CSP choice used in selecting the IMS NF that interacts with the Verification AS (IBCF Vs Telephony AS).

All the illustrations assume the default option (see TS 33.127 [3]) for IMS LI.

4.4.5.7.1.2 CSP choice is IBCF

In the illustrations, the S-CSCF (of B) provides the IMS-LI for target (Party B).

The ingress IBCF interacts with the Verification AS to validate the received PASSporT (s).

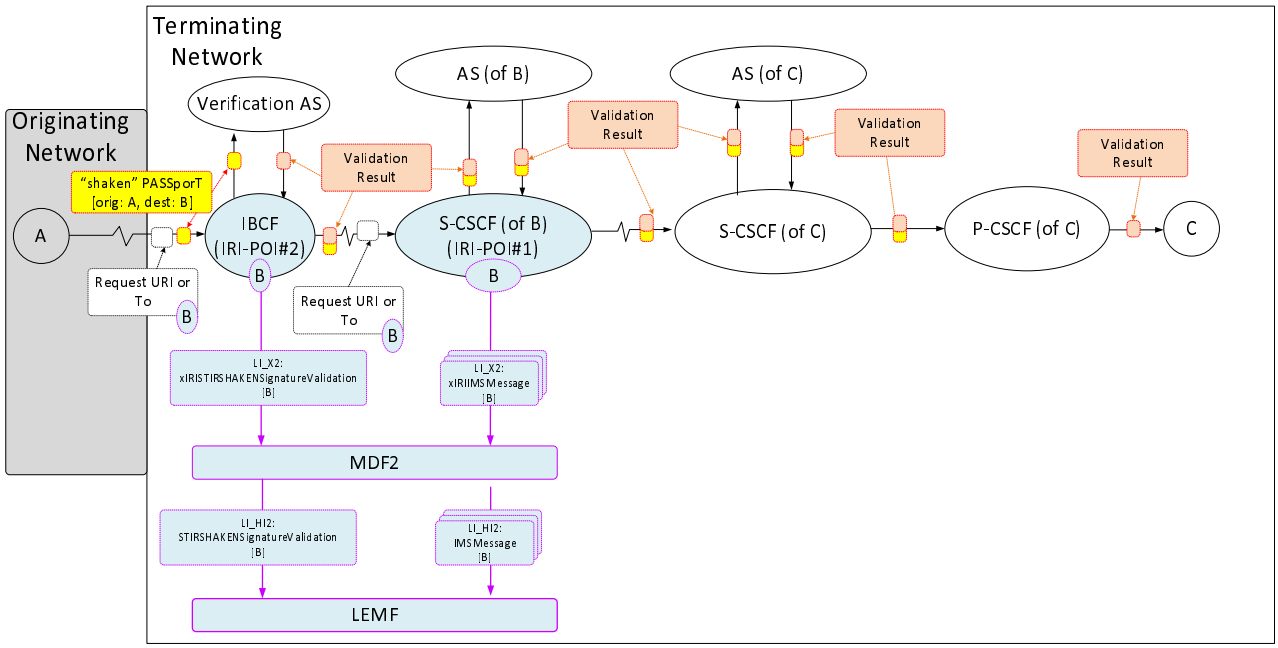


Figure 4.4.5.7.1.2-1: The LI aspects of scenario 1 (IBCF)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The ingress IBCF interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same ingress IBCF generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the ingress IBCF uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

4.4.5.7.1.3 CSP choice is Telephony AS

In the illustrations, the S-CSCF (of B) provides the IMS-LI for target (Party B).

The Telephony AS (of B) interacts with the Verification AS to validate the received PASSporT (s).

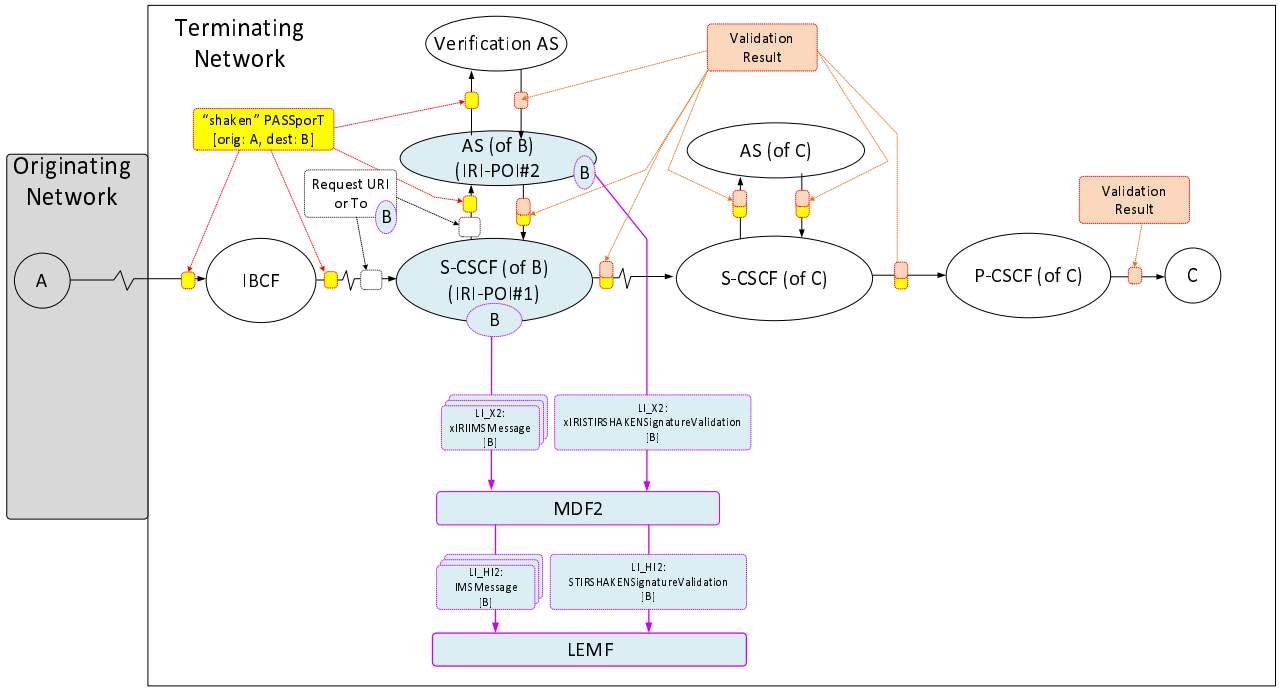


Figure 4.4.5.7.1.3-1: The LI aspects of scenario 1 (Telephony AS)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

4.4.5.7.2 Scenario 2: Party A calls Party B redirected to Party C (target)

4.4.5.7.2.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party A is in the originating network and Party B and Party C are in the terminating network. Party C is the target.

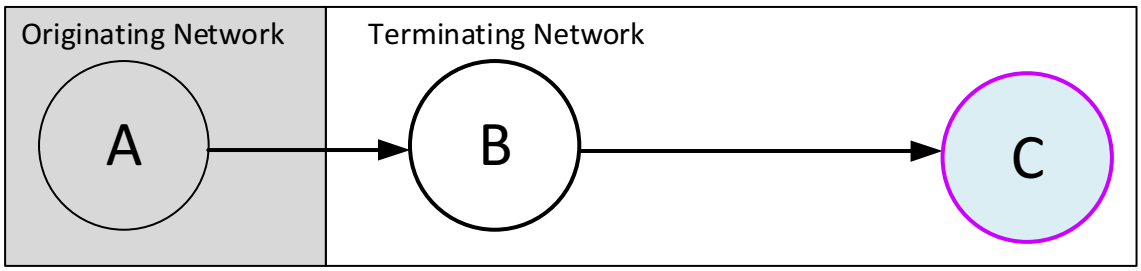


Figure 4.4.5.7.2.1-1: Party A calls Party B redirected to Party C (target)

The subsequent sub-clauses illustrate the LI aspects for this scenario for the CSP choice used in selecting the IMS NF that interacts with the Verification AS (IBCF Vs Telephony AS).

All the illustrations assume the default option (see TS 33.127 [3]) for IMS LI.

4.4.5.7.2.2 CSP choice is IBCF

In the illustrations, the S-CSCF (of C) provides the IMS-LI for target (Party C).

The ingress IBCF interacts with the Verification AS to validate the received PASSporT (s).

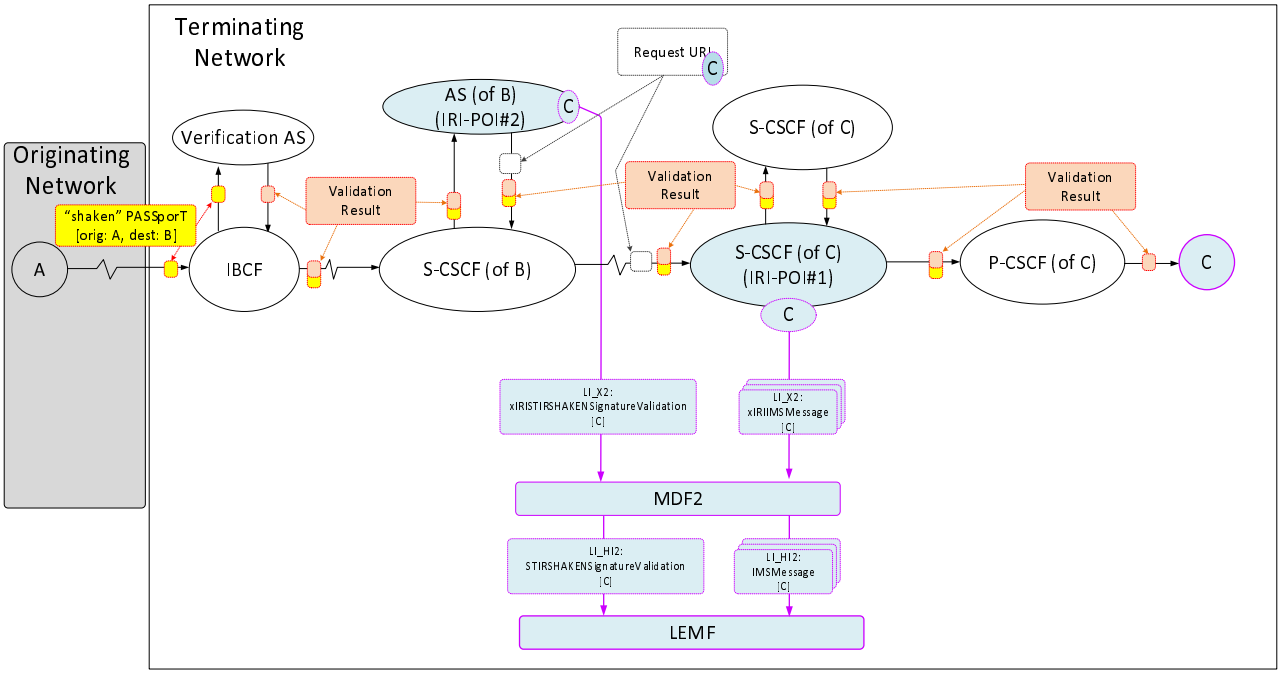


Figure 4.4.5.7.2.2-1: The LI aspects of scenario 2 (IBCF)

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The ingress IBCF interacts with the Verification AS to validate the received "shaken" PASSporT.

The IRI-POI present in the Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the S-CSCF (of C) uses Request URI header (pointing Party C) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI header (pointing Party C) of the outgoing SIP INVITE to perform a target match.

4.4.5.7.2.3 CSP choice is Telephony AS

In the illustrations, the S-CSCF (of C) provides the IMS-LI for target (Party C).

The Telephony AS (of B) interacts with the Verification AS to validate the received PASSporT (s).

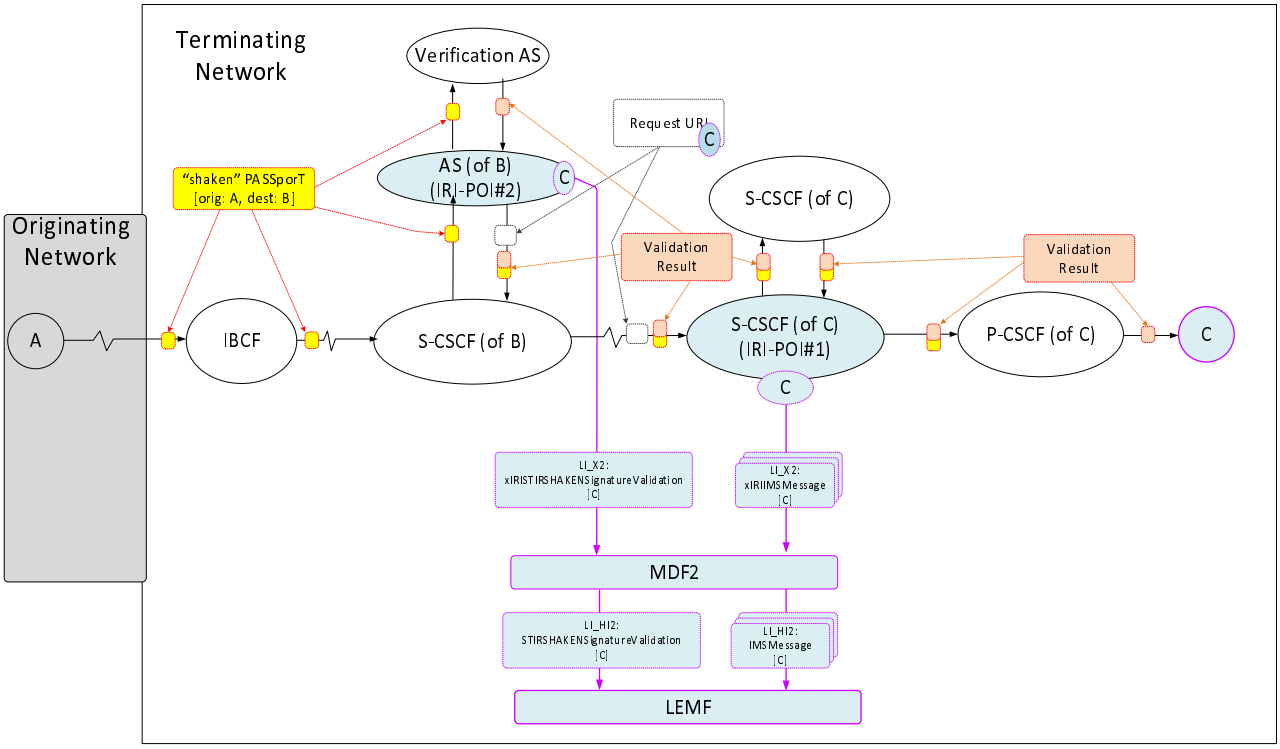


Figure 4.4.5.7.2.3-1: The LI aspects of scenario 2 (Telephony AS)

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to validate the received "shaken" PASSporT, and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the S-CSCF (of C) uses Request URI or the To headers (pointing Party B) of the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI header (pointing Party C) of the outgoing SIP INVITE to perform a target match.

4.4.5.8 Inter-network IMS sessions (intermediate network, basic scenarios)

4.4.5.8.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI for the intermediate network. Here, a basic scenario infers only one redirection in the intermediate network.

When the CSP choice is IBCF, the ingress IBCF in the intermediate network interacts with the Verification AS to validate the received PASSporTs. The egress IBCF would interact with the Signing AS to sign all of the redirecting party identity that is local to the intermediate network.

When the CSP choice is the Telephony AS, the Telephony AS of the party in the intermediate network interacts with the Verification AS to validate the received PASSporTs. The same Telephony AS (since there are no further redirections) in the intermediate network would also interact with the Signing AS to sign the redirecting party identity.

The IMS NFs that provide the LI functions for the STIR/SHAKEN related reporting depends on the call scenario and the CSP choice.

For the destination target non-local ID, with the option 1 (introduced in clause 4.4.4.4.3.1), the egress IBCF provides the LI functions for STIR/SHAKEN related reporting.

4.4.5.8.2 Scenario 1: Party A calls Party B (target) redirected to Party C

4.4.5.8.2.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party A is in the originating network and Part C is in the terminating network. Party B is in the intermediate network and is a target in the intermediate network.

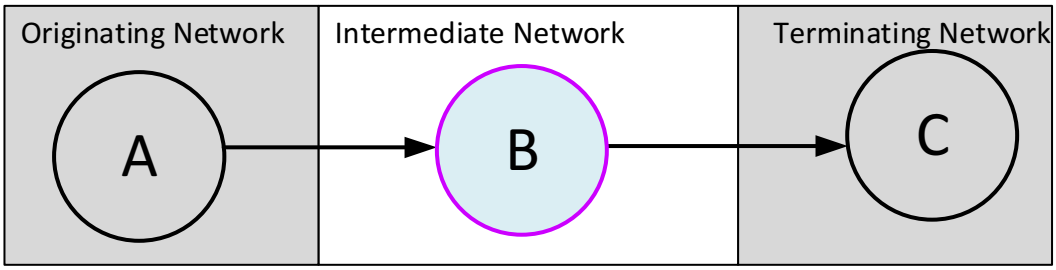


Figure 4.4.5.8.2.1-1: Party (originating network) calls Party B (target) redirected to Party C (terminating network)

Figures 4.4.5.8.2.2-1 and 4.4.5.8.2.3-1 illustrate the LI aspects for this scenario.

4.4.5.8.2.2 CSP choice is IBCF

The illustration shown in figure 4.4.5.8.2.2-1 assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI for target (Party B).

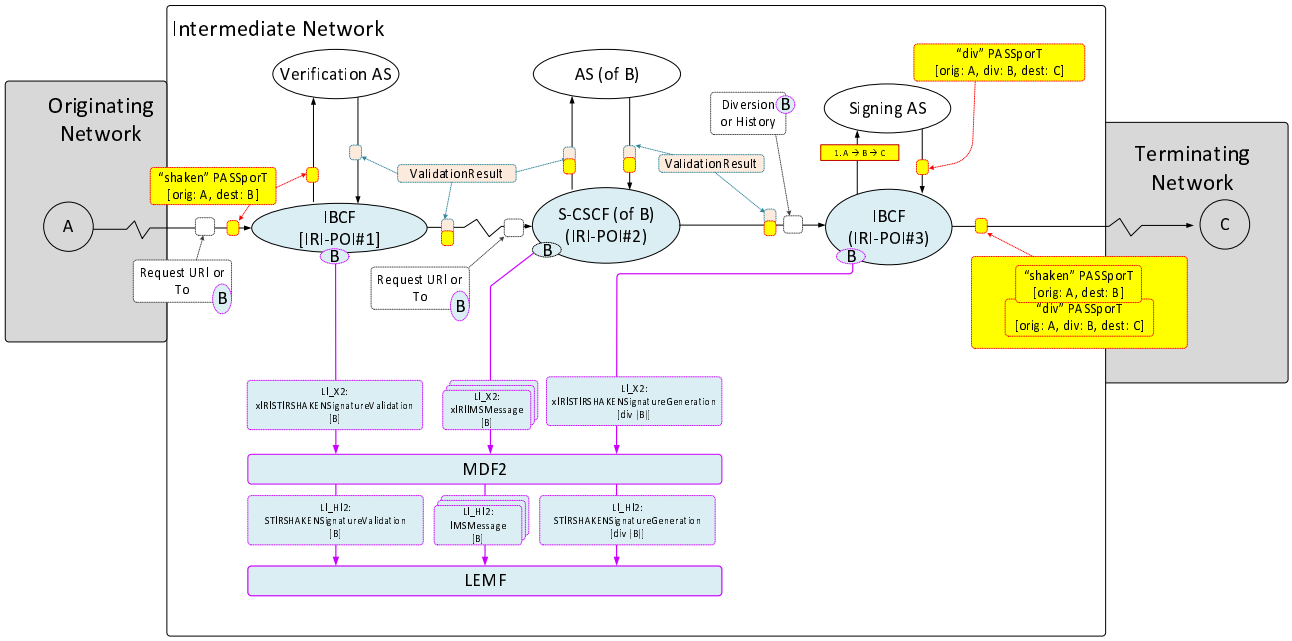


Figure 4.4.5.8.2.2-1: The LI aspects for scenario 1 (IBCF)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The ingress IBCF interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in that ingress IBCF generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The egress IBCF interacts with the Signing AS to authenticate/sign B's identity and the IRI-POI present in that egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for target Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the Diversion or the History Info headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

4.4.5.8.2.3 CSP choice is Telephony AS

The illustration shown in figure 4.4.5.8.2.3-1 assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI for target (Party B).

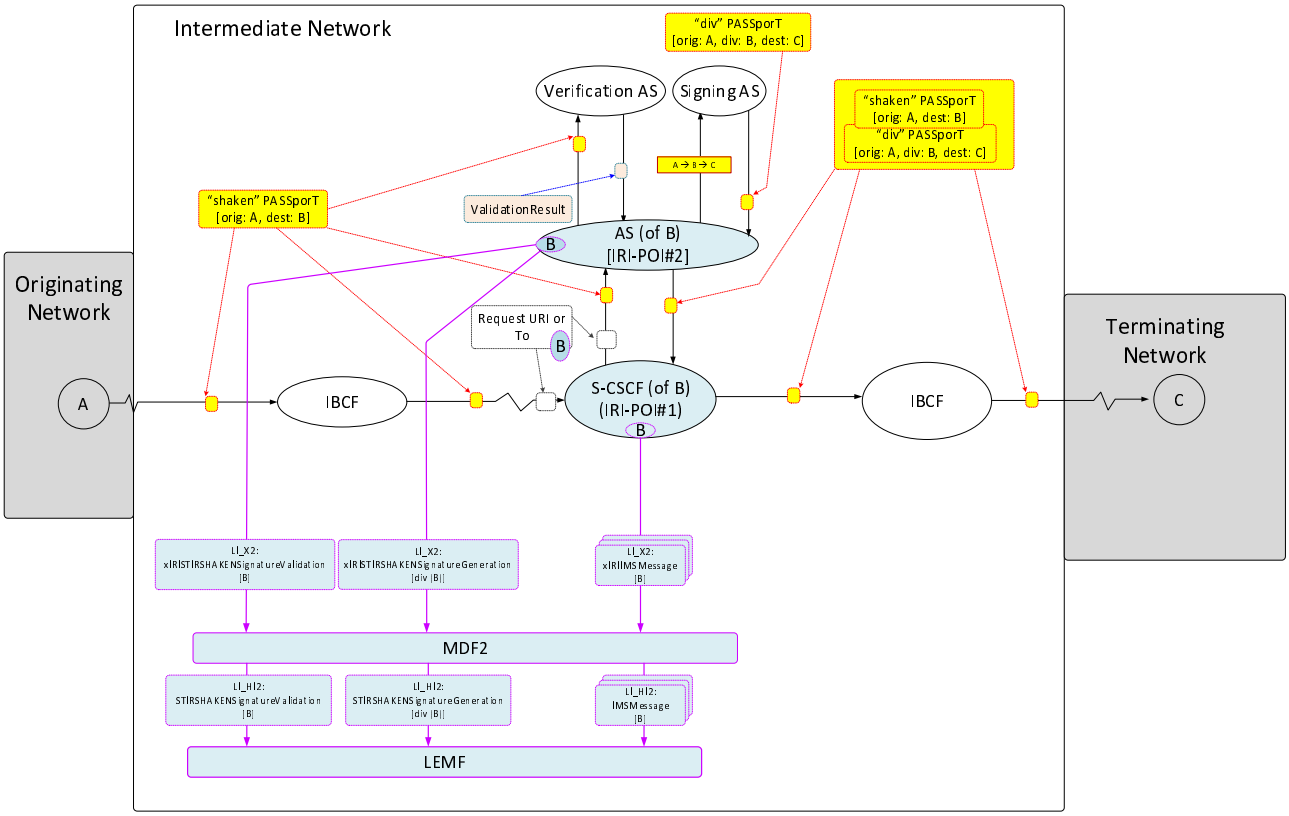


Figure 4.4.5.8.2.3-1: The LI aspects for scenario 1 (Telephony AS)

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The Telephony AS of B interacts with the Signing AS to authenticate/sign B's identity and the IRI-POI present in same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for target Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS of B uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

4.4.5.8.3 Scenario 2: Party A (target non-local ID) calls Party B redirected Party C (target non-local ID).

4.4.5.8.3.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party A is in the originating network and is target non-local IDs in the intermediate network. Party C is in the terminating network and is a target non-local ID in the intermediate network.

LEMF-1 is intercepting the target non-local ID Party A. The LEMF-2 is intercepting the target non-local ID (Party C).

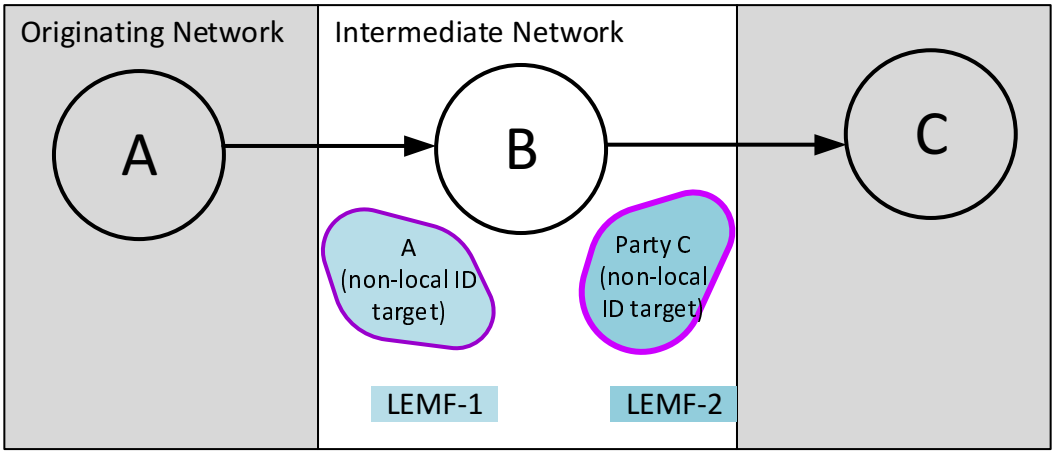


Figure 4.4.5.8.3.1-1: Party A (target non-local ID) calls Party B redirected to Party C (target non-local ID)

Figures in the subsequent clauses illustrate the LI aspects for this scenario.

4.4.5.8.3.2 CSP choice IBCF

4.4.5.8.3.2.1 ReportDiversionPASSporTInfo is set to "False"

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID (Party A) and the egress IBCF provides the IRI-POI for IMS LI for the target non-local ID (Party C).

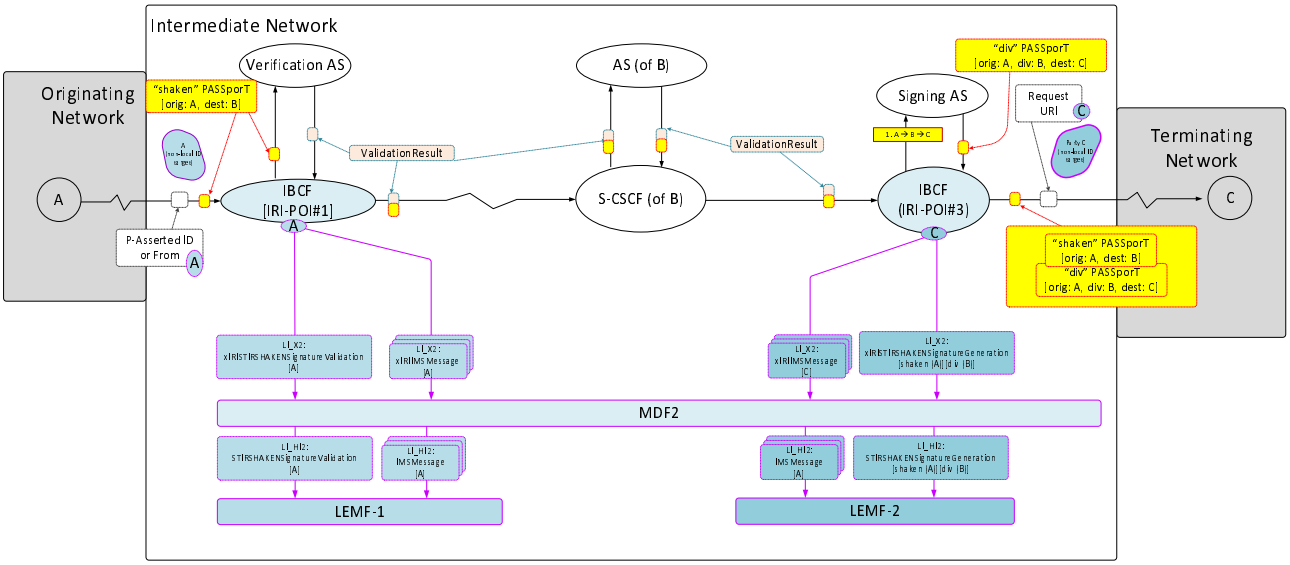


Figure 4.4.5.8.3.2.1-1: The LI aspects for scenario 2 (ReportDiversionPASSporTInfo is set to "False")

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The ingress IBCF interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in that ingress IBCF generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party C)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A and the "div" PASSporT for Party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the egress IBCF uses the Request URI (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

4.4.5.8.3.2.2 ReportDiversionPASSporTInfo is set to "True"

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID (Party A) and the egress IBCF provides the IRI-POI for IMS LI for the target non-local ID (Party C).

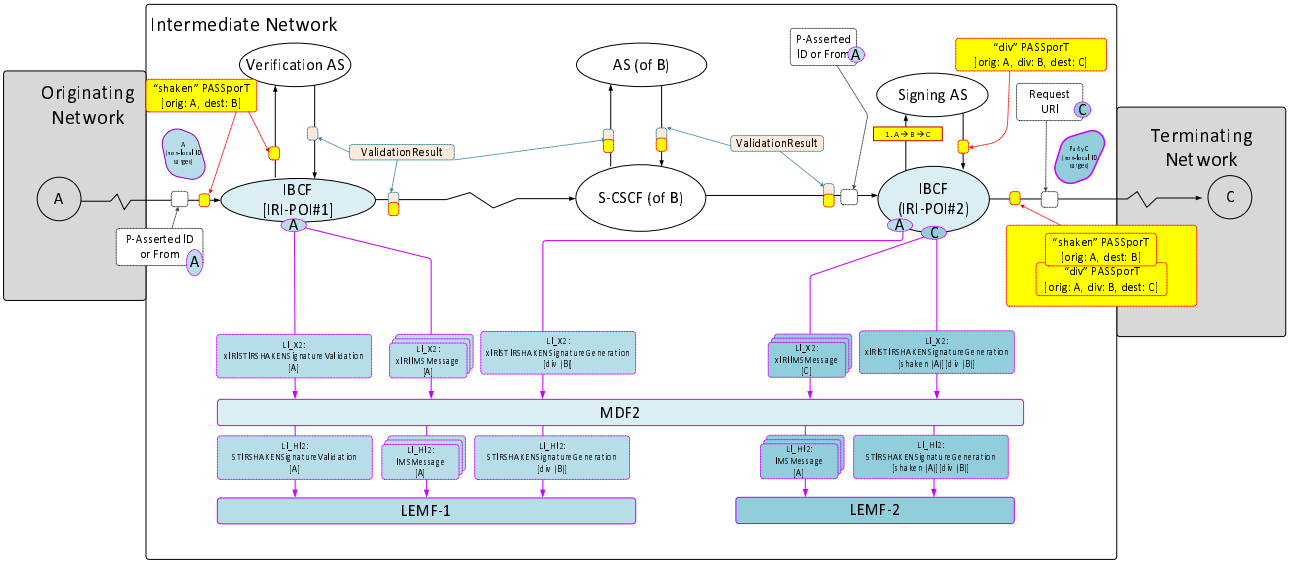


Figure 4.4.5.8.3.2.2-1: The LI aspects for scenario 2 (ReportDiversionPASSporTInfo is set to "True")

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The ingress IBCF interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in that ingress IBCF generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The egress IBCF interacts with the Signing AS to authenticate/sign B's identity and with ReportDiversionPASSporTInfo set to "True" for target non-local ID (Party A), the IRI-POI present egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party C)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A and the "div" PASSporT for Party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the egress IBCF uses the Request URI header (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

4.4.5.8.3.3 CSP choice Telephony AS

4.4.5.8.3.3.1 ReportDiversionPASSporTInfo is set to "False"

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID (Party A) and the egress IBCF provides the IRI-POI for IMS LI for the target non-local ID (Party C).

The illustration assumes option 1(introduced in clause 4.4.4.4.3.1) for the STIR/SHAKEN related LI reporting when the destination is a target non-local ID. Therefore, the Telephony AS (of B) provides the IRI-POI for STIR/SHAKEN related LI reporting for target non-local ID (Party C).

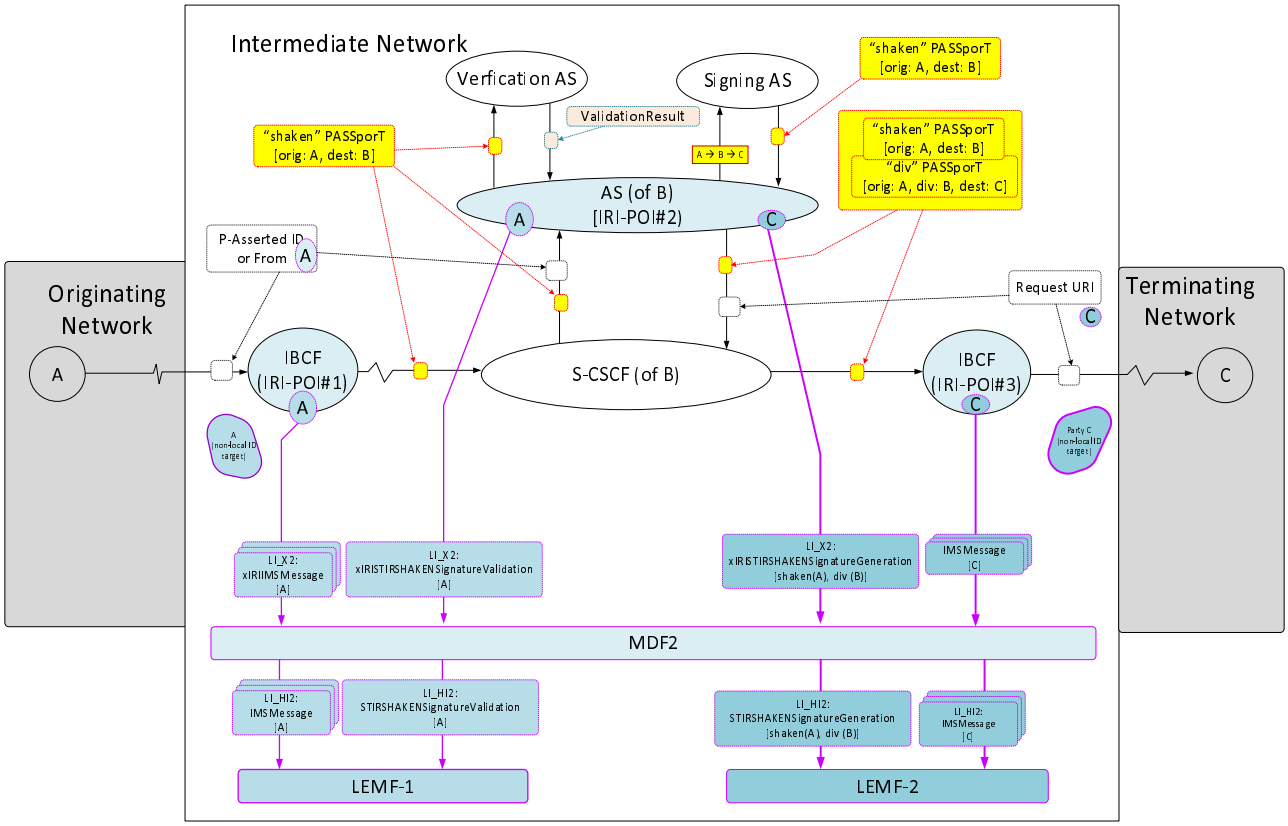


Figure 4.4.5.8.3.3.1-1: The LI aspects for scenario 2 (ReportDiversionPASSporTInfo is set to "False")

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in same Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party C)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in Telephony AS (of B), upon detecting that the destination (Party C) is a target non-local ID, generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A and the "div" PASSporT for Party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the egress IBCF uses the Request URI header (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the egress Telephony AS (of B) uses the Request URI header (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

4.4.5.8.3.3.2 ReportDiversionPASSporTInfo is set to "True"

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID (Party A) and the egress IBCF provides the IRI-POI for IMS LI for the target non-local ID (Party C).

The illustration assumes option 1 (introduced in clause 4.4.4.4.3.1) for the STIR/SHAKEN related LI reporting when the destination is a target non-local ID. Therefore, the Telephony AS (of B) provides the IRI-POI for STIR/SHAKEN related LI reporting for target non-local ID (Party C).

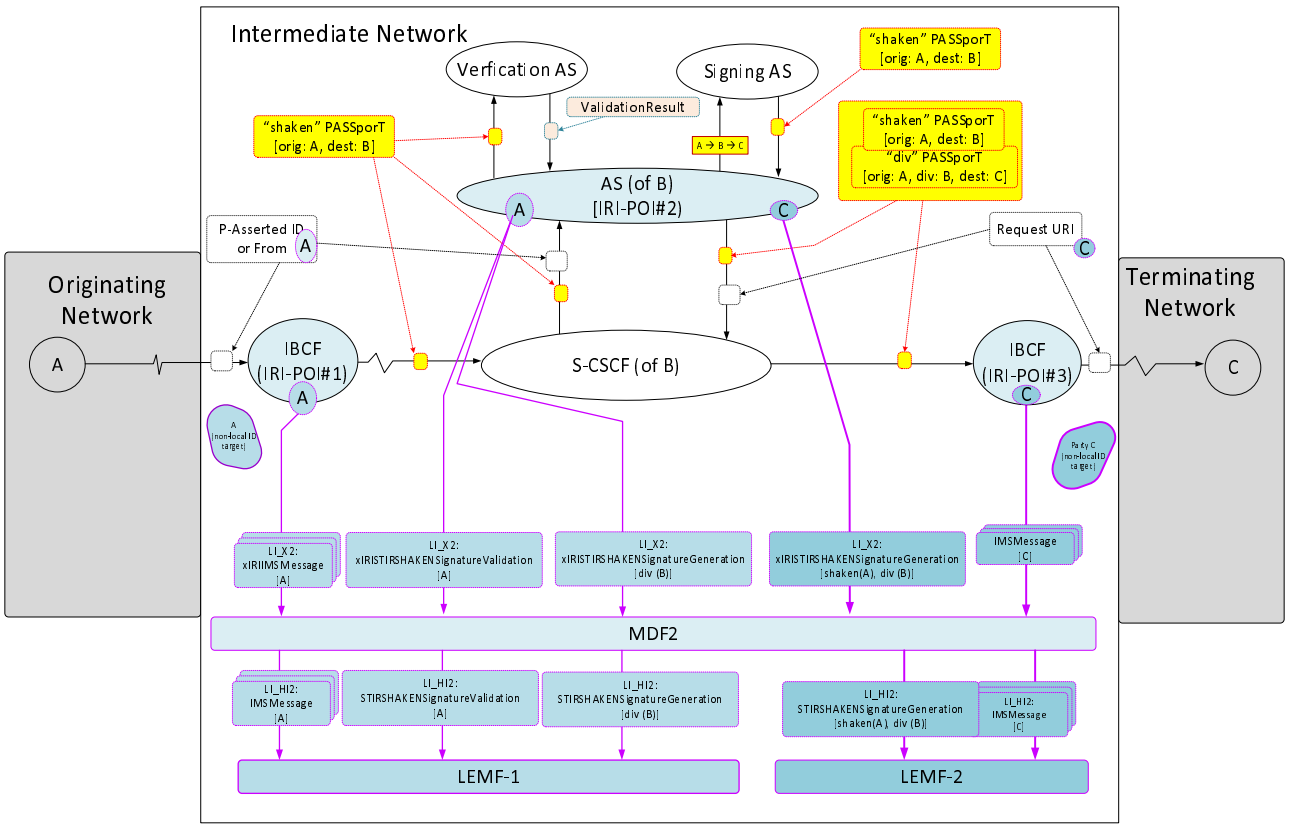


Figure 4.4.5.8.3.3.2-1: The LI aspects for scenario 2 (ReportDiversionPASSporTInfo is set to "True")

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in same Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The Telephony AS (of B) interacts with the Signing AS to authenticate/sign B's identity and with ReportDiversionPASSporTInfo set to "True" for target non-local ID (Party A), the IRI-POI present in the Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "div" PASSporT for Party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party C)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in Telephony AS (of B), upon detecting that the destination (Party C) is a target non-local ID, generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A and the "div" PASSporT for Party B) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the egress IBCF uses the Request URI (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the egress Telephony AS (of B) uses the Request URI header (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

4.4.5.9 Inter-network IMS sessions (intermediate network, redirecting scenarios)

4.4.5.9.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI for the intermediate network.

When the CSP choice is IBCF, the ingress IBCF in the intermediate network interacts with the Verification AS to validate the received PASSporTs. The egress IBCF in the intermediate network would interact with the Signing AS to sign all of the redirecting party identities that are local to the intermediate network.

When the CSP choice is the Telephony AS, the Telephony AS of the first party in the intermediate network interacts with the Verification AS to validate the received PASSporTs. The Telephony AS of the last party (in the case of further redirections) in the intermediate network would interact with the Signing AS to sign all of the redirecting party identities that are local to the intermediate network.

The IMS NFs that provide the LI functions for the STIR/SHAKEN related reporting depends on the call scenario and the CSP choice.

For the destination target non-local ID, all the illustrations assume the option 1 (introduced in clause 4.4.4.4.3.1), for STIR/SHAKEN related reporting.

4.4.5.9.2 Scenario 1: Party A calls Party B (target) redirected to Party C redirected to Party D

4.4.5.9.2.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C, who in turn, has call forwarding to Party D. Party A is in the originating network and Part D is in the terminating network. Party B and Party C are in the intermediate network and both Party B and Party C are the targets in the terminating network.

LEMF-1 is intercepting the target Party B and the LEMF-2 is intercepting the target Party C.

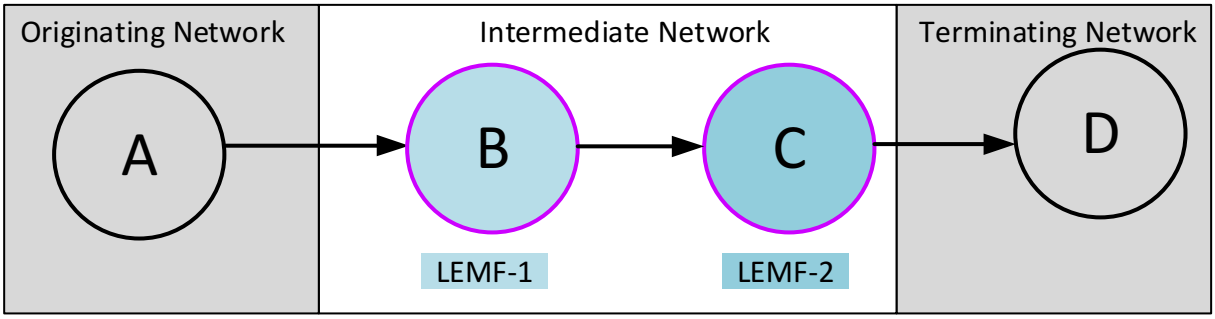


Figure 4.3.5.9.2.1-1: Party (originating network) calls Party B (target), redirected to Party C (target) redirected to Party D (terminating network)

The descriptions given in clause 4.4.3 apply to this scenario as well.

The subsequent clauses below illustrate the LI aspects for this scenario.

4.4.5.9.2.2 CSP choice is IBCF

4.4.5.9.2.2.1 ReportDiversionPASSporTInfo is set to "False"

The illustration shown in figure 4.4.5.10.2.2.1-1 assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI for the target (Party B). S-CSCF of C provides the IRI-POI functions for IMS LI for the target (Party C).

The egress IBCF interacts 1) with the Signing AS to authenticate/sign B's identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

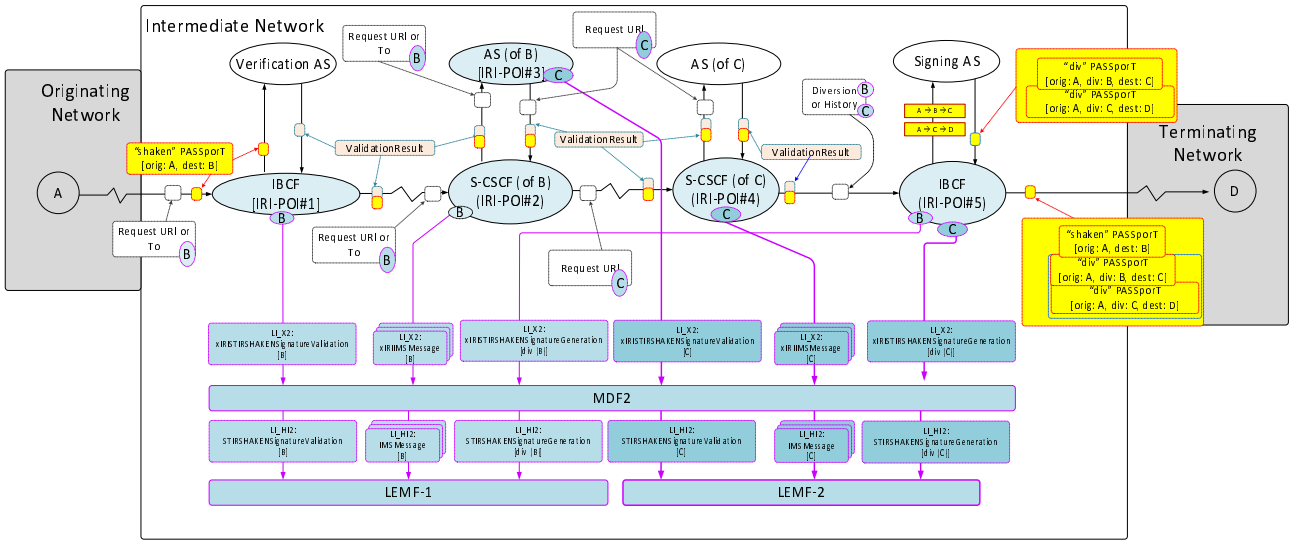


Figure 4.4.5.9.2.2.1-1: The LI aspects for scenario 1 (IBCF) with ReportDiversionPASSporTInfo is set to "False"

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The ingress IBCF interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in that ingress IBCF generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The IRI-POI present in the egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the ingress IBCF uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the Diversion or the History Info headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in the Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record for to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-2.

The egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for target Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the S-CSCF (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI header (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the Diversion or the History Info headers (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

4.4.5.9.2.2.2 ReportDiversionPASSporTInfo is set to "True"

The illustration shown in figure 4.4.5.10.2.2.2-1 assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI for target (Party B). S-CSCF of C provides the IRI-POI functions for IMS LI for target (Party C).

The egress IBCF interacts 1) with the Signing AS to authenticate/sign B’s identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

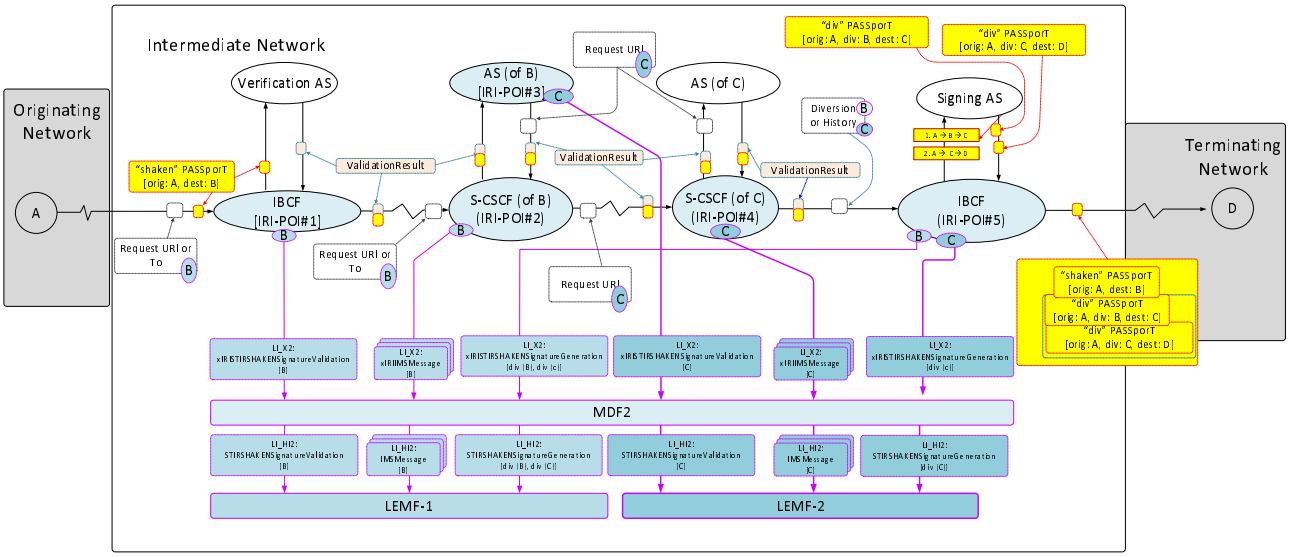


Figure 4.4.5.9.2.2.2-1: The LI aspects for scenario 1 (IBCF) with ReportDiversionPASSporTInfo is set to "True"

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The ingress IBCF interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in that ingress IBCF generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The IRI-POI present in the egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the two "div" PASSporTs (for Party B and for Party C) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the ingress IBCF uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the Diversion or the History Info headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in the Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record for to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-2.

The egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for target Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the S-CSCF (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI header (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the Diversion or the History Info headers (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

4.4.5.9.2.3 CSP choice is Telephony AS

4.4.5.9.2.3.1 ReportDiversionPASSporTInfo is set to "False"

The illustration shown in figure 4.4.5.10.2.3.1-1 assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI for target Party B. S-CSCF of C provides the IRI-POI functions for IMS LI for target Party C.

The Telephony AS (of C) interacts 1) with the Signing AS to authenticate/sign B's identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

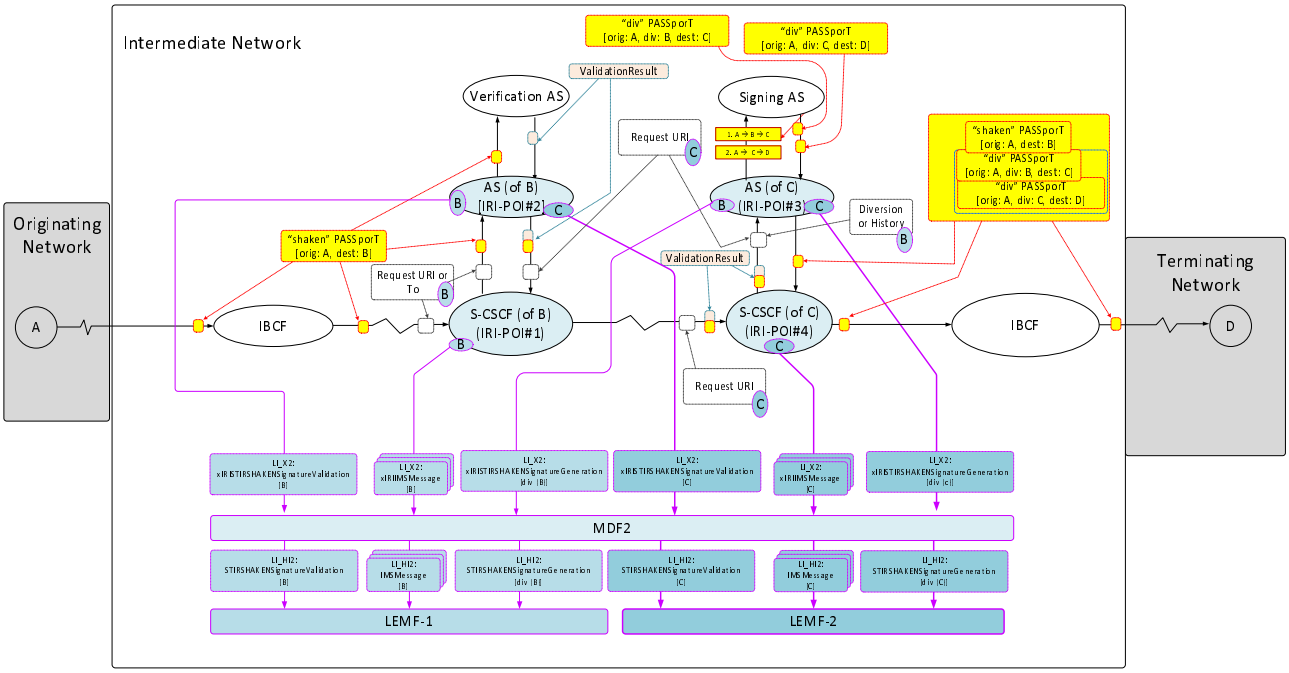


Figure 4.4.5.9.2.3.1-1: The LI aspects for scenario 1 (Telephony AS) with ReportDiversionPASSporTInfo is set to "False"

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The IRI-POI present in the Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Diversion or the History Info headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in the Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record for to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-2.

The IRI-POI present in the Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for target Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the S-CSCF (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI header (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

4.4.5.9.2.3.2 ReportDiversionPASSporTInfo is set to "True"

The illustration shown in figure 4.4.5.99.2.3.2-1 assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the S-CSCF of B provides the IRI-POI functions for IMS LI for target Party B. S-CSCF of C provides the IRI-POI functions for IMS LI for target Party C.

The Telephony AS (of C) interacts 1) with the Signing AS to authenticate/sign B's identity and, again interacts (2) with the Signing AS, to authenticate/sign C's identity.

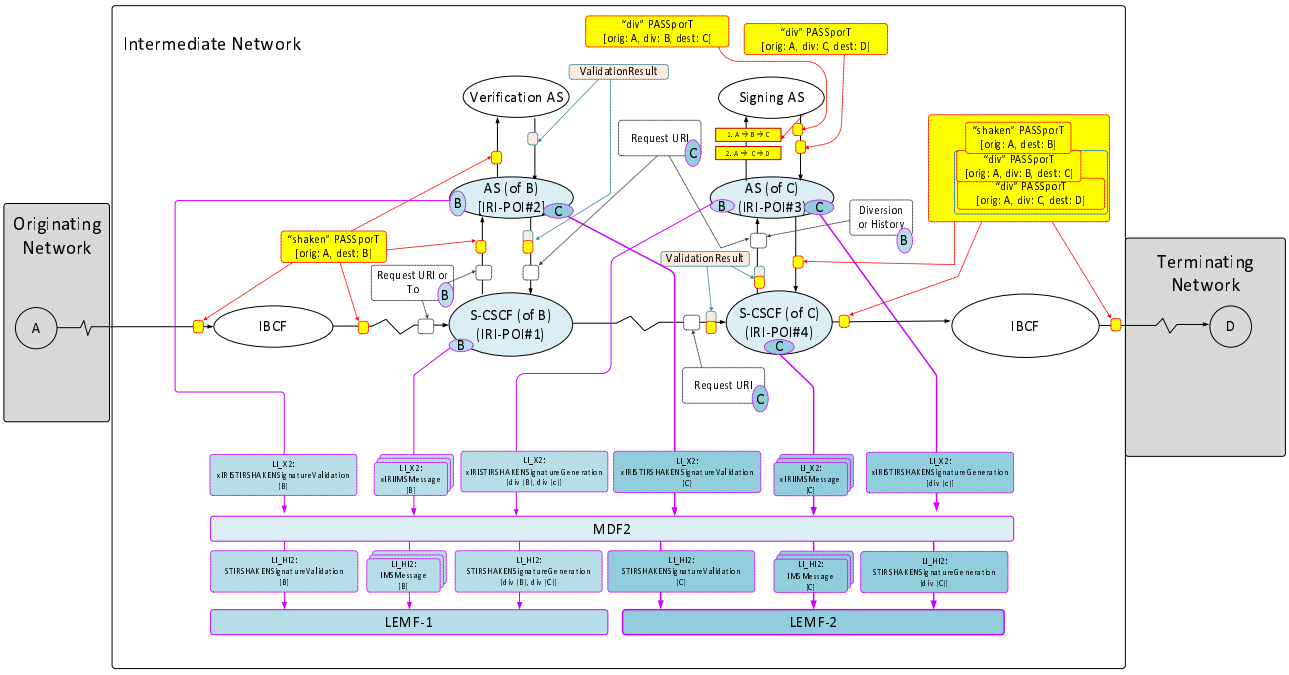


Figure 4.4.5.9.2.3.2-1: The LI aspects for scenario 1 (IBCF) with ReportDiversionPASSporTInfo is set to "True"

For target (Party B)

The IRI-POI present in the S-CSCF (of B) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in same Telephony AS generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The IRI-POI present in the Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the two "div" PASSporTs (for Party B and for Party C) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the S-CSCF (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI or the To headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Diversion or the History Info headers (pointing to Party B) present in the incoming SIP INVITE to perform a target match.

For target (Party C)

The IRI-POI present in the S-CSCF (of C) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in the Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record for to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-2.

The IRI-POI present in the Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for target Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the S-CSCF (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the Request URI header (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the IRI-POI present in the Telephony AS (of C) uses the Request URI header (pointing to Party C) present in the incoming SIP INVITE to perform a target match.

4.4.5.9.3 Scenario 2: Party A (target non-local ID) calls Party B redirected Party C redirected to Party D (target non-local ID)

4.4.5.9.3.1 The scenario

In this scenario, Party A calls Party B who has a call forwarding to Party C, who in turn, has call forwarding to Party D. Party A is in the originating network and is target non-local IDs in the intermediate network. Party D is in the terminating network and is a target non-local ID in the intermediate network. Party B and Party C are in the intermediate network.

LEMF-1 is intercepting the target non-local ID Party A. The LEMF-2 is intercepting the target non-local ID (Party C).

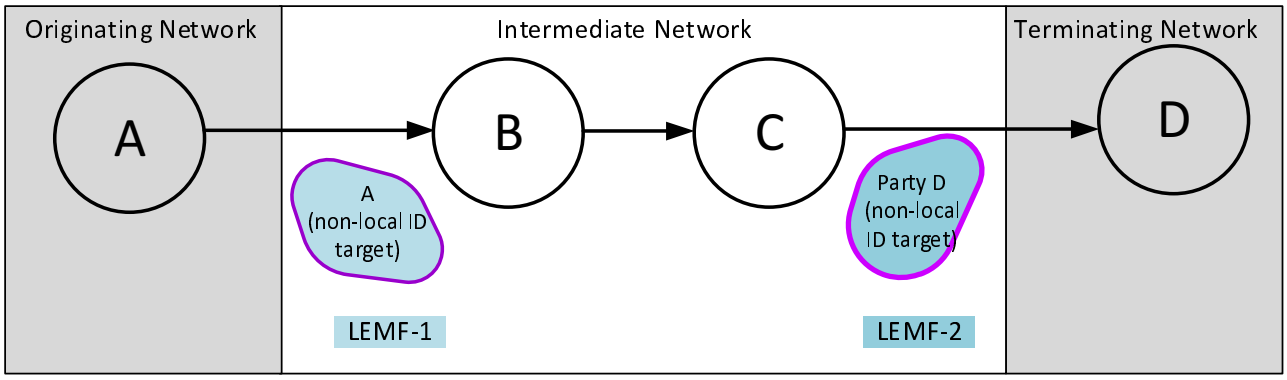


Figure 4.4.5.9.3.1-1: Party A (target non-local ID) calls Party B redirected to Party C (target non-local ID)

Figures subsequent clauses illustrate the LI aspects for this scenario.

4.4.5.9.3.2 CSP choice IBCF

4.4.5.9.3.2.1 ReportDiversionPASSporTInfo is set to "False".

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID Party A and the egress IBCF provides the IRI-POI for IMS LI for the target non-local ID Party D.

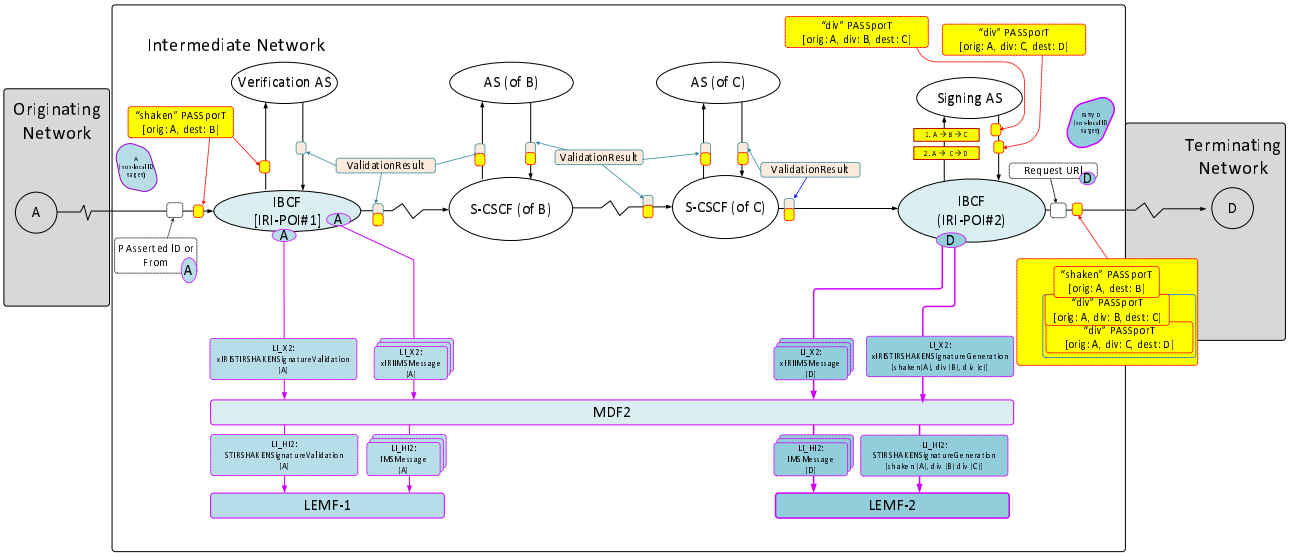


Figure 4.4.5.9.3.2.1-1: The LI aspects for scenario 2 (ReportDiversionPASSporTInfo is set to "False")

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The ingress IBCF interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in that ingress IBCF generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party D)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record to the MDF2. The xIRI includes the "shaken" PASSporT (for Party A) and the two "div" PASSporTs (for Party B and Party C). The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the egress IBCF uses the Request URI (pointing to Party D) present in the outgoing SIP INVITE to perform a target match.

4.4.5.9.3.2.2 ReportDiversionPASSporTInfo is set to "True"

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID (Party A) and the egress IBCF provides the IRI-POI for IMS LI for the target non-local ID (Party D).

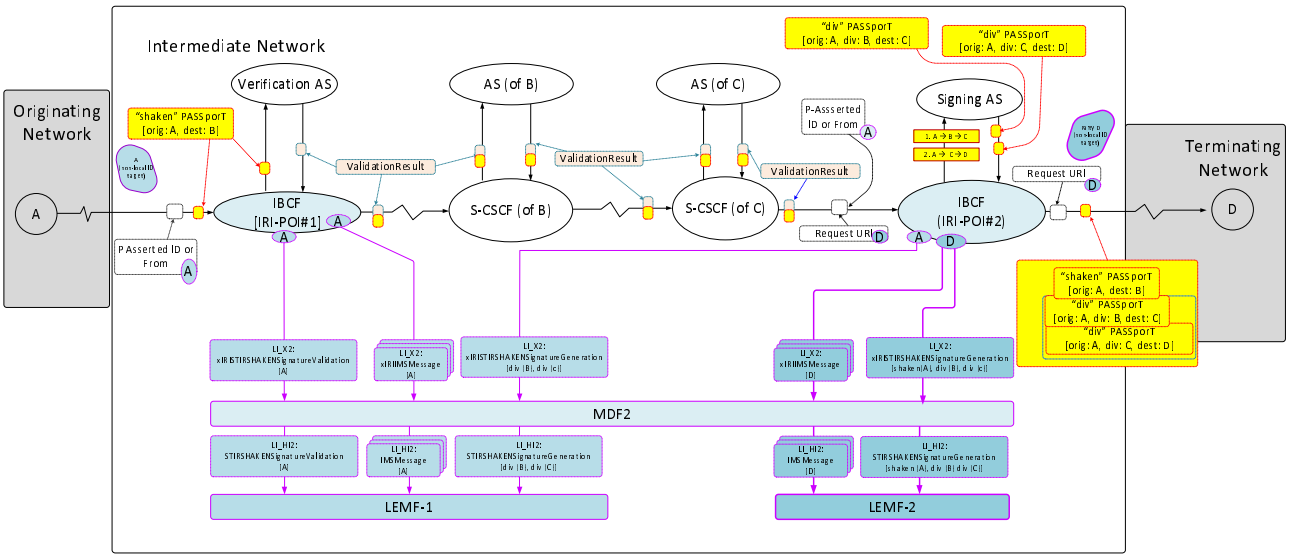


Figure 4.4.5.9.3.2.2-1: The LI aspects for scenario 2 (ReportDiversionPASSporTInfo is set to "True")

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The ingress IBCF interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in that ingress IBCF generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The egress IBCF interacts with the Signing AS to authenticate/sign B's identity and with ReportDiversionPASSporTInfo set to "True" for target non-local ID (Party A), the IRI-POI present egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "div" PASSporT for Party B and the "div" PASSporT for Party C to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party D)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record to the MDF2. The xIRI includes the "shaken" PASSporT (for Party A) and the two "div" PASSporTs (for Party B and Party C). The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the egress IBCF uses the Request URI (pointing to Party C) present in the outgoing SIP INVITE to perform a target match.

4.4.5.9.3.3 CSP choice Telephony AS

4.4.5.9.3.3.1 ReportDiversionPASSporTInfo is set to "False"

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID (Party A) and the egress IBCF provides the IRI-POI for IMS LI for the target non-local ID (Party D).

The illustration assumes option 1 (introduced in clause 4.4.4.4.3.1) for the STIR/SHAKEN related LI reporting when the destination is a target non-local ID. Therefore, the Telephony AS (of C) provides the IRI-POI for STIR/SHAKEN related LI reporting for target non-local ID (Party D).

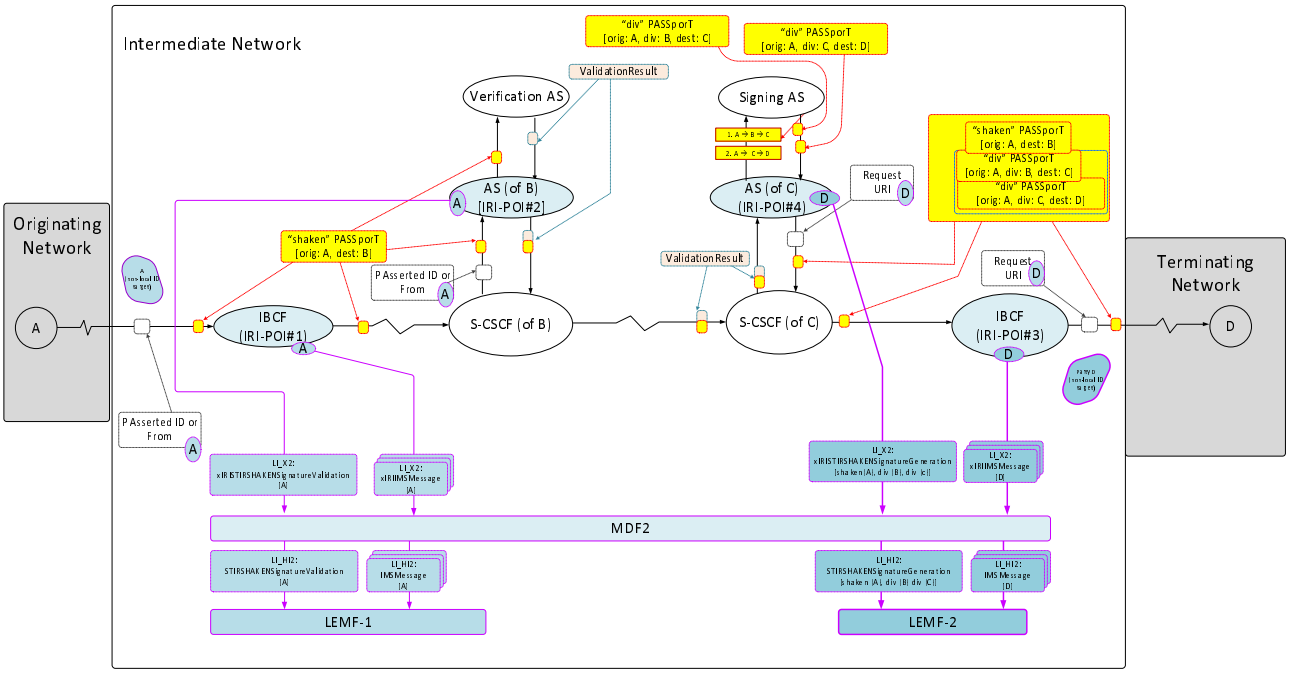


Figure 4.4.5.9.3.3.1-1: The LI aspects for scenario 2 (ReportDiversionPASSporTInfo is set to "False")

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in same Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party D)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in Telephony AS (of C) upon detecting the destination (Party D) is a target non-local ID generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A and the two "div" PASSporTs (for Party B and Party C)) to the MDF2. The xIRI The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the egress IBCF uses the Request URI (pointing to Party D) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Request URI (pointing to Party D) present in the outgoing SIP INVITE to perform a target match.

4.4.5.9.3.3.2 ReportDiversionPASSporTInfo is set to "True"

The illustration assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the ingress IBCF provides the IRI-POI functions for IMS LI for target non-local ID (Party A) and the egress IBCF provides the IRI-POI for IMS LI for the target non-local ID (Party C).

The illustration assumes option 1 (introduced in clause 4.4.4.4.3.1) for the STIR/SHAKEN related LI reporting when the destination is a target non-local ID. Therefore, the Telephony AS (of C) provides the IRI-POI for STIR/SHAKEN related LI reporting for target non-local ID (Party D).

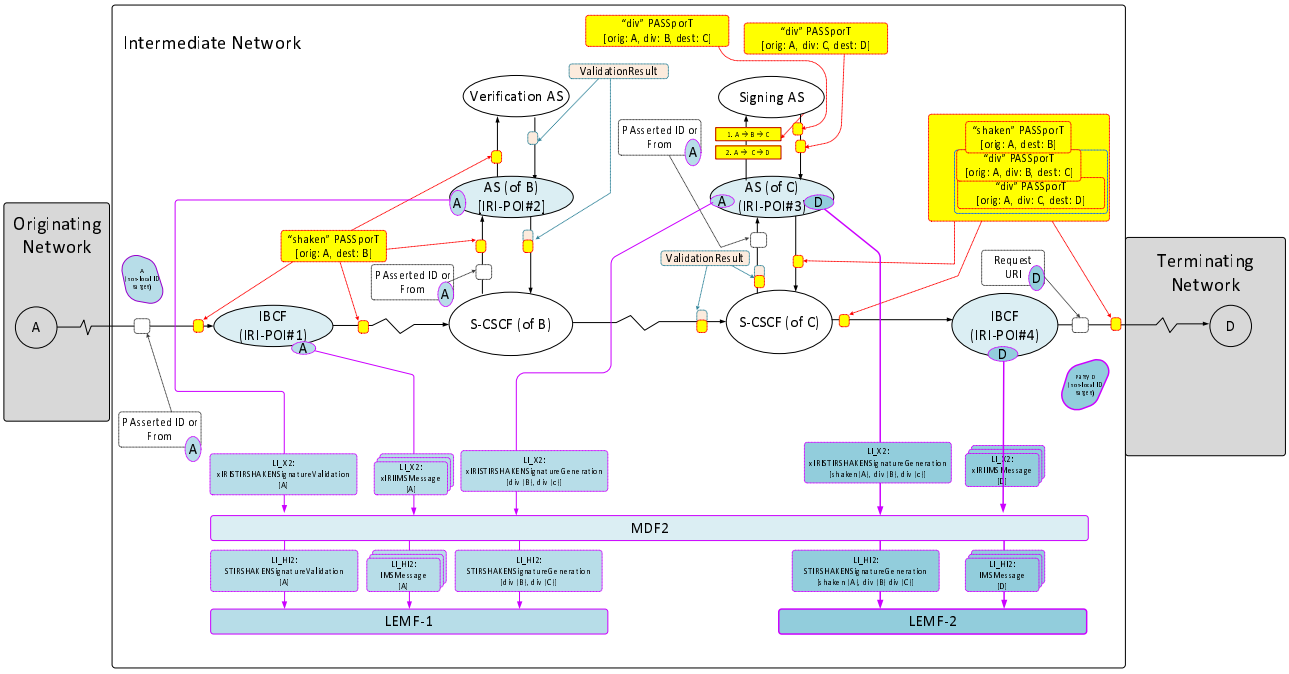


Figure 4.4.5.9.4.3.2-1: The LI aspects for scenario 2 (ReportDiversionPASSporTInfo is set to "True")

For target non-local ID (Party A)

The IRI-POI present in the ingress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-1.

The Telephony AS (of B) interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in same Telephony AS (of B) generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF-1.

The Telephony AS (of C) interacts with the Signing AS to authenticate/sign B's identity and with ReportDiversionPASSporTInfo set to "True" for target non-local ID (Party A), the IRI-POI present in the Telephony AS (of C) generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "div" PASSporT for Party B and the "div" PASSporT for Party C) to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-1.

The IRI-POI present in the ingress IBCF uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of B) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the P-Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

For target non-local ID (Party D)

The IRI-POI present in the egress IBCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF-2.

The IRI-POI present in Telephony AS (of C) upon detecting the destination (Party D) is a target non-local ID generates and delivers the xIRISTIRSHAKENSignatureGeneration record (that includes the "shaken" PASSporT for Party A and the two "div" PASSporTs (for Party B and Party C)) to the MDF2. The xIRI The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF-2.

The IRI-POI present in the egress IBCF uses the Request URI (pointing to Party D) present in the outgoing SIP INVITE to perform a target match.

The IRI-POI present in the Telephony AS (of C) uses the Request URI (pointing to Party D) present in the outgoing SIP INVITE to perform a target match.

4.4.6 Terminating VPLMN

4.4.6.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI in the VPLMN. The two roaming architectures defined in TS 33.127 [3] and TS 33.128 [4] are considered in these illustrations:

- Local Breakout (LBO).
- Home Routed (HR).

With LBO, the P-CSCF provides the LI functions in the VPLMN with the default option as defined in TS 33.127 [3] and TS 33.128 [4]. With HR, the LI specific function LMISF-IRI provides the LI functions as defined in the TS 33.127 [3] and TS 33.128 [8].

For LBO, the illustrations show only the default option.

For performing a target match on the inbound roaming UE, the IRI-POI present in the P-CSCF or the LMISF-IRI use the just the Request URI header when either of Diversion or History Info is present in the SIP INVITE. It may also use the To header when neither Diversion nor the History Info is present in the SIP INVITE.

With LBO, the incoming SIP INVITE includes the Validation Result with the PASSporT and outgoing SIP INVITE sent to the UE does not include the PASSporT information. With HR, the SIP INVITE scanned by the LMISF-IRI only sees the Validation Result and does not see any PASSporT (s) because it was presumed to be removed by the P-CSCF present in the HPLMN.

The STIR/SHAKEN related LI reporting in the terminating VPLMN is independent of whether, or not, an intra-network IMS session is signed.

4.4.6.2 Scenario 1: Party A calls Party B (target in VPLMN)

In this scenario, Party A calls Party B. Party A is in HPLMN and Party B is in VPLMN. Party B, an inbound roamer, is a target in the VPLMN.

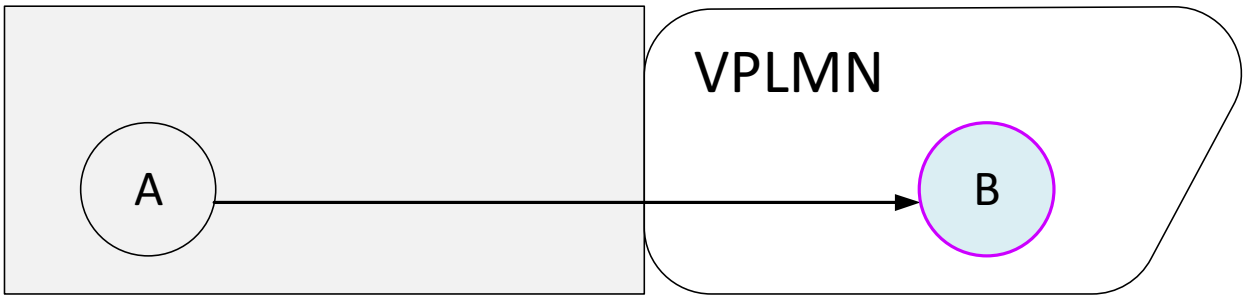


Figure 4.4.6.2-1: Party A calls Party B. Party B (inbound roamer) is target in VPLMN

Figure 4.4.6.2-2 illustrates the LI aspects in VPLMN with LBO roaming for target Party B and the figure 4.4.6.2-3 illustrates the LI aspects in VPLMN with HR roaming for target Party B.

In both illustrations, as can be seen, there is no interaction with the Verification AS in the VPLMN.

LBO

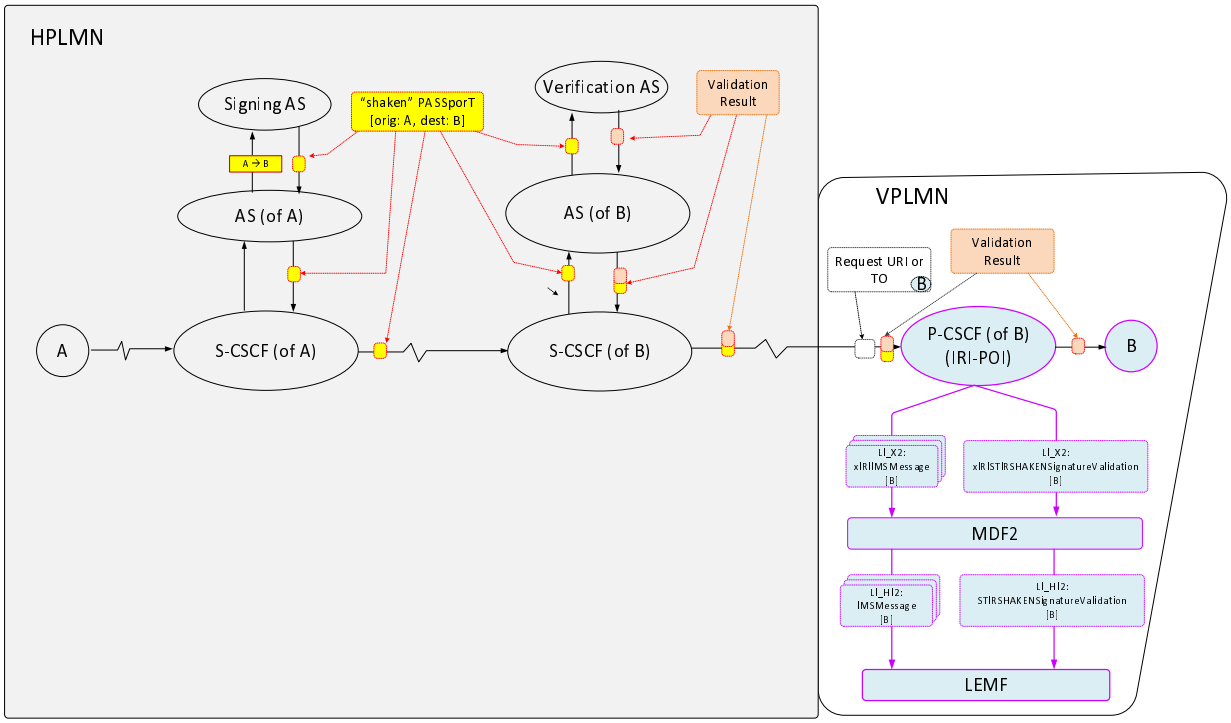


Figure 4.4.6.2-2: STIR/SHAKEN related LI reporting for scenario 1 (VPLMN LBO)

For target (Party B)

For STIR/SHAKEN related LI reporting, the IRI-POI present in the P-CSCF (of B) in the terminating VPLMN upon seeing the Validation Result in the SIP INVITE message generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

For IMS LI, the IRI-POI present in the P-CSCF (of B) also generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

NOTE: When the SIP INVITE carries either of the Diversion or the History Info, the IRI-POI present in the P-CSCF (of B) uses the Request URI header (pointing to target Party B's identity) present in the SIP INVITE to perform a target match.

In this illustration, the SIP INVITE carries neither the Diversion nor the History Info and therefore, the IRI-POI present in the P-CSCF (of B) uses the Request URI or the To headers (pointing to target Party B's identity) present in the SIP INVITE to perform a target match.

HR

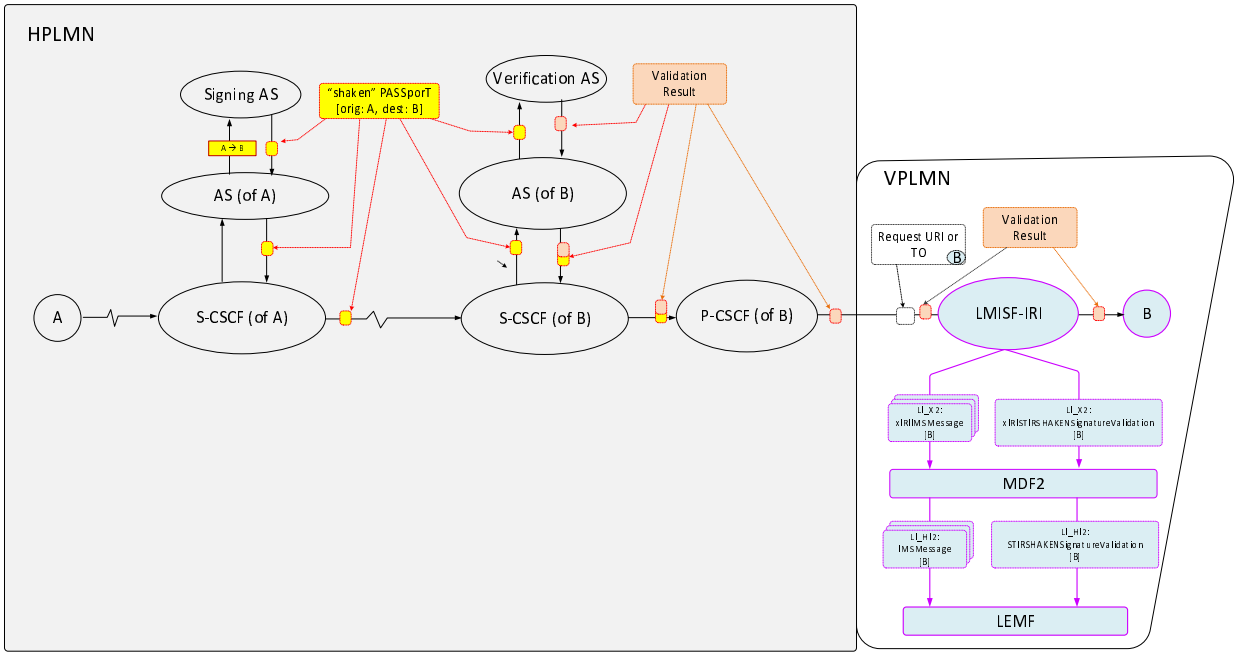


Figure 4.4.6.2-3: STIR/SHAKEN related LI reporting for scenario 1 (VPLMN HR)

For target (Party B)

For STIR/SHAKEN related LI reporting, the LMISF-IRI in the terminating VPLMN upon seeing the Validation Result in the SIP INVITE message generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

For IMS LI, the LMISF-IRI also generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

NOTE: When the SIP INVITE carries either of the Diversion or the History Info, the LMISF-IRI uses the Request URI header (pointing to target Party B's identity) present in the SIP INVITE to perform a target match.

In this illustration, the SIP INVITE carries neither the Diversion nor the History Info headers and therefore, the LMISF-IRI uses the Request URI or the To headers (pointing to target Party B's identity) present in the SIP INVITE to perform a target match.

4.4.6.3 Scenario 2: Party A (target non-local ID) calls Party B (VPLMN)

In this scenario, Party A calls Party B. Party A is in HPLMN and Party B is in VPLMN. Party A is a target non-local ID in the VPLMN.

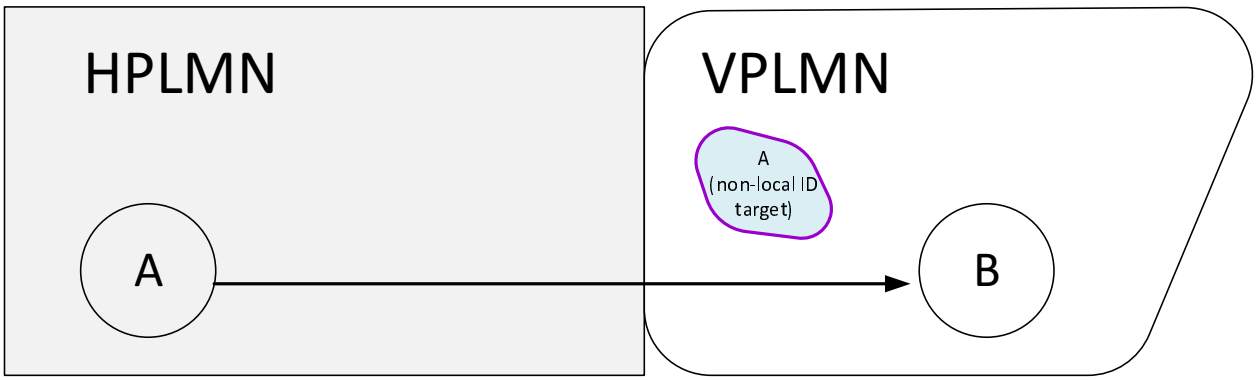


Figure 4.4.6.3-1: Party A calls Party B. Party A is a target non-local ID in VPLMN

Figure 4.4.6.3-2 illustrates the LI aspects in VPLMN with LBO roaming for target non-local ID Party A and the figure 4.4.6.3-3 illustrates the LI aspects in VPLMN with HR roaming for target non-local ID Party A.

In both illustrations, as can be seen, there is no interaction with the Verification AS in the VPLMN.

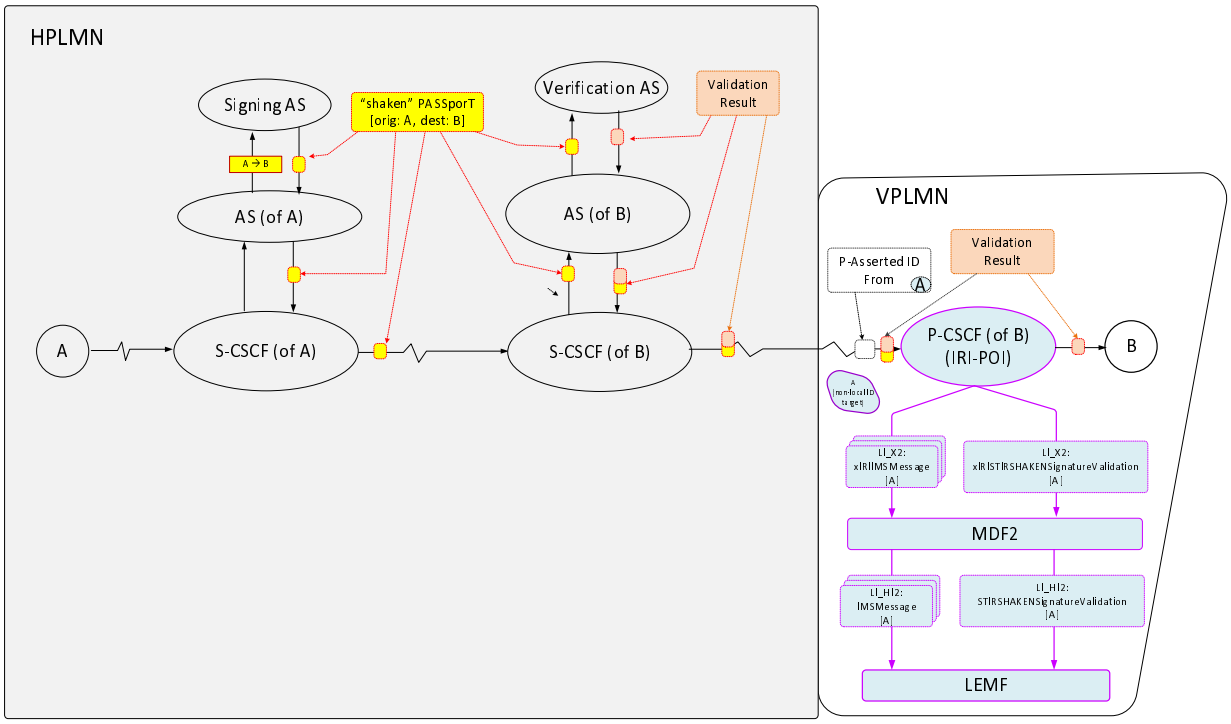


Figure 4.4.6.3-2: STIR/SHAKEN related LI reporting for scenario 2 (VPLMN LBO, target non-local ID)

For target non-local ID (Party A)

For STIR/SHAKEN related LI reporting, the IRI-POI present in the P-CSCF (of B) in the VPLMN upon seeing the Validation Result in the SIP INVITE message generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

For IMS LI, the IRI-POI present in the P-CSCF (of B) also generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The IRI-POI present in the P-CSCF (of B) uses the P-Asserted ID or From Header (pointing to Party A's identity) present in the SIP INVITE to perform a target match.

HR

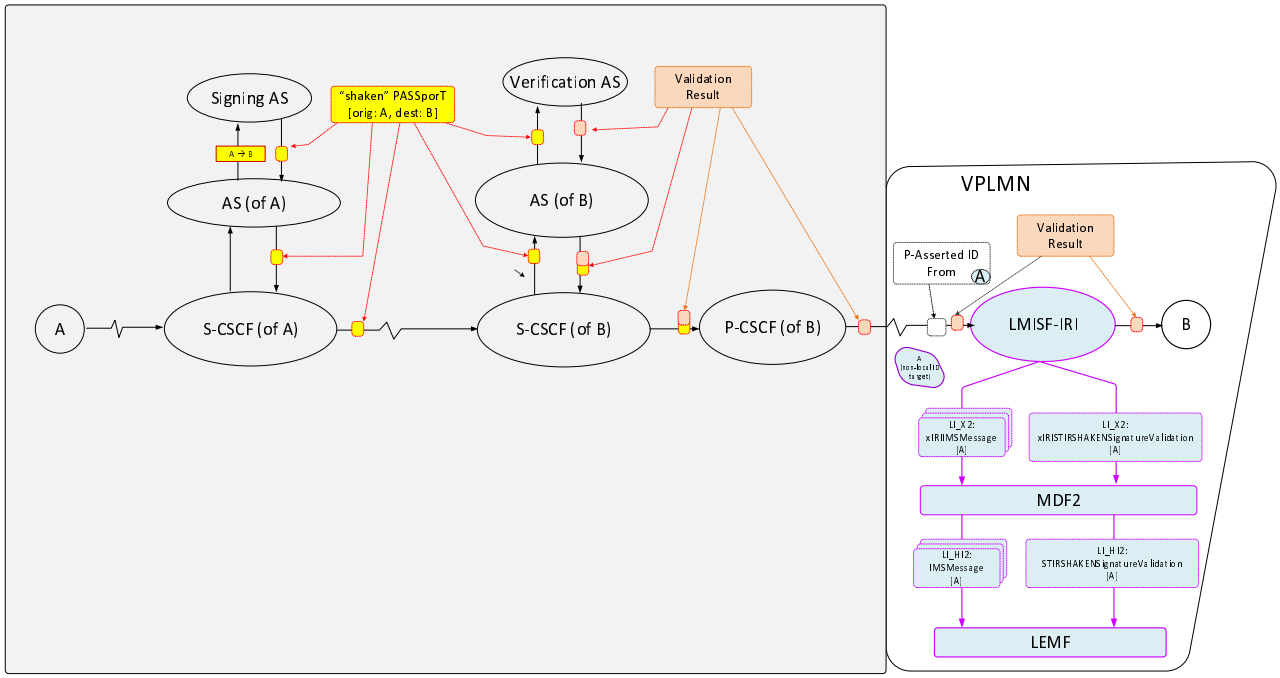


Figure 4.4.6.3-3: STIR/SHAKEN related LI reporting for scenario 2 (VPLMN HR, target non-local ID)

For target non-local ID (Party A)

For STIR/SHAKEN related LI reporting, the LMISF-IRI in the VPLMN upon seeing the Validation Result in the SIP INVITE message generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

For IMS LI, the LMISF-IRI also generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The LMISF-IRI uses the P-Asserted ID or From headers (pointing to target non-local ID Party A's identity) present in the SIP INVITE to perform a target match.

4.4.6.4 Scenario 3: Party A calls Party B (target non-local ID) redirected to Party C (VPLMN)

In this scenario, Party A calls Party B who has a call forwarding to Party C. Party A and Party B are in HPLMN and Party C is VPLMN. Party B happens to be a target non-local ID in the VPLMN.

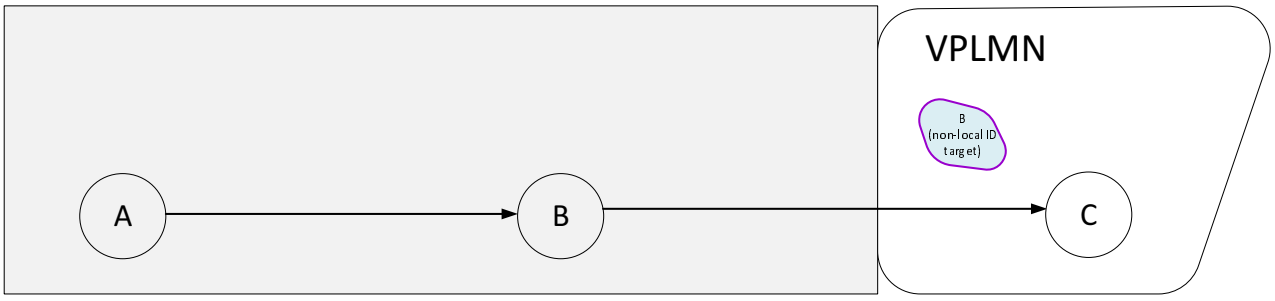


Figure 4.4.6.4-1: Party A calls Party B redirected to Party C. Party B is target non-local ID in VPLMN

Figure 4.4.6.4-2 illustrates the LI aspects in VPLMN with LBO for target non-local ID Party B and the figure 4.4.6.4-3 illustrates the LI aspects in VPLMN with HR for target non-local ID Party B.

In both illustrations, as can be seen, there is no interaction with the Verification AS in the VPLMN.

LBO

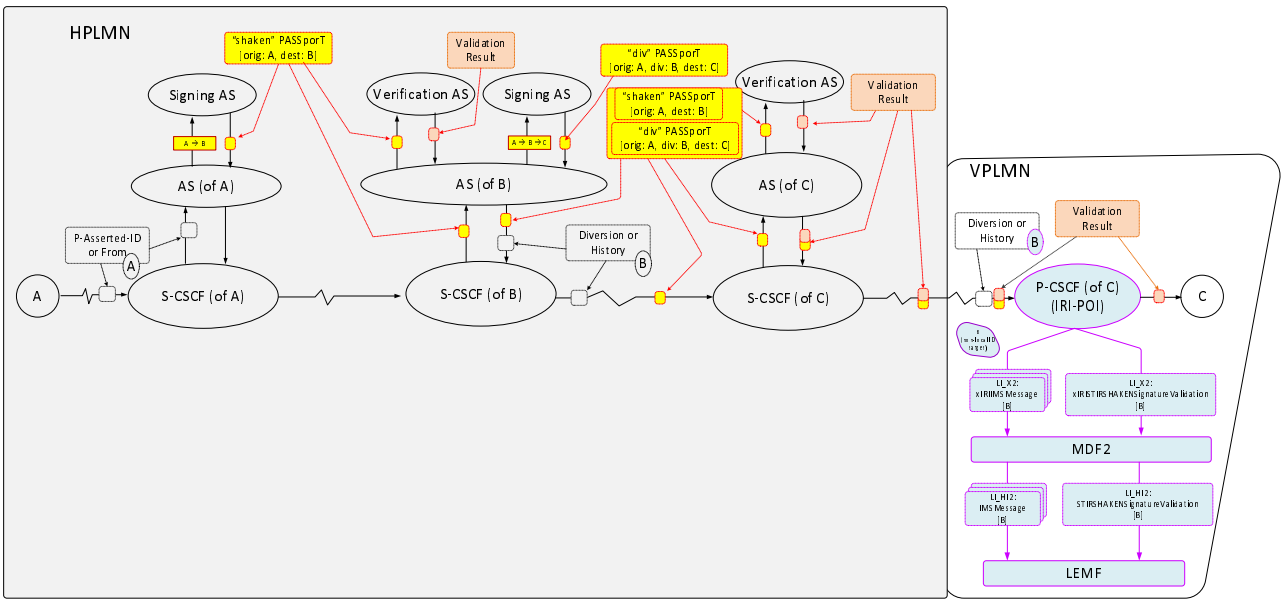


Figure 4.4.6.4-2: STIR/SHAKEN related LI reporting scenario 3 (VPLMN LBO, target non-local ID)

For target non-local ID (Party B)

For STIR/SHAKEN related LI reporting, the IRI-POI present in the P-CSCF (of C) in the VPLMN upon seeing the Validation Result in the SIP INVITE message generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

For IMS LI, the IRI-POI present in the P-CSCF (of C) also generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The IRI-POI present in the P-CSCF (of C) uses the Diversion or the History Info headers (pointing to target non-local ID Party B's identity) present in the SIP INVITE to perform a target match.

HR

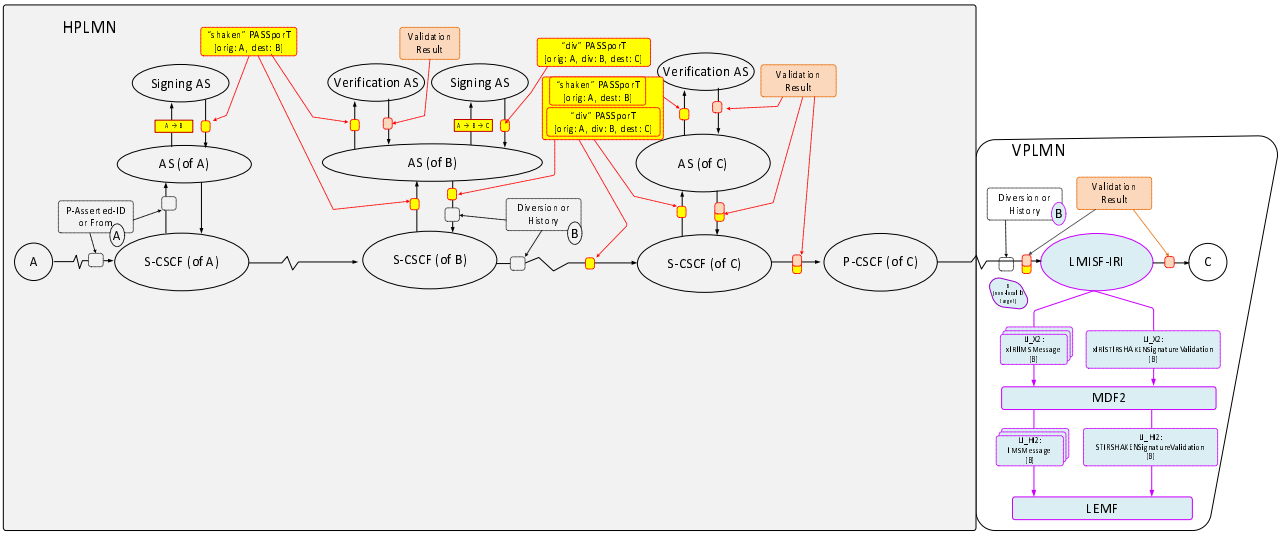


Figure 4.4.6.4-3: STIR/SHAKEN related LI reporting for scenario 3 (VPLMN HR, target non-local ID)

For target non-local ID (Party B)

For STIR/SHAKEN related LI reporting, the LMISF-IRI in the VPLMN upon seeing the Validation Result in the SIP INVITE message generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

For IMS LI, the LMISF-IRI also generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The LMISIF-IRI uses the Diversion or the History Info headers (which points to target non-local ID Party B's identity) present in the SIP INVITE to perform a target match.

4.4.6.5 Scenario 4: Party A calls Party B (target in VPLMN)

In this scenario, Party A calls Party B. Party A is in Originating Network and Party B is in VPLMN. Party B, an inbound roamer, is a target in the VPLMN.

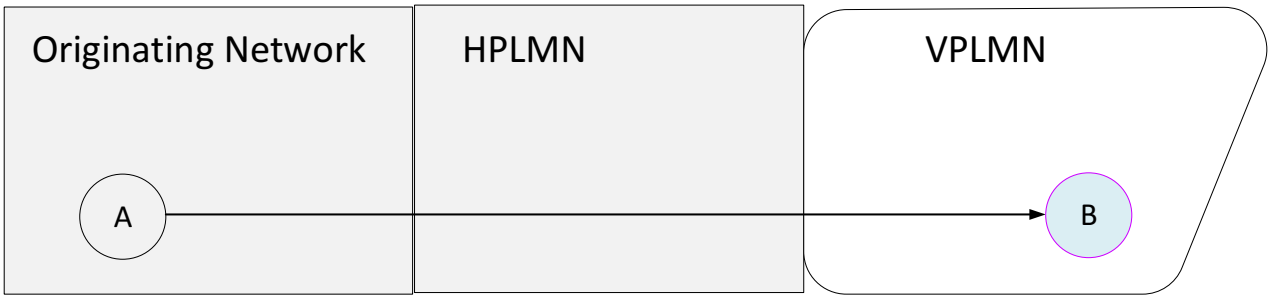


Figure 4.4.6.5-1: Party A calls Party B. Party B (inbound roamer) is target in VPLMN

Figure 4.4.6.5-2 illustrates the LI aspects in VPLMN with LBO roaming for target Party B and the figure 4.4.6.5-3 illustrates the LI aspects in VPLMN with HR roaming for target Party B.

In both illustrations, as can be seen, there is no interaction with the Verification AS in the VPLMN.

LBO

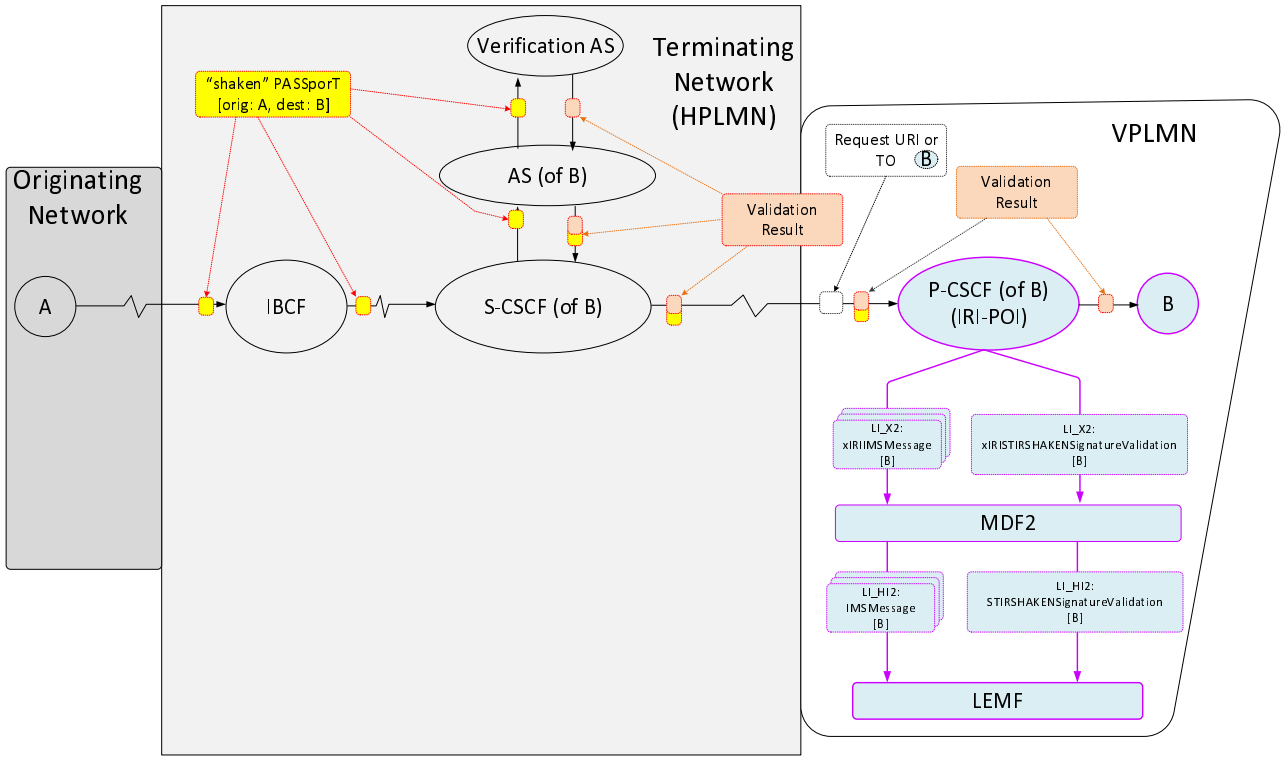


Figure 4.4.6.5-2: STIR/SHAKEN related LI reporting for scenario 4 (VPLMN LBO)

For target (Party B)

For STIR/SHAKEN related LI reporting, the IRI-POI present in the P-CSCF (of B) in the terminating VPLMN upon seeing the Validation Result in the SIP INVITE message generates and delivers the

xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

For IMS LI, the IRI-POI present in the P-CSCF (of B) also generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

NOTE: When the SIP INVITE carries either of the Diversion or the History Info, the IRI-POI present in the P-CSCF (of B) uses the Request URI header (pointing to target Party B's identity) present in the SIP INVITE to perform a target match.

In this illustration, the SIP INVITE carries neither the Diversion nor the History Info and therefore, the IRI-POI present in the P-CSCF (of B) uses the Request URI or the To headers (pointing to target Party B's identity) present in the SIP INVITE to perform a target match.

HR

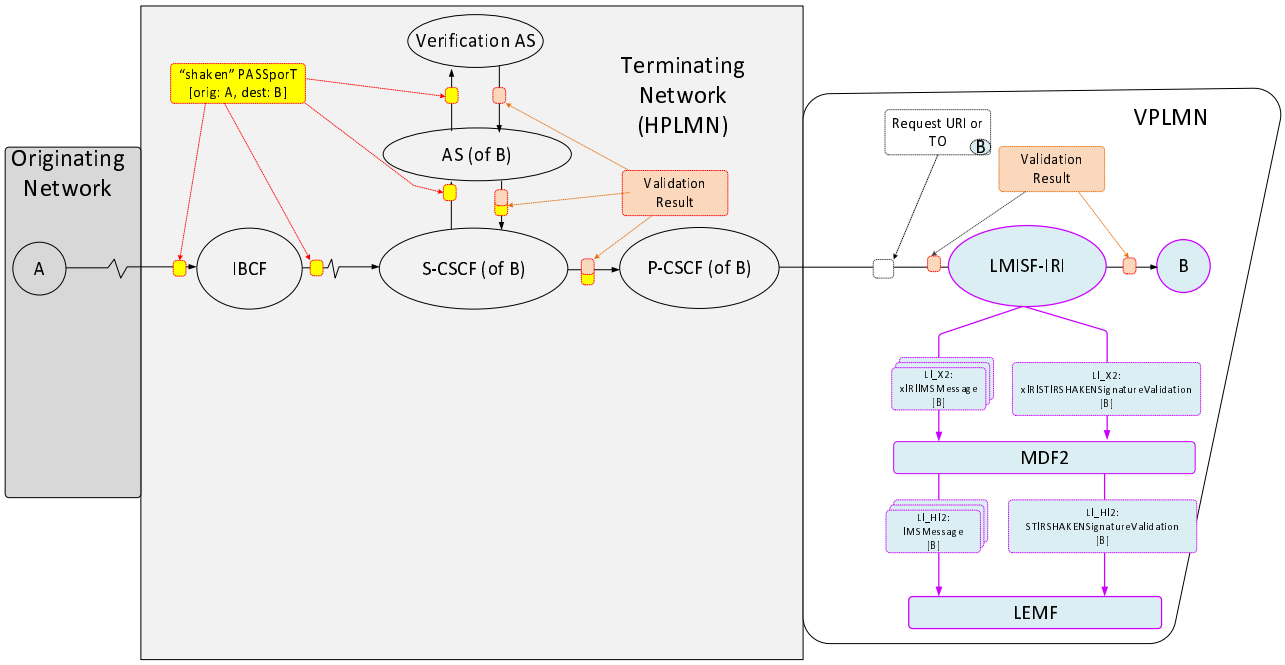


Figure 4.4.6.5-3: STIR/SHAKEN related LI reporting (VPLMN HR)

For target (Party B)

For STIR/SHAKEN related LI reporting, the LMISF-IRI in the terminating VPLMN upon seeing the Validation Result in the SIP INVITE message generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

For IMS LI, the LMISF-IRI also generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

NOTE: When the SIP INVITE carries either of the Diversion or History Info, the LMISF-IRI uses the Request URI header (pointing to target Party B's identity) present in the SIP INVITE to perform a target match.

In this illustration, the SIP INVITE carries neither the Diversion nor the History Info headers and therefore, the LMISF-IRI uses the Request URI or the To headers (pointing to target Party B's identity) present in the SIP INVITE to perform a target match.

4.4.6.6 Scenario 5: Party A (target non-local ID) calls Party B (VPLMN)

In this scenario, Party A calls Party B. Party A is in Originating Network and Party B is in VPLMN. Party A is a target non-local ID in the VPLMN.

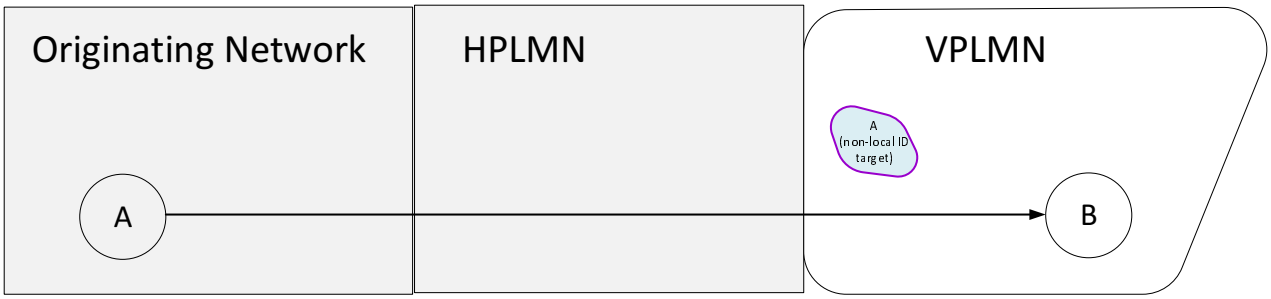


Figure 4.4.6.6-1: Party A calls Party B. Party A is a target non-local ID in VPLMN

Figure 4.4.6.6-2 illustrates the LI aspects in VPLMN with LBO roaming for target non-local ID Party A and the figure 4.4.6.6-3 illustrates the LI aspects in VPLMN with HR roaming for target non-local ID Party A.

In both illustrations, as can be seen, there is no interaction with the Verification AS in the VPLMN.

LBO

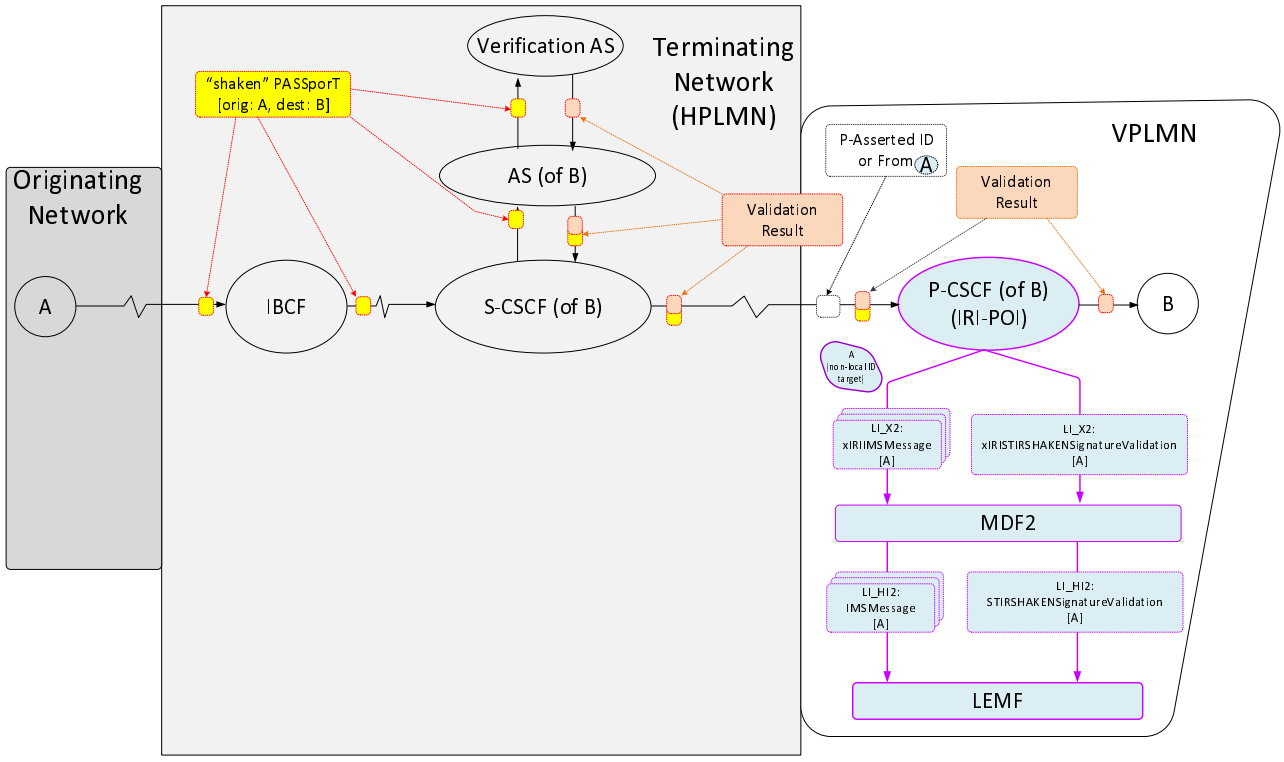


Figure 4.4.6.6-2: STIR/SHAKEN related LI reporting for scenario 5 (VPLMN LBO, target non-local ID)

For target non-local ID (Party A)

For STIR/SHAKEN related LI reporting, the IRI-POI present in the P-CSCF (of B) in the VPLMN upon seeing the Validation Result in the SIP INVITE message generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

For IMS LI, the IRI-POI present in the P-CSCF (of B) also generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The IRI-POI present in the P-CSCF (of B) uses the P-Asserted ID or the From Header (pointing to target non-local ID Party A's identity) present in the SIP INVITE to perform a target match.

HR

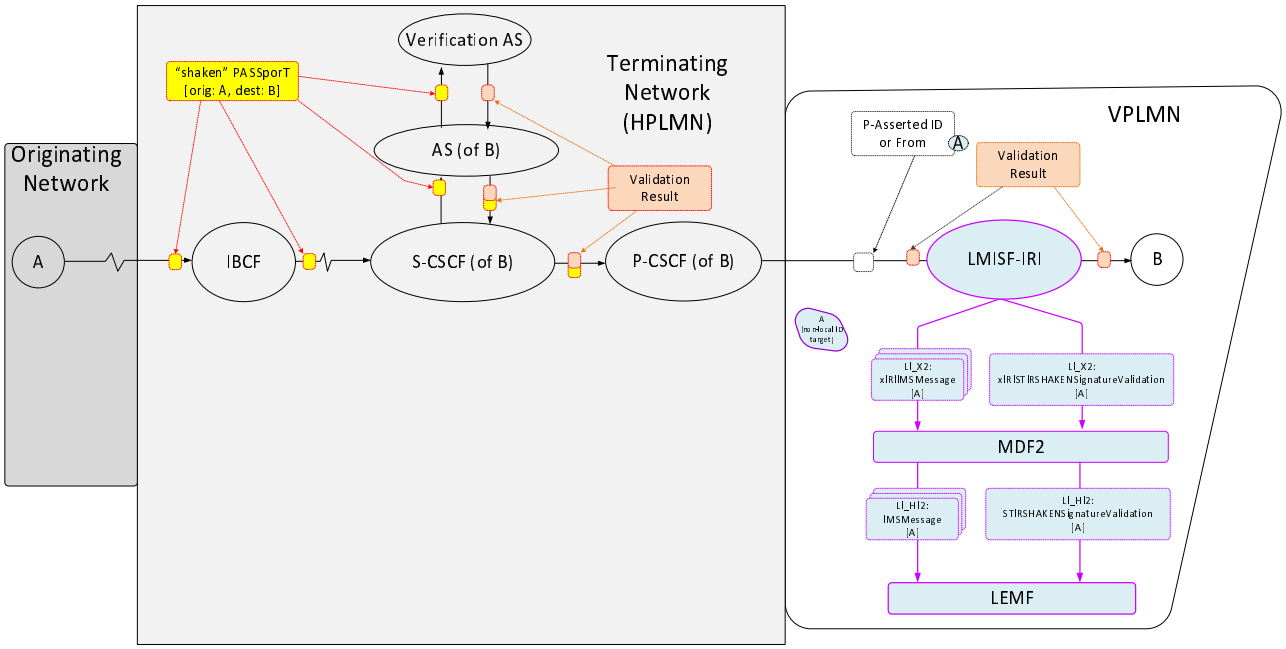


Figure 4.4.6.6-3: STIR/SHAKEN related LI reporting for scenario 5 (VPLMN HR, target non-local ID)

For target non-local ID (Party A)

For STIR/SHAKEN related LI reporting, the LMISF-IRI in the VPLMN upon seeing the Validation Result in the SIP INVITE message generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

For IMS LI, the LMISF-IRI also generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The LMISF-IRI uses the P-Asserted ID or the From headers (pointing to target non-local ID Party A's identity) present in the SIP INVITE to perform a target match.

4.4.6.7 Scenario 6: Party A calls Party B redirected to Party C (target in VPLMN)

In this scenario, Party A calls Party B redirected to Party C. Party A is in Originating Network and Party B is in HPLMN and Party C is in the VPLMN. Party C, an inbound roamer, is a target in the VPLMN.

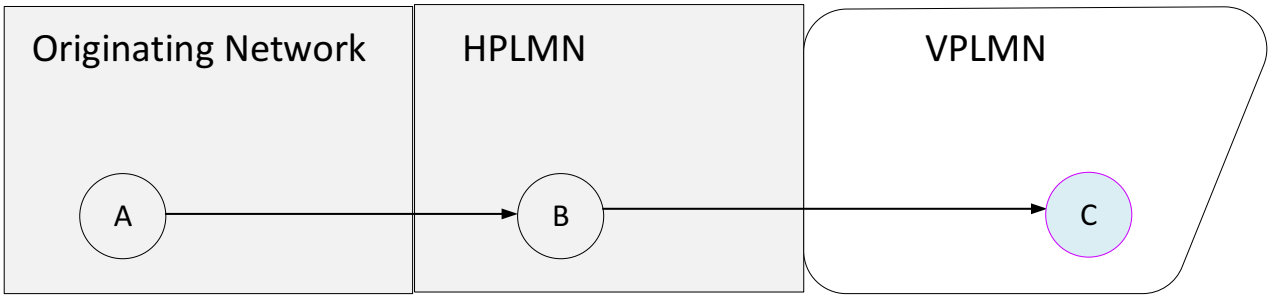


Figure 4.4.6.7-1: Party A calls Party B. Party B redirected to Party C (inbound roamer) is target in VPLMN

Figure 4.4.6.7-2 illustrates the LI aspects in VPLMN with LBO roaming for target Party C and the figure 4.4.6.7-3 illustrates the LI aspects in VPLMN with HR roaming for target Party C.

In both illustrations, as can be seen, there is no interaction with the Verification AS in the VPLMN.

LBO

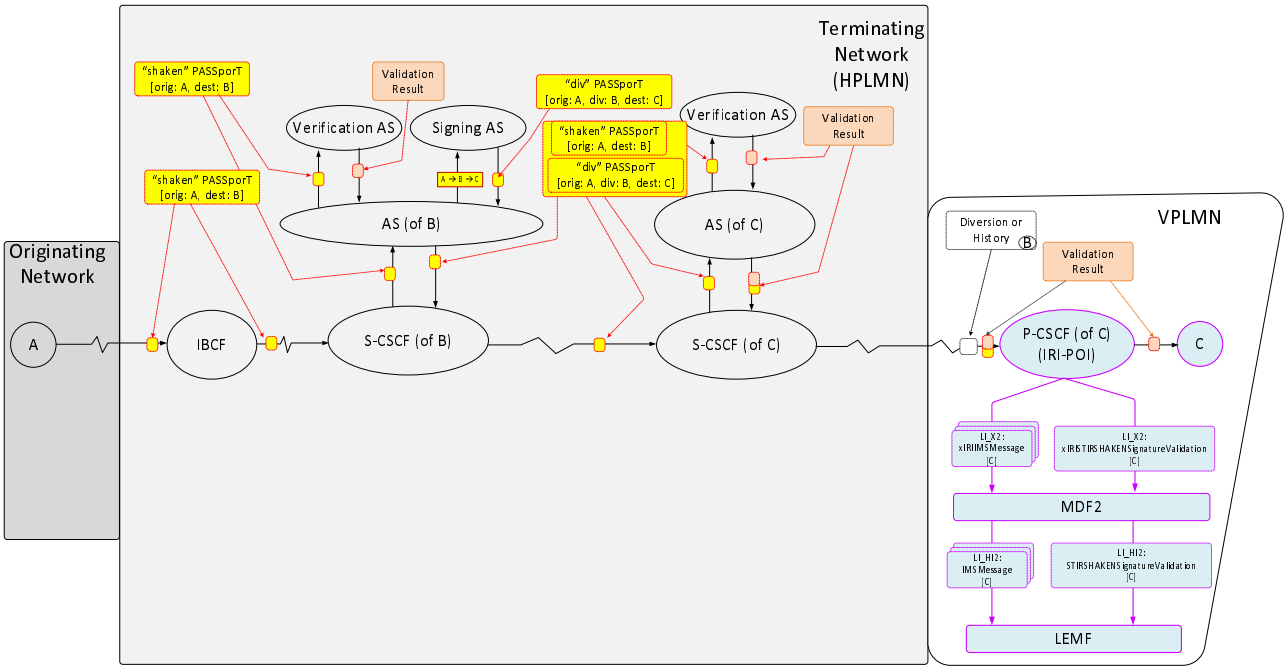


Figure 4.4.6.7-2: STIR/SHAKEN related LI reporting for scenario 6 (VPLMN LBO)

For target (Party C)

For STIR/SHAKEN related LI reporting, the IRI-POI present in the P-CSCF (of C) in the terminating VPLMN upon seeing the Validation Result in the SIP INVITE message generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

For IMS LI, the IRI-POI present in the P-CSCF (of C) also generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

NOTE: When the SIP INVITE carries either of the Diversion or History Info, the IRI-POI present in the P-CSCF (of C) uses the Request URI header (pointing to target Party C's identity) present in the SIP INVITE to perform a target match.

In this illustration, the SIP INVITE carries the Diversion or History Info headers and therefore, the IRI-POI present in the P-CSCF (of C) uses the Request URI header (pointing to target Party C's identity) present in the SIP INVITE to perform a target match.

HR

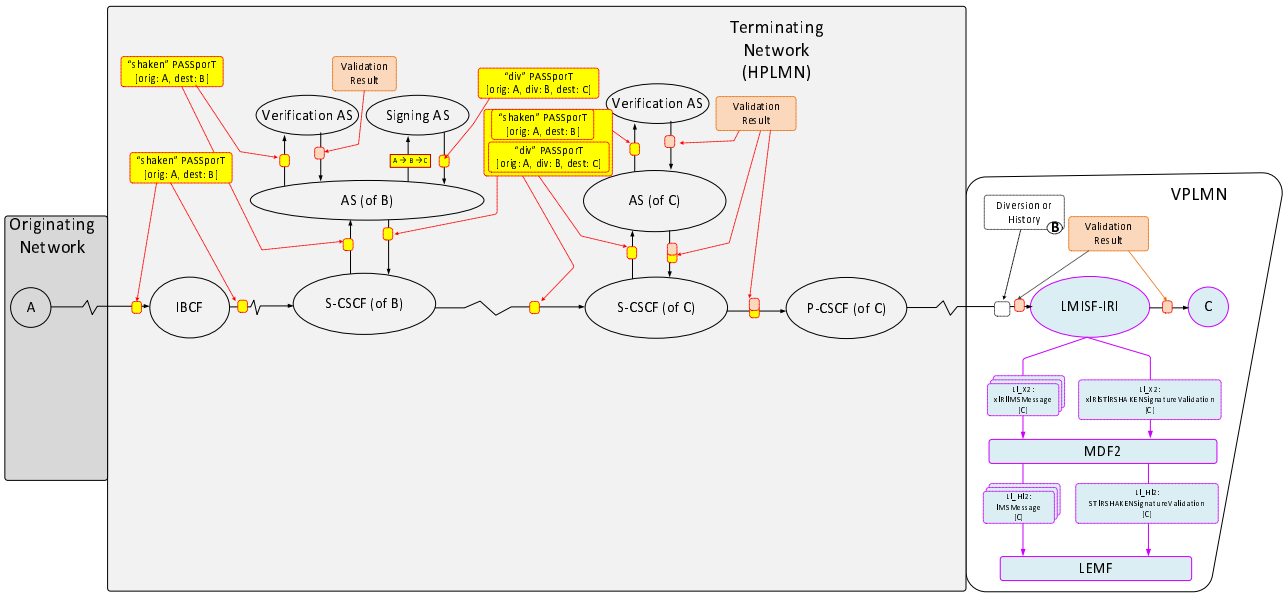


Figure 4.4.6.7-3: STIR/SHAKEN related LI reporting for scenario 6 (VPLMN HR)

For target (Party C)

For STIR/SHAKEN related LI reporting, the LMISF-IRI in the terminating VPLMN upon seeing the Validation Result in the SIP INVITE message generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

For IMS LI, the LMISF-IRI also generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

NOTE: When the SIP INVITE carries either of the Diversion or History Info, the LMISF-IRI uses the Request URI header (pointing to target Party C's identity) present in the SIP INVITE to perform a target match.

In this illustration, the SIP INVITE carries the Diversion or History Info headers and therefore, the LMISF-IRI uses the Request URI header (pointing to target Party C's identity) present in the SIP INVITE to perform a target match.

4.4.7 Emergency calls

4.4.7.1 General

The subsequent sub-clauses illustrate the STIR/SHAKEN related LI reporting along with the IMS LI for emergency calls and emergency call back scenarios.

For emergency calls, the CSP network (i.e. the originating network) signs/authenticates the calling party's identity and sends the "shaken" PASSporT with calling party's identity in the origination claim and the PSAP identity in the destination claim to the emergency services network.

For emergency call backs, the CSP network (i.e. the terminating network) that receives the "shaken" PASSporT would validate the received PASSporT before offering the call to the terminating party.

For emergency call and call back scenarios, the IMS NF that interacts with the Signing AS/Verification AS is the egress IBCF and the ingress IBCF, respectively.

4.4.7.2 Scenario 1: Party A (target) makes an emergency call

In this scenario, Party A calls the PSAP. Party A is in the originating network is a target.

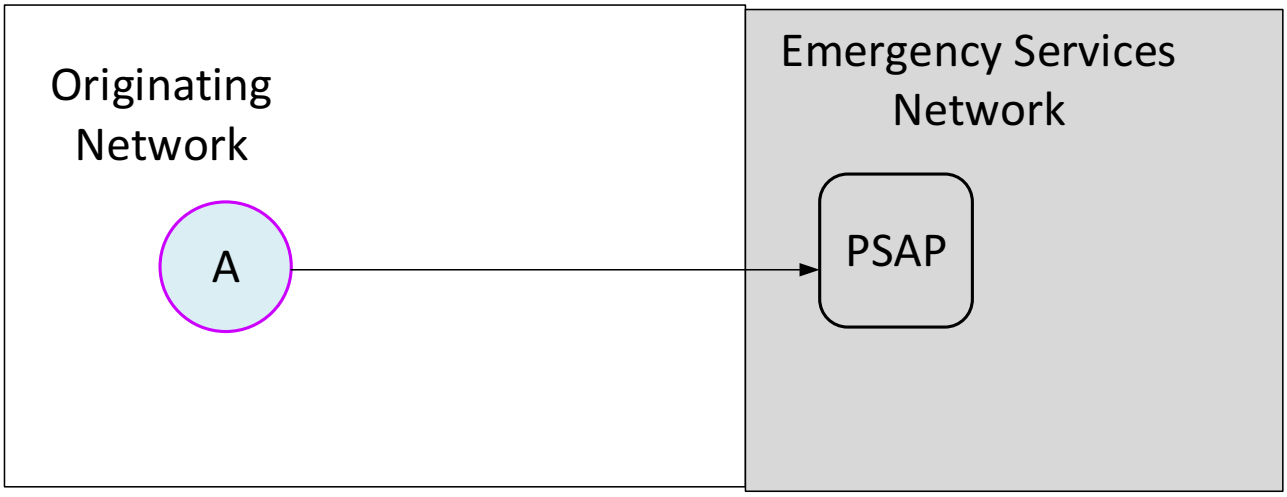


Figure 4.4.7.2-1: Party A (target) calls makes an emergency call

Figure 4.4.7.2-2 illustrates the LI aspects for this scenario.

The illustration shown in figure 4.4.5.11.2-2 assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the E-CSCF provides the IRI-POI functions for IMS LI for target Party A.

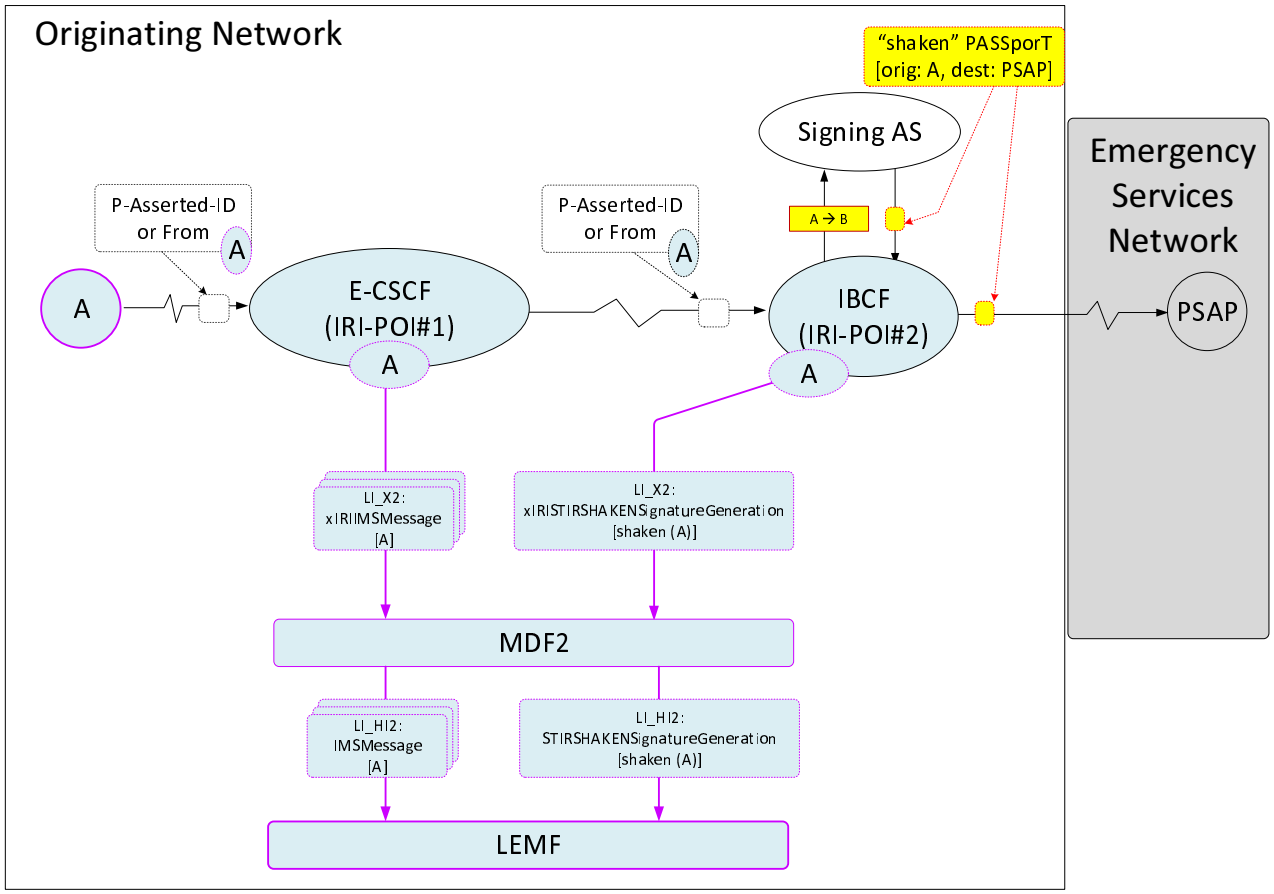


Figure 4.4.7.2-2: The LI aspects for an emergency call scenario (emergency call)

For target (Party A)

The IRI-POI present in the E-CSCF generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The egress IBCF interacts with the Signing AS to authenticate/sign A's identity and the IRI-POI present in that egress IBCF generates and delivers the xIRISTIRSHAKENSignatureGeneration record that includes the "shaken" PASSporT for target Party A to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureGeneration to the LEMF.

The IRI-POI present in the E-CSCF uses the P Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the egress IBCF uses the P Asserted ID or the From headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

4.4.7.3 Scenario 2: Emergency callback

In this scenario, the PSAP calls back the Party A who had made an emergency call earlier. Party A is in the terminating network is a target.

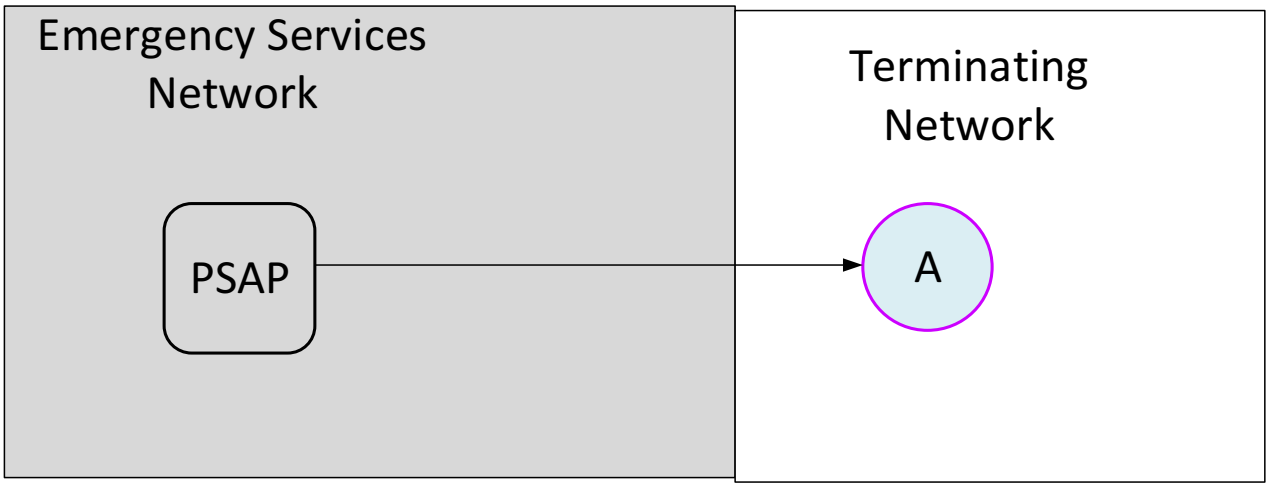


Figure 4.4.7.3-1: Party A (target) receives an emergency callback

Figure 4.4.7.3-2 illustrates the LI aspects for this scenario.

The illustration shown in figure 4.4.7.3-2 assumes the default option (see TS 33.127 [3]) for IMS LI. Therefore, the S-CSCF of A provides the IRI-POI functions for IMS LI for target Party A.

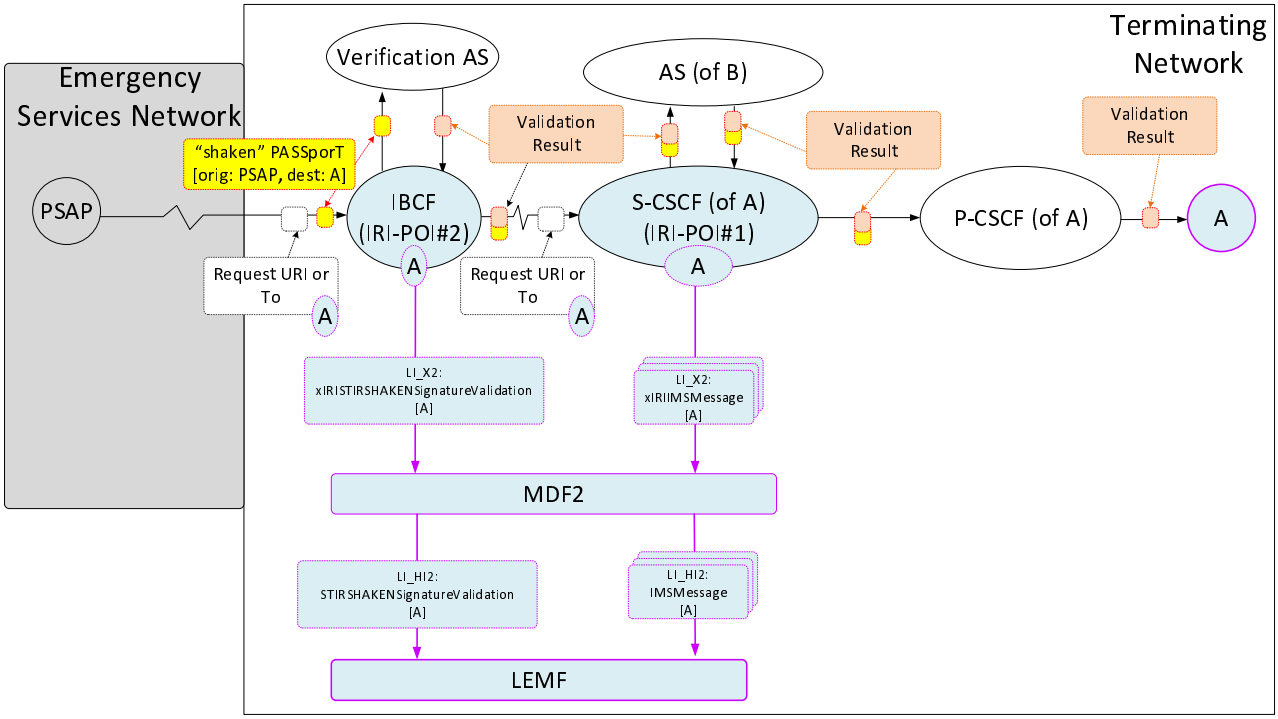


Figure 4.4.7.3-2: The LI aspects for scenario 2 (emergency call back)

For target (Party A)

The IRI-POI present in the S-CSCF (of A) generates and delivers the xIRIIMSMessage to the MDF2 and the MDF2 forwards it as an IRI message IMSMessage to the LEMF.

The ingress IBCF interacts with the Verification AS to verify the received "shaken" PASSporT and the IRI-POI present in the same ingress IBCF generates and delivers the xIRISTIRSHAKENSignatureValidation record to the MDF2. The MDF2 forwards it as an IRI message STIRSHAKENSignatureValidation to the LEMF.

The IRI-POI present in the S-CSCF (of A) uses the Request URI or the To headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

The IRI-POI present in the ingress IBCF uses the Request URI or the To headers (pointing to Party A) present in the incoming SIP INVITE to perform a target match.

Annex A:
Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2024-07	SA3#94-LI	S3i240505				Initial version	0.0.1
2026-07	SA3LI#102	S3i260357				Agreed for Change Control	0.0.2
2026-08	SA#113	SP-260728				Submitted to SA#113 Plenary for Approval	1.0.0
2026-09	SA#113	-				Approved by SA#113 Plenary for Change Control	19.0.0

History

Version	Date	Status
V19.0.0	October 2026	Publication